

TPO748-IN3x User Manual

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1 Introduction

1.1 Features

This document covers the TPO748_IN32 device(s).

This is a PLE transponder / aggregator that can be used for carrying mixes of 10-100 Gbps OTN and 10GE/100GE Ethernet traffic transparently over a packet-based network in accordance with the IETF PLE draft standard.

- Up to 16 x 10GE (XAUI) mapped to PLE
- Up to 16 x OTU2(HO) G.709 RS-FEC mapped to PLE
- Up to 8 x OTU2(LO) G.709 RS-FEC mapped to PLE
- Up to 16 x OTU2e G.709 RS-FEC mapped to PLE
- Up to 2 x OTU4(HO) (OTL4.4) mapped to PLE
- Up to 1 x OTU4(LO) (OTL4.4) mapped to PLE
- Up to 2 x 100GE (CAUI-4) mapped to PLE

Supported trunk protocols:

- 2 x 400GE (400GAUI-8) or 2 x 200GE (200GAUI-4) or 2 x 100GE (CAUI-4)
- Connection to 2 x QSFP-DD
- Optional per-client 1+1 protection

Other features

- Full OTN OH add/drop interface
- PCIe Gen3 x4 CPU interface

FPGA device

- Agilex I-series 041

1.2 Associated documentation

This document is the primary reference for the physical device and its functional modules, register interfaces etc. Associated documentation for included functionality is referenced in the Overview chapter. Further resources that are available for use in implementing systems using the TPO748_IN32 are:

- API release. Contains API code for use in configuration and monitoring the device, plus a command line debugger with register name interpretation for assistance in bringup and debug. This also integrates the header files and configuration data.

The intended audience for this document are those building hardware or developing API or management software for a system incorporating the TPO748_IN32 device.

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1.3 Programmers Model

1.3.1 Overview

This section describes how the device is documented from the programmer's point of view.

1.3.2 Access to internal registers

All accesses to the device from the CPU must be single 32-bit word accesses except for registers marked with "64-bit access only". The latter type of registers may only be accessed using 64-bit accesses from the CPU.

Any form of burst access is not supported.

Any side effects of a write (such as clearing an interrupt status bit) are guaranteed to be visible on the next read. Write posting can take place, but ordering of writes and reads will be maintained.

Register addresses specified in device documentation are physical addresses, which are the actual addresses used on the address bus for communicating with the device.

Where a core is instantiated in a device, registers in the documentation for that core are relative to the start address of each instantiation.

Where a module (containing registers) is instantiated multiple times as an array, the documented addresses of the registers within that module are relative to the start address of the module.

1.3.3 Access to undocumented addresses

Accesses to undocumented device addresses should never be performed, as they can have unpredictable side-effects that can severely affect system stability.

1.3.4 Reserved bits

Many registers contain reserved bits which are currently unused but which may be used in future versions of the device. The register documentation does not specifically mention reserved bits, so any bit within a register which is not included in a named field is to be considered reserved.

When reading from a register, the software must ignore the value of the reserved bits.

When writing to a register, all reserved bits must be written as zero, unless software has previously read the register, in which case the value previously read may be used for these bits if convenient.

By following these rules, software can achieve maximum compatibility with later versions of the device.

1.3.5 Bit ordering (endianness)

Bit, byte and word numbering is little endian, i.e. the first byte in a given 32-bit word is the 8 LSBs, bits [7..0].

1.3.6 Counters

Unless stated otherwise, all statistics counters wrap if they reach their maximum value, and do not saturate. Counter size is by design chosen that a counter cannot wrap within a period of 2 seconds under worst-case conditions.

Software that monitors the counters should therefore keep a copy of the last value read, and use the new value to calculate how many events have occurred since the last poll (if the new value is less than the previous, then it can be assumed that the counter has wrapped).

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By following the above, no events should ever be missed.

The counters need never be written to zero, although this can be done if desired. Writing to other values than zero is not supported.

Counters larger than 32 bits are presented for CPU access as multiple 32-bit words, with hardware support for atomic access. The LSBs are placed in the first (lowest address) 32-bit word, with the last word (highest address) holding the MSBs. Any unused bits in the last word will be the MSBs of this word. Reading and writing must follow the rules below:

- Reads: The 32-bit word at the lowest address is read first. This latches the values of all the other words comprising the field, which can then be read in sequence to give a single consistent read value for the whole field. Re-reading the lowest address will cause a new value to be latched.
- Writes: The 32-bit word at the highest address is written first. This is stored, as are subsequent writes to the other 32-bit words, until the 32-bit word at the lowest address is written. This will trigger the actual write of the complete value of the entire field.

Large counters are of type ROA (for read only) or RWA (if a write function is implemented).

1.3.6.1 PMTICK counters

In functional modules that include statistics counters, these counters are documented as being dual banked, i.e. as tables with two entries (the first for bank 0, the second for bank 1). One bank is active at any given time, meaning that events will increment the counters in the active bank only. The purpose of the dual bank counters is to support the PMTICK feature, where active bank swapping is controlled by a device pin, for applications that require this.

Dual bank support may not be available in a particular device. See the description of the `PMTICK_x` registers for further information, for devices with this support.

1.3.7 Large fields

Fields larger than 32 bits which are not classified as counters, are documented in a straightforward manner indicating the actual field size. They are placed onto 32-bit words following the same rules as everything else, i.e. bits [31:0] are placed in the first 32-bit word (lowest address), and the more significant bits in the next 32-bit word(s).

There is no mechanism to synchronize the values read from the individual 32-bit words (or 64-bit words in case of 64-bit access registers, see below). Software can avoid problems caused by race conditions on reads or writes by disabling the associated functions while updating or reading a multi-word field. The mechanism provided to do this may vary.

1.3.8 Registers with 64-bit access only

Registers marked with "64-bit access only" on the address line may only be accessed using 64-bit atomic access instructions from the CPU. Accessing such a register with 32-bit access instructions from the CPU will provide undefined results. Apart from the size of the access instructions used to read and write the words that make up the register, everything else regarding register functionality and documentation is unchanged compared to the normal registers which are using 32-bit access method.

1.3.9 Field types

The possible field access types are:

Table 1-1 Field types

Type	Description
RO	Read Only.

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Type	Description
ROA	Read Only Atomic. Used for large counters spanning multiple access words, see **Counters** on page 8.
RW	Read / Write.
RWA	Read / Write Atomic. Used for large counters that may be written to clear.
RWE	Read / Write Exclusive. Indicates that this field is the only one present in the register access word, and any reserved bits can be written without side-effects, both now and in the future (in order to guarantee backwards compatibility).
WO	Write only.
W1C	Write '1' to clear (on individual bits).

1.3.10 Register documentation examples

An example of documentation of a simple register is given below.

Name: REG_EXAMPLE
Description: Example register
Address: 0x0000.4028

Bits	Field	Type	Reset	Description
7:0	addr	RW	0x0	Example of an address.
10:8	ports	RW	0x0	Number of ports.
13	alarmForce	RW	0x0	Forces an alarm to be reported.
27:24	opMode	RW	0x0	Sets the operating mode: 0: norm. Normal mode. 1: leftMode. Left mode. 2: rightMode. Right mode.

The columns are:

- Bits: Lists the bits used for each field. For fields larger than 32 bits, see **Large fields** on page 9.
- Field: The name of the field.
- Type: The access type. See **Field types** on page 9 for all the possibilities.
- Reset: The value following a hardware reset of the device. If undefined, is shown as "-".
- Description: The usage of the field.

An example of an array (or table) of registers is given below:

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Name: ARRAY_EXAMPLE
Description: Packet configuration
Address: 0x0000.4080
Entries: 16
Entry size: 0x8
Index: Port number

Bits	Field	Type	Reset	Description
47:0	macAddr	RW	-	MAC address.
63:48	fieldData	RW	-	Updated every 47µs.

The `Entries` and `Entry Size` headings describe the number of entries in the array and the spacing between the array elements (in bytes) respectively. The `Index` heading explains what should be used as the index to the table. Index ranges start at 0.

When referring to a given index 'n' in the register array the following syntax is used: `IFTX_PKT_CFG[n]`.

Arrays may be more than one-dimensional. Below is an example of a two-dimensional array:

Name: REG_2D_EXAMPLE
Description: Example 2D register
Address: 0x0000.4800
Entries: 16, 32
Entry size: 0x80, 0x4
Index: Channel number, port number

Bits	Field	Type	Reset	Description
7:0	data	RW	-	Field in 2D register.

In this case the array is two dimensional:

- The inner dimension is an array of 32 entries each of size 0x4, indexed by the "port number".
- The outer dimension is an array of 16 entries, each of size 0x80, indexed by the "channel number".

More dimensions are possible. When referring to a given outer index 'n' and inner index 'm' in the register array the following syntax is used: `IFTX_2D_CFG[n][m]`.

In the example, the address offset of entry would be calculated as:

```
offset = (n * 0x80) + (m * 0x4)
```

Note: In this case, the size of the inner table rows is the same as the offset between each outer entry (32 × 0x4), but this need not always be the case.

1.3.11 Alarm monitoring registers

Alarms are presented to the system software in sets of registers, with the following naming convention:

- `BLOCK_FUNCTION_STAT.alarm`
- `BLOCK_FUNCTION_HOLD.alarm`
- `BLOCK_FUNCTION_CHLD.alarm`

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- BLOCK_FUNCTION_INTEN.alarm
- BLOCK_FUNCTION_INTSUM

The _STAT register holds the current value of the alarm signal. This may (depending on the nature of the alarm) be unstable, and should generally only be used for ad-hoc monitoring of the alarm state. It is always read-only. Note that in cases where it is not relevant, the _STAT register is sometimes omitted. This is always the case if the event in question has zero duration, for example detection of a CRC error, but can also be true if the state is random or noisy, for example a PLL out-of-lock state.

The _HOLD register contains latching bits that are set to '1' if the (filtered if relevant) alarm becomes active, even briefly. These can be read periodically to provide a stable alarm state for the system. They are of the type W1C, and should be written with a '1' to clear an alarm only if that alarm has been seen active, to avoid ever missing an alarm event. Following a clear, there may be a delay of up to 500 µs before the bit is set again even if the alarm is constantly active.

The _CHLD register contains latching bits that are set to '1' if the (filtered if relevant) alarm value changes, even briefly. They are only used for alarms for which a _STAT register is present. They are also W1C.

The _CHLD bits are the source of interrupts when present, when they are not present the _HOLD bits are the interrupt sources.

The _INTEN register (which will only be present if an interrupt is supported, and may not be present if only a single interrupt is supported) contains interrupt enable bits for each bit in the _HOLD or _CHLD register. Bits are set to '1' to enable the interrupt.

An interrupt enable bit can be cleared to disable further interrupts, once an alarm has been seen and acted upon. When enabling the interrupt again, care should be taken to avoid race conditions with interrupt routines that might simultaneously be clearing other bits in the same register. Read-modify-writes to the _INTEN register should in other words be guaranteed atomic.

The optional _INTSUM register(s) contain a bitwise summary of the active, enabled interrupts, to enable software to rapidly identify which locations need to be examined in greater detail once an interrupt has been received. See the individual register documentation for details, as these may vary.

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2 Overview

2.1 Block diagram

The TPO748_IN32 block diagram is shown below.

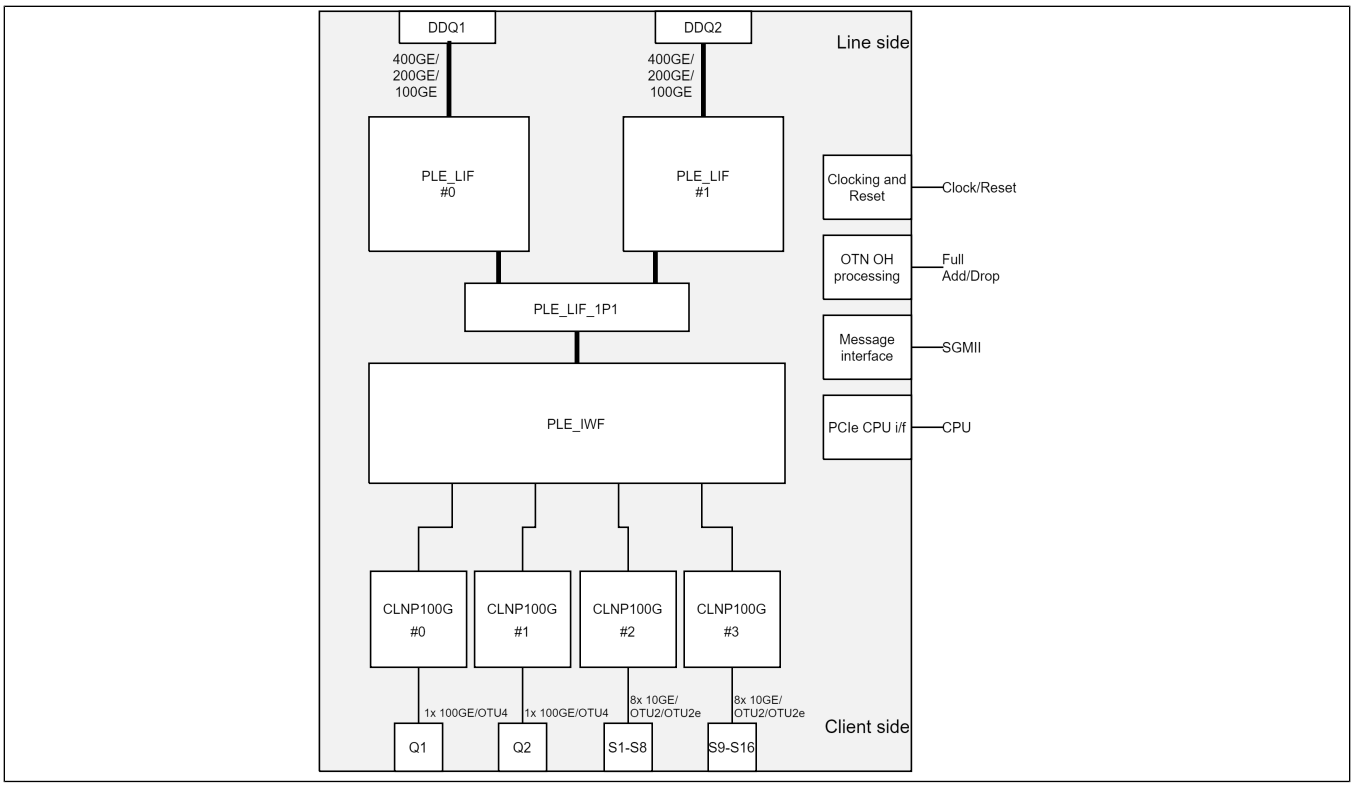


Figure 2-1 Block diagram

2.2 Functional description

The TPO748_IN32 maps OTN and Ethernet clients between client and line interfaces as shown in the block diagram above.

The interfaces are designated according to the intended optical pluggable, i.e.:

- DDQn - 8 lanes towards a QSFP-DD, operating at either 56Gbit/s PAM4 or 25Gbit/s NRZ
- Qn - 4 lanes towards a QSFP28, operating at 25/10 Gbit/s NRZ
- Sn - 1 lane towards an SFP, operating at 10Gbit/s NRZ

The client-side interfaces are connected to the client monitoring, processing and multiplexing functions, which are implemented in CLNP100G blocks.

See the following documents for details of the contents of the CLNP100G and its sub-blocks:

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- TD0494, CLNP100G User Manual (covers 100G clients and instantiation of multiple 10G clients in CL1P16G block)
- TD0384, CL1P16G User Manual (covers 10G clients)
- TD0376, OTN1PXG OTN Interface User Manual (100G and 10G OTN)

On the line side, there are two PLE_LIF blocks supporting 400GE/200GE/100GE, connected to DDQ1/2. The PLE_LIF_1P1 provides 1+1 protection per PLE client between the two line interfaces. The PLE_IWF packetizes client data from the 4 x CLNP100G devices into a single PLE packet stream. From the line interfaces, the PLE_IWF regenerates the client timing from timestamps in the PLE packets and forwards the client data to the 4 x CLNP100G devices.

2.2.1 CLNP100G client mapping blocks - TODO

The CLNP100G configuration options in this device are illustrated below.

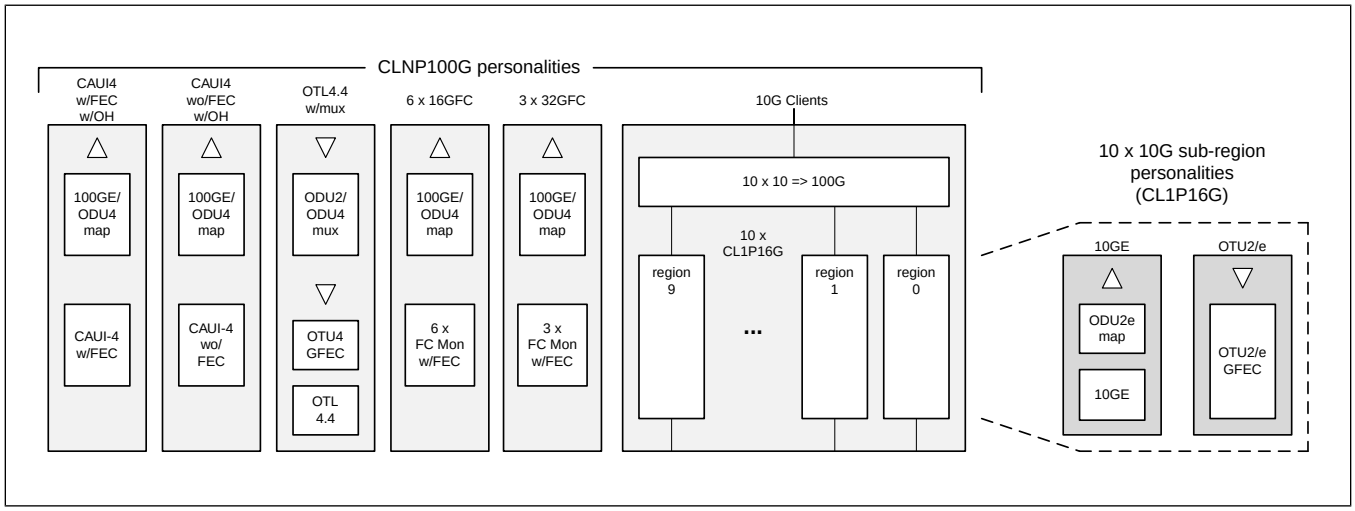


Figure 2-2 CLNP100G

Each CLNP100G block can be configured with the following personalities:

- PR4: CAUI4 w/FEC w/OH: Ethernet mapping with CAUI-4 client processing, including RS(528,514) FEC, ingress/egress MAC statistics and OTN OH
- PR16: CAUI4 wo/FEC w/OH: Ethernet mapping with CAUI-4 client processing (no FEC), ingress/egress MAC statistics and OTN OH
- PR7: OTL4.4 w/mux: OTL4.4 termination, optional ODU4/ODU2 multiplexing
- PR6: 8x10G Clients: clients mapped to ODU2/2e implemented in CL1P16G blocks

Each of the 10G client regions in a CLNP100G region in 8x10G (PR6) mode can be configured as:

- PR0: 10G Ethernet mapped to ODU2e
- PR1: 10G OTN (OTU2 or OTU2e) with GFEC termination

The 8 x ODU2(e) from the client regions are multiplexed together (fixed configuration) as a 100G datapath towards the PLE_IWF.

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2.2.2 Port connectivity

Table 2-1 Port connectivity TPO748-IN31

CLNP100G region	Ports
0	Q1 (100G)
1	Q2 (100G)
2	Q3 (100G)

Table 2-2 Port connectivity TPO748-IN32

CLNP100G region	Ports
0	Q1 (100G)
1	Q2 (100G)
2	S[1..8] (8x 10G)
3	S[9..16] (8x 10G)

Table 2-3 Port connectivity TPO748-IN33

CLNP100G region	Ports
0	Q2 (4x 10G), Q3 (4x 10G)
1	S[1..8] (8x 10G)
2	S[9..16] (8x 10G)

2.2.3 Supported protocols

The following table lists the client and line protocols supported by the device.

Table 2-4 Supported protocols

Protocol	Physical layer	Notes
100GE	CAUI-4	Line and client sides.
200GE	200GAUI-4	
400GE	400GAUI-8	
OTU4	OTL4.4	
10GE	XAUI	
OTU2	OTU2	
OTU2e	OTU2e	

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2.2.4 API / faceplate interface numbering

The mapping between interface designators, API port numbers and faceplate port numbers for the UTM2C card are given below. The mapping scheme between chip interface numbers and API numbering follows a scheme which is consistent across all devices in the TPO748_IN32 family:

- All API numbering is 1-based and follows the chip numbering order.
- QSFPs are in the range 1-39, SFPs are in the range 41-99, DDQ are in the range 101-199.
- Where a single interface can be divided up into multiple ports, the numbering is incremental from the interface base. As an example, a QSFP when used as 1 x 100GE has port number #1, and when used as 4 x 10GE these ports are numbered 1, 2, 3 and 4.

It should therefore be possible to convert between chip and API numbering with a simple formula.

Table 2-5 Interface, API and faceplate numbering (UTM2C)

Interface	API port#	Faceplate#	Notes
DDQ1	101	TBD	Lanes in reverse order
DDQ2	111	TBD	Lanes in reverse order
Q1	1	TBD	Lanes in reverse order
Q2	11	TBD	Lanes in reverse order
Q3	21-24	TBD	Lanes in reverse order
Q4	31-34	TBD	Lanes in reverse order
S1-S8	41-48	TBD	
S9-S16	51-58	TBD	

2.2.5 Clocking

See **Clocking and reset** on page 492 for details of the device clocking pins.

The internal core logic of the device is clocked from the G_CLK pins, which must be stable at all times from before device reset is de-asserted.

All OTN and client timing is derived from the G_CLK3 input, so this clock should be sourced from an origin with suitable long term stability and wander performance.

2.2.6 OTN overhead

OTN overhead is terminated at various places withing the device as described in the documentation for the individual blocks.

See **OTN Overhead** on page 395 for details of the OTN OH processing functions.

For details of how to address individual OTUk and ODUk termination points see **OTN overhead indexing** on page 389.

All GCC bytes for any of these ODU levels can be inserted and/or extracted via the OTN GCC Add/Drop interface. See for details of the interface protocol.

3 Technology Functions

3.1 Features

- PCIe CPU interface
- Interrupt signalling to CPU for fast software processing of events
- Chip information and control
- PLL monitoring and control
- SerDes monitoring and control

3.2 Overview

The CPU interface provides the CPU with read and write access to the programmable resources in the various functional blocks inside the device. A single combined interrupt output is provided for signalling certain events to the controlling CPU in order to reduce the response latency of the software.

These CPU register access and interrupt routing functions are shown below:

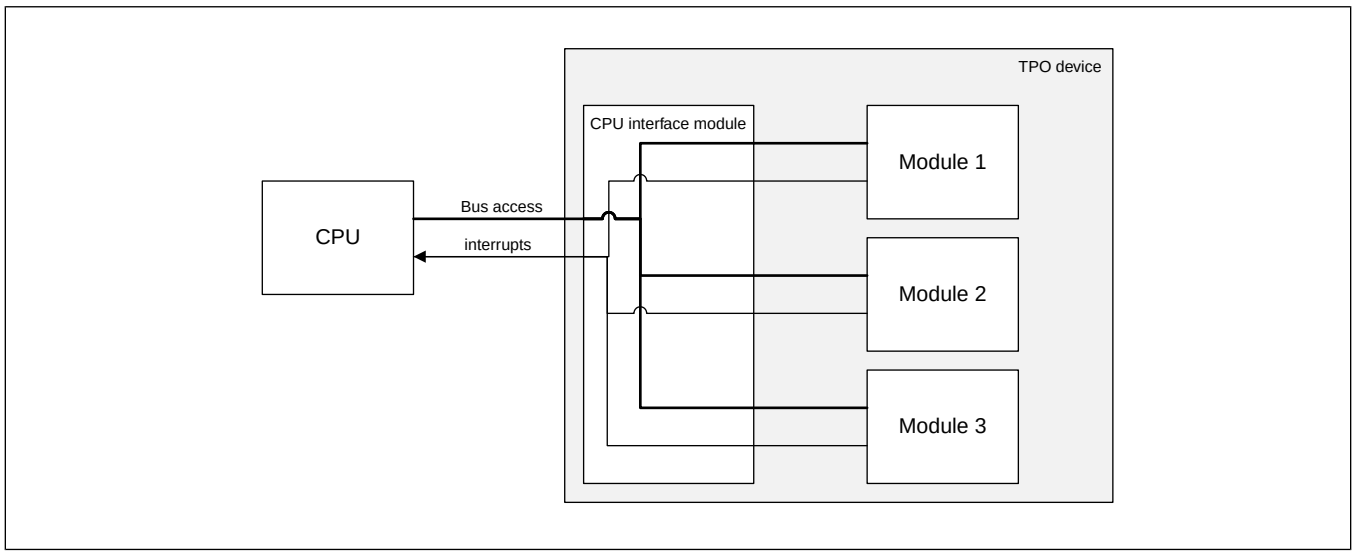


Figure 3-1 CPU register access and interrupt routing

The interrupt line shown above does not physically exist. The interrupt is carried across the PCI-Express interface as INTA or MSI interrupt messages.

A number of registers for monitoring and controlling the chip-level technology components such as PLLs and SerDes interfaces are provided, which are subsequently described in this chapter.

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3.3 Operation

3.3.1 Chip information and control

The CHIP_VER register holds a chip identifier field as well as major and minor version numbers.

3.3.2 PLL monitoring and control

The device has a number of PLLs, which are designed to lock to incoming clocks and provide a phase-locked output. A number of registers are provided to monitor and control these PLLs. If the PLLs lose lock, an auto-reset circuit applies a reset signal to the PLL counters to ensure that the output phase is maintained correctly. The register PLL_OOL_HOLD allow this feature to be monitored. During normal operation, software should not access or use these registers. They are only intended to be used for system debugging.

3.3.3 TODO - Recovered clocks

The device has two output pins, IF_REC_CLK[1:0], which can each be configured to output one of the internal clocks within the device.

Some of the clocks are fixed, others are based on the recovered clocks for signals arriving at the device ports. For the recovered clocks, the output clock frequency on IF_REC_CLK is given by LINE_RATE / 64 / 4.

The source clocks for these output pins are selected by the ifRecClkSel fields in CHIP_CFG. See the table below for details.

Table 3-1 IF_REC_CLK source selection

Value	Clock(s)	Notes
0	Core clock 357MHz	Fixed.
1	Core clock client 413MHz	Fixed.
2	Core clock line 439MHz	Fixed.
3	Core clock DTA 311.04MHz	Fixed.
32-35	q0_rec_rx_clks	
36-39	q0_rec_tx_clks	
40-43	q1_rec_rx_clks	
44-47	q1_rec_tx_clks	
48-51	q2_rec_rx_clks	
52-55	q2_rec_tx_clks	
56-59	q3_rec_rx_clks	
60-63	q3_rec_tx_clks	
64-67	s0_3_rec_rx_clks	
68-71	s0_3_rec_tx_clks	
72-75	s4_7_rec_rx_clks	
76-79	s4_7_rec_tx_clks	

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Value	Clock(s)	Notes
80-83	s8_11_rec_rx_clks	
84-87	s8_11_rec_tx_clks	
88-91	s12_15_rec_rx_clks	
92-95	s12_15_rec_tx_clks	
96-103	ddq0_rec_rx_clks	
104-111	ddq0_rec_tx_clks	
112-119	ddq1_rec_rx_clks	
120-127	ddq1_rec_tx_clks	

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3.3.4 SerDes monitoring and control

An overview of the SerDes present in the physical FPGA device can be obtained by reading the SERDES_INFO register. This details each tile position and type.

Configuration and monitoring of the individual SerDes is a two-step process:

- The appropriate SerDes tile, group and lane are addressed via SERDES_SEL
- The other registers documented here are used to perform the actual reads and writes

Refer to the Agilex User Guide and the guides for the transceivers for more details.

3.3.5 Agilex Mailbox Client

The Agilex Mailbox Client Registers are based on the unpublished version user guide UG-20087.

The following tables describe the available fields in the relevent Mailbox Client registers.

Table 3-2 Descriptions for cmd Field

Field	Bits	Description
noop	0	Sends an OK status response.
getIdcode	16	The response contains one argument which is the JTAG IDCODE for the device.
getChipId	18	The response contains 64-bit CHIPID value with the least significant word first.
getUserc	19	The response contains one argument which is the 32-bit JTAG USERCODE provided to the device by the configuration bitstream.
getVoltage	24	Command has a single argument which is the bitmask of which channels to read. Bit 0 is set to read channel 0, bit 1 to read channel 1 and so on. The response contains one word argument for each bit set in the bitmask. Each return value is 32-bit value. The voltage is an unsigned fixed point number with 16 bits below the binary point.
getTemp	25	Command has a single argument which is a bitmask indicating which temperature sensors to read. • Bits[31:28]: Reserved. • Bits[27:16]: Sensor Location. Specifies the TSD location. • Bits[15:0]: Sensor mask. Specifies the sensors to read for the sensor location specified. The response contains one word for each temperature requested. If omitted, the command reads channel 0. The least significant bit (lsb) corresponds to sensor 0. The most significant bit (msb) corresponds to channel 15. The response contains one word for each channel(3) temperature requested. The temperature returned as a signed fixed value with 8 bits below the binary point. Example: A Temperature of 10°C returns 0x00000A00 and a of temperature -1.5°C returns 0xFFFFFE80.
qspiOpen	50	Requests exclusive access to the quad-serial peripheral interface (QSPI). If the SDM accepts the request (if the QSPI is not already in use or the SDM is not in the process of a device configuration), it returns the OK response, else it returns the error response. Note: The exclusive access is granted only to the client using this mailbox. Other clients are not be able to access the QSPI until it is closed by this client.
qspiClose	51	Closes the exclusive access to the QSPI interface.

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Field	Bits	Description
qspiSetCs	52	Select which of the attached QSPI device via the chip select lines. Takes on one word argument as described below: • Bit [31:28]: Flash device to be selected. Bit setting 0000 is used to select flash that attached to nCSO[0]. • Bit [27:0]: Reserved (write as 0).
qspiRDevR	53	Reads registers from the attached QSPI device and takes two parameters. • The opcode for the read command. • The number of bytes to read (the maximum size is 8 bytes). A successful read returns an OK response code followed by the data read from the device. If the data is not an exact multiple of 4 bytes then it is padded with 0 bytes until the next word boundary.
qspiWrDevR	54	Writes to registers on the attached QSPI and takes three arguments: • The opcode for the write command. • The number of bytes to write (the maximum size is 8 bytes). • The data to write (maximum 2 words, padded with 0 to word boundary). A successful write returns an OK response code.
qspiSendOp	55	Sends a command opcode to the QSPI and takes one argument: • The opcode to send the attached QSPI device. A successful command returns an OK response code.
qspiErase	56	Erases sector on the attached QSPI device and takes two parameters: • The flash address offset within the device to start erasing from (one word). • The number of bytes to erase. The erase size is the multiple of 64K bytes. A successful erase returns an OK response code.
qspiWrite	57	Writes data on the attached QSPI device and takes three parameters: • The flash address offset from where you want the command to start writing to the QSPI device (one word). • The number of words to write (one word). • The data to be written (one or more words). A successful write returns an OK response code. The client may need to issue QSPI_ERASE command before issuing this command to prepare the memory for writing. Note: The maximum transfer size is limited to 4K bytes and cannot be called while a configuration is in progress.

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Field	Bits	Description
qspiRead	58	Reads the attached QSPI device and takes two parameters: • The flash address offset from where you want to start reading from the QSPI device (one word). • Number of words to read (one word). A successful response returns an OK response code followed by the data read from the QSPI device. A failure response is either: • Returns an error code. • Returns OK but partial of the data read from QSPI device is incorrect. Note: The maximum transfer size is limited to 4K bytes and cannot be called while a configuration is in progress.

Table 3-3 Descriptions for err Field

Field	Bits	Description
ok	0	Indicates that the command completed successfully. This does not mean that everything is necessarily ok, for example a pin to read status might return OK if it has successfully read the status. Further examination of the response arguments might be needed to see the error.
invalid	1	Indicates that the command is in an incorrect format.
unknownBr	2	Indicates that the command code is not understood. The device is running in the boot rom, load firmware for advanced command support.
unknown	3	Indicates that the command code is not understood by the currently loaded firmware.
notConf	256	Indicates that the device is not configured.
respBusy	511	Indicates that the device is busy.
respNoVldR	767	Indicates that there is no valid response available.
respErr	1023	General Error.

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3.4 Registers

Name: SEU_HOLD
Description: SEU status change
Address: 0x0200.0400

Bits	Field	Type	Reset	Description
0	dataValid	W1C	0x0	Valid data in the SEU cache.
1	sysErr	W1C	0x0	System error when retrieving SEU error. Requires CRAM reloading if asserted.

Name: SEU_STAT
Description: SEU status
Address: 0x0200.0404

Bits	Field	Type	Reset	Description
0	dataValid	RO	-	Valid data in the SEU cache.
1	sysErr	RO	-	System error when retrieving SEU error. Requires CRAM reloading if asserted.

Name: SEU_INTEN
Description: SEU interrupt enable
Address: 0x0200.0408

Bits	Field	Type	Reset	Description
0	dataValid	RW	-	Valid data in the SEU cache.
1	sysErr	RW	-	System error when retrieving SEU error. Requires CRAM reloading if asserted.

Name: SEU_DATA
Description: SEU Data
Address: 0x0200.0410

Bits	Field	Type	Reset	Description
63:0	data	ROA	-	SEU Data.

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3.4.1 Chip information and Controls registers

Name: CHIP_VER
Description: Chip ID and version
Address: 0x0200.0000

Bits	Field	Type	Reset	Description
7:0	buildId	RO	-	Internal manufacturer use only.
15:8	minVer	RO	-	Minor version number.
23:16	majVer	RO	-	Major version number.
31:24	id	RO	0x9c	Chip identifier.

Name: CHIP_STAT
Description: Chip status
Address: 0x0200.0004

Bits	Field	Type	Reset	Description
0	prPresent	RO	-	PR present in BASE image.
1	drEnabled	RO	-	DR enabled on XCVRs.
2	skeletonMode	RO	-	If true, no core exists in image.
3	pmTick	RO	-	Shows value of G_PMTICK input.
7:4	platform	RO	-	Target Platform. 0: platFm87A0. FM87 with A0 silicon. 1: platFm87B0. FM87 with B0 silicon. 2: platUtm2c. UTM2C. 3: platFm87B2. FM87 with B2 silicon. 4: platFm91. FM91.
8	devClksPgmd	RO	-	Dev board (FM87/FM91) clocks have been programmed by the Max10. Only relevant for FM87/FM91 board.
9	devClksErr	RO	-	Max10 reported an error while programming the FM87/FM91 clocks. Only relevant for FM87/FM91 board.

Name: CHIP_CFG
Description: Chip configuration
Address: 0x0200.0008

Bits	Field	Type	Reset	Description
63:32	scratch	RW	-	Scratchpad register.

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Name: **PR_FREEZE**
Description: **PR freeze control**
Address: **0x0200.0018**

Bits	Field	Type	Reset	Description
4:0	region	RW	0x0	Set to freeze PR region.

Name: **INTR_ENABLE**
Description: **Global interrupt enable**
Address: **0x0200.0020**

Bits	Field	Type	Reset	Description
0	mbClient	RW	0x0	Set to enable interrupt from Mailbox Client.
1	serdesMc	RW	0x0	Set to enable interrupt from Serdes MC.
2	prCtrl	RW	0x0	Set to enable interrupt from PR Controller.
3	seu	RW	0x0	Set to enable interrupt from SEU Controller.

Name: **TXDIS_SFP_CFG**
Description: **Debug Bringup SFP TXDIS Pin control. Only valid when CHIP_STAT.skeletonMode set**
Address: **0x0200.2000**

Bits	Field	Type	Reset	Description
31:0	txDis	RW	-	TX Disable per port.

Name: **TXDIS_QSFP_CFG**
Description: **Debug Bringup QSFP TXDIS Pin control. Only valid when CHIP_STAT.skeletonMode set**
Address: **0x0200.2008**

Bits	Field	Type	Reset	Description
31:0	txDis	RW	-	TX Disable per port.

Name: **TXDIS_DDQ_CFG**
Description: **Debug Bringup DDQ TXDIS Pin control. Only valid when CHIP_STAT.skeletonMode set**
Address: **0x0200.2020**

Bits	Field	Type	Reset	Description
31:0	txDis	RW	-	TX Disable per port.

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Name: **LOS_SFP_STAT**
Description: **Debug Bringup SFP LOS Pin monitor. Only valid when CHIP_STAT.skeletonMode set**
Address: **0x0200.2100**

Bits	Field	Type	Reset	Description
31:0	los	RO	-	Los Of Signal per port.

Name: **LOS_QSFP_STAT**
Description: **Debug Bringup QSFP LOS Pin monitor. Only valid when CHIP_STAT.skeletonMode set**
Address: **0x0200.2108**

Bits	Field	Type	Reset	Description
31:0	los	RO	-	Los Of Signal per port.

Name: **LOS_DDQ_STAT**
Description: **Debug Bringup DDQ LOS Pin monitor. Only valid when CHIP_STAT.skeletonMode set**
Address: **0x0200.2120**

Bits	Field	Type	Reset	Description
31:0	los	RO	-	Los Of Signal per port.

Name: **SPARE_CFG**
Description: **Debug Bringup G_SPARE Pin control. Only valid when CHIP_STAT.skeletonMode set**
Address: **0x0200.2200**

Bits	Field	Type	Reset	Description
31:0	spare	RW	-	G_SPARE control.

3.4.2 PLL monitoring and Controls registers

Name: **PLL_OOL_HOLD**
Description: **PLL out-of-lock status hold**
Address: **0x0200.0010**

Bits	Field	Type	Reset	Description
0	core0	W1C	-	Core PLL 0 (G_CLK0).
1	core1	W1C	-	Core PLL 1 (G_CLK1).
2	core2	W1C	-	Core PLL 2 (G_CLK2).

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Bits	Field	Type	Reset	Description
3	core3	W1C	-	Core PLL 3 (G_CLK3).
4	core4	W1C	-	Core PLL 4.
5	core5	W1C	-	Core PLL 5. Not available on Utm2c platform.
6	dtasync	W1C	-	DTASYNC_CLK. Not available on Utm2c platform.
7	ad	W1C	-	AD_REFCLK.
8	pcie	W1C	-	System PLL PCIE.
9	tile12a	W1C	-	System PLL Tile 12A.
10	tile12b	W1C	-	System PLL Tile 12B.
11	tile12c	W1C	-	System PLL Tile 12C.
12	tile13a	W1C	-	System PLL Tile 13A.
13	tile13b	W1C	-	System PLL Tile 13B.
14	tile13c	W1C	-	System PLL Tile 13C.

3.4.3 Partial reconfiguration control

Name: PR_DATA
Description: PR Data
Address: 0x0200.0200

Bits	Field	Type	Reset	Description
31:0	data	WO	-	Write PR bitstream data to this address. Byte ordering is little-endian i.e. the first byte from the RBF file should be placed on bits [7:0].

Name: PR_CFG
Description: PR config
Address: 0x0200.0204

Bits	Field	Type	Reset	Description
0	prStart	RW	-	Write this bit to '1' to initiate a PR download. It will automatically reset to '0' immediately.
1	prReset	RW	0x1	Write 1 to reset the PR controller.

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Name: PR_STAT
Description: PR state
Address: 0x0200.0208

Bits	Field	Type	Reset	Description
2:0	status	RO	-	Current status. Following assertion of prStart, status should be inProgress. When all data has been received, it should change to prDone. 0: `inReset`. PR system is in reset state. 1: `busy`. Configuration system busy. 2: `inProgress`. In progress. 3: `prDone`. Completed successfully. 4: `error`. Error detected. 5: `reserved`. Reserved.
3	fifoOflw	RO	-	PR data FIFO overflow.
17:8	fifoLevel	RO	-	PR data FIFO level.

Name: PR_HOLD
Description: PR state hold
Address: 0x0200.020C

Bits	Field	Type	Reset	Description
3	fifoOflw	W1C	-	PR data FIFO overflow.

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3.4.4 Agilex Mailbox Client registers

Name: MBCL_CMD_CFG
Description: Command configuration
Address: 0x0200.1000

Bits	Field	Type	Reset	Description
10:0	cmd	WO	-	Command. See Agilex Mailbox Client cmd Field Descriptions table. 0: noop. 16: getIdcode. 18: getChipId. 19: getUserc. 24: getVoltage. 25: getTemp. 50: qspiOpen. 51: qspiClose. 52: qspiSetCs. 53: qspiRDevR. 54: qspiWrDevR. 55: qspiSendOp. 56: qspiErase. 57: qspiWrite. 58: qspiRead.
11	i	WO	-	Must be set to to 0.
22:12	length	WO	-	Number of words of arguments following header.
27:24	id	WO	-	A response header returns the same ID value specified in the command header. If you set different ID values in each command, you can use the ID value to match responses with commands if they are out of order.

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Name: MBCL_LAST_CFG
Description: Last command configuration
Address: 0x0200.1004

Bits	Field	Type	Reset	Description
10:0	cmd	WO	-	Command. See Agilex Mailbox Client cmd Field Descriptions table. 0: noop. 16: getIdcode. 18: getChipId. 19: getUserc. 24: getVoltage. 25: getTemp. 50: qspiOpen. 51: qspiClose. 52: qspiSetCs. 53: qspiRDevR. 54: qspiWrDevR. 55: qspiSendOp. 56: qspiErase. 57: qspiWrite. 58: qspiRead.
11	i	WO	-	Must be set to to 0.
22:12	length	WO	-	Number of words of arguments following header.
27:24	id	WO	-	A response header returns the same ID value specified in the command header. If you set different ID values in each command, you can use the ID value to match responses with commands if they are out of order.

Name: MBCL_CFIFO_STAT
Description: Command FIFO status
Address: 0x0200.1008

Bits	Field	Type	Reset	Description
31:0	space	RO	-	Command FIFO empty space.

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Name: MBCL_UCMD_CFG
Description: Urgent command configuration
Address: 0x0200.100C

Bits	Field	Type	Reset	Description
10:0	cmd	WO	-	Urgent command. See Agilex Mailbox Client cmd Field Descriptions table. 0: noop. 16: getIdcode. 18: getChipId. 19: getUserc. 24: getVoltage. 25: getTemp. 50: qspiOpen. 51: qspiClose. 52: qspiSetCs. 53: qspiRDevR. 54: qspiWrDevR. 55: qspiSendOp. 56: qspiErase. 57: qspiWrite. 58: qspiRead.
11	i	WO	-	Must be set to to 0.
22:12	length	WO	-	Number of words of arguments following header.
27:24	id	WO	-	A response header returns the same ID value specified in the command header. If you set different ID values in each command, you can use the ID value to match responses with commands if they are out of order.

Name: MBCL_UFIFO_STAT
Description: Urgent FIFO status
Address: 0x0200.1010

Bits	Field	Type	Reset	Description
31:0	space	RO	-	Urgent FIFO empty space.

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Name: MBCL_RSP_DATA
Description: Response data
Address: 0x0200.1014

Bits	Field	Type	Reset	Description
10:0	err	RO	-	Error code. See Agilex Mailbox Client err Field Descriptions table. 0: ok. 1: invalid. 3: unknown. 256: notConf. 511: respBusy. 767: respNoVldR. 1023: respErr.
22:12	length	RO	-	Number of words of arguments following header.
27:24	id	RO	-	A response header returns the same ID value specified in the command header. If you set different ID values in each command, you can use the ID value to match responses with commands if they are out of order.

Name: MBCL_RFIFO_STAT
Description: Response FIFO status
Address: 0x0200.1018

Bits	Field	Type	Reset	Description
0	sop	RO	-	Start of packet.
1	eop	RO	-	End of packet.
31:2	level	RO	-	Response FIFO fill level.

Name: MBCL_INT_CFG
Description: Interrupt configuration
Address: 0x0200.101C

Bits	Field	Type	Reset	Description
0	dataValid	RW	-	The enable bit of data valid in response FIFO.
1	cmdFifoNotFull	RW	-	The enable bit interrupt of command FIFO is not full.

Name: MBCL_INT_STAT
Description: Interrupt status
Address: 0x0200.1020

Bits	Field	Type	Reset	Description
0	dataValid	RO	-	Data valid interrupt (data exists in the FIFO).

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Bits	Field	Type	Reset	Description
1	cmdFifoNotFull	RO	-	Command FIFO is not full interrupt.

3.4.5 CPU Interface Protection module

Name: CPU_PROT_ADDR
Description: Last accessed address
Address: 0x0200.0600

Bits	Field	Type	Reset	Description
31:0	lastAddr	RO	-	Last address accessed by the CPU (remember to disable the poller).
63:32	failAddr	RO	-	Address that failed to reply to the CPU.

Name: CPU_PROT_LATENCY
Description: Slave reply latency
Address: 0x0200.0608

Bits	Field	Type	Reset	Description
15:0	lastCycles	RO	-	Last access latency between request and response in clock cycles (remember to disable the poller).

Name: CPU_PROT_STAT
Description: Status
Address: 0x0200.060C

Bits	Field	Type	Reset	Description
0	timeout	RO	-	Slave didn't reply within configured time. Protection triggered.

Name: CPU_PROT_HOLD
Description: Status hold
Address: 0x0200.0610

Bits	Field	Type	Reset	Description
0	timeout	W1C	-	Slave didn't reply within configured time. Protection triggered.

Name: CPU_PROT_INTEN
Description: Interrupt enable
Address: 0x0200.0614

Bits	Field	Type	Reset	Description
0	timeout	RW	-	Slave didn't reply within configured time. Protection triggered.

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3.4.6 FGT Reference and Core Clocks measurement

Name: **CORE_CLK_NUM_INFO**
Description: **Number of Core clocks**
Address: **0x0200.0800**

Bits	Field	Type	Reset	Description
5:0	num	RO	-	Number of Core clocks.

Name: **CORE_CLK_FREQ_INFO**
Description: **Expected Core Clock Frequency**
Address: **0x0200.0880**
Entries: **32**
Entry size: **0x4**
Index: **Core Clock Index**

Bits	Field	Type	Reset	Description
4:0	freq	RO	-	Core clock index. 0: clk100. 100.00 MHz. 1: clk102. 102.30 MHz. 2: clk311. 311.04 MHz. 3: clk333. 333.00 MHz. 4: clk357. 357.00 MHz. 5: clk357x05. 178.50 MHz. 6: clk413. 413.00 MHz. 7: clk413x05. 206.50 MHz. 8: clk426. 426.00 MHz. 9: clk439. 439.00 MHz. 10: clk125. 125.00 MHz. 11: clk114. 114.2857 MHz. 12: clk415. 415.00 MHz. 31: clkUnused. 1 Hz.

Name: **CORE_CLK_SEL_CFG**
Description: **Core Clock Index Selector**
Address: **0x0200.0D00**

Bits	Field	Type	Reset	Description
4:0	ix	RW	-	Core clock index.

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Name: REF_CLK_SEL_CFG
Description: Reference Clock Index Selector
Address: 0x0200.0D04

Bits	Field	Type	Reset	Description
0	side	RW	-	Side of the device. 0: `left`. Left side. 1: `right`. Right side.
3:1	tile	RW	-	Transceiver tile number. The value 0 corresponds to the first (lower) tile on the selected side of the device, 1 to the next tile on that side.
7:4	clk	RW	-	FGT Tile reference clock index. Only FGT clocks can be measured.

Name: FREQ_MEAS
Description: Measured Frequency
Address: 0x0200.0D08

Bits	Field	Type	Reset	Description
31:0	refClk	RO	-	Selected reference clock measured frequency.
63:32	coreClk	RO	-	Selected core clock measured frequency.

3.4.7 SerDes Recovered Clocks & Rates measurement

Name: REC_CLK_MEAS
Description: SerDes Recovered Clocks & Rates measurement
Address: 0x0200.0E00
Entries: 2
Entry size: 0x10
Index: IF_REC_CLK Index

Register Name	Entries	Description
IF_REC_SEL_CFG	-	Serdes Interface Index Selector
REC_FREQ_MEAS	-	Measured Frequency

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Name: REC_CLK_MEAS.IF_REC_SEL_CFG
Description: Serdes Interface Index Selector
Address: 0x0000.0000

Bits	Field	Type	Reset	Description
3:0	prtType	RW	-	Port Type. 0: sfp. Small Form-factor Pluggable. 1: qsfp. Quad SFP. 2: cfp. C Form-factor Pluggable. 3: dds. SFP Double Density. 4: ddq. Quad SFP Double Density. 5: dta. DTA. 6: pcie. PCIe. 7: sgmii. SGMII. 8: ilk. Interlaken. 15: unused. Unused.
11:4	prtIx	RW	-	Port Index.
13:12	laneIx	RW	-	Lane Index.
14	dir	RW	-	Direction. 0: rx. Rx. 1: tx. Tx.

Name: REC_CLK_MEAS.REC_FREQ_MEAS
Description: Measured Frequency
Address: 0x0000.0008

Bits	Field	Type	Reset	Description
31:0	clk	RO	-	Selected clock measured frequency.
63:32	rate	RO	-	Selected rate measured frequency.

3.4.8 SerDes monitoring and Controls registers

Name: SERDES_INFO
Description: SerDes information
Address: 0x03F0.0FC0

Bits	Field	Type	Reset	Description
2:0	tileL0	RO	-	Type of tile at left 0. 0: none. No tile. 1: ftile. F-Tile.
3	presentL0	RO	-	Set if tile at left 0 is in use in the current device.
6:4	tileL1	RO	-	Type of tile at left 1. 0: none. No tile. 1: ftile. F-Tile.

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Bits	Field	Type	Reset	Description
7	presentL1	RO	-	Set if tile at left 1 is in use in the current device.
10:8	tileL2	RO	-	Type of tile at left 2. 0: none. No tile. 1: ftile. F-Tile.
11	presentL2	RO	-	Set if tile at left 2 is in use in the current device.
14:12	tileL3	RO	-	Type of tile at left 3. 0: none. No tile. 1: ftile. F-Tile.
15	presentL3	RO	-	Set if tile at left 3 is in use in the current device.
18:16	tileR0	RO	-	Type of tile at right 0. 0: none. No tile. 1: ftile. F-Tile.
19	presentR0	RO	-	Set if tile at right 0 is in use in the current device.
22:20	tileR1	RO	-	Type of tile at right 1. 0: none. No tile. 1: ftile. F-Tile.
23	presentR1	RO	-	Set if tile at right 1 is in use in the current device.
26:24	tileR2	RO	-	Type of tile at right 2. 0: none. No tile. 1: ftile. F-Tile.
27	presentR2	RO	-	Set if tile at right 2 is in use in the current device.
30:28	tileR3	RO	-	Type of tile at right 3. 0: none. No tile. 1: ftile. F-Tile.
31	presentR3	RO	-	Set if tile at right 3 is in use in the current device.

Name: SERDES_PHY_INFO
Description: SerDes PHY information
Address: 0x03F0.0FB0

Bits	Field	Type	Reset	Description
1:0	phyL0Q0	RO	-	Type of PHY at left 0 quad 0. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
3:2	phyL0Q1	RO	-	Type of PHY at left 0 quad 1. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.

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Bits	Field	Type	Reset	Description
5:4	phyL0Q2	RO	-	Type of PHY at left 0 quad 2. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
7:6	phyL0Q3	RO	-	Type of PHY at left 0 quad 3. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
9:8	phyL0Q4	RO	-	Type of PHY at left 0 quad 4. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
11:10	phyL0Q5	RO	-	Type of PHY at left 0 quad 5. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
13:12	phyL0Q6	RO	-	Type of PHY at left 0 quad 6. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
15:14	phyL0Q7	RO	-	Type of PHY at left 0 quad 7. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
17:16	phyL1Q0	RO	-	Type of PHY at left 1 quad 0. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
19:18	phyL1Q1	RO	-	Type of PHY at left 1 quad 1. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
21:20	phyL1Q2	RO	-	Type of PHY at left 1 quad 2. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
23:22	phyL1Q3	RO	-	Type of PHY at left 1 quad 3. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
25:24	phyL1Q4	RO	-	Type of PHY at left 1 quad 4. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.

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Bits	Field	Type	Reset	Description
27:26	phyL1Q5	RO	-	Type of PHY at left 1 quad 5. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
29:28	phyL1Q6	RO	-	Type of PHY at left 1 quad 6. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
31:30	phyL1Q7	RO	-	Type of PHY at left 1 quad 7. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
33:32	phyL2Q0	RO	-	Type of PHY at left 2 quad 0. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
35:34	phyL2Q1	RO	-	Type of PHY at left 2 quad 1. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
37:36	phyL2Q2	RO	-	Type of PHY at left 2 quad 2. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
39:38	phyL2Q3	RO	-	Type of PHY at left 2 quad 3. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
41:40	phyL2Q4	RO	-	Type of PHY at left 2 quad 4. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
43:42	phyL2Q5	RO	-	Type of PHY at left 2 quad 5. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
45:44	phyL2Q6	RO	-	Type of PHY at left 2 quad 6. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
47:46	phyL2Q7	RO	-	Type of PHY at left 2 quad 7. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.

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Bits	Field	Type	Reset	Description
49:48	phyL3Q0	RO	-	Type of PHY at left 3 quad 0. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
51:50	phyL3Q1	RO	-	Type of PHY at left 3 quad 1. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
53:52	phyL3Q2	RO	-	Type of PHY at left 3 quad 2. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
55:54	phyL3Q3	RO	-	Type of PHY at left 3 quad 3. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
57:56	phyL3Q4	RO	-	Type of PHY at left 3 quad 4. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
59:58	phyL3Q5	RO	-	Type of PHY at left 3 quad 5. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
61:60	phyL3Q6	RO	-	Type of PHY at left 3 quad 6. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
63:62	phyL3Q7	RO	-	Type of PHY at left 3 quad 7. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
65:64	phyR0Q0	RO	-	Type of PHY at right 0 quad 0. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
67:66	phyR0Q1	RO	-	Type of PHY at right 0 quad 1. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
69:68	phyR0Q2	RO	-	Type of PHY at right 0 quad 2. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.

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Bits	Field	Type	Reset	Description
71:70	phyR0Q3	RO	-	Type of PHY at right 0 quad 3. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
73:72	phyR0Q4	RO	-	Type of PHY at right 0 quad 4. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
75:74	phyR0Q5	RO	-	Type of PHY at right 0 quad 5. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
77:76	phyR0Q6	RO	-	Type of PHY at right 0 quad 6. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
79:78	phyR0Q7	RO	-	Type of PHY at right 0 quad 7. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
81:80	phyR1Q0	RO	-	Type of PHY at right 1 quad 0. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
83:82	phyR1Q1	RO	-	Type of PHY at right 1 quad 1. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
85:84	phyR1Q2	RO	-	Type of PHY at right 1 quad 2. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
87:86	phyR1Q3	RO	-	Type of PHY at right 1 quad 3. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
89:88	phyR1Q4	RO	-	Type of PHY at right 1 quad 4. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
91:90	phyR1Q5	RO	-	Type of PHY at right 1 quad 5. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.

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Bits	Field	Type	Reset	Description
93:92	phyR1Q6	RO	-	Type of PHY at right 1 quad 6. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
95:94	phyR1Q7	RO	-	Type of PHY at right 1 quad 7. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
97:96	phyR2Q0	RO	-	Type of PHY at right 2 quad 0. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
99:98	phyR2Q1	RO	-	Type of PHY at right 2 quad 1. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
101:100	phyR2Q2	RO	-	Type of PHY at right 2 quad 2. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
103:102	phyR2Q3	RO	-	Type of PHY at right 2 quad 3. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
105:104	phyR2Q4	RO	-	Type of PHY at right 2 quad 4. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
107:106	phyR2Q5	RO	-	Type of PHY at right 2 quad 5. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
109:108	phyR2Q6	RO	-	Type of PHY at right 2 quad 6. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
111:110	phyR2Q7	RO	-	Type of PHY at right 2 quad 7. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
113:112	phyR3Q0	RO	-	Type of PHY at right 3 quad 0. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.

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Bits	Field	Type	Reset	Description
115:114	phyR3Q1	RO	-	Type of PHY at right 3 quad 1. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
117:116	phyR3Q2	RO	-	Type of PHY at right 3 quad 2. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
119:118	phyR3Q3	RO	-	Type of PHY at right 3 quad 3. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
121:120	phyR3Q4	RO	-	Type of PHY at right 3 quad 4. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
123:122	phyR3Q5	RO	-	Type of PHY at right 3 quad 5. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
125:124	phyR3Q6	RO	-	Type of PHY at right 3 quad 6. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.
127:126	phyR3Q7	RO	-	Type of PHY at right 3 quad 7. 0: none. No Phy present. 1: fgt. FGT DirectPhy. 2: fht. FHT DirectPhy.

Name: SERDES_SEL
Description: SerDes selector
Address: 0x03F0.0FFC

Bits	Field	Type	Reset	Description
15:12	offset	RW	-	Address offset.
25:22	resId	RW	-	Resource identifier. 0: xcvr0. Channel 0 FGT. 1: xcvr1. Channel 1 FGT. 2: xcvr2. Channel 2 FGT. 3: xcvr3. Channel 3 FGT. 4: pdp0. Channel 0 PMA. 5: pdp1. Channel 1 PMA. 6: pdp2. Channel 2 PMA. 7: pdp3. Channel 3 PMA. 8: drctrl. DR controller for tile. Only 1 per tile (on grp 0).

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Bits	Field	Type	Reset	Description
28:26	group	RW	-	Transceiver group identifier. The resources in a tile is divided into groups, the value 0 corresponds to the first (lowest) group in the tile. For F-Tiles each group consists of 4 transceiver channels and the Tile associated DR controller. Only DR controllers on group 0 are used.
30:29	tile	RW	-	Transceiver tile number. The value 0 corresponds to the first (lower) tile on the selected side of the device, 1 to the next tile on that side.
31	side	RW	-	Side of the device. 0: `left`. Left side. 1: `right`. Right side.

Name: SERDES_SEL_INFO
Description: SerDes info
Address: 0x03F0.0000

Bits	Field	Type	Reset	Description
0	reservedCom	RO	-	Reserved for type.

Name: SERDES_SEL_CFG
Description: SerDes configuration
Address: 0x03F0.0040
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	recfgReq	RW	0x0	Set to request access to the reconfiguration registers.

Name: SERDES_SEL_STAT
Description: SerDes status
Address: 0x03F0.0060
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	presentRecfg	RO	-	Set if the _RECFG interface for the currently selected resource is in use in the current device.
1	recfgIfBusy	RO	-	Set when interface towards the reconfiguration registers is busy. For debug purposes only.
2	recfgReqBusy	RO	-	Set when a write operation to `.recfgReq` is being processed. For debug purposes only.
3	recfgIfWait	RO	-	Set when interface towards the reconfiguration registers indicates it is not ready for access. For debug purposes only.
4	recfgRdy	RO	-	When set indicates that the reconfiguration registers are ready for access.

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Bits	Field	Type	Reset	Description
5	present	RO	-	Set if the config/status interface for the currently selected resource is in use in the current device.

Name: SERDES_SEL_HOLD
Description: SerDes status hold
Address: 0x03F0.0080
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	recfgErr	W1C	-	When set indicates that an error has occured towards the reconfiguration registers.

Name: SERDES_F_XCVR_INFO
Description: SerDes F-Tile transceiver info
Address: 0x03F0.0100

Bits	Field	Type	Reset	Description
7:0	softportNo	RO	-	Softport Number assgined for SW.
11:8	portType	RO	-	Serdes Port Type. 0: sfp. Small Form-factor Pluggable. 1: qsfp. Quad SFP. 2: cfp. C Form-factor Pluggable. 3: dds. SFP Double Density. 4: ddq. Quad SFP Double Density. 5: dta. DTA. 6: pcie. PCIe. 7: sgmii. SGMII. 8: ilk. Interlaken. 15: unused. Unused.
35:32	portMode	RO	-	Mode XCVR instance is configured as. 0: mode100g4. 1x 100G-4 - for QSFP ports. Lanes are grouped in 1x4. 1: mode50g1. 4x 50G-1 - for SFP ports. Lanes are independent. 2: mode400g8. 1x 400G-8 - for DDQ ports. Lanes are grouped in 1x8 (FOIC 4.8) or 1x2 (FOIC 1.2). 15: unused. Unused.
41:36	xcvrIx	RO	-	XCVR index number.
47:44	laneIx	RO	-	XCVR lane number.
48	enabled	RO	-	Set if the XCVR (lane) is enabled.
49	drEnabled	RO	-	Set if the XCVR has Dynamic Reconfiguration capabilities.
50	fecIncluded	RO	-	Set if FEC is availalbe in the XCVR.
51	resetAckType	RO	0x0	Set if tx/rxResetDone reflects the actual status in the XCVR.

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Bits	Field	Type	Reset	Description
67:64	clkType	RO	-	Clock PLL Type. 0: devNa. Not applicable. 1: devZL30237. ZL30237. 2: devAD9578. AD9578. 3: devSI5341. SI5341. 4: devIntPll. Internal Pll. 5: devSi5345. Si5345.
75:72	busNo	RO	-	Physical bus number device resides on.
79:76	devNo	RO	-	Device number on bus.
83:80	pllNo	RO	-	PLL number.
87:84	outNo	RO	-	Output number.
91:88	outNo2	RO	-	Output number 2.
92	useOut2	RO	-	Use output 2?.
135:128	cl1p16gNo	RO	-	Default CL1P16G block for this lane.
143:136	clnp100gNo	RO	-	Default CLNP100G block for this lane.
151:144	clnpx00gNo	RO	-	Default CLNPX00G block for this lane.
175:160	drProfOff	RO	-	DR Start offset for DR profiles for this port.
183:176	drProf1st	RO	-	DR Mapping of first profile from IP.
191:184	drProfSt	RO	-	DR Startup profile.
220:216	reserved	RO	-	Reserved to preserve softport type. 0: none. Unknown. 1: ge10. 10G Ethernet (10GE LAN). 2: ge100G2. 100G Ethernet (GAUI-2). 3: ge100C4. 100G Ethernet (CAUI-4) with RSFEC. 4: ge100C4nF. 100G Ethernet (CAUI-4) no FEC. 5: e4c. 400G Ethernet. 6: fc16g. 16GFC. 7: fc32g. 32GFC. 8: otu2. OTU2. 9: otu2e. OTU2e. 10: otu4. OTU4 (OTL4.4) without ODU mux. 11: otu4Mux. OTU4 (OTL4.4) with ODU mux. 12: foic12. FOIC1.2 (B100G slice OTUC). 13: fc32gT. 32GFC Transparent. 14: stm64Sync. Synchronous STM64 / OC 192. 15: fc32gFec. 32GFCFec. 16: fc64g. 64GFC. 17: foic48. FOIC4.8.
255:224	spTypes	RO	-	Supported Softport Types by this port (one bit per type).

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Name: SERDES_F_XCVR_CFG
Description: SerDes F-Tile transceiver configuration
Address: 0x03F0.0140
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
32	txDigitalReset	RW	0x1	If toggled to '1' and then '0', performs a reset of the Tx digital logic on the specific lanes.
33	rxDigitalReset	RW	0x1	If toggled to '1' and then '0', performs a reset of the Rx digital logic on the specific lanes.
34	xcvrReset	RW	0x1	If toggled to '1' and then '0', performs a reset of both Tx and Rx logic on the specific lanes.
35	fecSnapshot	RW	0x0	If toggled to '1' and then '0', performs a snapshot of RS-FEC status to CSR. Traffic must be stopped prior to taking a snapshot. Only valid if SERDES_F_XCVR_INFO.fecIncluded = 1.
37:36	ddqProfileMode	RW	0x0	Set DDQ port DR profile mode. Only valid if SERDES_F_XCVR_INFO.drEnabled = 1. 0: mode0. 1x 400G PAM4 mode - for FOIC4.8 rate. 1: mode1. 4x 100G PAM4 mode - for FOIC1.2 rates. 2: mode2. 2x 100G NRZ mode - for 100G rates. 3: mode3. 8x 25G NRZ mode - for 10G rates.
39:38	qProfileMode	RW	0x0	Set Q port DR profile mode. Only valid if SERDES_F_XCVR_INFO.drEnabled = 1. 0: mode0. 1x 100G mode - for 100G rates. 1: mode1. 4x 25G mode - for 10G rates.
41:40	sProfileMode	RW	0x0	Set S port DR profile mode. Only valid if SERDES_F_XCVR_INFO.drEnabled = 1. 0: mode0. 1x 50G mode - for 64GFC rate. 1: mode1. 1x 25G mode - for 10G/32GFC/16GFC rates.
42	kCtrlTimeEn	RW	0x0	Enable maxKvalueWait k updating. Enabling this feature will make the serdes ignore k-value updates from the dcxo while tx reset is active. When tx reset is done and tx ready is active, a counter will start and when it hits the maxKvalueWait threshold the serdes will use the k-value update again.
43	kCtrlStepEn	RW	0x0	Enable maxKvalueStep updating. When this is enabled, k-value updates from the dcxo will be overwritten if they exceed the maxKvalueStep size.
53:44	maxKvalueWait	RW	0x0	Amount of avmm_clk counts before serdes will start k-value updating after TX reset.
63:54	maxKvalueStep	RW	0x0	Maximum step for kvalue updating.

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Name: SERDES_F_XCVR_STAT
Description: SerDes F-Tile transceiver status
Address: 0x03F0.0160
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
32	txResetDone	RO	-	If set, the Tx datapath has been reset successfully.
33	rxResetDone	RO	-	If set, the Rx datapath has been reset successfully.
34	pllOOL	RO	-	If set, the pll is out-of-lock to reference (inverted version of pll_locked status signal).
35	rxRefOOL	RO	-	If set, the receiver is out-of-lock to reference (inverted version of rx_is_lockedtoref status signal).
36	rxDataOOL	RO	-	If set, the receiver is out-of-lock to data (inverted version of rx_is_lockedtodata status signal).
37	rxSigDetect	RO	-	If set, the Rx datapath has detected data. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
38	rsfecRxSf	RO	-	Signal fail from FEC. Only valid if SERDES_F_XCVR_INFO.fecIncluded = 0.
39	rsfecRxNotAlign	RO	-	FEC not aligned. Only valid if SERDES_F_XCVR_INFO.fecIncluded = 0.
40	rsfecRxNotDeskew	RO	-	FEC deskew not working. Only valid if SERDES_F_XCVR_INFO.fecIncluded = 0.
41	rsfecRxNotLocked	RO	-	FEC not locked. Only valid if SERDES_F_XCVR_INFO.fecIncluded = 0.
42	txPhCompFull	RO	-	If set, the Tx phase compensation FIFO is full. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
43	txPhCompEmpty	RO	-	If set, the Tx phase compensation FIFO is empty. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
44	txPhCompPfull	RO	-	If set, the Tx phase compensation FIFO is partially full. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
45	txPhCompPempty	RO	-	If set, the Tx phase compensation FIFO is partially empty. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
46	rxPhCompFull	RO	-	If set, the Rx phase compensation FIFO is full. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
47	rxPhCompEmpty	RO	-	If set, the Rx phase compensation FIFO is empty. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
48	rxPhCompPfull	RO	-	If set, the Rx phase compensation FIFO is partially full. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
49	rxPhCompPempty	RO	-	If set, the Rx phase compensation FIFO is partially empty. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
50	txNotReady	RO	-	If set, the Tx is not ready (inverted version of tx_ready status signal).

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Bits	Field	Type	Reset	Description
51	rxNotReady	RO	-	If set, the Rx is not ready (inverted version of rx_ready status signal).
52	txPmaFifoOflw	RO	-	Tx PMA FIFO overflow. Only valid if SERDES_F_XCVR_INFO.drEnabled = 1.
53	txPmaFifoPfull	RO	-	If set, the Tx PMA FIFO is partially full.
54	txPmaFifoPempty	RO	-	If set, the Tx PMA FIFO is partially empty.
55	txPmaFifoEmpty	RO	-	If set, the Tx PMA FIFO is empty.
56	rxPmaFifoPfull	RO	-	If set, the Rx PMA FIFO is partially full.
57	rxPmaFifoPempty	RO	-	If set, the Rx PMA FIFO is partially empty.
58	rxPmaFifoEmpty	RO	-	If set, the Rx PMA FIFO is empty.
59	dllOOL	RO	-	If set, the tx dll is out-of-lock to reference (inverted version of dll_lock status signal).
60	txErr	RO	-	If set, the Tx datapath have had an error that have disturbed the datapath and possibly corrupted data.
61	rxErr	RO	-	If set, the Rx datapath have had an error that have disturbed the datapath and possibly corrupted data.

Name: SERDES_F_XCVR_HOLD
Description: SerDes F-Tile transceiver status hold
Address: 0x03F0.0180
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
32	txResetDone	W1C	-	If set, the Tx datapath has been reset successfully.
33	rxResetDone	W1C	-	If set, the Rx datapath has been reset successfully.
34	pllOOL	W1C	-	If set, the pll is out-of-lock to reference (inverted version of pll_locked status signal).
35	rxRefOOL	W1C	-	If set, the receiver is out-of-lock to reference (inverted version of rx_is_lockedtoref status signal).
36	rxDataOOL	W1C	-	If set, the receiver is out-of-lock to data (inverted version of rx_is_lockedtodata status signal).
37	rxSigDetect	W1C	-	If set, the Rx datapath has detected data. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
38	rsfecRxSf	W1C	-	Signal fail from FEC. Only valid if SERDES_F_XCVR_INFO.fecIncluded = 0.
39	rsfecRxNotAlign	W1C	-	FEC not aligned. Only valid if SERDES_F_XCVR_INFO.fecIncluded = 0.
40	rsfecRxNotDeskew	W1C	-	FEC deskew not working. Only valid if SERDES_F_XCVR_INFO.fecIncluded = 0.

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Bits	Field	Type	Reset	Description
41	rsfecRxNotLocked	W1C	-	FEC not locked. Only valid if SERDES_F_XCVR_INFO.fecIncluded = 0.
42	txPhCompFull	W1C	-	If set, the Tx phase compensation FIFO is full. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
43	txPhCompEmpty	W1C	-	If set, the Tx phase compensation FIFO is empty. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
44	txPhCompPfull	W1C	-	If set, the Tx phase compensation FIFO is partially full. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
45	txPhCompPempty	W1C	-	If set, the Tx phase compensation FIFO is partially empty. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
46	rxPhCompFull	W1C	-	If set, the Rx phase compensation FIFO is full. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
47	rxPhCompEmpty	W1C	-	If set, the Rx phase compensation FIFO is empty. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
48	rxPhCompPfull	W1C	-	If set, the Rx phase compensation FIFO is partially full. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
49	rxPhCompPempty	W1C	-	If set, the Rx phase compensation FIFO is partially empty. Only valid if SERDES_F_XCVR_INFO.drEnabled = 0.
50	txNotReady	W1C	-	If set, the Tx is not ready (inverted version of tx_ready status signal).
51	rxNotReady	W1C	-	If set, the Rx is not ready (inverted version of rx_ready status signal).
52	txPmaFifoOflw	W1C	-	Tx PMA FIFO overflow. Only valid if SERDES_F_XCVR_INFO.drEnabled = 1.
53	txPmaFifoPfull	W1C	-	If set, the Tx PMA FIFO is partially full.
54	txPmaFifoPempty	W1C	-	If set, the Tx PMA FIFO is partially empty.
55	txPmaFifoEmpty	W1C	-	If set, the Tx PMA FIFO is empty.
56	rxPmaFifoPfull	W1C	-	If set, the Rx PMA FIFO is partially full.
57	rxPmaFifoPempty	W1C	-	If set, the Rx PMA FIFO is partially empty.
58	rxPmaFifoEmpty	W1C	-	If set, the Rx PMA FIFO is empty.
59	dllOOL	W1C	-	If set, the tx dll is out-of-lock to reference (inverted version of dll_lock status signal).
60	txErr	W1C	-	If set, the Tx datapath have had an error that have disturbed the datapath and possibly corrupted data.
61	rxErr	W1C	-	If set, the Rx datapath have had an error that have disturbed the datapath and possibly corrupted data.

Name: SERDES_PDP_INFO
Description: SerDes PDP transceiver info
Address: 0x03F0.0200

Bits	Field	Type	Reset	Description
64	reservedPdP	RO	-	Reserved for PDP.

Name: SERDES_PDP_CFG
Description: SerDes PDP transceiver configuration
Address: 0x03F0.0240
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
64	reservedPdP	RW	0x0	Reserved for PDP.

Name: SERDES_PDP_STAT
Description: SerDes PDP transceiver status
Address: 0x03F0.0260
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
64	reservedPdP	RO	0x0	Reserved for PDP.

Name: SERDES_PDP_HOLD
Description: SerDes PDP transceiver status hold
Address: 0x03F0.0280
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
64	reservedPdP	W1C	0x0	Reserved for PDP.

Name: SERDES_DRCTRL_INFO
Description: SerDes F-Tile transceiver info
Address: 0x03F0.0300

Bits	Field	Type	Reset	Description
96	reservedDr	RO	-	Reserved for DRCTRL.

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Name: SERDES_DRCTRL_CFG
Description: SerDes F-Tile transceiver configuration
Address: 0x03F0.0340
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
96	reservedDr	RW	0x0	Reserved for DRCTRL.

Name: SERDES_DRCTRL_STAT
Description: SerDes F-Tile transceiver status
Address: 0x03F0.0360
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
96	drInProgress	RO	-	Dynamic reconfiguration is in progress.
97	drCfgApplied	RO	-	Dynamic reconfiguration config applied.
98	drErr	RO	-	Dynamic reconfiguration encountered an error.
126:112	drCurrProf	RO	-	Current profile ID.

Name: SERDES_DRCTRL_HOLD
Description: SerDes F-Tile transceiver status hold
Address: 0x03F0.0380
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
96	drInProgress	W1C	-	Dynamic reconfiguration is in progress.
97	drCfgApplied	W1C	-	Dynamic reconfiguration config applied.
98	drErr	W1C	-	Dynamic reconfiguration encountered an error.
126:112	drCurrProf	W1C	-	Current profile ID.

Name: SRDS_FGT_LC_MED0_0
Description: FGT 0 LC medium post divider
Address: 0x0304.0740
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31	cfgMedPostDivP5	RW	0x0	FGT lane 0 enable LC Med Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

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Name: SRDS_FGT_LC_MED1_0
Description: FGT 0 LC medium post divider
Address: 0x0304.0744
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
13:7	cfgMedPostDiv	RW	0x2	FGT lane 0 LC Med Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

Name: SRDS_FGT_LC_SLOW0_0
Description: FGT 0 LC slow post divider
Address: 0x0304.0840
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgSlowPostDivP5	RW	0x0	FGT lane0 enable LC Slow Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: SRDS_FGT_LC_SLOW1_0
Description: FGT 0 LC slow post divider
Address: 0x0304.0844
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
13:7	cfgSlowPostDiv	RW	0x0	FGT lane 0 LC Slow Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

Name: SRDS_FGT_LC_FAST0_0
Description: FGT 0 LC fast post divider
Address: 0x0304.0994
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgFastPostDivP5	RW	0x0	FGT lane 0 enable LC Fast Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.0.5 added into the post divider value.

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Name: SRDS_FGT_LC_FAST1_0
Description: FGT 0 LC fast post divider
Address: 0x0304.0998
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
27:21	cfgFastPostDiv	RW	0x2	FGT lane 0 LC Fast Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

Name: SRDS_FGT_LN0_0
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0304.1414
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
29	cfgLbParRx2TxCdr	RW	0x0	Direct top-level interface pin override for ictl_pcs_lb_cdrclk2txen_lx_nt. Enable clock loopback path from RX CDR VCO high-speed recovered clock to corresponding TX Lane.

Name: SRDS_FGT_LN1_0
Description: FGT 0 Serial Internal Loopback
Address: 0x0304.1418
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
29	cfgLbTx2RxOvr	RW	0x0	FGT lane 0 enable serial internal loopback LSB 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.
31	cfgLbOvrEn	RW	0x0	Override enable for registers with the name lb_*_locovr (required for all loopback mode). 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

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Name: SRDS_FGT_LN2_0
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0304.141C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgLbParRx2Tx	RW	0x0	Direct top-level interface pin override for ictl_pcs_lb_parrx2txtimeden_lx_nt. Enable RX parallel word to TX parallel word loopback.

Name: SRDS_FGT_LN3_0
Description: FGT 0 Serial Internal Loopback
Address: 0x0304.1420
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
25	cfgLbTx2Rxbuf	RW	0x0	FGT lane 0 enable serial internal loopback MSB 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

Name: SRDS_FGT_BIST0_0
Description: Override enable for PCS RX BIST locovr
Address: 0x0304.1424
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
26	cfgPcsRxBistOvr	RW	0x0	FGT 0 override enable for PCS RX BIST locovr: -1'b1: override enable -1'b0: override disable.

Name: SRDS_FGT_LN5_0
Description: Set RX PRBS Pattern and FGT 0 RX/TX polarity
Address: 0x0304.1428
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	cfgRxBistMode	RW	0x0	FGT lane 0 RX PRBS Pattern Selection: -UDP : 0x0 -PRBS7: 0x1 -PRBS9: 0x2 -PRBS10 : 0x3 - PRBS13 : 0x4 - PRBS15 : 0x5 - PRBS23 : 0x6 - PRBS28 : 0x7 - PRBS31 : 0x8 - QPRBS13 : 0x9 - PRBS13Q : 0xa - PRBS31Q : 0xb - SSPR : 0xc - SSPR1 : 0xd - SSPRQ : 0xe.

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Bits	Field	Type	Reset	Description
6	cfgPolarityRx	RW	0x0	FGT lane 0 enable RX polarity inversion. 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.
7	cfgPolarityTx	RW	0x0	FGT lane 0 enable TX polarity inversion. 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

Name: SRDS_FGT_BIST1_0
Description: RX BIST Overflow Check
Address: 0x0304.143C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
21	cfgRxBistErrOwfl	RW	0x0	RX BIST error counter overflow status: -1'b1: overflow detected -1'b0: overflow not detected.

Name: SRDS_FGT_BIST2_0
Description: RX BIST Error Check Count
Address: 0x0304.1444
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cfgRxBistErrCnt	RW	0x0	Error counts by RX BIST pattern checker direct mapping.

Name: SRDS_FGT_CDR_VCO0_0
Description: RX CDR VCO integral gain reg_34
Address: 0x0304.1484
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:10	cfgRxCdrGain10b	RW	0xd	RX CDR VCO integral gain_10b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x1. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
24:20	cfgRxCdrGain16b	RW	0xd	RX CDR VCO integral gain_16b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x2. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

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Name: SRDS_FGT_CDR_VCO1_0
Description: RX CDR VCO integral gain reg_35
Address: 0x0304.1488
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgRxCdrGain20b	RW	0xd	RX CDR VCO integral gain_20b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x3. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
14:10	cfgRxCdrGain32b	RW	0xd	RX CDR VCO integral gain_32b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x4. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
24:20	cfgRxCdrGain40b	RW	0xd	RX CDR VCO integral gain_40b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x5. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO2_0
Description: RX CDR VCO integral gain reg_36
Address: 0x0304.148C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgRxCdrGain64b	RW	0xd	RX CDR VCO integral gain_64b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x6. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
14:10	cfgRxCdrGain80b	RW	0xd	RX CDR VCO integral gain_80b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x7. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO5_0
Description: FSM current width
Address: 0x0304.1514
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
18:16	cfgRxFsmCurrWth	RW	0x0	FSM current width. Note: This register should not be overridden by users. It should be read-only to find out which RX CDR VCO intergal gain register takes effect.

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Name: SRDS_FGT_CDR_VCO3_0
Description: RX CDR VCO proportional gain reg_96
Address: 0x0304.157C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24:20	cfgRxCdrGainProp	RW	0x4	RX CDR VCO proportional gain. Larger value contributes larger bandwidth: 3~12 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO4_0
Description: RX CDR VCO integral gain reg_100
Address: 0x0304.158C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17:13	cfgRxCdrGain8b	RW	0xd	RX CDR VCO integral gain_8b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x0. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_RX_LN1_0
Description: FGT 0 RX bitflip
Address: 0x0304.15B0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
26:24	cfgRxpamBitorder	RW	0x1	FGT 0 enable RX bitflip 3'b1: enable bitflip. 3'b0: disable bitflip.
31	cfgRxPostDivP5	RW	0x0	FGT lane 0 enable CDR Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: SRDS_FGT_RX_LN0_0
Description: FGT 0 CDR post divider and RX PCS BIST Enable locovr and Override enable for RX PCS BIST locovr
Address: 0x0304.15B4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
6:0	cfgRxPcsPostDiv	RW	0x0	FGT lane 0 CDR Post Divider direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.
18	cfgRxPcsBistEn	RW	0x0	FGT 0 RX PCS BIST Enable locovr -1'b1: enable -1'b0: disable.
19	cfgRxPcsBistOvr	RW	0x0	FGT 0 override enable for RX PCS BIST locovr: -1'b1: override enable -1'b0: override disable.

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Name: SRDS_FGT_LN4_0
Description: FGT 0 Parallel Loopback
Address: 0x0304.16A4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	cfgTx2RxlbEn	RW	0x0	FGT lane 0 enable TX to RX parallel loopback 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_BIST3_0
Description: TX BIST Locovr
Address: 0x0304.16AC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
22	cfgTxBistErrInj	RW	0x0	Toggle bit to inject a single bit error on BIST pattern generator on TX side. Error is injected on every 0->1 transition of the register value.
23	cfgTxBistEn	RW	0x0	FGT 0 set TX BIST Locovr: -1'b1: Enable -1'b0: Disable.
31:28	cfgTxBistMode	RW	0x0	FGT lane 0 TX PRBS Pattern Selection: -UDP : 0x0 -PRBS7: 0x1 -PRBS9: 0x2 -PRBS10 : 0x3 -PRBS13 : 0x4 -PRBS15 : 0x5 -PRBS23 : 0x6 -PRBS28 : 0x7 -PRBS31 : 0x8 -QPRBS13 : 0x9 -PRBS13Q : 0xa -PRBS31Q : 0xb -SSPR : 0xc -SSPR1 : 0xd -SSPRQ : 0xe.

Name: SRDS_FGT_TX_MUTE0
Description: FGT 0 Mute TX Output
Address: 0x0304.1750
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24	cfgTxMuteOvr	RW	0x0	FGT lane 0 enable Mute TX output. 1'b1: enable 1'b0: disable.
25	cfgTxMuteOvrEn	RW	0x0	Override enable for registers with the name txdrvmute_*_locovr. 1'b1: enable 1'b0: disable.

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Name: SRDS_FGT_TX_LN_0
Description: FGT 0 TX bitflip
Address: 0x0304.1768
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24	cfgLbParRx2TxRpl	RW	0x0	This register is part of the RX to TX reverse parallel loopback enablement sequence: - To enable RX to TX reverse parallel loopback, you must set this register to 0x0. - To disable RX to TX reverse parallel loopback, you must set this register to 0x1.
30:28	cfgTxpamBitorder	RW	0x1	FGT 0 enable TX bitflip 3'b1: enable bitflip. 3'b0: disable bitflip.

Name: SRDS_FGT_BIST4_0
Description: TX PCS BIST Enable locovr
Address: 0x0304.176C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
27	cfgTxPcsBistEn	RW	0x0	FGT 0 TX PCS BIST Enable locovr -1'b1: Enable -1'b0: Disable.
28	cfgTxPcsBistOvr	RW	0x0	FGT 0 Override enable for TX PCS BIST locovr: -1'b1: override enable -1'b0: override disable.

Name: SRDS_FGT_TX_LN1_0
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0304.1830
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cfgLbParRx2TxIdl	RW	0x0	This register is part of the RX to TX reverse parallel loopback enablement sequence: - To enable RX to TX reverse parallel loopback, you must set this register to 0x0. - To disable RX to TX reverse parallel loopback, you must set this register to 0x3.

Name: SRDS_FGT_RX_EQ2_0
Description: FGT 0 RXEQ DFE Data Tap 1
Address: 0x0304.1BB0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
11:6	cfgRxEqDataTap	RW	0x0	FGT lane 0 RXEQ DFE Data Tap 1 0~63 direct decimal to binary mapping EX. 6'd63 = 6'b111111.

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Name: SRDS_FGT_RX_EQ1_0
Description: FGT 0 RX EQ High Freq Boost
Address: 0x0304.1BB8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24:19	cfgRxEqHFBoost	RW	0x0	FGT lane 0 RXEQ HF Boost 0~63 direct decimal to binary mapping EX. 6d'63 = 6'b111111.
31:25	cfgRxEqHFVgaGain	RW	0x0	FGT lane 0 RXEQ VGA Gain 0~127 direct decimal to binary mapping EX. 7'd127 = 7'b1111111.

Name: SRDS_FGT_SLOW_DIV0
Description: FGT 0 LC Slow M Counter
Address: 0x0304.4000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgSlowFbDivInt	RW	0x28	FGT lane 0 LC Slow M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgSlowFbDivFrac	RW	0x0	FGT lane 0 LC Slow K Counter 0~2^22-1 direct decimal to binary mapping.

Name: SRDS_FGT_SLOW_LCK0
Description: FGT 0 Slow K Strobe
Address: 0x0304.400C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgSlowFbDivStrb	RW	0x0	FGT lane 0 Slow K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

Name: SRDS_FGT_MED_DIV0
Description: FGT 0 LC Medium M Counter
Address: 0x0304.4100
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgMedFbDivInt	RW	0x28	FGT lane 0 LC Med M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.

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Bits	Field	Type	Reset	Description
30:9	cfgMedFbDivFrac	RW	0x0	FGT lane 0 LC Med K Counter 0~2^22-1 direct decimal to binary mapping.

Name: SRDS_FGT_MED_LCK0
Description: FGT 0 Medium K Strobe
Address: 0x0304.410C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgMedFbDivStrb	RW	0x0	FGT lane 0 Medium K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

Name: SRDS_FGT_FAST_DIV0
Description: FGT 0 LC Fast M Counter
Address: 0x0304.4200
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgFastFbDivInt	RW	0x28	FGT lane 0 LC Fast M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgFastFbDivFrac	RW	0x0	FGT lane 0 LC Fast K Counter 0~2^22-1 direct decimal to binary mapping.

Name: SRDS_FGT_FAST_LCK0
Description: FGT 0 Fast K Strobe
Address: 0x0304.420C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgFastFbDivStrb	RW	0x0	FGT lane 0 Fast K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

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Name: SRDS_FGT_IF_TX_0
Description: FGT 0 TX precode
Address: 0x0304.7808
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2	cfgTxPamPrecode	RW	0x0	FGT lane 0 TX pre code enable 1'b1: enable 1'b0: disable.
3	cfgTxPamGrayEn	RW	0x0	FGT lane 0 TX gray code enable 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_IF_RX_0
Description: FGT 0 RX graycode
Address: 0x0304.7810
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2	cfgRxPamGrayEn	RW	0x0	FGT lane 0 enable gray coding 1'b1: enable gray coding. 1'b0: diable gray coding.
3	cfgRxPamPrecode	RW	0x0	FGT lane 0 enable precoding 1'b1: enable precoding. 1'b0: diable precoding.

Name: SRDS_FGT_IF_DBG0
Description: FGT 0 Serial Internal Loopback
Address: 0x0304.781C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
1	statLbTx2Rxbuf	RO	0x0	FGT lane 0 serial internal loopback status 1'b1: serial internal loopback is enabled. 1'b0: serial internal loopback is not enabled.

Name: SRDS_FGT_TX_EQ0
Description: FGT 0 TX Equalization
Address: 0x0304.7830
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgTxEqPostTap	RW	0x0	FGT lane 0 TX post_tap_1: 0~19 direct mapping decimal to hex.
9:5	cfgTxEqPreTap	RW	0x5	FGT lane 0 TX pre tap 1 0~15 direct mapping decimal to hex.
15:10	cfgTxEqMainTap	RW	0x28	FGT lane 0 TX main tap 0~47 direct mapping decimal to hex.
18:16	cfgTxEqPreTap2	RW	0x0	FGT lane 0 TX pre_tap_2: 0~7 direct mapping decimal to hex.

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Name: SRDS_FGT_LC_MED0_1
Description: FGT 0 LC medium post divider
Address: 0x0304.8740
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31	cfgMedPostDivP5	RW	0x0	FGT lane 0 enable LC Med Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: SRDS_FGT_LC_MED1_1
Description: FGT 0 LC medium post divider
Address: 0x0304.8744
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
13:7	cfgMedPostDiv	RW	0x2	FGT lane 0 LC Med Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

Name: SRDS_FGT_LC_SLOW0_1
Description: FGT 0 LC slow post divider
Address: 0x0304.8840
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgSlowPostDivP5	RW	0x0	FGT lane0 enable LC Slow Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: SRDS_FGT_LC_SLOW1_1
Description: FGT 0 LC slow post divider
Address: 0x0304.8844
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
13:7	cfgSlowPostDiv	RW	0x0	FGT lane 0 LC Slow Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

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Name: SRDS_FGT_LC_FAST0_1
Description: FGT 0 LC fast post divider
Address: 0x0304.8994
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgFastPostDivP5	RW	0x0	FGT lane 0 enable LC Fast Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.0.5 added into the post divider value.

Name: SRDS_FGT_LC_FAST1_1
Description: FGT 0 LC fast post divider
Address: 0x0304.8998
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
27:21	cfgFastPostDiv	RW	0x2	FGT lane 0 LC Fast Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

Name: SRDS_FGT_LN0_1
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0304.9414
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
29	cfgLbParRx2TxCdr	RW	0x0	Direct top-level interface pin override for ictl_pcs_lb_cdrclk2txen_lx_nt. Enable clock loopback path from RX CDR VCO high-speed recovered clock to corresponding TX Lane.

Name: SRDS_FGT_LN1_1
Description: FGT 0 Serial Internal Loopback
Address: 0x0304.9418
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
29	cfgLbTx2RxOvr	RW	0x0	FGT lane 0 enable serial internal loopback LSB 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

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Bits	Field	Type	Reset	Description
31	cfgLbOvrEn	RW	0x0	Override enable for registers with the name lb_*_locovr (required for all loopback mode). 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

Name: SRDS_FGT_LN2_1
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0304.941C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgLbParRx2Tx	RW	0x0	Direct top-level interface pin override for ictl_pcs_lb_parrx2txtimeden_lx_nt. Enable RX parallel word to TX parallel word loopback.

Name: SRDS_FGT_LN3_1
Description: FGT 0 Serial Internal Loopback
Address: 0x0304.9420
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
25	cfgLbTx2Rxbuf	RW	0x0	FGT lane 0 enable serial internal loopback MSB 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

Name: SRDS_FGT_BIST0_1
Description: Override enable for PCS RX BIST locovr
Address: 0x0304.9424
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
26	cfgPcsRxBistOvr	RW	0x0	FGT 0 override enable for PCS RX BIST locovr: -1'b1: override enable -1'b0: override disable.

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Name: SRDS_FGT_LN5_1
Description: Set RX PRBS Pattern and FGT 0 RX/TX polarity
Address: 0x0304.9428
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	cfgRxBistMode	RW	0x0	FGT lane 0 RX PRBS Pattern Selection: -UDP : 0x0 -PRBS7: 0x1 -PRBS9: 0x2 -PRBS10 : 0x3 - PRBS13 : 0x4 - PRBS15 : 0x5 - PRBS23 : 0x6 - PRBS28 : 0x7 - PRBS31 : 0x8 - QPRBS13 : 0x9 - PRBS13Q : 0xa - PRBS31Q : 0xb - SSPR : 0xc - SSPR1 : 0xd - SSPRQ : 0xe.
6	cfgPolarityRx	RW	0x0	FGT lane 0 enable RX polarity inversion. 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.
7	cfgPolarityTx	RW	0x0	FGT lane 0 enable TX polarity inversion. 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

Name: SRDS_FGT_BIST1_1
Description: RX BIST Overflow Check
Address: 0x0304.943C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
21	cfgRxBistErrOwfl	RW	0x0	RX BIST error counter overflow status: -1'b1: overflow detected -1'b0: overflow not detected.

Name: SRDS_FGT_BIST2_1
Description: RX BIST Error Check Count
Address: 0x0304.9444
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cfgRxBistErrCnt	RW	0x0	Error counts by RX BIST pattern checker direct mapping.

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Name: SRDS_FGT_CDR_VCO0_1
Description: RX CDR VCO integral gain reg_34
Address: 0x0304.9484
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:10	cfgRxCdrGain10b	RW	0xd	RX CDR VCO integral gain_10b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x1. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
24:20	cfgRxCdrGain16b	RW	0xd	RX CDR VCO integral gain_16b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x2. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO1_1
Description: RX CDR VCO integral gain reg_35
Address: 0x0304.9488
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgRxCdrGain20b	RW	0xd	RX CDR VCO integral gain_20b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x3. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
14:10	cfgRxCdrGain32b	RW	0xd	RX CDR VCO integral gain_32b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x4. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
24:20	cfgRxCdrGain40b	RW	0xd	RX CDR VCO integral gain_40b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x5. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO2_1
Description: RX CDR VCO integral gain reg_36
Address: 0x0304.948C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgRxCdrGain64b	RW	0xd	RX CDR VCO integral gain_64b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x6. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

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Bits	Field	Type	Reset	Description
14:10	cfgRxCdrGain80b	RW	0xd	RX CDR VCO integral gain_80b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x7. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO5_1
Description: FSM current width
Address: 0x0304.9514
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
18:16	cfgRxFsmCurrWth	RW	0x0	FSM current width. Note: This register should not be overridden by users. It should be read-only to find out which RX CDR VCO intergal gain register takes effect.

Name: SRDS_FGT_CDR_VCO3_1
Description: RX CDR VCO proportional gain reg_96
Address: 0x0304.957C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24:20	cfgRxCdrGainProp	RW	0x4	RX CDR VCO proportional gain. Larger value contributes larger bandwidth: 3~12 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO4_1
Description: RX CDR VCO integral gain reg_100
Address: 0x0304.958C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17:13	cfgRxCdrGain8b	RW	0xd	RX CDR VCO integral gain_8b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x0. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_RX_LN1_1
Description: FGT 0 RX bitflip
Address: 0x0304.95B0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
26:24	cfgRxpamBitorder	RW	0x1	FGT 0 enable RX bitflip 3'b1: enable bitflip. 3'b0: disable bitflip.

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Bits	Field	Type	Reset	Description
31	cfgRxPostDivP5	RW	0x0	FGT lane 0 enable CDR Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: SRDS_FGT_RX_LN0_1
Description: FGT 0 CDR post divider and RX PCS BIST Enable locovr and Override enable for RX PCS BIST locovr
Address: 0x0304.95B4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
6:0	cfgRxPcsPostDiv	RW	0x0	FGT lane 0 CDR Post Divider direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.
18	cfgRxPcsBistEn	RW	0x0	FGT 0 RX PCS BIST Enable locovr -1'b1: enable -1'b0: disable.
19	cfgRxPcsBistOvr	RW	0x0	FGT 0 override enable for RX PCS BIST locovr: -1'b1: override enable -1'b0: override disable.

Name: SRDS_FGT_LN4_1
Description: FGT 0 Parallel Loopback
Address: 0x0304.96A4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	cfgTx2RxlbEn	RW	0x0	FGT lane 0 enable TX to RX parallel loopback 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_BIST3_1
Description: TX BIST Locovr
Address: 0x0304.96AC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
22	cfgTxBistErrInj	RW	0x0	Toggle bit to inject a single bit error on BIST pattern generator on TX side. Error is injected on every 0->1 transition of the register value.
23	cfgTxBistEn	RW	0x0	FGT 0 set TX BIST Locovr: -1'b1: Enable -1'b0: Disable.
31:28	cfgTxBistMode	RW	0x0	FGT lane 0 TX PRBS Pattern Selection: -UDP : 0x0 -PRBS7: 0x1 -PRBS9: 0x2 -PRBS10 : 0x3 -PRBS13 : 0x4 -PRBS15 : 0x5 -PRBS23 : 0x6 -PRBS28 : 0x7 -PRBS31 : 0x8 -QPRBS13 : 0x9 -PRBS13Q : 0xa -PRBS31Q : 0xb -SSPR : 0xc -SSPR1 : 0xd -SSPRQ : 0xe.

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Name: SRDS_FGT_TX_MUTE1
Description: FGT 0 Mute TX Output
Address: 0x0304.9750
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24	cfgTxMuteOvr	RW	0x0	FGT lane 0 enable Mute TX output. 1'b1: enable 1'b0: disable.
25	cfgTxMuteOvrEn	RW	0x0	Override enable for registers with the name txdrvmute_*_locovr. 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_TX_LN_1
Description: FGT 0 TX bitflip
Address: 0x0304.9768
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24	cfgLbParRx2TxRpl	RW	0x0	This register is part of the RX to TX reverse parallel loopback enablement sequence: - To enable RX to TX reverse parallel loopback, you must set this register to 0x0. - To disable RX to TX reverse parallel loopback, you must set this register to 0x1.
30:28	cfgTxpamBitorder	RW	0x1	FGT 0 enable TX bitflip 3'b1: enable bitflip. 3'b0: disable bitflip.

Name: SRDS_FGT_BIST4_1
Description: TX PCS BIST Enable locovr
Address: 0x0304.976C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
27	cfgTxPcsBistEn	RW	0x0	FGT 0 TX PCS BIST Enable locovr -1'b1: Enable -1'b0: Disable.
28	cfgTxPcsBistOvr	RW	0x0	FGT 0 Override enable for TX PCS BIST locovr: -1'b1: override enable -1'b0: override disable.

Name: SRDS_FGT_TX_LN1_1
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0304.9830
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cfgLbParRx2TxIdl	RW	0x0	This register is part of the RX to TX reverse parallel loopback enablement sequence: - To enable RX to TX reverse parallel loopback, you must set this register to 0x0. - To disable RX to TX reverse parallel loopback, you must set this register to 0x3.

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Name: SRDS_FGT_RX_EQ2_1
Description: FGT 0 RXEQ DFE Data Tap 1
Address: 0x0304.9BB0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
11:6	cfgRxEqDataTap	RW	0x0	FGT lane 0 RXEQ DFE Data Tap 1 0~63 direct decimal to binary mapping EX. 6'd63 = 6'b111111.

Name: SRDS_FGT_RX_EQ1_1
Description: FGT 0 RX EQ High Freq Boost
Address: 0x0304.9BB8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24:19	cfgRxEqHFBoost	RW	0x0	FGT lane 0 RXEQ HF Boost 0~63 direct decimal to binary mapping EX. 6d'63 = 6'b111111.
31:25	cfgRxEqHFVgaGain	RW	0x0	FGT lane 0 RXEQ VGA Gain 0~127 direct decimal to binary mapping EX. 7'd127 = 7'b1111111.

Name: SRDS_FGT_SLOW_DIV1
Description: FGT 0 LC Slow M Counter
Address: 0x0304.C000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgSlowFbDivInt	RW	0x28	FGT lane 0 LC Slow M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgSlowFbDivFrac	RW	0x0	FGT lane 0 LC Slow K Counter 0~2^22-1 direct decimal to binary mapping.

Name: SRDS_FGT_SLOW_LCK1
Description: FGT 0 Slow K Strobe
Address: 0x0304.C00C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgSlowFbDivStrb	RW	0x0	FGT lane 0 Slow K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

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Name: SRDS_FGT_MED_DIV1
Description: FGT 0 LC Medium M Counter
Address: 0x0304.C100
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgMedFbDivInt	RW	0x28	FGT lane 0 LC Med M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgMedFbDivFrac	RW	0x0	FGT lane 0 LC Med K Counter 0~2^22-1 direct decimal to binary mapping.

Name: SRDS_FGT_MED_LCK1
Description: FGT 0 Medium K Strobe
Address: 0x0304.C10C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgMedFbDivStrb	RW	0x0	FGT lane 0 Medium K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

Name: SRDS_FGT_FAST_DIV1
Description: FGT 0 LC Fast M Counter
Address: 0x0304.C200
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgFastFbDivInt	RW	0x28	FGT lane 0 LC Fast M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgFastFbDivFrac	RW	0x0	FGT lane 0 LC Fast K Counter 0~2^22-1 direct decimal to binary mapping.

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Name: SRDS_FGT_FAST_LCK1
Description: FGT 0 Fast K Strobe
Address: 0x0304.C20C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgFastFbDivStrb	RW	0x0	FGT lane 0 Fast K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

Name: SRDS_FGT_IF_TX_1
Description: FGT 0 TX precode
Address: 0x0304.F808
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2	cfgTxPamPrecode	RW	0x0	FGT lane 0 TX pre code enable 1'b1: enable 1'b0: disable.
3	cfgTxPamGrayEn	RW	0x0	FGT lane 0 TX gray code enable 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_IF_RX_1
Description: FGT 0 RX graycode
Address: 0x0304.F810
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2	cfgRxPamGrayEn	RW	0x0	FGT lane 0 enable gray coding 1'b1: enable gray coding. 1'b0: diable gray coding.
3	cfgRxPamPrecode	RW	0x0	FGT lane 0 enable precoding 1'b1: enable precoding. 1'b0: diable precoding.

Name: SRDS_FGT_IF_DBG1
Description: FGT 0 Serial Internal Loopback
Address: 0x0304.F81C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
1	statLbTx2Rxbuf	RO	0x0	FGT lane 0 serial internal loopback status 1'b1: serial internal loopback is enabled. 1'b0: serial internal loopback is not enabled.

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Name: SRDS_FGT_TX_EQ1
Description: FGT 0 TX Equalization
Address: 0x0304.F830
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgTxEqPostTap	RW	0x0	FGT lane 0 TX post_tap_1: 0~19 direct mapping decimal to hex.
9:5	cfgTxEqPreTap	RW	0x5	FGT lane 0 TX pre tap 1 0~15 direct mapping decimal to hex.
15:10	cfgTxEqMainTap	RW	0x28	FGT lane 0 TX main tap 0~47 direct mapping decimal to hex.
18:16	cfgTxEqPreTap2	RW	0x0	FGT lane 0 TX pre_tap_2: 0~7 direct mapping decimal to hex.

Name: SRDS_FGT_LC_MED0_2
Description: FGT 0 LC medium post divider
Address: 0x0305.0740
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31	cfgMedPostDivP5	RW	0x0	FGT lane 0 enable LC Med Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: SRDS_FGT_LC_MED1_2
Description: FGT 0 LC medium post divider
Address: 0x0305.0744
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
13:7	cfgMedPostDiv	RW	0x2	FGT lane 0 LC Med Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

Name: SRDS_FGT_LC_SLOW0_2
Description: FGT 0 LC slow post divider
Address: 0x0305.0840
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgSlowPostDivP5	RW	0x0	FGT lane0 enable LC Slow Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

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Name: SRDS_FGT_LC_SLOW1_2
Description: FGT 0 LC slow post divider
Address: 0x0305.0844
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
13:7	cfgSlowPostDiv	RW	0x0	FGT lane 0 LC Slow Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

Name: SRDS_FGT_LC_FAST0_2
Description: FGT 0 LC fast post divider
Address: 0x0305.0994
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgFastPostDivP5	RW	0x0	FGT lane 0 enable LC Fast Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.0.5 added into the post divider value.

Name: SRDS_FGT_LC_FAST1_2
Description: FGT 0 LC fast post divider
Address: 0x0305.0998
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
27:21	cfgFastPostDiv	RW	0x2	FGT lane 0 LC Fast Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

Name: SRDS_FGT_LN0_2
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0305.1414
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
29	cfgLbParRx2TxCdr	RW	0x0	Direct top-level interface pin override for ictl_pcs_lb_cdrclk2txen_lx_nt. Enable clock loopback path from RX CDR VCO high-speed recovered clock to corresponding TX Lane.

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Name: SRDS_FGT_LN1_2
Description: FGT 0 Serial Internal Loopback
Address: 0x0305.1418
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
29	cfgLbTx2RxOvr	RW	0x0	FGT lane 0 enable serial internal loopback LSB 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.
31	cfgLbOvrEn	RW	0x0	Override enable for registers with the name lb_*_locovr (required for all loopback mode). 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

Name: SRDS_FGT_LN2_2
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0305.141C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgLbParRx2Tx	RW	0x0	Direct top-level interface pin override for ictl_pcs_lb_parrx2txtimeden_lx_nt. Enable RX parallel word to TX parallel word loopback.

Name: SRDS_FGT_LN3_2
Description: FGT 0 Serial Internal Loopback
Address: 0x0305.1420
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
25	cfgLbTx2Rxbuf	RW	0x0	FGT lane 0 enable serial internal loopback MSB 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

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Name: SRDS_FGT_BIST0_2
Description: Override enable for PCS RX BIST locovr
Address: 0x0305.1424
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
26	cfgPcsRxBistOvr	RW	0x0	FGT 0 override enable for PCS RX BIST locovr: -1'b1: override enable -1'b0: override disable.

Name: SRDS_FGT_LN5_2
Description: Set RX PRBS Pattern and FGT 0 RX/TX polarity
Address: 0x0305.1428
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	cfgRxBistMode	RW	0x0	FGT lane 0 RX PRBS Pattern Selection: -UDP : 0x0 -PRBS7: 0x1 -PRBS9: 0x2 -PRBS10 : 0x3 - PRBS13 : 0x4 - PRBS15 : 0x5 - PRBS23 : 0x6 - PRBS28 : 0x7 - PRBS31 : 0x8 - QPRBS13 : 0x9 - PRBS13Q : 0xa - PRBS31Q : 0xb - SSPR : 0xc - SSPR1 : 0xd - SSPRQ : 0xe.
6	cfgPolarityRx	RW	0x0	FGT lane 0 enable RX polarity inversion. 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.
7	cfgPolarityTx	RW	0x0	FGT lane 0 enable TX polarity inversion. 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

Name: SRDS_FGT_BIST1_2
Description: RX BIST Overflow Check
Address: 0x0305.143C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
21	cfgRxBistErrOwfl	RW	0x0	RX BIST error counter overflow status: -1'b1: overflow detected -1'b0: overflow not detected.

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Name: SRDS_FGT_BIST2_2
Description: RX BIST Error Check Count
Address: 0x0305.1444
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cfgRxBistErrCnt	RW	0x0	Error counts by RX BIST pattern checker direct mapping.

Name: SRDS_FGT_CDR_VCO0_2
Description: RX CDR VCO integral gain reg_34
Address: 0x0305.1484
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:10	cfgRxCdrGain10b	RW	0xd	RX CDR VCO integral gain_10b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x1. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
24:20	cfgRxCdrGain16b	RW	0xd	RX CDR VCO integral gain_16b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x2. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO1_2
Description: RX CDR VCO integral gain reg_35
Address: 0x0305.1488
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgRxCdrGain20b	RW	0xd	RX CDR VCO integral gain_20b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x3. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
14:10	cfgRxCdrGain32b	RW	0xd	RX CDR VCO integral gain_32b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x4. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
24:20	cfgRxCdrGain40b	RW	0xd	RX CDR VCO integral gain_40b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x5. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

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Name: SRDS_FGT_CDR_VCO2_2
Description: RX CDR VCO integral gain reg_36
Address: 0x0305.148C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgRxCdrGain64b	RW	0xd	RX CDR VCO integral gain_64b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x6. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
14:10	cfgRxCdrGain80b	RW	0xd	RX CDR VCO integral gain_80b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x7. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO5_2
Description: FSM current width
Address: 0x0305.1514
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
18:16	cfgRxFsmCurrWth	RW	0x0	FSM current width. Note: This register should not be overridden by users. It should be read-only to find out which RX CDR VCO intergal gain register takes effect.

Name: SRDS_FGT_CDR_VCO3_2
Description: RX CDR VCO proportional gain reg_96
Address: 0x0305.157C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24:20	cfgRxCdrGainProp	RW	0x4	RX CDR VCO proportional gain. Larger value contributes larger bandwidth: 3~12 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO4_2
Description: RX CDR VCO integral gain reg_100
Address: 0x0305.158C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17:13	cfgRxCdrGain8b	RW	0xd	RX CDR VCO integral gain_8b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x0. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

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Name: **SRDS_FGT_RX_LN1_2**
Description: **FGT 0 RX bitflip**
Address: **0x0305.15B0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
26:24	cfgRxpamBitorder	RW	0x1	FGT 0 enable RX bitflip 3'b1: enable bitflip. 3'b0: disable bitflip.
31	cfgRxPostDivP5	RW	0x0	FGT lane 0 enable CDR Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: **SRDS_FGT_RX_LN0_2**
Description: **FGT 0 CDR post divider and RX PCS BIST Enable locovr and Override enable for RX PCS BIST locovr**
Address: **0x0305.15B4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
6:0	cfgRxPcsPostDiv	RW	0x0	FGT lane 0 CDR Post Divider direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.
18	cfgRxPcsBistEn	RW	0x0	FGT 0 RX PCS BIST Enable locovr -1'b1: enable -1'b0: disable.
19	cfgRxPcsBistOvr	RW	0x0	FGT 0 override enable for RX PCS BIST locovr: -1'b1: override enable -1'b0: override disable.

Name: **SRDS_FGT_LN4_2**
Description: **FGT 0 Parallel Loopback**
Address: **0x0305.16A4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
8	cfgTx2RxlbEn	RW	0x0	FGT lane 0 enable TX to RX parallel loopback 1'b1: enable 1'b0: disable.

Name: **SRDS_FGT_BIST3_2**
Description: **TX BIST Locovr**
Address: **0x0305.16AC**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
22	cfgTxBistErrInj	RW	0x0	Toggle bit to inject a single bit error on BIST pattern generator on TX side. Error is injected on every 0->1 transition of the register value.
23	cfgTxBistEn	RW	0x0	FGT 0 set TX BIST Locovr: -1'b1: Enable -1'b0: Disable.

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Bits	Field	Type	Reset	Description
31:28	cfgTxBistMode	RW	0x0	FGT lane 0 TX PRBS Pattern Selection: -UDP : 0x0 -PRBS7: 0x1 -PRBS9: 0x2 -PRBS10 : 0x3 -PRBS13 : 0x4 -PRBS15 : 0x5 -PRBS23 : 0x6 -PRBS28 : 0x7 -PRBS31 : 0x8 -QPRBS13 : 0x9 -PRBS13Q : 0xa -PRBS31Q : 0xb -SSPR : 0xc -SSPR1 : 0xd -SSPRQ : 0xe.

Name: SRDS_FGT_TX_MUTE2
Description: FGT 0 Mute TX Output
Address: 0x0305.1750
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24	cfgTxMuteOvr	RW	0x0	FGT lane 0 enable Mute TX output. 1'b1: enable 1'b0: disable.
25	cfgTxMuteOvrEn	RW	0x0	Override enable for registers with the name txdrvmute_*_locovr. 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_TX_LN_2
Description: FGT 0 TX bitflip
Address: 0x0305.1768
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24	cfgLbParRx2TxRpl	RW	0x0	This register is part of the RX to TX reverse parallel loopback enablement sequence: - To enable RX to TX reverse parallel loopback, you must set this register to 0x0. - To disable RX to TX reverse parallel loopback, you must set this register to 0x1.
30:28	cfgTxpamBitorder	RW	0x1	FGT 0 enable TX bitflip 3'b1: enable bitflip. 3'b0: disable bitflip.

Name: SRDS_FGT_BIST4_2
Description: TX PCS BIST Enable locovr
Address: 0x0305.176C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
27	cfgTxPcsBistEn	RW	0x0	FGT 0 TX PCS BIST Enable locovr -1'b1: Enable -1'b0: Disable.
28	cfgTxPcsBistOvr	RW	0x0	FGT 0 Override enable for TX PCS BIST locovr: -1'b1: override enable -1'b0: override disable.

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Name: SRDS_FGT_TX_LN1_2
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0305.1830
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cfgLbParRx2TxIdl	RW	0x0	This register is part of the RX to TX reverse parallel loopback enablement sequence: - To enable RX to TX reverse parallel loopback, you must set this register to 0x0. - To disable RX to TX reverse parallel loopback, you must set this register to 0x3.

Name: SRDS_FGT_RX_EQ2_2
Description: FGT 0 RXEQ DFE Data Tap 1
Address: 0x0305.1BB0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
11:6	cfgRxEqDataTap	RW	0x0	FGT lane 0 RXEQ DFE Data Tap 1 0~63 direct decimal to binary mapping EX. 6'd63 = 6'b111111.

Name: SRDS_FGT_RX_EQ1_2
Description: FGT 0 RX EQ High Freq Boost
Address: 0x0305.1BB8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24:19	cfgRxEqHFBoost	RW	0x0	FGT lane 0 RXEQ HF Boost 0~63 direct decimal to binary mapping EX. 6d'63 = 6'b111111.
31:25	cfgRxEqHFVgaGain	RW	0x0	FGT lane 0 RXEQ VGA Gain 0~127 direct decimal to binary mapping EX. 7'd127 = 7'b1111111.

Name: SRDS_FGT_SLOW_DIV2
Description: FGT 0 LC Slow M Counter
Address: 0x0305.4000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgSlowFbDivInt	RW	0x28	FGT lane 0 LC Slow M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgSlowFbDivFrac	RW	0x0	FGT lane 0 LC Slow K Counter 0~2^22-1 direct decimal to binary mapping.

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Name: SRDS_FGT_SLOW_LCK2
Description: FGT 0 Slow K Strobe
Address: 0x0305.400C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgSlowFbDivStrb	RW	0x0	FGT lane 0 Slow K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

Name: SRDS_FGT_MED_DIV2
Description: FGT 0 LC Medium M Counter
Address: 0x0305.4100
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgMedFbDivInt	RW	0x28	FGT lane 0 LC Med M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgMedFbDivFrac	RW	0x0	FGT lane 0 LC Med K Counter 0~2^22-1 direct decimal to binary mapping.

Name: SRDS_FGT_MED_LCK2
Description: FGT 0 Medium K Strobe
Address: 0x0305.410C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgMedFbDivStrb	RW	0x0	FGT lane 0 Medium K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

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Name: SRDS_FGT_FAST_DIV2
Description: FGT 0 LC Fast M Counter
Address: 0x0305.4200
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgFastFbDivInt	RW	0x28	FGT lane 0 LC Fast M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgFastFbDivFrac	RW	0x0	FGT lane 0 LC Fast K Counter 0~2^22-1 direct decimal to binary mapping.

Name: SRDS_FGT_FAST_LCK2
Description: FGT 0 Fast K Strobe
Address: 0x0305.420C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgFastFbDivStrb	RW	0x0	FGT lane 0 Fast K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

Name: SRDS_FGT_IF_TX_2
Description: FGT 0 TX precode
Address: 0x0305.7808
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2	cfgTxPamPrecode	RW	0x0	FGT lane 0 TX pre code enable 1'b1: enable 1'b0: disable.
3	cfgTxPamGrayEn	RW	0x0	FGT lane 0 TX gray code enable 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_IF_RX_2
Description: FGT 0 RX graycode
Address: 0x0305.7810
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2	cfgRxPamGrayEn	RW	0x0	FGT lane 0 enable gray coding 1'b1: enable gray coding. 1'b0: diable gray coding.
3	cfgRxPamPrecode	RW	0x0	FGT lane 0 enable precoding 1'b1: enable precoding. 1'b0: diable precoding.

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Name: SRDS_FGT_IF_DBG2
Description: FGT 0 Serial Internal Loopback
Address: 0x0305.781C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
1	statLbTx2Rxbuf	RO	0x0	FGT lane 0 serial internal loopback status 1'b1: serial internal loopback is enabled. 1'b0: serial internal loopback is not enabled.

Name: SRDS_FGT_TX_EQ2
Description: FGT 0 TX Equalization
Address: 0x0305.7830
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgTxEqPostTap	RW	0x0	FGT lane 0 TX post_tap_1: 0~19 direct mapping decimal to hex.
9:5	cfgTxEqPreTap	RW	0x5	FGT lane 0 TX pre tap 1 0~15 direct mapping decimal to hex.
15:10	cfgTxEqMainTap	RW	0x28	FGT lane 0 TX main tap 0~47 direct mapping decimal to hex.
18:16	cfgTxEqPreTap2	RW	0x0	FGT lane 0 TX pre_tap_2: 0~7 direct mapping decimal to hex.

Name: SRDS_FGT_LC_MED0_3
Description: FGT 0 LC medium post divider
Address: 0x0305.8740
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31	cfgMedPostDivP5	RW	0x0	FGT lane 0 enable LC Med Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: SRDS_FGT_LC_MED1_3
Description: FGT 0 LC medium post divider
Address: 0x0305.8744
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
13:7	cfgMedPostDiv	RW	0x2	FGT lane 0 LC Med Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

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Name: SRDS_FGT_LC_SLOW0_3
Description: FGT 0 LC slow post divider
Address: 0x0305.8840
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgSlowPostDivP5	RW	0x0	FGT lane0 enable LC Slow Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: SRDS_FGT_LC_SLOW1_3
Description: FGT 0 LC slow post divider
Address: 0x0305.8844
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
13:7	cfgSlowPostDiv	RW	0x0	FGT lane 0 LC Slow Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

Name: SRDS_FGT_LC_FAST0_3
Description: FGT 0 LC fast post divider
Address: 0x0305.8994
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgFastPostDivP5	RW	0x0	FGT lane 0 enable LC Fast Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.0.5 added into the post divider value.

Name: SRDS_FGT_LC_FAST1_3
Description: FGT 0 LC fast post divider
Address: 0x0305.8998
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
27:21	cfgFastPostDiv	RW	0x2	FGT lane 0 LC Fast Post Divider range 12 to 139 direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.

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Name: SRDS_FGT_LN0_3
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0305.9414
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
29	cfgLbParRx2TxCdr	RW	0x0	Direct top-level interface pin override for ictl_pcs_lb_cdrclk2txen_lx_nt. Enable clock loopback path from RX CDR VCO high-speed recovered clock to corresponding TX Lane.

Name: SRDS_FGT_LN1_3
Description: FGT 0 Serial Internal Loopback
Address: 0x0305.9418
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
29	cfgLbTx2RxOvr	RW	0x0	FGT lane 0 enable serial internal loopback LSB 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.
31	cfgLbOvrEn	RW	0x0	Override enable for registers with the name lb_*_locovr (required for all loopback mode). 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

Name: SRDS_FGT_LN2_3
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0305.941C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
30	cfgLbParRx2Tx	RW	0x0	Direct top-level interface pin override for ictl_pcs_lb_parrx2txtimeden_lx_nt. Enable RX parallel word to TX parallel word loopback.

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Name: SRDS_FGT_LN3_3
Description: FGT 0 Serial Internal Loopback
Address: 0x0305.9420
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
25	cfgLbTx2Rxbuf	RW	0x0	FGT lane 0 enable serial internal loopback MSB 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

Name: SRDS_FGT_BIST0_3
Description: Override enable for PCS RX BIST locovr
Address: 0x0305.9424
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
26	cfgPcsRxBistOvr	RW	0x0	FGT 0 override enable for PCS RX BIST locovr: -1'b1: override enable -1'b0: override disable.

Name: SRDS_FGT_LN5_3
Description: Set RX PRBS Pattern and FGT 0 RX/TX polarity
Address: 0x0305.9428
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	cfgRxBistMode	RW	0x0	FGT lane 0 RX PRBS Pattern Selection: -UDP : 0x0 -PRBS7: 0x1 -PRBS9: 0x2 -PRBS10 : 0x3 - PRBS13 : 0x4 - PRBS15 : 0x5 - PRBS23 : 0x6 - PRBS28 : 0x7 - PRBS31 : 0x8 - QPRBS13 : 0x9 - PRBS13Q : 0xa - PRBS31Q : 0xb - SSPR : 0xc - SSPR1 : 0xd - SSPRQ : 0xe.
6	cfgPolarityRx	RW	0x0	FGT lane 0 enable RX polarity inversion. 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.
7	cfgPolarityTx	RW	0x0	FGT lane 0 enable TX polarity inversion. 1'b1: enable 1'b0: disable Note : The sequence is valid only when RX manual tuning is used (RX auto adaptation is bypassed). If RX auto adaptation is used, refer to 0x9003c[15:0], 0x9003c[31:16], 0x90040[31:16], and 0x90040[15:0] registers for FGT attribute access method.

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Name: SRDS_FGT_BIST1_3
Description: RX BIST Overflow Check
Address: 0x0305.943C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
21	cfgRxBistErrOwfl	RW	0x0	RX BIST error counter overflow status: -1'b1: overflow detected -1'b0: overflow not detected.

Name: SRDS_FGT_BIST2_3
Description: RX BIST Error Check Count
Address: 0x0305.9444
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cfgRxBistErrCnt	RW	0x0	Error counts by RX BIST pattern checker direct mapping.

Name: SRDS_FGT_CDR_VCO0_3
Description: RX CDR VCO integral gain reg_34
Address: 0x0305.9484
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:10	cfgRxCdrGain10b	RW	0xd	RX CDR VCO integral gain_10b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x1. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
24:20	cfgRxCdrGain16b	RW	0xd	RX CDR VCO integral gain_16b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x2. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO1_3
Description: RX CDR VCO integral gain reg_35
Address: 0x0305.9488
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgRxCdrGain20b	RW	0xd	RX CDR VCO integral gain_20b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x3. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

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Bits	Field	Type	Reset	Description
14:10	cfgRxCdrGain32b	RW	0xd	RX CDR VCO integral gain_32b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x4. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
24:20	cfgRxCdrGain40b	RW	0xd	RX CDR VCO integral gain_40b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x5. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO2_3
Description: RX CDR VCO integral gain reg_36
Address: 0x0305.948C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgRxCdrGain64b	RW	0xd	RX CDR VCO integral gain_64b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x6. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.
14:10	cfgRxCdrGain80b	RW	0xd	RX CDR VCO integral gain_80b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x7. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_CDR_VCO5_3
Description: FSM current width
Address: 0x0305.9514
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
18:16	cfgRxFsmCurrWth	RW	0x0	FSM current width. Note: This register should not be overridden by users. It should be read-only to find out which RX CDR VCO intergal gain register takes effect.

Name: SRDS_FGT_CDR_VCO3_3
Description: RX CDR VCO proportional gain reg_96
Address: 0x0305.957C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24:20	cfgRxCdrGainProp	RW	0x4	RX CDR VCO proportional gain. Larger value contributes larger bandwidth: 3~12 direct mapping decimal to hex.

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Name: SRDS_FGT_CDR_VCO4_3
Description: RX CDR VCO integral gain reg_100
Address: 0x0305.958C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17:13	cfgRxCdrGain8b	RW	0xd	RX CDR VCO integral gain_8b. This register only takes effect when cfg_rxfsm_main_cur_width is 0x0. Larger value contributes larger bandwidth: 10~14 direct mapping decimal to hex.

Name: SRDS_FGT_RX_LN1_3
Description: FGT 0 RX bitflip
Address: 0x0305.95B0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
26:24	cfgRxpamBitorder	RW	0x1	FGT 0 enable RX bitflip 3'b1: enable bitflip. 3'b0: disable bitflip.
31	cfgRxPostDivP5	RW	0x0	FGT lane 0 enable CDR Post Divider Fractional 1'b1: 0.5 added into the post divider value. 1'b0: not adding 0.5 into the post divider value.

Name: SRDS_FGT_RX_LN0_3
Description: FGT 0 CDR post divider and RX PCS BIST Enable locovr and Override enable for RX PCS BIST locovr
Address: 0x0305.95B4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
6:0	cfgRxPcsPostDiv	RW	0x0	FGT lane 0 CDR Post Divider direct map 7'h0 -> 7'd12 7'h7F -> 7'd139.
18	cfgRxPcsBistEn	RW	0x0	FGT 0 RX PCS BIST Enable locovr -1'b1: enable -1'b0: disable.
19	cfgRxPcsBistOvr	RW	0x0	FGT 0 override enable for RX PCS BIST locovr: -1'b1: override enable -1'b0: override disable.

Name: SRDS_FGT_LN4_3
Description: FGT 0 Parallel Loopback
Address: 0x0305.96A4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	cfgTx2RxlbEn	RW	0x0	FGT lane 0 enable TX to RX parallel loopback 1'b1: enable 1'b0: disable.

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Name: SRDS_FGT_BIST3_3
Description: TX BIST Locovr
Address: 0x0305.96AC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
22	cfgTxBistErrInj	RW	0x0	Toggle bit to inject a single bit error on BIST pattern generator on TX side. Error is injected on every 0->1 transition of the register value.
23	cfgTxBistEn	RW	0x0	FGT 0 set TX BIST Locovr: -1'b1: Enable -1'b0: Disable.
31:28	cfgTxBistMode	RW	0x0	FGT lane 0 TX PRBS Pattern Selection: -UDP : 0x0 -PRBS7: 0x1 -PRBS9: 0x2 -PRBS10 : 0x3 -PRBS13 : 0x4 -PRBS15 : 0x5 -PRBS23 : 0x6 -PRBS28 : 0x7 -PRBS31 : 0x8 -QPRBS13 : 0x9 -PRBS13Q : 0xa -PRBS31Q : 0xb -SSPR : 0xc -SSPR1 : 0xd -SSPRQ : 0xe.

Name: SRDS_FGT_TX_MUTE3
Description: FGT 0 Mute TX Output
Address: 0x0305.9750
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24	cfgTxMuteOvr	RW	0x0	FGT lane 0 enable Mute TX output. 1'b1: enable 1'b0: disable.
25	cfgTxMuteOvrEn	RW	0x0	Override enable for registers with the name txdrvmute_*_locovr. 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_TX_LN_3
Description: FGT 0 TX bitflip
Address: 0x0305.9768
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24	cfgLbParRx2TxRpl	RW	0x0	This register is part of the RX to TX reverse parallel loopback enablement sequence: - To enable RX to TX reverse parallel loopback, you must set this register to 0x0. - To disable RX to TX reverse parallel loopback, you must set this register to 0x1.
30:28	cfgTxpamBitorder	RW	0x1	FGT 0 enable TX bitflip 3'b1: enable bitflip. 3'b0: disable bitflip.

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Name: SRDS_FGT_BIST4_3
Description: TX PCS BIST Enable locovr
Address: 0x0305.976C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
27	cfgTxPcsBistEn	RW	0x0	FGT 0 TX PCS BIST Enable locovr -1'b1: Enable -1'b0: Disable.
28	cfgTxPcsBistOvr	RW	0x0	FGT 0 Override enable for TX PCS BIST locovr: -1'b1: override enable -1'b0: override disable.

Name: SRDS_FGT_TX_LN1_3
Description: FGT 0 Reversed Parallel Loopback
Address: 0x0305.9830
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cfgLbParRx2TxIdl	RW	0x0	This register is part of the RX to TX reverse parallel loopback enablement sequence: - To enable RX to TX reverse parallel loopback, you must set this register to 0x0. - To disable RX to TX reverse parallel loopback, you must set this register to 0x3.

Name: SRDS_FGT_RX_EQ2_3
Description: FGT 0 RXEQ DFE Data Tap 1
Address: 0x0305.9BB0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
11:6	cfgRxEqDataTap	RW	0x0	FGT lane 0 RXEQ DFE Data Tap 1 0~63 direct decimal to binary mapping EX. 6'd63 = 6'b111111.

Name: SRDS_FGT_RX_EQ1_3
Description: FGT 0 RX EQ High Freq Boost
Address: 0x0305.9BB8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
24:19	cfgRxEqHFBoost	RW	0x0	FGT lane 0 RXEQ HF Boost 0~63 direct decimal to binary mapping EX. 6d'63 = 6'b111111.
31:25	cfgRxEqHFVgaGain	RW	0x0	FGT lane 0 RXEQ VGA Gain 0~127 direct decimal to binary mapping EX. 7'd127 = 7'b1111111.

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Name: SRDS_FGT_SLOW_DIV3
Description: FGT 0 LC Slow M Counter
Address: 0x0305.C000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgSlowFbDivInt	RW	0x28	FGT lane 0 LC Slow M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgSlowFbDivFrac	RW	0x0	FGT lane 0 LC Slow K Counter 0~2^22-1 direct decimal to binary mapping.

Name: SRDS_FGT_SLOW_LCK3
Description: FGT 0 Slow K Strobe
Address: 0x0305.C00C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgSlowFbDivStrb	RW	0x0	FGT lane 0 Slow K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

Name: SRDS_FGT_MED_DIV3
Description: FGT 0 LC Medium M Counter
Address: 0x0305.C100
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgMedFbDivInt	RW	0x28	FGT lane 0 LC Med M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgMedFbDivFrac	RW	0x0	FGT lane 0 LC Med K Counter 0~2^22-1 direct decimal to binary mapping.

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Name: SRDS_FGT_MED_LCK3
Description: FGT 0 Medium K Strobe
Address: 0x0305.C10C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgMedFbDivStrb	RW	0x0	FGT lane 0 Medium K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

Name: SRDS_FGT_FAST_DIV3
Description: FGT 0 LC Fast M Counter
Address: 0x0305.C200
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8:0	cfgFastFbDivInt	RW	0x28	FGT lane 0 LC Fast M Counter 12~267 direct decimal to binary mapping EX. 9'd267 = 9'b100001011.
30:9	cfgFastFbDivFrac	RW	0x0	FGT lane 0 LC Fast K Counter 0~2^22-1 direct decimal to binary mapping.

Name: SRDS_FGT_FAST_LCK3
Description: FGT 0 Fast K Strobe
Address: 0x0305.C20C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17	cfgFastFbDivStrb	RW	0x0	FGT lane 0 Fast K Strobe Certain applications that use the LC PLL in fractional mode require IP to monitor PPM and adjust the PLL frequency up or down slightly by modifying the fractional counter K in real time through reconfiguration. Modification of the K counter is accomplished by writing to the 22 bit K value register of the active LC PLL followed by writing first a 1 and then a 0 to the associated strobe register.

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Name: SRDS_FGT_IF_TX_3
Description: FGT 0 TX precode
Address: 0x0305.F808
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2	cfgTxPamPrecode	RW	0x0	FGT lane 0 TX pre code enable 1'b1: enable 1'b0: disable.
3	cfgTxPamGrayEn	RW	0x0	FGT lane 0 TX gray code enable 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_IF_RX_3
Description: FGT 0 RX graycode
Address: 0x0305.F810
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2	cfgRxPamGrayEn	RW	0x0	FGT lane 0 enable gray coding 1'b1: enable gray coding. 1'b0: diable gray coding.
3	cfgRxPamPrecode	RW	0x0	FGT lane 0 enable precoding 1'b1: enable precoding. 1'b0: diable precoding.

Name: SRDS_FGT_IF_DBG3
Description: FGT 0 Serial Internal Loopback
Address: 0x0305.F81C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
1	statLbTx2Rxbuf	RO	0x0	FGT lane 0 serial internal loopback status 1'b1: serial internal loopback is enabled. 1'b0: serial internal loopback is not enabled.

Name: SRDS_FGT_TX_EQ3
Description: FGT 0 TX Equalization
Address: 0x0305.F830
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	cfgTxEqPostTap	RW	0x0	FGT lane 0 TX post_tap_1: 0~19 direct mapping decimal to hex.
9:5	cfgTxEqPreTap	RW	0x5	FGT lane 0 TX pre tap 1 0~15 direct mapping decimal to hex.
15:10	cfgTxEqMainTap	RW	0x28	FGT lane 0 TX main tap 0~47 direct mapping decimal to hex.
18:16	cfgTxEqPreTap2	RW	0x0	FGT lane 0 TX pre_tap_2: 0~7 direct mapping decimal to hex.

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Name: SRDS_FGT_CAR_CLKEN0
Description: Lane CAR clock enable
Address: 0x0306.2000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
16	cfgDspRxClken	RW	0x1	SERDES DSP RX Clock Enable 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_CAR_SWRST0
Description: Lane CAR software reset
Address: 0x0306.2004
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
12	cfgDspRxSwrstn	RW	0x1	SERDES DSP RX Soft reset bar 1'b1: not reset 1'b0: reset.

Name: SRDS_FGT_CAR_CLKEN1
Description: Lane CAR clock enable
Address: 0x0306.A000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
16	cfgDspRxClken	RW	0x1	SERDES DSP RX Clock Enable 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_CAR_SWRST1
Description: Lane CAR software reset
Address: 0x0306.A004
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
12	cfgDspRxSwrstn	RW	0x1	SERDES DSP RX Soft reset bar 1'b1: not reset 1'b0: reset.

Name: SRDS_FGT_CAR_CLKEN2
Description: Lane CAR clock enable
Address: 0x0307.2000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
16	cfgDspRxClken	RW	0x1	SERDES DSP RX Clock Enable 1'b1: enable 1'b0: disable.

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Name: SRDS_FGT_CAR_SWRST2
Description: Lane CAR software reset
Address: 0x0307.2004
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
12	cfgDspRxSwrstn	RW	0x1	SERDES DSP RX Soft reset bar 1'b1: not reset 1'b0: reset.

Name: SRDS_FGT_CAR_CLKEN3
Description: Lane CAR clock enable
Address: 0x0307.A000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
16	cfgDspRxClken	RW	0x1	SERDES DSP RX Clock Enable 1'b1: enable 1'b0: disable.

Name: SRDS_FGT_CAR_SWRST3
Description: Lane CAR software reset
Address: 0x0307.A004
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
12	cfgDspRxSwrstn	RW	0x1	SERDES DSP RX Soft reset bar 1'b1: not reset 1'b0: reset.

Name: SRDS_FGT_LINK_CPI
Description: FGT 0 Serial Loopback, PRBS select. BER test control and polarity change
Address: 0x0309.003C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	cfgLinkCpiCmd	RW	0x0	Refer to F-tile Architecture and PMA and FEC Direct PHY IP User Guide, FGT Attribute Access Method Example topic for usage.
31:16	cfgLinkCpiData	RW	0x0	

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Name: **SRDS_FGT_PHY_CPI**
Description: **FGT 0 Serial Loopback, PRBS select. BER test control and polarity change**
Address: **0x0309.0040**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
15:0	cfgPhyCpiCmd	RW	0x0	Refer to F-tile Architecture and PMA and FEC Direct PHY IP User Guide, FGT Attribute Access Method Example topic for usage.
31:16	cfgPhyCpiData	RW	0x0	

Name: **SRDS_FGT_DL_CTRL_L0**
Description: **Deterministic Latency Control**
Address: **0x030F.0010**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
17:0	cfgRxLatBitAsync	RW	0x0	FGT lane 0 Rx Deterministic Latency counter value for async pulse generation.
20:18	cfgSelRxbitAdder	RW	0x0	FGT lane 0 Rx Deterministic Latency counter increment, which is equal to data bus width.
31:21	cfgCtrlReserved	RW	0x0	Reserved.

Name: **SRDS_FGT_DL_CTRL_L1**
Description: **Deterministic Latency Control**
Address: **0x030F.0018**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
17:0	cfgRxLatBitAsync	RW	0x0	FGT lane 1 Rx Deterministic Latency counter value for async pulse generation.
20:18	cfgSelRxbitAdder	RW	0x0	FGT lane 1 Rx Deterministic Latency counter increment, which is equal to data bus width.
31:21	cfgCtrlReserved	RW	0x0	Reserved.

Name: **SRDS_FGT_DL_CTRL_L2**
Description: **Deterministic Latency Control**
Address: **0x030F.0020**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
17:0	cfgRxLatBitAsync	RW	0x0	FGT lane 2 Rx Deterministic Latency counter value for async pulse generation.

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Bits	Field	Type	Reset	Description
20:18	cfgSelRxbitAdder	RW	0x0	FGT lane 2 Rx Deterministic Latency counter increment, which is equal to data bus width.
31:21	cfgCtrlReserved	RW	0x0	Reserved.

Name: SRDS_FGT_DL_CTRL_L3
Description: Deterministic Latency Control
Address: 0x030F.0028
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
17:0	cfgRxLatBitAsync	RW	0x0	FGT lane 3 Rx Deterministic Latency counter value for async pulse generation.
20:18	cfgSelRxbitAdder	RW	0x0	FGT lane 3 Rx Deterministic Latency counter increment, which is equal to data bus width.
31:21	cfgCtrlReserved	RW	0x0	Reserved.

Name: SRDS_FGT_NUM
Description: Quad and Lane Number
Address: 0x030F.FFFC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
1:0	cfgLaneNumber	RO	0x0	Returns FGT Lane number: - Lane Number: - 2b11-Lane3 - 2b10-Lane2 - 2b01-Lane1 - 2b00-Lane0 Note: You can use 0xFFFFC register to read out all the 16 FGT PMA physical quad and lane information. You do not need to add offset 0x8000h to read out lane 1, lane 2, lane 3 information.
3:2	cfgQuadNumber	RO	0x0	Returns FGT Quad number: - Lane Number: - 2b11-Lane3 - 2b10-Lane2 - 2b01-Lane1 - 2b00-Lane0 Note: You can use 0xFFFFC register to read out all the 16 FGT PMA physical quad and lane information. You do not need to add offset 0x8000h to read out lane 1, lane 2, lane 3 information.

Name: SRDS_ETH_GUI_OPT
Description: Device and IP variant. Includes the following fields: - device_name - tile_name - eth_rate - anlt_enable - modulation_type - rsfec_type - ptp_enable - flow_control_mode - client_intf - xcvr_type - num_lanes
Address: 0x0320.0100
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
1:0	deviceName	RO	0x1	Device name 0:Reserved, 1:Intel Agilex.

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Bits	Field	Type	Reset	Description
4:2	tileName	RO	0x3	Tile name 0-2:Reserved,3:F-tile, 4-7:Reserved.
7:5	ethRate	RO	0x1	Ethernet rate 0:10G, 1:25G, 2:40G, 3:50G, 4:100G, 5:200G, 6:400G, 7:Reserved.
8	anltEnable	RO	0x0	Enable AN/LT 0:AN/LT is disabled, 1:AN/LT is enabled.
9	modulationType	RO	0x0	Modulation type 0:NRZ, 1:PAM-4.
12:10	rsfecType	RO	0x0	FEC mode 0:None 1:Firecode (Clause 74), 2:RS(528,514) (Clause 91), 3:RS(544,514) (Clause 134), 4:Ethernet Technology Consortium RS(272,258), 5-7:Reserved.
13	ptpEnable	RO	0x0	1588 PTP Enabled? 0:1588 PTP is disabled, 1:1588 PTP is enabled.
16:14	flowControlMode	RO	0x0	Flow control mode 0:Disabled, 1:SFC, 2:SFC with no XOFF, 3:PFC, 4:PFC with no XOFF, 5:Both SFC and PFC, 6:Both SFC and PFC wit no XOFF, 7:RX SFC Disabled.
19:17	clientIntf	RO	0x0	Client interface 0: MAC segemented interface, 1: MAC Avalon Streaming interface, 2: PCS interface, 3: OTN interface, 4: FlexE interface, 5-7:Reserved.
20	xcvrType	RO	0x0	PMA type 0: FGT, 1: FHT.
24:21	numLanes	RO	0x1	Number of lanes 1: 1 lane, 2: 2 lanes, 4: 4 lanes, 8: 8 lanes.

Name: SRDS_ETH_SCRATCH
Description: Scratch Register
Address: 0x0320.0104
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	scratch	RW	0x0	32 bits of scratch register space for testing.

Name: SRDS_ETH_RST
Description: IP Soft Reset Register Includes the following: - eio_sys_rst (Soft Global Reset) - soft_tx_rst (Soft TX Reset) - soft_rx_rst (Soft RX Reset) - tx_clear_alarm (Clear TX Reset Alarm) - rx_clear_alarm (Clear RX Reset Alarm)
Address: 0x0320.0108
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	eioSysRst	RW	0x0	Soft Global Reset 1: Global Reset (Reset datapath and all CSR registers).
1	softTxRst	RW	0x0	Soft TX Reset 1:Reset the TX datapath.
2	softRxRst	RW	0x0	Soft RX Reset 1:Reset the RX datapath.

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Bits	Field	Type	Reset	Description
23:16	txClearAlarm	RW	0x0	Clear TX Reset Alarm 1:Clear TX Reset Alarm indication for transceiver lane.
31:24	rxClearAlarm	RW	0x0	Clear RX reset alarm 1:Clear RX reset alarm indication for transceiver lane.

Name: SRDS_ETH_RST_STAT
Description: IP Reset Status Register. Not sticky bit. Includes the following fields: - rst_ack_n - tx_rst_ack_n - rx_rst_ack_n - tx_lane_current_state - rx_lane_current_state - tx_alarm - rx_alarm
Address: 0x0320.010C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rstAckN	RO	0x0	Global reset acknowledge. Active low (hard reset or the global soft reset) 1:Acknowledge the reset completed.
1	txRstAckN	RO	0x0	TX Reset acknowledge. Active low (either the hard tx reset or the soft_tx_rst) 1:Acknowledge the reset completed.
2	rxRstAckN	RO	0x0	RX Reset acknowledge. Active high (either the hard rx reset or the soft_rx_rst) 1:acknowledge the reset completed.
10:8	txLaneCurrState	RO	0x0	TX Lane Current State Current reset state for TX PMA and datapath: - [0]: Fully in reset state - [1]: Fully out of reset state (operational) - Reset control initialized (system clock stable) If not fully in reset or out of reset, TX reset is currently transitioning.
14:12	rxLaneCurrState	RO	0x0	RX Lane Current State Current reset state for RX PMAs and datapath: - [0]: Fully in reset state - [1]: Fully out of reset state (operational) - Reset control initialized (system clock stable) If not fully in reset or out of reset, RX reset is currently transitioning.
23:16	txAlarm	RO	0x0	TX Reset Alarm Indicates an unexpected loss-off-PLL lock caused a change in the TX reset state.
31:24	rxAlarm	RO	0x0	RX Reset Alarm Indicates an unexpected loss-off-PLL lock caused a change in the RX reset state.

Name: SRDS_ETH_TX_PLL_LCKD
Description: TX PLL locked TX PLL used by the corresponding physical lane is locked
Address: 0x0320.0110
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	txPllLocked	RO	0x0	TX PLL Locked 1:TX PLL used by this lane physical lane is locked.

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Name: **SRDS_ETH_RX_PLL_LCKD**
Description: **RX CDR PLL locked One field per physical lane to indicate whether or not each CDR successfully locks to the incoming data stream. FHT and FGT PMAs share the same register bits**
Address: **0x0320.0114**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	eioFreqLock	RO	0x0	CDR PLL locked 1:Corresponding physical lane's CDR has locked to reference.

Name: **SRDS_ETH_PCS_STAT**
Description: **PCS status Includes the following fields: - dskew_status - dskew_chng - tx_lanes_stable - rx_pcs_ready - kr_mode - kr_fec_mode**
Address: **0x0320.0118**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
0	dskewStatus	RO	-	Deskewed status 1:RX EMIB is deskewed 0:RX EMIB is not currently deskewed. Note: There is some latency between this status bit and the actual state - Not valid for single lane channels (10G/25G) without PTP - Note: This bit is not a sticky bit. Intel recommends using a soft logic replacement for this bit that can be made sticky based on the deskew_done port.
1	dskewChng	RO	-	Change in deskewed status 1:RX EMIB went from deskewed to not deskewed, or from not deskewed to deskewed - Not valid for single lane channels (10G/25G) without PTP - This bit is sticky - use pcs_control.clr_dskew_chng to set this bit back to 0 - Resetting the RX datapath, or the entire core will also clear the bit.
2	txLanesStable	RO	0x0	TX Lanes Stable 1:TX datapath is ready.
3	rxPcsReady	RO	0x0	RX PCS Ready 1:RX datapath is ready.
4	krMode	RO	0x0	AN/LT mode 1:Port is currently executing in AN/LT mode.
5	krFecMode	RO	0x0	AN/LT FEC mode 1:AN has negotiated FEC enable mode.

Name: **SRDS_ETH_PCS_CTRL**
Description: **PCS Control Includes the following field: - clr_dskew_chng**
Address: **0x0320.011C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
0	clrDskewChng	RW	0x0	Clear EMIB deskew Not a self clearing bit. You must write 1'b1 and then 1'b0 to clear this bit.

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Name: SRDS_ETH_FAULT_STAT
Description: Link Fault Status Status registers to report Local and Remote Link Faults
Address: 0x0320.0120
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lfault	RO	0x0	Local Fault detected 1:Ethernet Hard IP detected a local fault.
1	rfault	RO	0x0	Remote Fault detected 1:Ethernet Hard IP detected a remote fault.

Name: SRDS_ETH_CLK_TX
Description: i_clk_tx Clock Frequency (in kHz)
Address: 0x0320.0128
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	clkTxKhz	RO	0x0	i_clk_tx clock frequency in KHz.

Name: SRDS_ETH_CLK_RX
Description: i_clk_rx Clock Frequency (in kHz)
Address: 0x0320.012C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	clkRxKhz	RO	0x0	i_clk_rx clock frequency in KHz.

Name: SRDS_ETH_CLK_PLL
Description: o_clk_pll Clock Frequency (in kHz)
Address: 0x0320.0130
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	clkPllKhz	RO	0x0	o_clk_pll clock frequency in KHz.

Name: SRDS_ETH_CLK_TX_DIV
Description: o_clk_tx_div Clock Frequency (in kHz)
Address: 0x0320.0134
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	clkTxDivKhz	RO	0x0	o_clk_tx_div clock frequency in KHz.

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Name: SRDS_ETH_CLK_REC_D64
Description: o_clk_rec_div64 Clock Frequency (in kHz)
Address: 0x0320.0138
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	clkRecDiv64Khz	RO	0x0	o_clk_rec_div64 clock frequency in KHz.

Name: SRDS_ETH_CLK_REC_DIV
Description: o_clk_rec_div Clock Frequency (in kHz)
Address: 0x0320.013C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	clkRecDivKhz	RO	0x0	o_clk_rec_div clock frequency in KHz.

Name: SRDS_ETH_DROP_CNTL
Description: RX MAC adapter dropped frame counter lower bits
Address: 0x0320.0140
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	rxDropLo	RO	0x0	Bits [31:0] of RX MAC adapter dropped frame counter.

Name: SRDS_ETH_DROP_CNTH
Description: RX MAC adapter dropped frame counter upper bits
Address: 0x0320.0144
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	rxDropHi	RO	0x0	Bits [63:32] of RXMAC adapter dropped frame counter.

Name: SRDS_ETH_DROP_CTRL
Description: RX MAC Adapter Control
Address: 0x0320.0148
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxDropClr	RW	0x0	Clear RXMAC adapter dropped counter.
1	rxDropSnapshot	RW	0x0	Snapshot RXMAC adapter dropped counter.

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Name: SRDS_ETH_CWBIN_CTRL
Description: Soft cwbin control register 1 : resets the cwbin module
Address: 0x0320.0400
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	cwbinCtrl	RW	0x0	Soft cwbin control register 1 : resets the cwbin module.

Name: SRDS_ETH_CWBIN0_A
Description: Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 0 Errors are accumulated to 32 bit counter Block A
Address: 0x0320.0404
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin0A	RO	0x0	Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 0 Errors are accumulated to 32 bit counter Block A.

Name: SRDS_ETH_CWBIN1_A
Description: Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 1 Errors are accumulated to 32 bit counter Block A
Address: 0x0320.0408
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin1A	RO	0x0	Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 1 Errors are accumulated to 32 bit counter Block A.

Name: SRDS_ETH_CWBIN2_A
Description: Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 2 Errors are accumulated to 32 bit counter Block A
Address: 0x0320.040C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin2A	RO	0x0	Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 2 Errors are accumulated to 32 bit counter Block A.

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Name: SRDS_ETH_CWBIN3_A
Description: Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 3 Errors are accumulated to 32 bit counter Block A
Address: 0x0320.0410
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin3A	RO	0x0	Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 3 Errors are accumulated to 32 bit counter Block A.

Name: SRDS_ETH_CWBIN0_B
Description: Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 0 Errors are accumulated to 32 bit counter BlockB
Address: 0x0320.0414
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin0B	RO	0x0	Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 0 Errors are accumulated to 32 bit counter BlockB.

Name: SRDS_ETH_CWBIN1_B
Description: Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 1 Errors are accumulated to 32 bit counter Block B
Address: 0x0320.0418
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin1B	RO	0x0	Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 1 Errors are accumulated to 32 bit counter Block B.

Name: SRDS_ETH_CWBIN2_B
Description: Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 2 Errors are accumulated to 32 bit counter Block B
Address: 0x0320.041C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin2B	RO	0x0	Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 2 Errors are accumulated to 32 bit counter Block B.

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Name: **SRDS_ETH_CWBIN3_B**
Description: **Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 3 Errors are accumulated to 32 bit counter Block B**
Address: **0x0320.0420**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	cwbin3B	RO	0x0	Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 3 Errors are accumulated to 32 bit counter Block B.

Name: **SRDS_PMA_GUI_OPT**
Description: **Device and IP variant. Includes the following fields: - line_rate - PMA_type - modulation_type - PMA_mode - number of PMA lanes - RS-FEC_enable - number of EMIBs**
Address: **0x0320.0800**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
11:0	lineRate	RO	-	PMA data rate. Applicable to both FGT and FHT. Possible range of values are from 1Gbps to 116Gbps. Every bit increment has a 0.1Gbps increase. -12'h00A = 1Gbps -12'h00B = 1.1Gbps to - 12'h487 = 115.9Gbps - 12'h488 = 116Gbps.
12	xcvrType	RO	-	PMA type -0:FGT -1:FHT.
13	modulationType	RO	-	Modulation type -0:NRZ -1:PAM-4.
15:14	duplexMode	RO	-	PMA mode -2'b00:Not used -2'b01:RX simplex -2'b10:TX simplex -2'b11:Duplex.
20:16	numXcvr	RO	-	Number of PMA lanes Defines the number of PMA per system. The valid values are: 1,2,4,6,8,12,16. -5'b00001 = 1 -5'b00010 = 2 -5'b00100 = 4 -5'b00110 = 6 -5'b01000 = 8 -5'b01100 = 12 -5'b10000 = 16.
21	fecEnable	RO	-	RS-FEC Enabled: -0:RS-FEC is disabled -1:RS-FEC is enabled.
28:24	numEmibs	RO	-	Number of EMIBs Number of EMIBs per system. The valid values are from 1 to 16: -5'b00001 = 1 -5'b00010 = 2 -5'b00011 = 3 -5'b00100 = 4 -5'b00101 = 5 -5'b00110 = 6 -5'b00111 = 7 -5'b01000 = 8 -5'b01001 = 9 -5'b01010 = 10 -5'b01011 = 11 -5'b01100 = 12 -5'b01101 = 13 -5'b01110 = 14 -5'b01111 = 15 -5'b10000 = 16.

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Name: SRDS_PMA_SCRATCH
Description: Scratch Register Provides 32 bits of scratch register space for testing
Address: 0x0320.0804
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	scratch	RW	0x0	Register to test the basic read and write functionality. Provides a full 32-bit writable/readable register to confirm the data bus connection in soft CSR address space.

Name: SRDS_PMA_RST
Description: IP Soft Reset Register Includes the following: - soft_tx_rst (Soft TX Reset) - soft_rx_rst (Soft RX Reset) - tx_rst_ovr (TX Reset Override) - rx_rst_ovr (RX Reset Override)
Address: 0x0320.0808
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	softTxRst	RW	0x0	Soft TX Reset, applicable only if tx_rst_ovr is set: -0: De-assert soft TX reset request -1: Assert soft TX reset request, this will reset the TX datapath.
1	softRxRst	RW	0x0	Soft RX Reset, applicable only if rx_rst_ovr is set: -0: De-assert soft RX reset request -1: Assert soft RX reset request, this will reset the RX datapath.
4	txRstOvr	RW	0x0	TX Reset Override -0:Use TX Reset port from PMA/FEC Direct PHY IP -1:Use Soft TX Reset.
5	rxRstOvr	RW	0x0	RX Reset Override -0:Use RX Reset port from PMA/FEC Direct PHY IP -1:Use Soft RX Reset.

Name: SRDS_PMA_RST_STAT
Description: IP Reset, tx_ready, rx_ready, and AVMM ready Status Register None of the bits are sticky. Includes the following fields: - tx_rst_ack_n - rx_rst_ack_n - tx_ready - rx_ready - avmm_ready
Address: 0x0320.080C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txRstAckN	RO	0x0	TX Reset acknowledge. Active high (for either controlling with tx_reset port or soft_tx_rst) -0: No action. -1: Acknowledge the reset request. When the signal is high, it holds reset sequence. When signal de-asserts to low, the reset sequence starts.

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Bits	Field	Type	Reset	Description
1	rxRstAckN	RO	0x0	RX Reset acknowledge. Active high (for either controlling with rx_reset port or soft_rx_rst) -0: No action. -1: Acknowledge the reset request. When the signal is high, it holds reset sequence. When the signal de-asserts to low, the reset sequence starts.
4	txReady	RO	0x0	tx_ready state. Current state of TX side: -0: TX link not ready to transmit data -1: TX link ready to send user data. Indicates TX PLL locked and data path reset done.
5	rxReady	RO	0x0	rx_ready state. Current state of RX side: -0: RX link not ready to recover incoming data to paralell interface. -1:RX link ready to deliver recovered data to parallel interface. Indicates RX CDR locked and data path reset done.
7	avmmReady	RO	0x0	Soft CSR bit is from datapath Avalon Interface port, which indicates if the tile is ready to accept datapath or PMA Avalon memory-mapped access. -0: Low means tile is not ready and will not respond to datapath or PMA Avalon memory-mapped commands. -1: High indicates the tile is ready. The status is for whole tile although it comes from an EMIB Avalon memory-mapped port.

Name: SRDS_PMA_TX_PLL_LCKD
Description: TX PLL locked TX PLL used by the corresponding physical lane is locked
Address: 0x0320.0810
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	txPllLocked	RO	0x0	TX PLL Locked status bit. Applicable to both FGT and FHT. -0:TX PLL unlocked -1:Corresponding physical lane TX PLL is locked.

Name: SRDS_PMA_RX_CDR_LCKD
Description: RX CDR PLL locked One bit per physical lane to indicate whether or not each CDR has succesfully locked to the reference clock or data. FGT and FHT share with the same register bits
Address: 0x0320.0814
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	rxCdrLocked	RO	0x0	RX FGT CDR locked status bit. Applicable to FGT only. - 0: RX FGT CDR unlocked -1: Corresponding physical lane RX FGT CDR is locked to reference.

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Bits	Field	Type	Reset	Description
31:16	rxCdrLocked2Data	RO	0x0	RX CDR PLL locked to data. Applicable to both FGT and FHT. -0: RX CDR PLL is unlocked. -1: Corresponding physical lane RX CDR is locked to data.

Name: SRDS_PMA_SRC_CTRL
Description: Control to soft reset controller (SRC) to ignore locked2data status
Address: 0x0320.0818
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxIgnLckd2dat	RW	0x0	This bit controls the soft reset controller to monitor the RX CDR LTD status. -0:Reset controller will monitor LTD status -1:Reset controller will ignore LTD status.

Name: SRDS_PMA_DR_FRAC
Description: Control fracture count and rate per fracture
Address: 0x0320.0820
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	ratePerPma	RW	-	Set the rate per PMA lane. - 1: 25G - 2: 50G - 3: 100G.
8:4	lanePerFrac	RW	-	Set the number of PMA lanes per fracture. - 1: 1 PMA lane - 2: 2 PMA lanes - 4: 4 PMA lanes - 8: 8 PMA lanes - 12: 12 PMA lanes - 16: 16 PMA lanes.
13:9	numFrac	RW	-	Set the number of fractures. - 1: 1 fractures - 2: 2 fractures - 3: 3 fractures - 4: 4 fractures - 6: 6 fractures - 8: 8 fractures - 12: 12 fractures.

Name: SRDS_PMA_DR_FEC1
Description: Controls the FEC mode per fracture (fractures 0 to 7). - 0: Disable - 1: IEEE 802.3 RS(528,514) (CL 91,KR) - 2: IEEE 802.3 RS(544,514) (CL 91,KP) - 3: Ethernet Tech. Cons. LL RS(272,258) - 4: Ethernet Consortium RS(528,514) - 5: Ethernet Consortium RS(544,514) - 6: Interlaken RS(544,514) - 7: FC RS(544,514) - 8: FC RS(528,514) - 9: FlexO RS(544,514) - 10: FlexO RS(528,514) - 11-15: Reserved
Address: 0x0320.0824
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecModeFrac0	RW	-	Set FEC mode for fracture 0.

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Bits	Field	Type	Reset	Description
7:4	fecModeFrac1	RW	-	Set FEC mode for fracture 1.
11:8	fecModeFrac2	RW	-	Set FEC mode for fracture 2.
15:12	fecModeFrac3	RW	-	Set FEC mode for fracture 3.
19:16	fecModeFrac4	RW	-	Set FEC mode for fracture 4.
23:20	fecModeFrac5	RW	-	Set FEC mode for fracture 5.
27:24	fecModeFrac6	RW	-	Set FEC mode for fracture 6.
31:28	fecModeFrac7	RW	-	Set FEC mode for fracture 7.

Name: SRDS_PMA_DR_FEC2
Description: Controls the FEC mode per fracture (fractures 8 to 11). - 0: Disable - 1: IEEE 802.3 RS(528,514) (CL 91,KR) - 2: IEEE 802.3 RS(544,514) (CL 91,KP) - 3: Ethernet Tech. Cons. LL RS(272,258) - 4: Ethernet Consortium RS(528,514) - 5: Ethernet Consortium RS(544,514) - 6: Interlaken RS(544,514) - 7: FC RS(544,514) - 8: FC RS(528,514) - 9: FlexO RS(544,514) - 10: FlexO RS(528,514) - 11-15: Reserved
Address: 0x0320.0828
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecModeFrac8	RW	-	Set FEC mode for fracture 8.
7:4	fecModeFrac9	RW	-	Set FEC mode for fracture 9.
11:8	fecModeFrac10	RW	-	Set FEC mode for fracture 10.
15:12	fecModeFrac11	RW	-	Set FEC mode for fracture 11.

Name: SRDS_PMA_DR_STAT
Description: Status indicating if errors occured when setting the fractures
Address: 0x0320.082C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	numFracErr	RO	0x0	For example if a reconfiguration group has at the most 4 sub-fractures, but value written indicates 8, that is an error.
1	lanesPerFracErr	RO	0x0	For example, if a reconfiguration group does not have 4 transceiver fractures, but value written to this register indicates 4, that is an error.
2	ratePerPmaErr	RO	0x0	For example if a reconfiguration group does not have 100G base transceiver rate but write to this register happens for 100G, that is an error.

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Bits	Field	Type	Reset	Description
3	fecEnErr	RO	0x0	For example, trying to enable 5th fracture in a 2 fracture reconfiguration group is an error.

Name: SRDS_PMA_CWBIN_CTRL
Description: Soft cwbin control register Resets the cwbin module
Address: 0x0320.0900
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	cwbinCtrl	RW	0x0	Soft cwbin control register -0:Default -1: Resets the cwbin module.

Name: SRDS_PMA_CWBIN0_A
Description: cwbin0_A Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 0 register
Address: 0x0320.0904
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin0A	RO	0x0	cwbin0_A Errors are accumulated to 32-bit counter Block A.

Name: SRDS_PMA_CWBIN1_A
Description: cwbin1_A Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 1 register
Address: 0x0320.0908
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin1A	RO	0x0	cwbin1_A Errors are accumulated to 32-bit counter Block A.

Name: SRDS_PMA_CWBIN2_A
Description: cwbin2_A Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 2 register
Address: 0x0320.090C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin2A	RO	0x0	cwbin2_A Errors are accumulated to 32-bit counter Block A.

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Name: **SRDS_PMA_CWBIN3_A**
Description: **cwbin3_A Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 3 register**
Address: **0x0320.0910**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	cwbin3A	RO	0x0	cwbin3_A Errors are accumulated to 32-bit counter Block A.

Name: **SRDS_PMA_CWBIN0_B**
Description: **cwbin0_B Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 0 register**
Address: **0x0320.0914**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	cwbin0B	RO	0x0	cwbin0_B Errors are accumulated to 32-bit counter Block B.

Name: **SRDS_PMA_CWBIN1_B**
Description: **cwbin1_B Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 1 register**
Address: **0x0320.0918**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	cwbin1B	RO	0x0	cwbin1_B Errors are accumulated to 32-bit counter Block B.

Name: **SRDS_PMA_CWBIN2_B**
Description: **cwbin2_B Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 2 register**
Address: **0x0320.091C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	cwbin2B	RO	0x0	cwbin2_B Errors are accumulated to 32-bit counter Block B.

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Name: SRDS_PMA_CWBIN3_B
Description: cwbin3_B Hard IP rsfec_corr_cwbin_cnt_0_3 - cwbin 3 register
Address: 0x0320.0920
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbin3B	RO	0x0	cwbin3_B Errors are accumulated to 32-bit counter Block B.

Name: SRDS_PMA_CWBIN_TIMER
Description: cwbin_timer Soft cwbin timer timeout value register
Address: 0x0320.0924
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	cwbinTimer	RW	0x0	cwbin_timer Soft cwbin timer timeout value, in each loop the hard cwbin registers will be read and accumulated in soft 32-bit registers.

Name: SRDS_ETH_STAT_CFG
Description: config configuration config control register
Address: 0x0320.1000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txShadowReq	RW	0x0	TX Shadow Request 1:Freeze stats CSRs so that all TX Stats values read from the registers will be from the same moment. Note that the actual stats collection counters are not frozen, but because they are all read at the time of the freeze, they are cleared.
1	rxShadowReq	RW	0x0	RX Shadow Request 1:Freeze stats CSRs so that all RX Stats values read from the registers will be from the same moment. Note: The actual stats collection counters are not frozen, but because they are all read at the time of the freeze, they are cleared.

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Name: SRDS_ETH_TXPLD_CFG
Description: Defines the configuration fields for TX PLD
Address: 0x0320.1010
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2:0	txEhipMode	RW	0x0	Specify the Internal EHIP Interface to which EMIB Data will be routed This applies to the EMIB Avalon ST signal routing 3h0: MAC 3h1: MAC/PTP 3h2: PCS (MII) 3h3: OTN 3h4: PCS66 3h5: PMA_DIR 3h6: PTP_ONLY 3h7: FEC_DIR.

Name: SRDS_ETH_RXPLD_CFG
Description: Configuration bits for the RX PLD block
Address: 0x0320.1014
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2:0	rxEhipMode	RW	0x0	Specify the internal EHIP Interface to which EMIB Data will be routed This applies to the EMIB Avalon ST signal routing 3h0: MAC 3h1: MAC/PTP 3h2: PCS (MII) 3h3: OTN 3h4: PCS66 3h5: PMA_DIR 3h7: FEC_DIR.
5	rxpldmuxPcs	RW	0x0	Select RXPLD PCS input mode 0:Use RX PCS data 1:Use TX MAC data.
6	rxpldmuxPcs66	RW	0x0	Select RXPLD PCS66 input mode 0:Use RX PCS data 1:Use TX PCS data.
10:8	rxpldmuxSel	RW	0x0	Select the input to the RX PLD Interface 0:RX MAC 1:RX PCS 2:RX FEC 3:RX PMA 4:TX FEC 5:TX PCS 6:TX MAC 7:TX EMIB Interface.

Name: SRDS_ETH_EHIP_MODES
Description: EHIP PCS Mode Configuration Registers to control whether the TX and RX PCS use the 66b interface (skip encoder/decoder), and whether they use the scrambler and descrambler
Address: 0x0320.1048
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
12	sclrFrameError	RW	0x0	Clear phy_frame_error bits 1:Return all sticky frame error bits to 0.

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Name: **SRDS_ETH_TX_AM0**
Description: **Programmable Alignment Marker Encoding 0 Defines the 24b encoding used for a single alignment marker. Only applies to 40GE and 50GE variants**
Address: **0x0320.1050**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
23:0	am	RW	0x90.7647	This code and its inverse are combined with a BIP value to create an alignment marked for the corresponding PCS Virtual lane 0.

Name: **SRDS_ETH_TX_AM1**
Description: **Programmable Alignment Marker Encoding 1 Defines the 24b encoding used for a single alignment marker. Only applies to 40GE and 50GE variants**
Address: **0x0320.1054**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
23:0	am	RW	0xf0.c4e6	This code and its inverse are combined with a BIP value to create an alignment marked for the corresponding PCS Virtual lane 1.

Name: **SRDS_ETH_TX_AM2**
Description: **Programmable Alignment Marker Encoding 2 Defines the 24b encoding used for a single alignment marker. Only applies to 40GE and 50GE variants**
Address: **0x0320.1058**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
23:0	am	RW	0xc5.659b	This code and its inverse are combined with a BIP value to create an alignment marked for the corresponding PCS Virtual lane 2.

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Name: **SRDS_ETH_TX_AM3**
Description: **Programmable Alignment Marker Encoding 3 Defines the 24b encoding used for a single alignment marker. Only applies to 40GE and 50GE variants**
Address: **0x0320.105C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
23:0	am	RW	0xa2.793d	This code and its inverse are combined with a BIP value to create an alignment marked for the corresponding PCS Virtual lane 3.

Name: **SRDS_ETH_RX_AM0**
Description: **Programmable Alignment Marker Encoding 0 Defines the 24b encoding used for a single alignment marker. Only applies to 40GE and 50GE variants**
Address: **0x0320.1060**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
23:0	am	RW	0x90.7647	This code and its inverse are combined with a BIP value to create an alignment marked for the corresponding PCS Virtual lane 0.

Name: **SRDS_ETH_RX_AM1**
Description: **Programmable Alignment Marker Encoding 1 Defines the 24b encoding used for a single alignment marker. Only applies to 40GE and 50GE variants**
Address: **0x0320.1064**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
23:0	am	RW	0xf0.c4e6	This code and its inverse are combined with a BIP value to create an alignment marked for the corresponding PCS Virtual lane 1.

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Name: SRDS_ETH_RX_AM2
Description: Programmable Alignment Marker Encoding 2 Defines the 24b encoding used for a single alignment marker. Only applies to 40GE and 50GE variants
Address: 0x0320.1068
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
23:0	am	RW	0xc5.659b	This code and its inverse are combined with a BIP value to create an alignment marked for the corresponding PCS Virtual lane 2.

Name: SRDS_ETH_RX_AM3
Description: Programmable Alignment Marker Encoding 3 Defines the 24b encoding used for a single alignment marker. Only applies to 40GE and 50GE variants
Address: 0x0320.106C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
23:0	am	RW	0xa2.793d	This code and its inverse are combined with a BIP value to create an alignment marked for the corresponding PCS Virtual lane 3.

Name: SRDS_ETH_BIP_TIMER
Description: Timer Window for BER Measurements
Address: 0x0320.1070
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
20:0	cycles	RW	0x3.12c7	Sets the timer window for BER measurements in clock cycles.

Name: SRDS_ETH_HI_BER_CFG
Description: Hi-BER Frame Errors
Address: 0x0320.1074
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
6:0	count	RW	0x61	Sets the BER count that triggers hi_ber.

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Name: SRDS_ETH_ERR_INJ
Description: PCS Error Injection
Address: 0x0320.1078
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	injErr	RW	0x0	Allows errors injection in the corresponding virtual lane. 0->1:Flip bits to inject encoding errors in corresponding virtual lane. 0:Clear all error injection settings.

Name: SRDS_ETH_FRM_ERR
Description: Frame Errors Detected
Address: 0x0320.1080
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	frmErr	RO	0x0	1:A frame error was detected on the corresponding physical lane. This bit is sticky, and must be cleared by asserting sclr_frame_error.

Name: SRDS_ETH_RX_PCS_STAT
Description: RX PCS Status
Address: 0x0320.1084
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxAligned	RO	0x0	RX PCS fully aligned 1:The RX PCS is fully aligned and ready to start decoding data.
1	hiBer	RO	0x0	Status bits to monitor Hi-BER and PCS fully aligned. 1:One or more virtual lanes are in the HI BER state defined in the Ethernet specification.

Name: SRDS_ETH_AM_LCKD
Description: Alignment Marker Lock Achieved
Address: 0x0320.1088
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	amLock	RO	0x0	1: Indicates that Alignment Marker lock has been achieved.

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Name: SRDS_ETH_DSK_CHG
Description: Change in RX PCS Deskew Status
Address: 0x0320.108C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	dskewChng	RO	0x0	A status register asserted whenever there is a change to the deskewed status of the RX PCS. 1:RX PCS went from deskewed to not deskewed, or from not deskewed to deskewed. Register clears on read.

Name: SRDS_ETH_BER_CNT
Description: BER Count
Address: 0x0320.1090
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	count	RO	0x0	-32 bit counter increments each time when it detects an invalid sync header -Rolls over when maximum count is reached -Clears when the channel is reset -Can be captured using snapshot.

Name: SRDS_ETH_PCS_VLANE0
Description: PCS Virtual Lane 0 Mapping Indicates the remote alignment markers that were received on virtual lanes 0 to 5
Address: 0x0320.1094
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	vlane0	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 0.
9:5	vlane1	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 1.
14:10	vlane2	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 2.
19:15	vlane3	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 3.
24:20	vlane4	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 4.
29:25	vlane5	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 5.

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Name: **SRDS_ETH_PCS_VLANE1**
Description: **PCS Virtual Lane 1 Mapping Indicates the remote alignment markers that were received on virtual lanes 6 to 11**
Address: **0x0320.1098**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
4:0	vlane6	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 6.
9:5	vlane7	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 7.
14:10	vlane8	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 8.
19:15	vlane9	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 9.
24:20	vlane10	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 10.
29:25	vlane11	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 11.

Name: **SRDS_ETH_PCS_VLANE2**
Description: **PCS Virtual Lane 2 Mapping Indicates the remote alignment markers that were received on virtual lanes 12 to 17**
Address: **0x0320.109C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
4:0	vlane12	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 12.
9:5	vlane13	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 13.
14:10	vlane14	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 14.
19:15	vlane15	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 15.
24:20	vlane16	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 16.
29:25	vlane17	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 17.

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Name: SRDS_ETH_PCS_VLANE3
Description: PCS Virtual Lane 3 Mapping Indicates the remote alignment markers that were received on virtual lanes 18 to 19
Address: 0x0320.10A0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	vlane18	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 18.
9:5	vlane19	RO	0x1f	Original virtual lane position of the data mapped to the PCS lane with this index 19.

Name: SRDS_ETH_BIP_CNT0
Description: BIP Counter 0
Address: 0x0320.10A4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT1
Description: BIP Counter 1
Address: 0x0320.10A8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT2
Description: BIP Counter 2
Address: 0x0320.10AC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

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Name: SRDS_ETH_BIP_CNT3
Description: BIP Counter 3
Address: 0x0320.10B0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT4
Description: BIP Counter 4
Address: 0x0320.10B4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT5
Description: BIP Counter 5
Address: 0x0320.10B8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT6
Description: BIP Counter 6
Address: 0x0320.10BC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

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Name: SRDS_ETH_BIP_CNT7
Description: BIP Counter 7
Address: 0x0320.10C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT8
Description: BIP Counter 8
Address: 0x0320.10C4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT9
Description: BIP Counter 9
Address: 0x0320.10C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT10
Description: BIP Counter 10
Address: 0x0320.10CC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

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Name: SRDS_ETH_BIP_CNT11
Description: BIP Counter 11
Address: 0x0320.10D0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT12
Description: BIP Counter 12
Address: 0x0320.10D4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT13
Description: BIP Counter 13
Address: 0x0320.10D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT14
Description: BIP Counter 14
Address: 0x0320.10DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

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Name: SRDS_ETH_BIP_CNT15
Description: BIP Counter 15
Address: 0x0320.10E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT16
Description: BIP Counter 16
Address: 0x0320.10E4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT17
Description: BIP Counter 17
Address: 0x0320.10E8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_BIP_CNT18
Description: BIP Counter 18
Address: 0x0320.10EC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

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Name: SRDS_ETH_BIP_CNT19
Description: BIP Counter 19
Address: 0x0320.10F0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	count	RO	0x0	Shows current BIP count for corresponding PCS lane. Rolls over at max count. Can be captured by snapshot.

Name: SRDS_ETH_ERR_BLK_CNT
Description: Error Block Count
Address: 0x0320.10F4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	count	RO	0x0	Counts the number of visits to the RX_E state when the RX Decoder is in normal mode. Rolls over when max count is reached. Can be captured by snapshot.

Name: SRDS_ETH_DSK_DEPTH0
Description: Deskew Depth 0
Address: 0x0320.10FC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
5:0	depth0	RO	0x0	Deskew depth for one of the PCS Virtual lanes.
11:6	depth1	RO	0x0	
17:12	depth2	RO	0x0	
23:18	depth3	RO	0x0	
29:24	depth4	RO	0x0	

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Name: SRDS_ETH_DSK_DEPTH1
Description: Deskew Depth 1
Address: 0x0320.1100
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
5:0	depth0	RO	0x0	Deskew depth for one of the PCS Virtual lanes.
11:6	depth1	RO	0x0	
17:12	depth2	RO	0x0	
23:18	depth3	RO	0x0	
29:24	depth4	RO	0x0	

Name: SRDS_ETH_DSK_DEPTH2
Description: Deskew Depth 2
Address: 0x0320.1104
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
5:0	depth0	RO	0x0	Deskew depth for one of the PCS Virtual lanes.
11:6	depth1	RO	0x0	
17:12	depth2	RO	0x0	
23:18	depth3	RO	0x0	
29:24	depth4	RO	0x0	

Name: SRDS_ETH_DSK_DEPTH3
Description: Deskew Depth 3
Address: 0x0320.1108
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
5:0	depth0	RO	0x0	Deskew depth for one of the PCS Virtual lanes.
11:6	depth1	RO	0x0	
17:12	depth2	RO	0x0	
23:18	depth3	RO	0x0	
29:24	depth4	RO	0x0	

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Name: SRDS_ETH_FAULT_CFG
Description: Link Fault Configuration Configures EHIP link fault behavior
Address: 0x0320.1200
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	enLf	RW	0x0	Enable Link Fault Reporting 0:The TX PCS does not respond to link faults 1:The TX PCS transmits link fault messages based on link faults detected by the RX.
1	enUnidir	RW	0x0	Enable Unidirectional Link Fault 1:EHIP enables support for unidirectional link fault signaling as described in Clause 66 Remote faults have no impact on TX data, and Local faults will cause the TX to transmit Remote fault Ordered sets between frames.
2	disableRf	RW	0x0	Send idles instead of remote faults for local faults in unidirectional mode 1:In unidirectional mode, local faults cause the TX to transmit Idles 0:In unidirectional mode, local faults cause the TX to transmit Remote Faults (spec default).
3	forceRf	RW	0x0	Force the TX MAC to transmit Remote Faults when link fault signaling is on 1:TX MAC transmits Remote Faults 0:TX MAC operates normally.

Name: SRDS_ETH_IPG_REM_CFG
Description: IPG Words to remove per Alignment Marker Period
Address: 0x0320.1204
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	ipgColRem	RW	0x14	16b value that sets the number of idle words to remove from IPG in an AM period to make space for alignment markers. This parameter can also be used to scale IPG in PPM increments for rate balance.

Name: SRDS_ETH_TX_MAX_CFG
Description: Max TX Frame Size
Address: 0x0320.1208
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	maxTx	RW	0x5ee	16b value that sets the maximum TX frame size. When TX frames exceed this size, the CNTR_TX_OVERSIZE statistic is incremented.

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Name: SRDS_ETH_TX_MAC_CFG
Description: TX MAC Configuration Configuration register for TX MAC features
Address: 0x0320.120C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
1	disableTxvlan	RW	0x0	Disable VLAN detection for TX Stats 0:TX frames with VLAN headers will be counted as VLAN frames in the TX stats 1:VLAN headers will not be considered by the TX stats block.
2	disableTxmac	RW	0x0	Disable TX MAC 0:TX MAC operates normally 1:TX MAX is disabled. It behaves as though it has been PAUSED by the remote link until disable is turned off.
3	enSaddrInsert	RW	0x0	Enable Source Address Insertion 0:Client provides Source Address 1:TX MAC inserts source addresses stored in CSRs.

Name: SRDS_ETH_TX_EHIP_CFG
Description: EHIP TX MAC Feature Configuration Configures EHIP specific TX MAC features
Address: 0x0320.1210
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	enPp	RW	0x0	Enable TX Preamble Passthrough 1:Preable-passthrough mode enabled - bytes 1 to 7 of each SOP word will be used as preamble bytes at the start of the Ethernet packet 0:A standard Ethernet preamble will be used for TX packets.
2:1	ipg	RW	0x0	DIC Average Min IPG Sets the average minimum IPG enforced by the Deficit Idle Counter 2'd0: 12 bytes (default) 2'd1:10 bytes 2'd2:8 bytes 2'd3:1 byte.
4	txPlenEn	RW	0x0	Enable TX Packet Length Check 1: TX stats checks TX packet length error 0:TX stats does not check TX packet length error.
9	txCrcPreambleEn	RW	0x0	Enable CRC Over Preamble 0:TX CRC calculated over Ethernet Frame (default) 1:TX CRC calculated over frame plus preamble.

Name: SRDS_ETH_TX_SMAC_LSB
Description: Source Address Insertion Source Address Lower Bytes
Address: 0x0320.1214
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	saddrl	RW	0x2233.4455	Lower 32 bits of the TX MAC Source Address, which is used when Source Address Insertion is enabled.

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Name: SRDS_ETH_TX_SMAC_MSB
Description: Source Address Insertion Source Address Upper Bytes
Address: 0x0320.1218
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	saddrh	RW	0x11	Upper 16 bits of the TX MAC Source Address, which is used when Source Address Insertion is enabled.

Name: SRDS_ETH_RX_MAX_CFG
Description: Max RX Frame Size 16b value that sets the maximum RX frame size. When RX frames exceed this size, the CNTR_RX_OVERSIZE statistic is incremented, and the appropriate rx_error bit is asserted with EOP on the frame to indicate the frame is oversize
Address: 0x0320.121C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
15:0	maxRx	RW	0x5ee	MAX_RX_SIZE_CONFIG Sets the maximum size of a RX frame in octets before it will be counted as an oversize frame.

Name: SRDS_ETH_RX_CRC_CFG
Description: RX CRC Forwarding Controls whether EHIP forwards FCS with each frame, or removes it
Address: 0x0320.1220
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	forwardRxCrc	RW	0x0	Forward RX CRC values 0:Remove CRC from RX frames 1:Leave CRC in RX frames and forward it to RX Client logic.

Name: SRDS_ETH_RX_MAC_CFG
Description: RX MAC Configuration Configuration register for RX MAC features
Address: 0x0320.1224
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	enPlen	RW	0x1	Enable Packet Length Checking 1:EHIP will assert the length error bit of rx_error at EOP for Frames where the Type/Length field is a length, and the length advertized is greater than the length of the frame that was received.

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Bits	Field	Type	Reset	Description
1	disableRxvlan	RW	0x0	Disable RX VLAN detection 0:EHIP detects VLAN frames, counts them separately in stats, and marks them at EOP 1:EHIP ignores VLAN in RX data, and treats VLAN headers as payload bytes.
3	enCheckSfd	RW	0x0	Enable Start Frame Delimiter Checking 0:Custom SFD bytes are allowed in preambles 1:Packets will be dropped if they do not have a standard Start Frame Delimiter.
4	enStrictPreamble	RW	0x0	Enable Strict Preamble Checking 0:Custom Preamble bytes are allowed between SOP and SFD 1:Packets will be dropped if they do not have standard preamble bytes.
7	enforceMaxRx	RW	0x0	Enforce Maximum frame size on RX packets 0:Ovesized frames are not altered 1:Frames are ended with FCS error if they exceed the programmed rx max frame size.
8	removeRxPad	RW	0x0	Remove PADs from padded frames 0:Padded frames are not altered 1:Pads are removed from padded frames.

Name: SRDS_ETH_RX_EHIP_CFG
Description: EHIP RX MAC Feature Configuration Configures EHIP specific RX MAC features
Address: 0x0320.1228
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	enPp	RW	0x0	Enable RX Preamble Passthrough 1:Preable-passthrough mode enabled - the preamble received with each packet will be passed to the user 0:RX preamble will not be passed to the user.
1	rxCrcPreambleEn	RW	0x0	Enable CRC over preamble 0:RX CRC calculated over Ethernet Frame (default) 1:RX CRC calculated over frame plus preamble.
3:2	macLbMode	RW	0x0	Enable MAC Loopback Mode 0:Normal operation PCS RX to MAC RX(default) 1:MAC Tx to MAC Rx 2: AIB Tx to MAC Rx.

Name: SRDS_ETH_TX_CNT_CTRL
Description: Configure TX Statistics Counters Configuration bits to control the behavior of the TX Statistics counters
Address: 0x0320.1274
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rstTxStats	RW	0x0	Reset TX Statistics 1:Reset all TX Stats counters.
1	rstTxParity	RW	0x0	Reset the TX Statistics Parity Error bit 1:Reset the parity error bit in CNTR_TX_STATUS.

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Name: **SRDS_ETH_RX_CNT_CTRL**
Description: **Configure RX Statistics Counters Configuration bits to control the behavior of the RX Statistics counters**
Address: **0x0320.1278**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
0	rstRxStats	RW	0x0	Reset RX Statistics 1:Reset all RX Stats counters.
1	rstRxParity	RW	0x0	Reset the RX Statistics Parity Error bit 1:Reset the parity error bit in CNTR_RX_STATUS.

Name: **SRDS_ETH_TXFRG_CNT**
Description: **Frames less than 64 bytes with CRC error Number of frames with less than 64 bytes and a CRC error This CSR can be mapped to etherStatsFragments**
Address: **0x0320.1800**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXJAB_CNT**
Description: **Oversized frames with CRC error Number of frames with more octets than the programmed maximum and a CRC error**
Address: **0x0320.1808**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXFCS_CNT**
Description: **Frames of any size with a CRC error on OK packet Number of frames of any size with a CRC error on OK packet**
Address: **0x0320.1818**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_TXMCDE_CNT**
Description: **Multicast data frames with CRC error Number of data frames -not a control type- with a multicast destination address and a CRC error**
Address: **0x0320.1820**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXBCDE_CNT**
Description: **Broadcast data frames with CRC error Number of data frames with a broadcast destination address and a CRC error**
Address: **0x0320.1828**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXUCDE_CNT**
Description: **Unicast data frames with CRC error Number of data frames with a unicast destination address and a CRC error**
Address: **0x0320.1830**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXMCCE_CNT**
Description: **Multicast control frames with CRC error Number of control frames with a multicast destination address and a CRC error**
Address: **0x0320.1838**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_TXBCCE_CNT**
Description: **Broadcast control frames with CRC error Number of control frames with a broadcast destination address and a CRC error**
Address: **0x0320.1840**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXUCCE_CNT**
Description: **Unicast control frames with CRC error Number of control frames with a unicast destination address and a CRC error**
Address: **0x0320.1848**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXPAUE_CNT**
Description: **Pause frame with CRC error Number of PAUSE (SFC) frames with a CRC error**
Address: **0x0320.1850**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN0_CNTL**
Description: **64 byte frames Number of frames of any type with a frame length of 64 bytes**
Address: **0x0320.1858**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_TXLEN0_CNTH**
Description: **64 byte frames Number of frames of any type with a frame length of 64 bytes**
Address: **0x0320.185C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN1_CNTL**
Description: **65 to 127 byte frames Number of frames of any type with a frame length of 65 to 127 bytes**
Address: **0x0320.1860**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN1_CNTH**
Description: **65 to 127 byte frames Number of frames of any type with a frame length of 65 to 127 bytes**
Address: **0x0320.1864**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN2_CNTL**
Description: **128 to 257 byte frames Number of frames of any type with a frame length of 128 to 257 bytes**
Address: **0x0320.1868**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: **SRDS_ETH_TXLEN2_CNTH**
Description: **128 to 257 byte frames Number of frames of any type with a frame length of 128 to 257 bytes**
Address: **0x0320.186C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN3_CNTL**
Description: **256 to 511 byte frames Number of frames of any type with a frame length of 258 to 511 bytes**
Address: **0x0320.1870**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN3_CNTH**
Description: **256 to 511 byte frames Number of frames of any type with a frame length of 258 to 511 bytes**
Address: **0x0320.1874**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN4_CNTL**
Description: **512 to 1023 byte frames Number of frames of any type with a frame length of 512 to 1023 bytes**
Address: **0x0320.1878**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_TXLEN4_CNTH**
Description: **512 to 1023 byte frames Number of frames of any type with a frame length of 512 to 1023 bytes**
Address: **0x0320.187C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN5_CNTL**
Description: **1024 to 1518 byte frames Number of frames of any type with a frame length of 1024 to 1518 bytes**
Address: **0x0320.1880**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN5_CNTH**
Description: **1024 to 1518 byte frames Number of frames of any type with a frame length of 1024 to 1518 bytes**
Address: **0x0320.1884**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXLEN6_CNTL**
Description: **1519 to max size frames Number of frames of any type with a frame length of 1519 to the programmed maximum size in bytes**
Address: **0x0320.1888**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: **SRDS_ETH_TXLEN6_CNTH**
Description: **1519 to max size frames Number of frames of any type with a frame length of 1519 to the programmed maximum size in bytes**
Address: **0x0320.188C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXOSZ_CNT**
Description: **Oversize frames Number of frames of any type with a frame length greater than the programmed maximum size in bytes**
Address: **0x0320.1890**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXMCD_CNTL**
Description: **Multicast data frames without error Number of data frames with a multicast destination address and no errors**
Address: **0x0320.1898**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXMCD_CNTH**
Description: **Multicast data frames without error Number of data frames with a multicast destination address and no errors**
Address: **0x0320.189C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_TXBCD_CNTL**
Description: **Broadcast data frames without error Number of data frames with a broadcast destination address and no errors**
Address: **0x0320.18A0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXBCD_CNTH**
Description: **Broadcast data frames without error Number of data frames with a broadcast destination address and no errors**
Address: **0x0320.18A4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXUCD_CNTL**
Description: **Unicast data frames without error Number of data frames with a unicast destination address and no errors**
Address: **0x0320.18A8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXUCD_CNTH**
Description: **Unicast data frames without error Number of data frames with a unicast destination address and no errors**
Address: **0x0320.18AC**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_TXMCC_CNTL**
Description: **Multicast control frames without error Number of control frames with a multicast destination address and no errors**
Address: **0x0320.18B0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXMCC_CNTH**
Description: **Multicast control frames without error Number of control frames with a multicast destination address and no errors**
Address: **0x0320.18B4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXBCC_CNTL**
Description: **Broadcast control frames without error Number of control frames with a broadcast destination address and no errors**
Address: **0x0320.18B8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXBCC_CNTH**
Description: **Broadcast control frames without error Number of control frames with a broadcast destination address and no errors**
Address: **0x0320.18BC**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: SRDS_ETH_TXUCC_CNTL
Description: Unicast control frames without error Number of control frames with unicast destination address and no errors
Address: 0x0320.18C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXUCC_CNTH
Description: Unicast control frames without error Number of control frames with unicast destination address and no errors
Address: 0x0320.18C4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXPAU_CNTL
Description: Pause frames without error Number of PAUSE (SFC) frames without error
Address: 0x0320.18C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXPAU_CNTH
Description: Pause frames without error Number of PAUSE (SFC) frames without error
Address: 0x0320.18CC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: SRDS_ETH_TXRUNT_CNT
Description: Frames with less than 64 bytes and a CRC error Frames with less than 64 bytes and a CRC error
Address: 0x0320.18D0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXST_CNTL
Description: Number of frame starts Number of frame starts
Address: 0x0320.18D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXST_CNTH
Description: Number of frame starts Number of frame starts
Address: 0x0320.18DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXLENE_CNT
Description: Number of length error Number of length error
Address: 0x0320.18E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXPFCE_CNT
Description: PFC frame with CRC error Number of PFC frames with a CRC error
Address: 0x0320.18E8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: **SRDS_ETH_TXPFC_CNTL**
Description: **PFC frames without error Number of PFC frames without error**
Address: **0x0320.18F0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXPFC_CNTH**
Description: **PFC frames without error Number of PFC frames without error**
Address: **0x0320.18F4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXPOCT_CNTL**
Description: **Data and padding octets with no errors of any type, including length error, undersize, oversize, malformed, or bad length/type field**
Address: **0x0320.18F8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXPOCT_CNTH**
Description: **Data and padding octets with no errors of any type, including length error, undersize, oversize, malformed, or bad length/type field**
Address: **0x0320.18FC**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: SRDS_ETH_TXOCT_CNTL
Description: Counts every octet in every packet with or without error (including CRC error)
Address: 0x0320.1900
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXOCT_CNTH
Description: Counts every octet in every packet with or without error (including CRC error)
Address: 0x0320.1904
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXMAL_CNT
Description: Malformed packets received Malformed packets received. Malformed packets contain a control character other than error and do not end on a term. For the Ethernet Hard IP lane, frames with a start block that does not contain at least 5 bytes of IPG from the preceding frame are considered invalid. Note that the RX client interface indicates a 3rd type of malformed frame, which is not counted by the malformed statistic packets at the RX Client interface of the RX MAC can also be malformed if they contain insufficient bytes to include a preamble, header, and CRC
Address: 0x0320.1908
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_TXDROP_CNT
Description: Packets dropped due to error
Address: 0x0320.1910
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: **SRDS_ETH_TXBADTL_CNT**
Description: **Frames with an L/T field that is neither a length nor an ethertype**
Address: **0x0320.1918**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXPTP_CNT**
Description: **Counts the packets which are timestamped**
Address: **0x0320.1920**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXPTP1S_CNT**
Description: **Counts the packets for which the timestamp information (TS,CF or EB) is updated in line with the packet transmission**
Address: **0x0320.1924**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXPTP2S_CNT**
Description: **Counts the packets for which the timestamp information is updated out of the line with the packet transmission**
Address: **0x0320.1928**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_TXPTPV1_CNT**
Description: **Counts the packets for which the timestamp information is updated in 1588 v1 mode Counts the packets for which the timestamp information is updated in 1588 v1 mode**
Address: **0x0320.192C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_TXPTPV2_CNT**
Description: **Counts the packets for which the timestamp information is updated in 1588 v2 mode (1 step or 2 step) Counts the packets for which the timestamp information is updated in 1588 v2 mode (1 step or 2 step)**
Address: **0x0320.1930**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXFRG_CNT**
Description: **Frames less than 64 bytes with CRC error Number of frames with less than 64 bytes and a CRC error This CSR can be mapped to etherStatsFragments**
Address: **0x0320.1934**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXJAB_CNT**
Description: **Oversized frames with CRC error Number of frames with more octets than the programmed maximum and a CRC error**
Address: **0x0320.193C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: **SRDS_ETH_RXFCS_CNT**
Description: **Frames of any size with a CRC error on OK packet Number of frames of any size with a CRC error on OK packet**
Address: **0x0320.194C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXMCDE_CNT**
Description: **Multicast data frames with CRC error Number of data frames -not a control type- with a multicast destination address and a CRC error**
Address: **0x0320.1954**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXBCDE_CNT**
Description: **Broadcast data frames with CRC error Number of data frames with a broadcast destination address and a CRC error**
Address: **0x0320.195C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXUCDE_CNT**
Description: **Unicast data frames with CRC error Number of data frames with a unicast destination address and a CRC error**
Address: **0x0320.1964**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: **SRDS_ETH_RXMCCE_CNT**
Description: **Multicast control frames with CRC error Number of control frames with a multicast destination address and a CRC error**
Address: **0x0320.196C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXBCCE_CNT**
Description: **Broadcast control frames with CRC error Number of control frames with a broadcast destination address and a CRC error**
Address: **0x0320.1974**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXUCCE_CNT**
Description: **Unicast control frames with CRC error Number of control frames with a unicast destination address and a CRC error**
Address: **0x0320.197C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXPAUE_CNT**
Description: **Pause frame with CRC error Number of PAUSE (SFC) frames with a CRC error**
Address: **0x0320.1984**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: **SRDS_ETH_RXLEN0_CNTL**
Description: **64 byte frames Number of frames of any type with a frame length of 64 bytes**
Address: **0x0320.198C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXLEN0_CNTH**
Description: **64 byte frames Number of frames of any type with a frame length of 64 bytes**
Address: **0x0320.1990**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXLEN1_CNTL**
Description: **65 to 127 byte frames Number of frames of any type with a frame length of 65 to 127 bytes**
Address: **0x0320.1994**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXLEN1_CNTH**
Description: **65 to 127 byte frames Number of frames of any type with a frame length of 65 to 127 bytes**
Address: **0x0320.1998**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: **SRDS_ETH_RXLEN2_CNTL**
Description: **128 to 257 byte frames Number of frames of any type with a frame length of 128 to 257 bytes**
Address: **0x0320.199C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXLEN2_CNTH**
Description: **128 to 257 byte frames Number of frames of any type with a frame length of 128 to 257 bytes**
Address: **0x0320.19A0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXLEN3_CNTL**
Description: **256 to 511 byte frames Number of frames of any type with a frame length of 258 to 511 bytes**
Address: **0x0320.19A4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXLEN3_CNTH**
Description: **256 to 511 byte frames Number of frames of any type with a frame length of 258 to 511 bytes**
Address: **0x0320.19A8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

INTEL CONFIDENTIAL

Name: SRDS_ETH_RXLEN4_CNTL
Description: 512 to 1023 byte frames Number of frames of any type with a frame length of 512 to 1023 bytes
Address: 0x0320.19AC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXLEN4_CNTH
Description: 512 to 1023 byte frames Number of frames of any type with a frame length of 512 to 1023 bytes
Address: 0x0320.19B0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXLEN5_CNTL
Description: 1024 to 1518 byte frames Number of frames of any type with a frame length of 1024 to 1518 bytes
Address: 0x0320.19B4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXLEN5_CNTH
Description: 1024 to 1518 byte frames Number of frames of any type with a frame length of 1024 to 1518 bytes
Address: 0x0320.19B8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_RXLEN6_CNTL**
Description: **1519 to max size frames Number of frames of any type with a frame length of 1519 to the programmed maximum size in bytes**
Address: **0x0320.19BC**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXLEN6_CNTH**
Description: **1519 to max size frames Number of frames of any type with a frame length of 1519 to the programmed maximum size in bytes**
Address: **0x0320.19C0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXOSZ_CNT**
Description: **Oversize frames Number of frames of any type with a frame length greater than the programmed maximum size in bytes**
Address: **0x0320.19C4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXMCD_CNTL**
Description: **Multicast data frames without error Number of data frames with a multicast destination address and no errors**
Address: **0x0320.19CC**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_RXMCD_CNTH**
Description: **Multicast data frames without error Number of data frames with a multicast destination address and no errors**
Address: **0x0320.19D0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXBCD_CNTL**
Description: **Broadcast data frames without error Number of data frames with a broadcast destination address and no errors**
Address: **0x0320.19D4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXBCD_CNTH**
Description: **Broadcast data frames without error Number of data frames with a broadcast destination address and no errors**
Address: **0x0320.19D8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXUCD_CNTL**
Description: **Unicast data frames without error Number of data frames with a unicast destination address and no errors**
Address: **0x0320.19DC**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_RXUCD_CNTH**
Description: **Unicast data frames without error Number of data frames with a unicast destination address and no errors**
Address: **0x0320.19E0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXMCC_CNTL**
Description: **Multicast control frames without error Number of control frames with a multicast destination address and no errors**
Address: **0x0320.19E4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXMCC_CNTH**
Description: **Multicast control frames without error Number of control frames with a multicast destination address and no errors**
Address: **0x0320.19E8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXBCC_CNTL**
Description: **Broadcast control frames without error Number of control frames with a broadcast destination address and no errors**
Address: **0x0320.19EC**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: SRDS_ETH_RXBCC_CNTH
Description: Broadcast control frames without error Number of control frames with a broadcast destination address and no errors
Address: 0x0320.19F0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXUCC_CNTL
Description: Unicast control frames without error Number of control frames with unicast destination address and no errors
Address: 0x0320.19F4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXUCC_CNTH
Description: Unicast control frames without error Number of control frames with unicast destination address and no errors
Address: 0x0320.19F8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXPAU_CNTL
Description: Pause frames without error Number of PAUSE (SFC) frames without error
Address: 0x0320.19FC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: SRDS_ETH_RXPAU_CNTH
Description: Pause frames without error Number of PAUSE (SFC) frames without error
Address: 0x0320.1A00
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXRUNT_CNT
Description: Frames with less than 64 bytes and a CRC error Frames with less than 64 bytes and a CRC error
Address: 0x0320.1A04
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXST_CNTL
Description: Number of frame starts Number of frame starts
Address: 0x0320.1A0C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXST_CNTH
Description: Number of frame starts Number of frame starts
Address: 0x0320.1A10
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_ETH_RXLENE_CNT
Description: Number of length error Number of length error
Address: 0x0320.1A14
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_RXPFCE_CNT**
Description: **PFC frame with CRC error Number of PFC frames with a CRC error**
Address: **0x0320.1A1C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXPFC_CNTL**
Description: **PFC frames without error Number of PFC frames without error**
Address: **0x0320.1A24**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXPFC_CNTH**
Description: **PFC frames without error Number of PFC frames without error**
Address: **0x0320.1A28**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXPOCT_CNTL**
Description: **Data and padding octets with no errors of any type, including length error, undersize, oversize, malformed, or bad length/type field Data and padding octets with no errors of any type, including length error, undersize, oversize, malformed, or bad length/type field**
Address: **0x0320.1A2C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_RXPOCT_CNTH**
Description: **Data and padding octets with no errors of any type, including length error, undersize, oversize, malformed, or bad length/type field Data and padding octets with no errors of any type, including length error, undersize, oversize, malformed, or bad length/type field**
Address: **0x0320.1A30**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXOCT_CNTL**
Description: **Counts every octet in every packet with or without error (including CRC error) Counts every octet in every packet with or without error (including CRC error)**
Address: **0x0320.1A34**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXOCT_CNTH**
Description: **Counts every octet in every packet with or without error (including CRC error) Counts every octet in every packet with or without error (including CRC error)**
Address: **0x0320.1A38**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: **SRDS_ETH_RXMAL_CNT**
Description: **Malformed packets received Malformed packets contain a control character other than error and do not end on a term. For the Ethernet Hard IP lane, frames with a start block that does not contain at least 5 bytes of IPG from the preceding frame are considered invalid. Note that the RX client interface indicates a 3rd type of malformed frame, which is not counted by the malformed statistic packets at the RX Client interface of the RX MAC can also be malformed if they contain insufficient bytes to include a preamble, header, and CRC**
Address: **0x0320.1A3C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXDROP_CNT**
Description: **Packets dropped due to error Packets dropped due to error**
Address: **0x0320.1A44**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: **SRDS_ETH_RXBADTL_CNT**
Description: **Frames with an L/T field that is neither a length nor an ethertype Frames with an L/T field that is neither a length nor an ethertype**
Address: **0x0320.1A4C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	statsPcnt32	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

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Name: SRDS_ETH_RXPTPTS_CNT
Description: Lower 4 bytes of the Destination address for RX Pause Frames Sets the lower 4 bytes of the 6 byte Destination MAC Address that must be found in incoming SFC and PFC frames for them to be processed
Address: 0x0320.1A54
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	statsPcnt	RO	0x0	Statistics word 4 bytes of an 8 byte EHIP Statistics.

Name: SRDS_PMA_CLK_SEL
Description: Clock selection
Address: 0x0320.6000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
2:0	reserved0	RO	-	Reserved.
6:3	txClkSel2	RW	-	Tx clock 2nd stage selection Values below are not applicable to every stream BOND_TX_E25G = 4'd0 BOND_TX_E50G = 4'd1 BOND_TX_E100G = 4'd2 BOND_TX_E200G = 4'd3 BOND_TX_E400G = 4'd4 BOND_TX_X6_LEFT = 4'd5 BOND_TX_X6_RIGHT = 4'd6 BOND_TX_X12_LEFT = 4'd7 BOND_TX_X12_RIGHT = 4'd8.
31:7	reserved1	RO	-	Reserved.

Name: SRDS_FEC1_IF0
Description: xcvrif_reg_12
Address: 0x0320.6670
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

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Name: SRDS_FEC1_RSFEC0
Description: RS-FEC Settings
Address: 0x0320.66C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

Name: SRDS_FEC1_LN_CFG0
Description: RSFEC per lane configuration
Address: 0x0320.66C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC1_F0_STAT0
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.6710
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.

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Bits	Field	Type	Reset	Description
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

Name: SRDS_FEC1_F0_HOLD0
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.6714
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC1_F1_STAT0
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.6720
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

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Name: SRDS_FEC1_F1_HOLD0
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.6724
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC1_ERR_INJ0
Description: RS-FEC Error Injection Mode
Address: 0x0320.6738
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.

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Bits	Field	Type	Reset	Description
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

Name: SRDS_FEC1_TX_STAT0
Description: RS-FEC Status per TX Lane
Address: 0x0320.6748
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

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Name: SRDS_FEC1_TX_HOLD0
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.674C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

Name: SRDS_FEC1_RX_STAT0
Description: RS-FEC Status per RX Lane
Address: 0x0320.6754
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).

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Bits	Field	Type	Reset	Description
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

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Bits	Field	Type	Reset	Description
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC1_RX_HOLD0
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.6758
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.

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Bits	Field	Type	Reset	Description
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_AGR_STAT0
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.6760
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_AGR_HOLD0
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.6764
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_LN_MAP0
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.676C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

Name: SRDS_FEC1_LN_SKEW0
Description: RS-FEC Lane Skew
Address: 0x0320.6770
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC1_CW_POS0
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.6774
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

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Name: SRDS_FEC1_ECC0
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.6778
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

Name: SRDS_FEC1_TX_ERR0
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.677C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

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Name: **SRDS_FEC1_CR_CNTL0**
Description: **RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]**
Address: **0x0320.6784**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: **SRDS_FEC1_CR_CNTH0**
Description: **RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]**
Address: **0x0320.6788**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: **SRDS_FEC1_UCR_CNTL0**
Description: **RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]**
Address: **0x0320.678C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: **SRDS_FEC1_UCR_CNTH0**
Description: **RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]**
Address: **0x0320.6790**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: **SRDS_FEC1_CRS_CNTL0**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]**
Address: **0x0320.6794**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: **SRDS_FEC1_CRS_CNTH0**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]**
Address: **0x0320.6798**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: **SRDS_FEC1_CR0_CNTL0**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]**
Address: **0x0320.679C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

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Name: SRDS_FEC1_CR0_CNTH0
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]
Address: 0x0320.67A0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC1_CR1_CNTL0
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]
Address: 0x0320.67A4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC1_CR1_CNTH0
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]
Address: 0x0320.67A8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

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Name: SRDS_FEC1_CWBN0_CNT0
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.67D0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: SRDS_FEC1_CWBN1_CNT0
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.67D4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC1_CWBN2_CNT0
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.67D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC1_CWBN3_CNT0
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.67DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC1_DBG_CFG0
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.67E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC1_IF1
Description: xcvrif_reg_12
Address: 0x0320.6870
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC1_RSFEC1
Description: RS-FEC Settings
Address: 0x0320.68C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

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Name: SRDS_FEC1_LN_CFG1
Description: RSFEC per lane configuration
Address: 0x0320.68C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC1_F0_STAT1
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.6910
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC1_F0_HOLD1
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.6914
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC1_F1_STAT1
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.6920
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC1_F1_HOLD1
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.6924
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC1_ERR_INJ1
Description: RS-FEC Error Injection Mode
Address: 0x0320.6938
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC1_TX_STAT1
Description: RS-FEC Status per TX Lane
Address: 0x0320.6948
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC1_TX_HOLD1
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.694C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC1_RX_STAT1
Description: RS-FEC Status per RX Lane
Address: 0x0320.6954
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC1_RX_HOLD1
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.6958
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

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Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_AGR_STAT1
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.6960
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_AGR_HOLD1
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.6964
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_LN_MAP1
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.696C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC1_LN_SKEW1
Description: RS-FEC Lane Skew
Address: 0x0320.6970
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC1_CW_POS1
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.6974
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC1_ECC1
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.6978
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

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Name: SRDS_FEC1_TX_ERR1
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.697C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC1_CR_CNTL1
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.6984
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC1_CR_CNTH1
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.6988
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: **SRDS_FEC1_UCR_CNTL1**
Description: **RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]**
Address: **0x0320.698C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: **SRDS_FEC1_UCR_CNTH1**
Description: **RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]**
Address: **0x0320.6990**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: **SRDS_FEC1_CRS_CNTL1**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]**
Address: **0x0320.6994**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: **SRDS_FEC1_CRS_CNTH1**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]**
Address: **0x0320.6998**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: **SRDS_FEC1_CR0_CNTL1**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]**
Address: **0x0320.699C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC1_CR0_CNTH1**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]**
Address: **0x0320.69A0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC1_CR1_CNTL1**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]**
Address: **0x0320.69A4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

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Name: **SRDS_FEC1_CR1_CNTH1**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]**
Address: **0x0320.69A8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC1_CWBN0_CNT1**
Description: **RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols**
Address: **0x0320.69D0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: **SRDS_FEC1_CWBN1_CNT1**
Description: **RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols**
Address: **0x0320.69D4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC1_CWBN2_CNT1
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.69D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC1_CWBN3_CNT1
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.69DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC1_DBG_CFG1
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.69E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC1_IF2
Description: xcvrif_reg_12
Address: 0x0320.6A70
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC1_RSFEC2
Description: RS-FEC Settings
Address: 0x0320.6AC0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

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Name: SRDS_FEC1_LN_CFG2
Description: RSFEC per lane configuration
Address: 0x0320.6AC8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC1_F0_STAT2
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.6B10
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC1_F0_HOLD2
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.6B14
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC1_F1_STAT2
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.6B20
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC1_F1_HOLD2
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.6B24
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC1_ERR_INJ2
Description: RS-FEC Error Injection Mode
Address: 0x0320.6B38
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC1_TX_STAT2
Description: RS-FEC Status per TX Lane
Address: 0x0320.6B48
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC1_TX_HOLD2
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.6B4C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC1_RX_STAT2
Description: RS-FEC Status per RX Lane
Address: 0x0320.6B54
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC1_RX_HOLD2
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.6B58
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

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Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_AGR_STAT2
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.6B60
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_AGR_HOLD2
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.6B64
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_LN_MAP2
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.6B6C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC1_LN_SKEW2
Description: RS-FEC Lane Skew
Address: 0x0320.6B70
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC1_CW_POS2
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.6B74
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC1_ECC2
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.6B78
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

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Name: SRDS_FEC1_TX_ERR2
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.6B7C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC1_CR_CNTL2
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.6B84
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC1_CR_CNTH2
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.6B88
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: SRDS_FEC1_UCR_CNTL2
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]
Address: 0x0320.6B8C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC1_UCR_CNTH2
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]
Address: 0x0320.6B90
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC1_CRS_CNTL2
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]
Address: 0x0320.6B94
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: SRDS_FEC1_CRS_CNTH2
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]
Address: 0x0320.6B98
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: SRDS_FEC1_CR0_CNTL2
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]
Address: 0x0320.6B9C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC1_CR0_CNTH2
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]
Address: 0x0320.6BA0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC1_CR1_CNTL2
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]
Address: 0x0320.6BA4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

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Name: SRDS_FEC1_CR1_CNTH2
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]
Address: 0x0320.6BA8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC1_CWBN0_CNT2
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.6BD0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: SRDS_FEC1_CWBN1_CNT2
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.6BD4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC1_CWBN2_CNT2
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.6BD8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC1_CWBN3_CNT2
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.6BDC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC1_DBG_CFG2
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.6BE0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC1_IF3
Description: xcvrif_reg_12
Address: 0x0320.6C70
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC1_RSFEC3
Description: RS-FEC Settings
Address: 0x0320.6CC0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

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Name: SRDS_FEC1_LN_CFG3
Description: RSFEC per lane configuration
Address: 0x0320.6CC8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC1_F0_STAT3
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.6D10
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC1_F0_HOLD3
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.6D14
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC1_F1_STAT3
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.6D20
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC1_F1_HOLD3
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.6D24
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC1_ERR_INJ3
Description: RS-FEC Error Injection Mode
Address: 0x0320.6D38
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC1_TX_STAT3
Description: RS-FEC Status per TX Lane
Address: 0x0320.6D48
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC1_TX_HOLD3
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.6D4C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC1_RX_STAT3
Description: RS-FEC Status per RX Lane
Address: 0x0320.6D54
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC1_RX_HOLD3
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.6D58
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_AGR_STAT3
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.6D60
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_AGR_HOLD3
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.6D64
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC1_LN_MAP3
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.6D6C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC1_LN_SKEW3
Description: RS-FEC Lane Skew
Address: 0x0320.6D70
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC1_CW_POS3
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.6D74
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC1_ECC3
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.6D78
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

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Name: SRDS_FEC1_TX_ERR3
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.6D7C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC1_CR_CNTL3
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.6D84
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC1_CR_CNTH3
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.6D88
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: **SRDS_FEC1_UCR_CNTL3**
Description: **RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]**
Address: **0x0320.6D8C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: **SRDS_FEC1_UCR_CNTH3**
Description: **RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]**
Address: **0x0320.6D90**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: **SRDS_FEC1_CRS_CNTL3**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]**
Address: **0x0320.6D94**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: **SRDS_FEC1_CRS_CNTH3**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]**
Address: **0x0320.6D98**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: **SRDS_FEC1_CR0_CNTL3**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]**
Address: **0x0320.6D9C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC1_CR0_CNTH3**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]**
Address: **0x0320.6DA0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC1_CR1_CNTL3**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]**
Address: **0x0320.6DA4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

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Name: SRDS_FEC1_CR1_CNTH3
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]
Address: 0x0320.6DA8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC1_CWBN0_CNT3
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.6DD0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: SRDS_FEC1_CWBN1_CNT3
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.6DD4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC1_CWBN2_CNT3
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.6DD8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC1_CWBN3_CNT3
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.6DDC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC1_DBG_CFG3
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.6DE0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC4_IF0
Description: xcvrif_reg_12
Address: 0x0320.7E70
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC4_RSFEC0
Description: RS-FEC Settings
Address: 0x0320.7EC0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

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Name: SRDS_FEC4_LN_CFG0
Description: RSFEC per lane configuration
Address: 0x0320.7EC8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC4_F0_STAT0
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.7F10
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC4_F0_HOLD0
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.7F14
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC4_F1_STAT0
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.7F20
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC4_F1_HOLD0
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.7F24
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC4_ERR_INJ0
Description: RS-FEC Error Injection Mode
Address: 0x0320.7F38
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC4_TX_STAT0
Description: RS-FEC Status per TX Lane
Address: 0x0320.7F48
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC4_TX_HOLD0
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.7F4C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC4_RX_STAT0
Description: RS-FEC Status per RX Lane
Address: 0x0320.7F54
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC4_RX_HOLD0
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.7F58
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

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Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_STAT0
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.7F60
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_HOLD0
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.7F64
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_LN_MAP0
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.7F6C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: **SRDS_FEC4_LN_SKEW0**
Description: **RS-FEC Lane Skew**
Address: **0x0320.7F70**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: **SRDS_FEC4_CW_POS0**
Description: **RS-FEC Codeword Bit Position on RX**
Address: **0x0320.7F74**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: **SRDS_FEC4_ECC0**
Description: **RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only**
Address: **0x0320.7F78**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for `.sbe` above.

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Name: SRDS_FEC4_TX_ERR0
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.7F7C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC4_CR_CNTL0
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.7F84
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CR_CNTH0
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.7F88
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: SRDS_FEC4_UCR_CNTL0
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]
Address: 0x0320.7F8C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_UCR_CNTH0
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]
Address: 0x0320.7F90
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CRS_CNTL0
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]
Address: 0x0320.7F94
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: SRDS_FEC4_CRS_CNTH0
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]
Address: 0x0320.7F98
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: SRDS_FEC4_CR0_CNTL0
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]
Address: 0x0320.7F9C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR0_CNTH0
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]
Address: 0x0320.7FA0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR1_CNTL0
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]
Address: 0x0320.7FA4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CR1_CNTH0**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]**
Address: **0x0320.7FA8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CWBN0_CNT0**
Description: **RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols**
Address: **0x0320.7FD0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: **SRDS_FEC4_CWBN1_CNT0**
Description: **RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols**
Address: **0x0320.7FD4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC4_CWBN2_CNT0
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.7FD8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC4_CWBN3_CNT0
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.7FDC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC4_DBG_CFG0
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.7FE0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC4_IF1
Description: xcvrif_reg_12
Address: 0x0320.8070
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC4_RSFEC1
Description: RS-FEC Settings
Address: 0x0320.80C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

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Name: SRDS_FEC4_LN_CFG1
Description: RSFEC per lane configuration
Address: 0x0320.80C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC4_F0_STAT1
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.8110
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC4_F0_HOLD1
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8114
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC4_F1_STAT1
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.8120
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC4_F1_HOLD1
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8124
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC4_ERR_INJ1
Description: RS-FEC Error Injection Mode
Address: 0x0320.8138
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC4_TX_STAT1
Description: RS-FEC Status per TX Lane
Address: 0x0320.8148
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC4_TX_HOLD1
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.814C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC4_RX_STAT1
Description: RS-FEC Status per RX Lane
Address: 0x0320.8154
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC4_RX_HOLD1
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.8158
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_STAT1
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8160
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_HOLD1
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8164
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_LN_MAP1
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.816C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC4_LN_SKEW1
Description: RS-FEC Lane Skew
Address: 0x0320.8170
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC4_CW_POS1
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.8174
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC4_ECC1
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.8178
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

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Name: SRDS_FEC4_TX_ERR1
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.817C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC4_CR_CNTL1
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.8184
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CR_CNTH1
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.8188
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: SRDS_FEC4_UCR_CNTL1
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]
Address: 0x0320.818C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_UCR_CNTH1
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]
Address: 0x0320.8190
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CRS_CNTL1
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]
Address: 0x0320.8194
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: **SRDS_FEC4_CRS_CNTH1**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]**
Address: **0x0320.8198**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: **SRDS_FEC4_CR0_CNTL1**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]**
Address: **0x0320.819C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CR0_CNTH1**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]**
Address: **0x0320.81A0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CR1_CNTL1**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]**
Address: **0x0320.81A4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR1_CNTH1
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]
Address: 0x0320.81A8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CWBN0_CNT1
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.81D0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: SRDS_FEC4_CWBN1_CNT1
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.81D4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC4_CWBN2_CNT1
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.81D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC4_CWBN3_CNT1
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.81DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC4_DBG_CFG1
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.81E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC4_IF2
Description: xcvrif_reg_12
Address: 0x0320.8270
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC4_RSFEC2
Description: RS-FEC Settings
Address: 0x0320.82C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

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Name: SRDS_FEC4_LN_CFG2
Description: RSFEC per lane configuration
Address: 0x0320.82C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC4_F0_STAT2
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.8310
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC4_F0_HOLD2
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8314
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC4_F1_STAT2
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.8320
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC4_F1_HOLD2
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8324
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC4_ERR_INJ2
Description: RS-FEC Error Injection Mode
Address: 0x0320.8338
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

Name: SRDS_FEC4_TX_STAT2
Description: RS-FEC Status per TX Lane
Address: 0x0320.8348
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC4_TX_HOLD2
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.834C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC4_RX_STAT2
Description: RS-FEC Status per RX Lane
Address: 0x0320.8354
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC4_RX_HOLD2
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.8358
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_STAT2
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8360
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_HOLD2
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8364
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_LN_MAP2
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.836C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC4_LN_SKEW2
Description: RS-FEC Lane Skew
Address: 0x0320.8370
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC4_CW_POS2
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.8374
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC4_ECC2
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.8378
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

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Name: SRDS_FEC4_TX_ERR2
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.837C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC4_CR_CNTL2
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.8384
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CR_CNTH2
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.8388
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: SRDS_FEC4_UCR_CNTL2
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]
Address: 0x0320.838C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_UCR_CNTH2
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]
Address: 0x0320.8390
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CRS_CNTL2
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]
Address: 0x0320.8394
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: **SRDS_FEC4_CRS_CNTH2**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]**
Address: **0x0320.8398**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: **SRDS_FEC4_CR0_CNTL2**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]**
Address: **0x0320.839C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CR0_CNTH2**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]**
Address: **0x0320.83A0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CR1_CNTL2**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]**
Address: **0x0320.83A4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

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Name: **SRDS_FEC4_CR1_CNTH2**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]**
Address: **0x0320.83A8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CWBN0_CNT2**
Description: **RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols**
Address: **0x0320.83D0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: **SRDS_FEC4_CWBN1_CNT2**
Description: **RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols**
Address: **0x0320.83D4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC4_CWBN2_CNT2
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.83D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC4_CWBN3_CNT2
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.83DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC4_DBG_CFG2
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.83E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC4_IF3
Description: xcvrif_reg_12
Address: 0x0320.8470
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC4_RSFEC3
Description: RS-FEC Settings
Address: 0x0320.84C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

Name: SRDS_FEC4_LN_CFG3
Description: RSFEC per lane configuration
Address: 0x0320.84C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC4_F0_STAT3
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.8510
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC4_F0_HOLD3
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8514
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC4_F1_STAT3
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.8520
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC4_F1_HOLD3
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8524
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC4_ERR_INJ3
Description: RS-FEC Error Injection Mode
Address: 0x0320.8538
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC4_TX_STAT3
Description: RS-FEC Status per TX Lane
Address: 0x0320.8548
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC4_TX_HOLD3
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.854C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC4_RX_STAT3
Description: RS-FEC Status per RX Lane
Address: 0x0320.8554
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC4_RX_HOLD3
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.8558
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_STAT3
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8560
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_HOLD3
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8564
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_LN_MAP3
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.856C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC4_LN_SKEW3
Description: RS-FEC Lane Skew
Address: 0x0320.8570
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC4_CW_POS3
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.8574
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC4_ECC3
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.8578
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for `.sbe` above.

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Name: SRDS_FEC4_TX_ERR3
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.857C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC4_CR_CNTL3
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.8584
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CR_CNTH3
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.8588
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: **SRDS_FEC4_UCR_CNTL3**
Description: **RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]**
Address: **0x0320.858C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: **SRDS_FEC4_UCR_CNTH3**
Description: **RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]**
Address: **0x0320.8590**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: **SRDS_FEC4_CRS_CNTL3**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]**
Address: **0x0320.8594**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: SRDS_FEC4_CRS_CNTH3
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]
Address: 0x0320.8598
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: SRDS_FEC4_CR0_CNTL3
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]
Address: 0x0320.859C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR0_CNTH3
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]
Address: 0x0320.85A0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR1_CNTL3
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]
Address: 0x0320.85A4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

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Name: **SRDS_FEC4_CR1_CNTH3**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]**
Address: **0x0320.85A8**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CWBN0_CNT3**
Description: **RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols**
Address: **0x0320.85D0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: **SRDS_FEC4_CWBN1_CNT3**
Description: **RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols**
Address: **0x0320.85D4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC4_CWBN2_CNT3
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.85D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC4_CWBN3_CNT3
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.85DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC4_DBG_CFG3
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.85E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC4_IF4
Description: xcvrif_reg_12
Address: 0x0320.8670
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC4_RSFEC4
Description: RS-FEC Settings
Address: 0x0320.86C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

Name: SRDS_FEC4_LN_CFG4
Description: RSFEC per lane configuration
Address: 0x0320.86C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC4_F0_STAT4
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.8710
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC4_F0_HOLD4
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8714
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC4_F1_STAT4
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.8720
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC4_F1_HOLD4
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8724
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC4_ERR_INJ4
Description: RS-FEC Error Injection Mode
Address: 0x0320.8738
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC4_TX_STAT4
Description: RS-FEC Status per TX Lane
Address: 0x0320.8748
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC4_TX_HOLD4
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.874C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC4_RX_STAT4
Description: RS-FEC Status per RX Lane
Address: 0x0320.8754
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC4_RX_HOLD4
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.8758
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

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Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_STAT4
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8760
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_HOLD4
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8764
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_LN_MAP4
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.876C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC4_LN_SKEW4
Description: RS-FEC Lane Skew
Address: 0x0320.8770
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC4_CW_POS4
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.8774
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC4_ECC4
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.8778
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

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Name: SRDS_FEC4_TX_ERR4
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.877C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC4_CR_CNTL4
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.8784
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CR_CNTH4
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.8788
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: SRDS_FEC4_UCR_CNTL4
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]
Address: 0x0320.878C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_UCR_CNTH4
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]
Address: 0x0320.8790
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CRS_CNTL4
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]
Address: 0x0320.8794
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: **SRDS_FEC4_CRS_CNTH4**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]**
Address: **0x0320.8798**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: **SRDS_FEC4_CR0_CNTL4**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]**
Address: **0x0320.879C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CR0_CNTH4**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]**
Address: **0x0320.87A0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CR1_CNTL4**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]**
Address: **0x0320.87A4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

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Name: SRDS_FEC4_CR1_CNTH4
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]
Address: 0x0320.87A8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CWBN0_CNT4
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.87D0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: SRDS_FEC4_CWBN1_CNT4
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.87D4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC4_CWBN2_CNT4
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.87D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC4_CWBN3_CNT4
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.87DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC4_DBG_CFG4
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.87E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC4_IF5
Description: xcvrif_reg_12
Address: 0x0320.8870
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC4_RSFEC5
Description: RS-FEC Settings
Address: 0x0320.88C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

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Name: SRDS_FEC4_LN_CFG5
Description: RSFEC per lane configuration
Address: 0x0320.88C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC4_F0_STAT5
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.8910
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC4_F0_HOLD5
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8914
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC4_F1_STAT5
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.8920
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC4_F1_HOLD5
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8924
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC4_ERR_INJ5
Description: RS-FEC Error Injection Mode
Address: 0x0320.8938
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC4_TX_STAT5
Description: RS-FEC Status per TX Lane
Address: 0x0320.8948
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC4_TX_HOLD5
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.894C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC4_RX_STAT5
Description: RS-FEC Status per RX Lane
Address: 0x0320.8954
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC4_RX_HOLD5
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.8958
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

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Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_STAT5
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8960
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_HOLD5
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8964
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_LN_MAP5
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.896C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC4_LN_SKEW5
Description: RS-FEC Lane Skew
Address: 0x0320.8970
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC4_CW_POS5
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.8974
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC4_ECC5
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.8978
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

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Name: SRDS_FEC4_TX_ERR5
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.897C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC4_CR_CNTL5
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.8984
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CR_CNTH5
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.8988
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: SRDS_FEC4_UCR_CNTL5
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]
Address: 0x0320.898C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_UCR_CNTH5
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]
Address: 0x0320.8990
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CRS_CNTL5
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]
Address: 0x0320.8994
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: SRDS_FEC4_CRS_CNTH5
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]
Address: 0x0320.8998
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: SRDS_FEC4_CR0_CNTL5
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]
Address: 0x0320.899C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR0_CNTH5
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]
Address: 0x0320.89A0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR1_CNTL5
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]
Address: 0x0320.89A4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

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Name: SRDS_FEC4_CR1_CNTH5
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]
Address: 0x0320.89A8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CWBN0_CNT5
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.89D0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: SRDS_FEC4_CWBN1_CNT5
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.89D4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC4_CWBN2_CNT5
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.89D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC4_CWBN3_CNT5
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.89DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC4_DBG_CFG5
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.89E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC4_IF6
Description: xcvrif_reg_12
Address: 0x0320.8A70
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC4_RSFEC6
Description: RS-FEC Settings
Address: 0x0320.8AC0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

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Name: SRDS_FEC4_LN_CFG6
Description: RSFEC per lane configuration
Address: 0x0320.8AC8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC4_F0_STAT6
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.8B10
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC4_F0_HOLD6
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8B14
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC4_F1_STAT6
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.8B20
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC4_F1_HOLD6
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8B24
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC4_ERR_INJ6
Description: RS-FEC Error Injection Mode
Address: 0x0320.8B38
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC4_TX_STAT6
Description: RS-FEC Status per TX Lane
Address: 0x0320.8B48
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC4_TX_HOLD6
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.8B4C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC4_RX_STAT6
Description: RS-FEC Status per RX Lane
Address: 0x0320.8B54
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC4_RX_HOLD6
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.8B58
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

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Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

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Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_STAT6
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8B60
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state `.not_align` shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_HOLD6
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8B64
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_LN_MAP6
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.8B6C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC4_LN_SKEW6
Description: RS-FEC Lane Skew
Address: 0x0320.8B70
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC4_CW_POS6
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.8B74
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC4_ECC6
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.8B78
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

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Name: SRDS_FEC4_TX_ERR6
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.8B7C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC4_CR_CNTL6
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.8B84
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CR_CNTH6
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.8B88
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: SRDS_FEC4_UCR_CNTL6
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]
Address: 0x0320.8B8C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_UCR_CNTH6
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]
Address: 0x0320.8B90
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CRS_CNTL6
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]
Address: 0x0320.8B94
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: **SRDS_FEC4_CRS_CNTH6**
Description: **RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]**
Address: **0x0320.8B98**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: **SRDS_FEC4_CR0_CNTL6**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]**
Address: **0x0320.8B9C**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CR0_CNTH6**
Description: **RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]**
Address: **0x0320.8BA0**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: **SRDS_FEC4_CR1_CNTL6**
Description: **RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]**
Address: **0x0320.8BA4**
Index: **Indirectly addressed via SERDES_SEL**

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR1_CNTH6
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]
Address: 0x0320.8BA8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CWBN0_CNT6
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.8BD0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: SRDS_FEC4_CWBN1_CNT6
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.8BD4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC4_CWBN2_CNT6
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.8BD8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC4_CWBN3_CNT6
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.8BDC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC4_DBG_CFG6
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.8BE0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_FEC4_IF7
Description: xcvrif_reg_12
Address: 0x0320.8C70
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	lpbkEn	RW	0x0	TX-RX loopback enable. This does not include UX/BK mapping.
1	forceSignalOk	RW	0x0	RX force signal_ok high - Unused.
2	signalOkSel	RW	0x0	RX select: sync'd or unsync'd - Unused.

Name: SRDS_FEC4_RSFEC7
Description: RS-FEC Settings
Address: 0x0320.8CC0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	fecDataMux	RW	0x0	Select FEC TX data source from: 1: EMIB Adapter 0: EHIP.
1	fecLpbkEn	RW	0x0	FEC Loopback enable FEC TX output to FEC RX input.
2	fecBypassEn	RW	0x0	FEC bypass enable for RX.

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Name: SRDS_FEC4_LN_CFG7
Description: RSFEC per lane configuration
Address: 0x0320.8CC8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
8	indicByp	RW	0x0	Set to bypass error indication (to reduce latency): 0: Synchronization headers of selected 66b PCS symbols extracted from uncorrectable FEC codewords are deliberately invalidated. 1: 66b PCS symbols extracted from uncorrectable codewords are not explicitly marked bad. When the number of FEC symbol errors in a block of 8192 consecutive codewords exceeds a threshold, then synchronization header errors will be generated towards the PCS layer for a period of 60ms to 75ms. Fire code: If .indicByp = 0, .fc = 0 then a subset of the 66b PCS symbols extracted from an uncorrectable codeword are invalidated. If .indicByp = 0, .fc = 1 then all the 32 PCS symbols extracted from an uncorrectable codeword are invalidated. This is required with 40GE. If .indicByp = 1 then no error indication occurs (hiSer is not defined with Fire code).
9	corrByp	RW	0x0	Set to bypass error correction (to reduce latency). Rx data is forwarded to the client directly. If the data was extracted from an error???ed codeword the data is NOT explicitly invalidated. Note that this differs from the behavior described in Clause 91. The error statistics are maintained regardless of this setting; i.e. as if data was corrected. Only the counting of bit errors is affected as no polarity information is available in this configuration. Fire code:.

Name: SRDS_FEC4_F0_STAT7
Description: xcvrif_stat_0: TX FIFO flags
Address: 0x0320.8D10
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RO	0x0	TX FIFO mode partially full flag.
1	txFifoFull	RO	0x0	TX FIFO mode full flag.
2	txFifoPempty	RO	0x0	TX FIFO mode partially empty flag.
3	txFifoEmpty	RO	0x0	TX FIFO mode empty flag.
4	txFifoUnderrun	RO	0x0	TX FIFO underrun flag.
5	txFifoOverflow	RO	0x0	TX FIFO overflow flag.
6	txGbRestarted	RO	0x0	TX DV GB restarted flag.
7	txGbReserved	RO	0x0	TX DV GB reserved flag.

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Name: SRDS_FEC4_F0_HOLD7
Description: xcvrif_stat_hold_1: TX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8D14
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	txFifoPfull	RW	0x0	TX FIFO mode partially full sticky flag.
1	txFifoFull	RW	0x0	TX FIFO mode full sticky flag.
2	txFifoPempty	RW	0x0	TX FIFO mode partially empty sticky flag.
3	txFifoEmpty	RW	0x0	TX FIFO mode empty sticky flag.
4	txFifoUnderrun	RW	0x0	TX FIFO underrun sticky flag.
5	txFifoOverflow	RW	0x0	TX FIFO overflow sticky flag.
6	txGbRestarted	RW	0x0	TX DV GB restarted sticky flag.
7	txGbReserved	RW	0x0	TX DV GB reserved sticky flag.

Name: SRDS_FEC4_F1_STAT7
Description: xcvrif_stat_3: RX FIFO flags
Address: 0x0320.8D20
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RO	0x0	RX FIFO mode partially full flag.
1	rxFifoFull	RO	0x0	RX FIFO mode full flag.
2	rxFifoPempty	RO	0x0	RX FIFO mode partially empty flag.
3	rxFifoEmpty	RO	0x0	RX FIFO mode empty flag.
4	rxFifoUnderrun	RO	0x0	RX FIFO underrun flag.
5	rxFifoOverflow	RO	0x0	RX FIFO overflow flag.
6	rxGbRestarted	RO	0x0	RX GB restarted flag.
7	rxGbReserved	RO	0x0	RX GB reserved flag.

Name: SRDS_FEC4_F1_HOLD7
Description: xcvrif_stat_hold_4: RX FIFO Sticky flags - Write 1 to Clear
Address: 0x0320.8D24
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	rxFifoPfull	RW	0x0	RX FIFO mode partially full sticky flag.
1	rxFifoFull	RW	0x0	RX FIFO mode full sticky flag.
2	rxFifoPempty	RW	0x0	RX FIFO mode partially empty sticky flag.

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Bits	Field	Type	Reset	Description
3	rxFifoEmpty	RW	0x0	RX FIFO mode empty sticky flag.
4	rxFifoUnderrun	RW	0x0	RX FIFO underrun sticky flag.
5	rxFifoOverflow	RW	0x0	RX FIFO overflow sticky flag.
6	rxGbRestarted	RW	0x0	RX GB restarted sticky flag.
7	rxGbReserved	RW	0x0	RX GB reserved sticky flag.

Name: SRDS_FEC4_ERR_INJ7
Description: RS-FEC Error Injection Mode
Address: 0x0320.8D38
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	rate	RW	0x0	TX error injection rate for each PMA interface lane. Data is output towards the PMA 40 bits at a time. The value specifies the fraction of such 40b words to hit. The unit is 1/256th so a value of, say, 7 causes 7/256th of the 40b words being sent on the lane to be hit. The distance between words being hit is approximately the same. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries and each PMA interface lane is manipulated separately and regardless of all other configuration. Fire code: Data is output towards the PMA 33 bits at a time. The value specifies the fraction of such 33b words to hit. The unit is 1/8K so a value of, say, 89 causes 89/8192th of the 33b words being sent on the lane to be hit and the distance bwteen words being hit will be 92 or 93. The rate is perturbed slightly on the different lanes to increase randomness. This is done by occasionally pausing the sequencer one cycle. The error inject mechanism is agnostic of FEC codeword boundaries.
15:8	pat	RW	0x0	Tx error injection pattern for each lane. The value specifies which bits are being toggled on each lane, when the lane is hit. There is an 8b pattern per lane. When a 40b word on a lane is hit, 8 consecutive bits starting in that 40b word (possibly extending into the next 40b) are XOR'ed with the pattern. The starting position of the 8 affected bits is varied pseudo randomly. Fire code: Only bits 3:0 are used. These specify the maximum burst length blen. A pseudo-random pattern of length blen is xor'ed onto the word being output. The offset within the word is incremented with each hit and with offsets larger than 33-blen the error pattern may extend into the next 33b word. The Fire code can correct bursts up to 11 bits. The error pattern is pseudo-random and setting blen to, say, 12 will cause about 1/4th of the actual bursts to have a length of 12, which is uncorrectable.

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Name: SRDS_FEC4_TX_STAT7
Description: RS-FEC Status per TX Lane
Address: 0x0320.8D48
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RO	-	PCS Tx 66b invalid synchronization header. Fire code: Same function.
1	blkInv	RO	-	PCS TX 66b Invalid Block Type.
2	resync	RO	-	PCS TX Alignment/Codeword Marker Resync. When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RO	-	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544, 514) or RS(272,258): .pace_inv is set when the layer above presents valid TX data in more than 33 consecutive cycles.
6:4	amSfTxIn	RO	-	am_sf presented on TX by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. Firecode: Not applicable.

Name: SRDS_FEC4_TX_HOLD7
Description: RS-FEC per TX Lane Hold Status Write 1 to clear
Address: 0x0320.8D4C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	hdrInv	RW	0x0	PCS TX 66b invalid synchronization header.
1	blkInv	RW	0x0	PCS TX 66b Invalid Block Type.
2	resync	RW	0x0	PCS TX Alignment/Codeword Marker Resync When markers are used it is set when the distance between sync inputs does not match. When transcoding is active it is also checked if the width of the sync pulse matches. If markers are not used, then the bit is set if the distance between two syncs is not a multiple of 640 transfers.
3	paceInv	RW	0x0	PCS TX Pacing Violation: -For RS(528,514): .pace_inv is never set. -For RS(544,514) or RS(272,258): .pace_inv is set when the layer above presents valid Tx data in more than 33 consecutive cycles.
6:4	amSfTxIn	RW	0x0	am_sf presented on Tx by upper layer Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}.

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Name: SRDS_FEC4_RX_STAT7
Description: RS-FEC Status per RX Lane
Address: 0x0320.8D54
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RO	-	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RO	-	RX lane not locked. Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
2	fec3Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RO	-	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers.
4	hiSer	RO	-	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RO	-	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RO	-	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used.

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Bits	Field	Type	Reset	Description
15:8	corrCwBinsLo	RO	-	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
23:16	corrCwBinsHi	RO	-	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RO	-	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
27:25	amSfRxIn	RO	-	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RS-FEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.

Name: SRDS_FEC4_RX_HOLD7
Description: RS-FEC per RX Lane Hold Status Write 1 to clear
Address: 0x0320.8D58
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	sf	RW	0x0	Incoming Signal Fail (PMA unable to lock to signal) One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.
1	notLocked	RW	0x0	RX lane not locked Not locked to alignment/codeword markers (400GE/200GE/100GE/128GFC/50GE/25GE) or to FEC codewords (when not using markers). One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac.

Bits	Field	Type	Reset	Description
2	fec3Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 3 consecutive FEC codewords could not be corrected. Restarts the synchronization. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Set upon reception of 8 consecutive err'd codewords (even if correctable).
3	am5Bad	RW	0x0	RX was locked (and aligned if multiple logical FEC lanes) but 5 consecutive alignment/codeword markers were not valid. Restarts the synchronization. Only applicable with protocols using markers. Fire code: Not applicable.
4	hiSer	RW	0x0	High symbol error rate. -Set when the number of symbol errors in a window exceeds a threshold. -The window is 8K consecutive codewords with RS(528,514), RS(544,514) and 16K codewords with RS(272,258). -The bit remains set for 60ms to 75ms thereafter. -If SRDS_FECx_LN_CFGy.indic_byp = 1, then synchronization header errors will be generated towards the client while .hi_ser is set. -For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used.
5	corrCw	RW	0x0	Set when a FEC codeword had one or more errors that could be corrected. For an 802.3ck 100GE client entry number 0 and 2 are used. For 8-lane clients entry number 0 in each of the two participating submodules are used. For 16-lane clients entry number 0 in submodules 0 and 2 are used. Fire code: Same function.
6	uncorrCw	RW	0x0	Set when a FEC codeword could not be corrected due to too many errors. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. Fire code: Same function.
15:8	corrCwBinsLo	RW	0x0	Set when 0-7 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n,n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.

Bits	Field	Type	Reset	Description
23:16	corrCwBinsHi	RW	0x0	Set when 8-15 symbols were corrected in a codeword. LSB set if no err in a CW, MSB set if 15 symbols were corrected. For an 802.3ck 100GE client the entries for lane number 0 and 2 are used. For a 8-lane client the entries for lane number 0 in each of the two participating submodules are used. For a 16-lane client the entries for lane number 0 in submodules 0 and 2 are used. If correction is bypassed, then the output is still driven as if errors were corrected. Fire code: Bit 0 set if no error in CW. Bit n, n is greater than 0 and less than 12, set if the distance between the first and last corrected bit was n-1.
24	degrSer	RW	0x0	Set when the FEC degraded SER status is set. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.
27:25	amSfRxIn	RW	0x0	am_sf received from peer. Only valid with 8- and 16-lane clients and only in entry 0 in the first RSFEC submodule for the client. These are the accepted values, i.e. a bit is updated when the received value matches the previous. Order is {RD bit (Remote Degraded), LD bit (Local Degraded), reserved bit}. For 8- and 16-lane clients entry number 0 in the lowest numbered participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_STAT7
Description: RS-FEC per Aggregate RX Status. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8D60
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RO	-	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RO	-	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

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Bits	Field	Type	Reset	Description
5:2	lastLane	RO	-	Last Rx PMA interface lane for the client. Only applicable when the client has multiple logical FEC lanes. When all the Rx lanes become aligned this field shows which PMA interface lane was the latest. For an N-lane client the number is in the range 0..n-1, where 0 is for the lowest numbered PMA interface lane in the aggregation. Regardless of dynamic skew variations it will not change while the link is up. It shows which PMA interface lane to use as reference when determining the receive path data delay to be used by timing protocols. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered, participating submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_AGR_HOLD7
Description: RS-FEC per Aggregate RX Hold. Write 1 to clear. Valid on Master Streams Only such as 50G/100G/200G/400G streams
Address: 0x0320.8D64
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	notAlign	RW	0x0	Client's Rx FEC lanes not aligned. Incoming signal fail, not all participating Rx lanes locked, alignment markers not unique or skew too large. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.
1	notDeskew	RW	0x0	Client's Rx lanes were all locked but the alignment markers were not unique or the skew was too large. This is an event that restarts the synchronization and the status will deassert immediately. The hold register shows if it occurred. To determine the current alignment state .not_align shall be used instead. Only applicable when the client has multiple logical FEC lanes. For a 4-lane client entry number 0 is used. For an 8- or 16-lane client entry number 0 in the lowest numbered submodule is used. Fire code: Not applicable.

Name: SRDS_FEC4_LN_MAP7
Description: RS-FEC FEC Lane Mapping
Address: 0x0320.8D6C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
3:0	fecLane	RO	-	FEC lane number received on the PMA interface lane. Only applicable when the lane is locked and the client has multiple logical FEC lanes. Fire code: Not applicable.

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Name: SRDS_FEC4_LN_SKEW7
Description: RS-FEC Lane Skew
Address: 0x0320.8D70
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	skew	RO	-	Lane skew value (unit is 40 bits). Only applicable when the client has multiple logical FEC lanes and only valid when the Rx lanes are aligned. The value shows approximately how much a lane is delayed relative to the first arriving (i.e. earliest) lane for the client. The latest arriving lane of a client thus has the largest value. Fire code: Not applicable.

Name: SRDS_FEC4_CW_POS7
Description: RS-FEC Codeword Bit Position on RX
Address: 0x0320.8D74
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	num	RO	-	Distance from bit marked on PMA interface input lane to first bit in group of FEC codewords. The same value is output on a port towards the PMA logic. May be used to inspect the deterministic latency mechanism. Fire code: Same function.

Name: SRDS_FEC4_ECC7
Description: RS-FEC SRAM ECC Status Hold Write 1 to clear. Applicable to LPHY 100G stream 0 fractured space only
Address: 0x0320.8D78
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	sbe	RW	0x0	SRAM ECC correctable (single bit) error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Should not become set during normal operation. One bit per SRAM. Bits 0-3 covers the deskew buffers for PMA interface lanes 0-3 (only used for clients with multiple logical FEC lanes). Bits 4-7 covers the data delay buffers within the FEC decoder. Fire code: Same function, i.e. bits 4-7 used.
15:8	mbe	RW	0x0	SRAM ECC uncorrectable error detected. This is only valid in the fractured register space LPHY 100G Stream 0. Same bit ordering as for .sbe above.

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Name: SRDS_FEC4_TX_ERR7
Description: RS-FEC per Lane Error Injection Status
Address: 0x0320.8D7C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	inj0S	RO	-	Number of bits (modulo 256) that were changed from 1 to 0 on each PMA interface lane. Cleared when the corresponding RSFEC_ERR_INJ_TX.rate is written with a non-zero value after being all zero, i.e. when a test is initiated. A value read from this register is not reliable while injecting. A value read is reliable when the test is completed, i.e. after the lane's RSFEC_ERR_INJ_TX.rate or RSFEC_ERR_INJ_TX.pat has been cleared. One entry per PMA interface lane, regardless of RSFEC_CORE_CFG.frac. Fire code: Same function.
15:8	inj1S	RO	-	Same for bits changed from 0 to 1 on each PMA interface lane.

Name: SRDS_FEC4_CR_CNTL7
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [31:0]
Address: 0x0320.8D84
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CR_CNTH7
Description: RS-FEC number of FEC Codewords with Errors that were corrected, bits [63:32]
Address: 0x0320.8D88
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

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Name: SRDS_FEC4_UCR_CNTL7
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [31:0]
Address: 0x0320.8D8C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_UCR_CNTH7
Description: RS-FEC number of FEC Codewords that cannot be corrected due to too many errors, bits [63:32]
Address: 0x0320.8D90
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	Statistics value. For an N lane client the entry for the 1st lane assigned to the client is used. For a client using two codewords A and B in parallel, two entries are used: An 802.3ck 100GE client uses entries 0 and 2 in the submodule for A and B respectively. An 8-lane client uses entry 0 in the even submodule for A and entry 0 in the odd submodule for B. A 16-lane client uses entry 0 in submodule 0 for A and entry 0 in submodule 2 for B. Fire code: Same function.

Name: SRDS_FEC4_CRS_CNTL7
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [31:0]
Address: 0x0320.8D94
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

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Name: SRDS_FEC4_CRS_CNTH7
Description: RS-FEC number of 10b symbols corrected for the FEC lane, bits [63:32]
Address: 0x0320.8D98
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter still counts the number of error'ed 10b symbols detected.

Name: SRDS_FEC4_CR0_CNTL7
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [31:0]
Address: 0x0320.8D9C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR0_CNTH7
Description: RS-FEC number of bits corrected 0->1 for the FEC lane, bits [63:32]
Address: 0x0320.8DA0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR1_CNTL7
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [31:0]
Address: 0x0320.8DA4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CR1_CNTH7
Description: RS-FEC number of bits corrected 1->0 for the FEC lane, bits [63:32]
Address: 0x0320.8DA8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	stat	RO	0x0	If correction is bypassed, then this counter counts the total number of bit errors detected regardless of polarity.

Name: SRDS_FEC4_CWBN0_CNT7
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.8DD0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin0Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin1Stat	RO	0x0	
23:16	cwbin2Stat	RO	0x0	
31:24	cwbin3Stat	RO	0x0	

Name: SRDS_FEC4_CWBN1_CNT7
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.8DD4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin4Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin5Stat	RO	0x0	
23:16	cwbin6Stat	RO	0x0	
31:24	cwbin7Stat	RO	0x0	

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Name: SRDS_FEC4_CWBN2_CNT7
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.8DD8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin8Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin9Stat	RO	0x0	
23:16	cwbin10Stat	RO	0x0	
31:24	cwbin11Stat	RO	0x0	

Name: SRDS_FEC4_CWBN3_CNT7
Description: RS-FEC number of corrected codewords having Errs=0,1,..,15 errored symbols
Address: 0x0320.8DDC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	cwbin12Stat	RO	0x0	-Firecode: Number of CWs with error burst length N, N=0,1..11. - For an N lane client the 16 counters for the 1st lane assigned to the client is used. - For a client using two codewords in parallel, two sets are used: - An 802.3ck 100GE client uses the 16 counters for lane 0 and the 16 counters for lane 2 in the submodule. - An 8-lane client uses the 16 counters for lane 0 in each of the two participating submodules. - A 16-lane client uses the 16 counters for lane 0 in each of submodules 0 and 2. -Fire code: - Error burst length is 0 if no error, else pos(last corrected bit) - pos(first corrected bit) + 1.
15:8	cwbin13Stat	RO	0x0	
23:16	cwbin14Stat	RO	0x0	
31:24	cwbin15Stat	RO	0x0	

Name: SRDS_FEC4_DBG_CFG7
Description: Extra Configuration and Debug Register for fec_clock
Address: 0x0320.8DE0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	shadowReq	RW	0x0	Snapshot or Shadow Request Freeze rsfec counter CSRs so that all counter values read from the registers will be from the same moment. Note that the actual collection counters are not frozen, but because they are all 'read' at the time of the freeze, they are cleared.

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Bits	Field	Type	Reset	Description
4	shadowClear	RW	0x0	Clear RS-FEC Counters. Clear the collection and shadow counters so that the next shadow request or snapshot will start from 0. If the counters are not cleared, they will continue counting and rollover.

Name: SRDS_DR_TRIGGER
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0000
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	drTrigger	RW	0x0	Trigger Reconfig When set to 1 by host software, NIOS shall initiate reconfiguration sequences NIOS clear it to 0 when reconfiguration sequences completes.
1	drNextTrigger	RW	0x0	Ready For New Trigger Indicate whether the DR IP is ready to take in a new Trigger Reconfig request Host software must poll for the readiness status before triggering a new request through Trigger Reconfig.
31:2	reserved	RO	0x0	Reserved.

Name: SRDS_DR_NXT_PRFL_0
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0004
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile	RW	0x0	Next Profile 0 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd	RW	0x0	Next Profile 0 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 0 1: Apply active/enable profile settings to Next Profile 0 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
16	finalNeturalInd	RW	0x0	.
17	startupRestrInd	RW	0x0	Start Up IP restore settings Indicator Host can use this field to load startup profile settings. When DR firmware terminates with error This required new DR trigger request (Trigger Reconfig) DR Controller doesn't handle it automatically. Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Bits	Field	Type	Reset	Description
18	disableCpiSeq	RW	0x0	Disable CPI sequence When set to 1, DR controller firmware bypass CPI sequence for UX XCVR channels adaptation Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
19	bypassMaibRecfg	RW	0x0	Bypass MAIB reconfiguration When set to 1, DR controller firmware bypass MAIB reconfiguration Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
20	srdrmifMode	RW	0x0	SRDRMIF mode When set to 1, DR controller firmware treats the profiles as SRDRMIF profiles Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31:21	reserved	RO	0x0	Reserved.

Name: SRDS_DR_NXT_PRFL_1
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0008
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile1	RW	0x0	Next Profile 1 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd1	RW	0x0	Next Profile 1 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 1 1: Apply active/enable profile settings to Next Profile 1 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile2	RW	0x0	Next Profile 2 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd2	RW	0x0	Next Profile 2 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 2 1: Apply active/enable profile settings to Next Profile 2 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Name: SRDS_DR_NXT_PRFL_2
Description: Dynamic reconfiguration control and status register
Address: 0x0330.000C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile3	RW	0x0	Next Profile 3 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd3	RW	0x0	Next Profile 3 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 3 1: Apply active/enable profile settings to Next Profile 3 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile4	RW	0x0	Next Profile 4 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd4	RW	0x0	Next Profile 4 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 4 1: Apply active/enable profile settings to Next Profile 4 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_3
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0010
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile5	RW	0x0	Next Profile 5 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd5	RW	0x0	Next Profile 5 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 5 1: Apply active/enable profile settings to Next Profile 5 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Bits	Field	Type	Reset	Description
30:16	nextProfile6	RW	0x0	Next Profile 6 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd6	RW	0x0	Next Profile 6 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 6 1: Apply active/enable profile settings to Next Profile 6 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_4
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0014
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile7	RW	0x0	Next Profile 7 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd7	RW	0x0	Next Profile 7 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 7 1: Apply active/enable profile settings to Next Profile 7 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile8	RW	0x0	Next Profile 8 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd8	RW	0x0	Next Profile 8 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 8 1: Apply active/enable profile settings to Next Profile 8 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Name: SRDS_DR_NXT_PRFL_5
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0018
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile9	RW	0x0	Next Profile 9 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd9	RW	0x0	Next Profile 9 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 9 1: Apply active/enable profile settings to Next Profile 9 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile10	RW	0x0	Next Profile 10 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd10	RW	0x0	Next Profile 10 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 10 1: Apply active/enable profile settings to Next Profile 10 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_6
Description: Dynamic reconfiguration control and status register
Address: 0x0330.001C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile11	RW	0x0	Next Profile 11 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd11	RW	0x0	Next Profile 11 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 11 1: Apply active/enable profile settings to Next Profile 11 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Bits	Field	Type	Reset	Description
30:16	nextProfile12	RW	0x0	Next Profile 12 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd12	RW	0x0	Next Profile 12 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 12 1: Apply active/enable profile settings to Next Profile 12 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_7
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0020
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile13	RW	0x0	Next Profile 13 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd13	RW	0x0	Next Profile 13 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 13 1: Apply active/enable profile settings to Next Profile 13 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile14	RW	0x0	Next Profile 14 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd14	RW	0x0	Next Profile 14 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 14 1: Apply active/enable profile settings to Next Profile 14 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Name: SRDS_DR_NXT_PRFL_8
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0024
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile15	RW	0x0	Next Profile 15 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd15	RW	0x0	Next Profile 15 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 15 1: Apply active/enable profile settings to Next Profile 15 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile16	RW	0x0	Next Profile 16 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd16	RW	0x0	Next Profile 16 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 16 1: Apply active/enable profile settings to Next Profile 16 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_9
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0028
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile17	RW	0x0	Next Profile 17 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd17	RW	0x0	Next Profile 17 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 17 1: Apply active/enable profile settings to Next Profile 17 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Bits	Field	Type	Reset	Description
30:16	nextProfile18	RW	0x0	Next Profile 18 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd18	RW	0x0	Next Profile 18 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 18 1: Apply active/enable profile settings to Next Profile 18 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_10
Description: Dynamic reconfiguration control and status register
Address: 0x0330.002C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile19	RW	0x0	Next Profile 19 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd19	RW	0x0	Next Profile 19 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 19 1: Apply active/enable profile settings to Next Profile 19 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile20	RW	0x0	Next Profile 20 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd20	RW	0x0	Next Profile 20 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 20 1: Apply active/enable profile settings to Next Profile 20 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Name: SRDS_DR_NXT_PRFL_11
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0030
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile21	RW	0x0	Next Profile 21 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd21	RW	0x0	Next Profile 21 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 21 1: Apply active/enable profile settings to Next Profile 21 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile22	RW	0x0	Next Profile 22 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd22	RW	0x0	Next Profile 22 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 22 1: Apply active/enable profile settings to Next Profile 22 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_12
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0034
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile23	RW	0x0	Next Profile 23 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd23	RW	0x0	Next Profile 23 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 23 1: Apply active/enable profile settings to Next Profile 23 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Bits	Field	Type	Reset	Description
30:16	nextProfile24	RW	0x0	Next Profile 24 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd24	RW	0x0	Next Profile 24 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 24 1: Apply active/enable profile settings to Next Profile 24 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_13
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0038
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile25	RW	0x0	Next Profile 25 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd25	RW	0x0	Next Profile 25 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 25 1: Apply active/enable profile settings to Next Profile 25 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile26	RW	0x0	Next Profile 26 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd26	RW	0x0	Next Profile 26 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 26 1: Apply active/enable profile settings to Next Profile 26 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Name: SRDS_DR_NXT_PRFL_14
Description: Dynamic reconfiguration control and status register
Address: 0x0330.003C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile27	RW	0x0	Next Profile 27 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd27	RW	0x0	Next Profile 27 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 27 1: Apply active/enable profile settings to Next Profile 27 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile28	RW	0x0	Next Profile 28 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd28	RW	0x0	Next Profile 28 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 28 1: Apply active/enable profile settings to Next Profile 28 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_15
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0040
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile29	RW	0x0	Next Profile 29 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd29	RW	0x0	Next Profile 29 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 29 1: Apply active/enable profile settings to Next Profile 29 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Bits	Field	Type	Reset	Description
30:16	nextProfile30	RW	0x0	Next Profile 30 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd30	RW	0x0	Next Profile 30 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 30 1: Apply active/enable profile settings to Next Profile 30 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_16
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0044
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile31	RW	0x0	Next Profile 31 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd31	RW	0x0	Next Profile 31 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 31 1: Apply active/enable profile settings to Next Profile 31 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile32	RW	0x0	Next Profile 32 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd32	RW	0x0	Next Profile 32 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 32 1: Apply active/enable profile settings to Next Profile 32 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Name: SRDS_DR_NXT_PRFL_17
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0048
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile33	RW	0x0	Next Profile 33 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd33	RW	0x0	Next Profile 33 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 33 1: Apply active/enable profile settings to Next Profile 33 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile34	RW	0x0	Next Profile 34 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd34	RW	0x0	Next Profile 34 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 34 1: Apply active/enable profile settings to Next Profile 34 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_18
Description: Dynamic reconfiguration control and status register
Address: 0x0330.004C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile35	RW	0x0	Next Profile 35 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd35	RW	0x0	Next Profile 35 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 35 1: Apply active/enable profile settings to Next Profile 35 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Bits	Field	Type	Reset	Description
30:16	nextProfile36	RW	0x0	Next Profile 36 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd36	RW	0x0	Next Profile 36 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 36 1: Apply active/enable profile settings to Next Profile 36 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

Name: SRDS_DR_NXT_PRFL_19
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0050
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	nextProfile37	RW	0x0	Next Profile 37 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
15	nextTriggerInd37	RW	0x0	Next Profile 37 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 37 1: Apply active/enable profile settings to Next Profile 37 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
30:16	nextProfile38	RW	0x0	Next Profile 38 This field represents a user profile ID to be DR into from the current one Host software set this user provided ID before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.
31	nextTriggerInd38	RW	0x0	Next Profile 38 Indicator This field determines whether corresponding Next Profile is the active profile or neutral profile 0: Apply Neutral/powerdown profile settings to Next Profile 38 1: Apply active/enable profile settings to Next Profile 38 Host software set this bit before Trigger Reconfig Host software write access is permitted only when Ready For New Trigger is 1.

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Name: SRDS_DR_AVMM_TO
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0054
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
7:0	reserved0	RO	0x0	.
27:8	avmmTimeout	RW	0x1fff	AVMM Access Time-Out Tick Program this field to modify the time-out value of an AVMM access cycle This target DR CSR address space and Tile CSR address space (via DR IP access path). The time-out resolution is in 256 DR CSR clock ticks. 0 = Disable time-out 1 = 1*256 clock ticks ..... N = N*256 clock ticks Host software write access is permitted only when Ready For New Trigger is 1.
31:28	reserved1	RO	0x0	.

Name: SRDS_DR_TXCH_BUSY
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0058
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	txCh2RecfgBusy	RW	0x0	TXChannel2ReconfigureBusy Bit 0 of this field maps to XCVR channel 0, Bit 1 maps to XCVR channel 1 and so on. When 1: Indicates that a given TX channel is currently busy and must not be set up for another reconfiguration. Upon write busy bit for all XCVR will be clear and DR SIP will be ready for another transaction. Host software write access is permitted only when Ready For New Trigger is 1.
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_RXCH_BUSY
Description: Dynamic reconfiguration control and status register
Address: 0x0330.005C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	rxCh2RecfgBusy	RW	0x0	RXChannel2ReconfigureBusy Bit 0 of this field maps to XCVR channel 0, Bit 1 maps to XCVR channel 1 and so on. When 1: Indicates that a given RX channel is currently busy and must not be set up for another reconfiguration. Upon write busy bit for all XCVR will be clear and DR SIP will be ready for another transaction. Host software write access is permitted only when Ready For New Trigger is 1.
31:20	reserved	RO	0x0	Reserved.

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Name: SRDS_DR_TX_RST_ACK
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0060
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	resetAck	RO	0x0	TX Fully in reset Acknowledge TX Current reset state for SRC xcvr 0 to 19 & datapath fully in reset state.
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_TX_OUT_RST
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0064
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	resetOperational	RO	0x0	TX Fully out of reset TX Current reset state for SRC xcvr 0 to 19 & datapath fully out of reset state (operational).
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_TX_RST_INIT
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0068
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	resetInit	RO	0x0	TX Fully out of reset TX Current reset state for SRC xcvr 0 to 19 & datapath reset control initialized (system clock stable).
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_TX_SRC_ALARM
Description: Dynamic reconfiguration control and status register
Address: 0x0330.006C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	srcAlarm	RO	0x0	TX SRC Alarm TX Current alarm state for SRC xcvr 0 to 19 & datapath TX SRC alarm.
31:20	reserved	RO	0x0	Reserved.

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Name: SRDS_DR_RX_RST_ACK
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0070
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	resetAck	RO	0x0	RX Fully in reset Acknowledge RX Current reset state for SRC xcvr 0 to 19 & datapath fully in reset state.
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_RX_OUT_RST
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0074
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	resetOperational	RO	0x0	RX Fully out of reset RX Current reset state for SRC xcvr 0 to 19 & datapath fully out of reset state (operational).
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_RX_RST_INIT
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0078
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	resetInit	RO	0x0	RX Fully out of reset RX Current reset state for SRC xcvr 0 to 19 & datapath reset control initialized (system clock stable).
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_RX_SRC_ALARM
Description: Dynamic reconfiguration control and status register
Address: 0x0330.007C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	srcAlarm	RO	0x0	RX SRC Alarm RX Current alarm state for SRC xcvr 0 to 19 & datapath RX SRC alarm.
31:20	reserved	RO	0x0	Reserved.

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Name: SRDS_DR_TXCH
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0180
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	txCh2Recfg	RW	0x0	TXChannel2Reconfigure Bit 0 of this field maps to XCVR channel 0, Bit 1 maps to XCVR channel 1 and so on This field must be configured correctly prior to setting the Trigger Reconfig bit.
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_RXCH
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0184
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	rxCh2Recfg	RW	0x0	RXChannel2Reconfigure Bit 0 of this field maps to XCVR channel 0, Bit 1 maps to XCVR channel 1 and so on This field must be configured correctly prior to setting the Trigger Reconfig bit.
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_TX_CLR_ALARM
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0188
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	clearAlarm	RW	0x0	TX SRC clear alarm Clears alarm state for SRC TX lane.
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_RX_CLR_ALARM
Description: Dynamic reconfiguration control and status register
Address: 0x0330.018C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	clearAlarm	RW	0x0	RX SRC clear alarm Clears alarm state for SRC RX lane.
31:20	reserved	RO	0x0	Reserved.

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Name: SRDS_DR_APPLY
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0190
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	applyReset	RW	0x0	Apply Channel Reset Program this field to 1 to apply pre-programmed Apply TX Channel and RX Channel Reset bit settings to Soft Reset Controller.
1	applyMuxSel	RW	0x0	Apply Channel Mux Select Program this field to 1 to apply pre-programmed Apply TX Channel and RX Channel Mux Select bit settings to DR MUX.
2	apply1HotMuxSel	RW	0x0	Apply One-hot AIB Mux Select Program this field to 1 to apply pre-programmed Apply TX Channel and RX Channel ont-hot AIB Mux Selection.
31:3	reserved	RO	0x0	Reserved.

Name: SRDS_DR_PAUSE_REQ
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0194
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
0	pauseRequest	RW	0x0	Apply Channel Reset Request pause to reset service (required before dynamically modifying group configuration info).
1	pauseGrant	RO	0x0	Apply Channel Mux Select Reset service paused acknowledge (grant).
31:2	reserved	RO	0x0	Reserved.

Name: SRDS_DR_TX_RST
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0198
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	channelReset	RW	0x0	TX Channel Reset Program this field to assert or de-assert reset request for selected TX channel(s) to Soft Reset Controller This field must be configured correctly prior to setting the Apply Channel Reset bit Program this field to assert or de-assert reset request Bit 0 = TX channel 0 Bit 1 = TX channel 1 ..... Bit N = TX channel N 0 = no reset request 1 = reset request.
31:20	reserved	RO	0x0	Reserved.

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Name: SRDS_DR_RX_RST
Description: Dynamic reconfiguration control and status register
Address: 0x0330.019C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	channelReset	RW	0x0	RX Channel Reset Program this field to assert or de-assert reset request for selected RX channel(s) to Soft Reset Controller This field must be configured correctly prior to setting the Apply Channel Reset bit Program this field to assert or de-assert reset request Bit 0 = RX channel 0 Bit 1 = RX channel 1 ..... Bit N = RX channel N 0 = no reset request 1 = reset request.
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_MUX_SEL_0
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01A0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m0	RW	0x0	OneHot MUX Select 0 TX/RX Profile 1-32 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_1
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01A4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m1	RW	0x0	OneHot MUX Select 1 TX/RX Profile 33-64 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_2
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01A8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m2	RW	0x0	OneHot MUX Select 2 TX/RX Profile 65-96 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

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Name: SRDS_DR_MUX_SEL_3
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01AC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m3	RW	0x0	OneHot MUX Select 3 TX/RX Profile 97-128 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_4
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01B0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m4	RW	0x0	OneHot MUX Select 4 TX/RX Profile 129-160 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_5
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01B4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m5	RW	0x0	OneHot MUX Select 5 TX/RX Profile 161-192 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_6
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01B8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m6	RW	0x0	OneHot MUX Select 6 TX/RX Profile 193-224 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

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Name: SRDS_DR_MUX_SEL_7
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01BC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m7	RW	0x0	OneHot MUX Select 7 TX/RX Profile 225-256 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_8
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01C0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m8	RW	0x0	OneHot MUX Select 8 TX/RX Profile 257-288 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_9
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01C4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m9	RW	0x0	OneHot MUX Select 9 TX/RX Profile 289-320 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_10
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01C8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m10	RW	0x0	OneHot MUX Select 10 TX/RX Profile 321-352 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

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Name: SRDS_DR_MUX_SEL_11
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01CC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m11	RW	0x0	OneHot MUX Select 11 TX/RX Profile 353-384 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_12
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01D0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m12	RW	0x0	OneHot MUX Select 12 TX/RX Profile 385-416 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_13
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01D4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m13	RW	0x0	OneHot MUX Select 13 TX/RX Profile 417-448 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_14
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01D8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m14	RW	0x0	OneHot MUX Select 14 TX/RX Profile 449-480 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

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Name: SRDS_DR_MUX_SEL_15
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01DC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m15	RW	0x0	OneHot MUX Select 15 TX/RX Profile 481-512 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_16
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01E0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m16	RW	0x0	OneHot MUX Select 16 TX/RX Profile 513-544 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_17
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01E4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m17	RW	0x0	OneHot MUX Select 17 TX/RX Profile 545-576 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_18
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01E8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m18	RW	0x0	OneHot MUX Select 18 TX/RX Profile 577-608 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

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Name: SRDS_DR_MUX_SEL_19
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01EC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m19	RW	0x0	OneHot MUX Select 19 TX/RX Profile 609-640 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_20
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01F0
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m20	RW	0x0	OneHot MUX Select 20 TX/RX Profile 641-672 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_21
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01F4
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m21	RW	0x0	OneHot MUX Select 21 TX/RX Profile 673-704 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_MUX_SEL_22
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01F8
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m22	RW	0x0	OneHot MUX Select 22 TX/RX Profile 705-736 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

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Name: SRDS_DR_MUX_SEL_23
Description: Dynamic reconfiguration control and status register
Address: 0x0330.01FC
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
31:0	m23	RW	0x0	OneHot MUX Select 23 TX/RX Profile 737-768 Mux Select: Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on one hot selector port.

Name: SRDS_DR_ERR
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0200
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
4:0	reserved0	RW	0x1	Reserved.
7:5	reserved	RO	0x0	.
15:8	avmmTimeoutErr	RO	0x0	AVMM (GAVMM/MAIB) Timeout Presents the DR timeout error due to HIP AVMM (MAIB and GAVMM) accesses [0] = GAVMM timeout, [1] = MAIB timeout, [7:2] reserved.
23:16	firmwareErrState	RW	0x0	firmwareErrorState Presents the DR terminated and firmware flaged out error.
31:24	firmwareState	RW	0x0	firmwareState PPresents the Firmware current state.

Name: SRDS_DR_CURR_PRFL
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0204
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
14:0	currentProfile	RW	0x0	Current Profile This field represents a user defined profile. N = User Profile#N For example; 0=User Profile0, 1=User Profile1.
15	nextProfileInd	RW	0x0	Next profile Indicator Whether next profile is neutral or active profile.
31:16	reserved	RO	0x0	Reserved.

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Name: SRDS_DR_TX_SRC_SEL
Description: Dynamic reconfiguration control and status register
Address: 0x0330.0208
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	muxSel	RW	0x0	SRC TX Channel Mux Select Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on SRC TX Channel 0-19.
31:20	reserved	RO	0x0	Reserved.

Name: SRDS_DR_RX_SRC_SEL
Description: Dynamic reconfiguration control and status register
Address: 0x0330.020C
Index: Indirectly addressed via SERDES_SEL

Bits	Field	Type	Reset	Description
19:0	muxSel	RW	0x0	SRC RX Channel Mux Select Determine the profile/protocol being selected (i.e. muxed/demuxed), currently on SRC RX Channel 0-19.
31:20	reserved	RO	0x0	Reserved.

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4 General Configuration

4.1 Features

- Miscellaneous toplevel functions (scratchpad, interrupt control, CPU latency measurements etc.)
- Deterministic Timing Architecture (DTA)
- Indexing to address:
    - Rate and scheduling control
    - Clock synchronization
    - Filter control
    - OTN overhead configuration and monitoring

4.2 Overview

The functional blocks described in this chapter perform a number of toplevel functions related to the device operation.

A scratchpad register provides a means of testing simple R/W access to the device during bringup.

The interrupt controller provides top-level interrupt status from the main functional blocks inside the device, and allows individual interrupt sources (from entire functional blocks) to be disabled under software control.

The CPU interface latency measurement function provides continuous measurement on the CPU register access response time, and latches the longest access as well as the address used. This is primarily a test function.

Signal timing throughout the device is controlled using the Deterministic Timing Architecture (DTA). See **Deterministic Timing Architecture (DTA)** on page 385 for details.

4.2.1 Indexing

The Rate / Scheduler functions implement datapath scheduling and signal rate control required at various locations through the device. See **Rate and scheduling control** on page 475 for details of the Rate / Scheduler block.

The filter implements low pass filtering of the control signal to the client Tx clock control output. See **Filter** on page 440 for details of the filter block, and **Filter indexing** on page 388 for indexing of the relevant registers.

The OTN overhead termination satellite blocks are distributed through the device. See **OTN Overhead** on page 395 for details of the OTN OH processing functions, and **OTN overhead indexing** on page 389 for indexing of the relevant registers.

4.3 Operation

4.3.1 Scratchpad

The device supports two sets of general purpose storage, a 32-bit scratchpad register named SCRATCH, and a memory area (array of 32-bit words) named SCR_RAM.

Reads and writes to these areas have no other effect on the device operation.

4.3.2 Core configuration

The CORE_CFG register is used to configure any routing and other toplevel operational parameters.

If OTL is used for line interfaces, `.otlEn` shall be set to select this in the port muxes for the DDQ0 and DDQ1 ports individually. Setting `.client10g[6-5]` will set the port muxes to use 10G clients on the DDQ ports. Setting `.client10g[3-0]` will set the port muxes to use 10G or 16GFC clients on the Q and S ports. Setting `.client32gfc[3-0]` will set the port muxes to use 32GFC clients on the S ports. Setting `.client64gfc[3-0]` will set the port muxes to use 10G clients on the S ports. Note only one of the clientXX can be enabled per mux.

4.3.3 Interrupts

The device has a single interrupt output through which all internal interrupt conditions are signalled. The interrupts are collected from the functional blocks within the chip and combined into a single signal. A global enable bit, which must be set to '1' to allow any device interrupt, is provided in `INT_ENABLE.all`.

The combination and presentation of the interrupt status is done through the registers INT_STAT and INT_ENABLE. Each of the functional blocks capable of generating interrupts has one or more interrupt lines coming out of the block. Each of these interrupt lines has been allocated a bit position in both of the above registers. The INT_STAT register is a read-only register which always reflects the status of the on-chip interrupt lines. The INT_ENABLE register decides which interrupt lines are combined into the final interrupt sent to the CPU. Even if an interrupt line is disabled by the corresponding bit being cleared in INT_ENABLE, the bit in INT_STAT still reflects the actual on-chip interrupt line status. The interrupt scheme is shown in **Figure 4-1**.

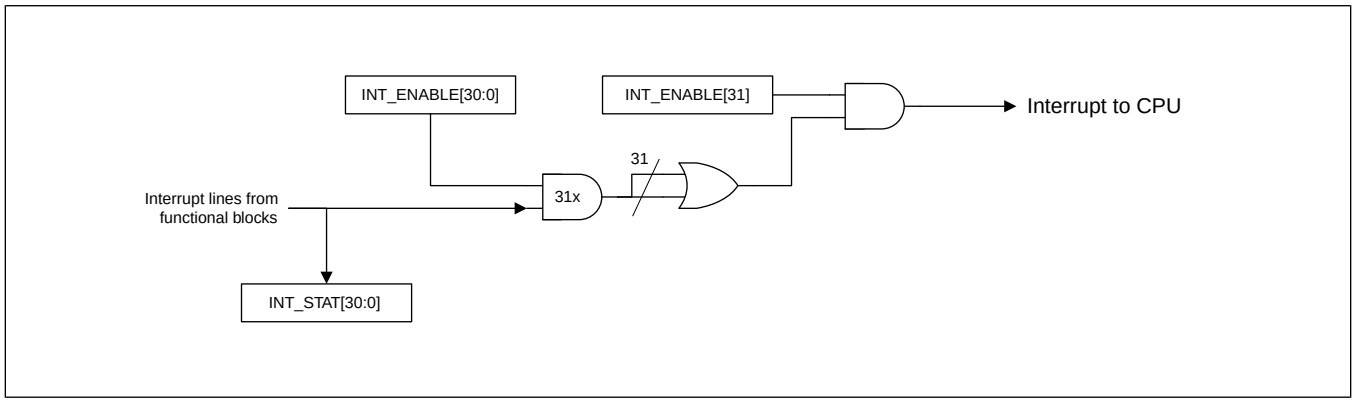


Figure 4-1 Global interrupt routing and enable control

The details of interrupt generation and enabling within each functional block that supports interrupts is documented in the chapters covering those specific blocks. This implies that interrupts must be enabled both in this global register as well as in the individual functional blocks.

4.3.4 CPU latency measurement

The CPU access latency is measured (in core clock units) for every access performed to any register within the device. If the access time is longer than the previously longest access seen, the new maximum latency is captured in `CPULAT_MAX.cycles`. The address of the slowest access is at the same time captured in `CPULAT_ADDR`. By writing to `CPULAT_MAX`, the maximum latency seen until now is reset to zero, effectively starting a new measurement period.

4.3.5 Deterministic Timing Architecture (DTA)

Each signal carried on a datapath within the device transports both data and timing information. The timing follows the signal, and there may therefore be local configuration and monitoring related to timing within each sub-block. This architecture is known as Deterministic Timing Architecture (DTA).

The following global constants are used for calculating local DTA configurations.

Table 4-1 DTA global constants

Name	Value	Description
`dtaPeriod`	6.25 µs	The period of time over which a DTA calculation is evaluated. Equal to 1944 311.04 MHz clock cycles. This value can be read in `DTA_PERIOD.period`.

Within a given context (a given datapath signal, with a bitrate equal to `clientBitrate` bit/s) the following signal-type dependent constants apply:

Table 4-2 DTA signal-dependent constants

Name	Value	Description
`dtaBitcount`	`clientBitrate * dtaPeriod`	Exact (floating point) number of bits at client nominal rate per `dtaPeriod`.

The above constants are applicable to any configurations that refer to them throughout the entire device.

4.3.6 Core information

The `CORE_INFO` register contains information about the configuration of the device, and can be used by software in order to adapt to devices with differing configurations while using a common code base.

4.3.7 Indexing

4.3.7.1 Rate / scheduler indexing

The central Rate / Scheduler module (see **Rate and scheduling control** on page 475) provides reference data rates for various endpoints in the device.

See the documentation for the relevant sub-blocks as listed in **Overview** on page 13 for details about the usage of the individual rate schedule interfaces.

The rate and schedule control module has 20 interfaces indexed by `rsIfIx`, see **Table 4-3** below which shows the indexing calculations.

Note that the different sub-blocks can have different feature sets, and therefore different rate generation requirements.

Table 4-3 Rate generation indexing

Block	rsIfIx	Type	maxDataWidth	maxScheduleLength	numPorts
PLE #n System IF #m	4n+m	Common	256	80	80
CLNP100G #n	4+4n	Common	330	1	1
CLNP100G #n	5+4n	Common	330	80	80
CLNP100G #n	6+4n	Common	32	1	1
CLNP100G #n	7+4n	Common	32	1	1

Each interface on each endpoint can have a different `dataWidth` parameter, and these can potentially vary depending on the endpoint configuration. Values for this device are given in the table below.

Table 4-4 `dataWidth` values per endpoint type

Endpoint		local index	dataWidth	Notes
Top	subblock			
PLE	IVF	0..(n-1)	256	Where n is the number of System Interfaces
CLNP100G	OT4 HO	0	320	
	OT4 LO	1	256	
	XGE client	0	330	5 × 66
	ODUk map	1	320	
	CL1P16G	0..3	32	4 common indexes shared between all CL1P16G blocks. See CLNP100G_CLIENT_CFG in the CLNP100G UM for details.
CL1P16G	HO/MUX/MAP	0	32	
	Client/LO	1	32	

See for information on the supported clients and their configurations in the CLNP100G blocks.

Global parameters needed to set up the rate and scheduling are given in the table below:

Table 4-5 Device parameters affecting rates and scheduling calculations

Parameter	Value
Common generator present?	Yes
Number of Dedicated rate generators.	0
Number of rates available in the common generator, indexed by `rateComIx`.	
`maxScheduleLengthPerIf`	80
`maxPortsPerIf`	80
Padding factor (selected by `paddingControl`) for all 320 bit ODUk.	3824/4080
Padding factor (selected by `paddingControl`) for OTUk, 32bit ODUk and all client signals.	1
Device core clock rate `coreClk`	413 MHz

4.3.7.2 Filter indexing

There are a number of filter satellites used for generating transmit reference clocks for the interfaces.

The table below lists the routing formula used to reach the satellite.

Table 4-6 Filter indexing (IN32)

Usage	routeId	#filter ports per satellite	Notes
UNUSED #n	`8 * lRouteId + n`	1,3,6,10	Reserved for OTN variants
CLNP100G #n	1+n	10	

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4.3.7.3 OTN overhead indexing

The various sub-blocks in the design have a number of OH terminations, as documented in the relevant User Manuals. The table below lists the routing formula used to reach OH satellites in each of these blocks, as well as any OH terminations at the device top level.

Table 4-7 TPO748_IN32 OTN OH indexing

Usage	routeId	Notes
PLE #n	`8 * lRouteId + n`	Insert the `lRouteId` documented in the PLE User Manual into this formula to calculate the routing ID seen from toplevel.
CLNP100G #n	`8 * lRouteId + 1 + n`	Insert the `lRouteId` documented in the CLNP100G User Manual into this formula to calculate the routing ID seen from toplevel.

Global parameters needed to set up the OH scheduling are given in the table below:

Table 4-8 TPO748_IN32: Device parameters affecting OH scheduling calculations

Parameter	Value
OH termination bandwidth in units of 100 Gbit/s.	24

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4.4 Registers

4.4.1 Core

Name: SCRATCH
Description: General purpose scratchpad register
Address: 0x0002.0004

Bits	Field	Type	Reset	Description
31:0	value	RW	-	Can be read and written by CPU.

Name: INT_STAT
Description: Global interrupt status
Address: 0x0002.0008

Bits	Field	Type	Reset	Description
0	ple	RO	-	PLE.
1	overhead	RO	-	Combined interrupt from OTN OH block.
5:2	clnp100g	RO	-	CLNP100G.
28	seu	RO	-	SEU.
29	pmtick	RO	-	Dual bank statistics interrupts.

Name: INT_ENABLE
Description: Global interrupt enable
Address: 0x0002.000C

Bits	Field	Type	Reset	Description
0	ple	RW	0x0	PLE.
1	overhead	RW	0x0	Combined interrupt from OTN OH block.
5:2	clnp100g	RW	0x0	CLNP100G.
28	seu	RW	0x0	SEU.
29	pmtick	RW	0x0	Dual bank statistics interrupts.
31	all	RW	0x0	Must be set to '1' to enable any interrupts from the device.

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Name: CPULAT_ADDR
Description: CPU latency measurement address
Address: 0x0002.0010

Bits	Field	Type	Reset	Description
27:0	addr	RO	-	Address used by the longest (slowest) access until now.

Name: CPULAT_MAX
Description: CPU latency measurement count
Address: 0x0002.0014

Bits	Field	Type	Reset	Description
11:0	cycles	RW	-	Shows the number of core clock cycles used by the longest (slowest) access until now. Whenever a new CPU access is performed, its duration is compared against the current value of `.cycles`. If the new access is slower, the `.cycles` field is updated with the actual number of clock cycles used, `.write` is updated with the type of operation and `CPULAT_ADDR.addr` is updated with the register address. The `.cycles` field is cleared by any write operation to this register, regardless of the value written.
31	write	RO	-	Set if the longest (slowest) access until now has been a write operation.

Name: CORE_CFG
Description: Core configuration
Address: 0x0002.0018

Bits	Field	Type	Reset	Description
0	reserved	RW	-	Reserved for future use.
2:1	otlEn	RW	-	Set when using OTL as the line interface.
8:3	client10g	RW	-	Set when using 10G clients in the associated client block.
12:9	client32gfc	RW	-	Set when using 32GFC clients in the associated client block.
16:13	client64gfc	RW	-	Set when using 64GFC clients in the associated client block.
17	enableDataGen	RW	-	Enable/disable Data gen in simple core. Debug only.
18	patternSelect	RW	-	0 = PRBS, 1 = count seq. Debug only.
19	databw	RW	-	Data width 0 = 32, 1 = 64. Debug only.

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Name: CORE_STAT
Description: Core status
Address: 0x0002.001C

Bits	Field	Type	Reset	Description
0	filtErr	RO	-	Filter error, too many updates towards DCXO interface.
2:1	dcxoOflw	RO	-	Overflow in FIFO towards DCXO interface. One bit per bus.
27:8	prbsErr	RO	-	PRBS error status from Simple Core. Debug only.
28	clDataSeen	RO	-	PRBS error hold from Simple Core. Debug only.

Name: CORE_HOLD
Description: Core status hold
Address: 0x0002.0020

Bits	Field	Type	Reset	Description
0	filtErr	W1C	-	Filter error, too many updates towards DCXO interface.
2:1	dcxoOflw	W1C	-	Overflow in FIFO towards DCXO interface. One bit per bus.
27:8	prbsErr	W1C	-	PRBS error status from Simple Core. Debug only.
28	clDataSeen	W1C	-	PRBS error hold from Simple Core. Debug only.

Name: DTA_PERIOD
Description: DTA period
Address: 0x0002.0028

Bits	Field	Type	Reset	Description
10:0	period	RW	0x798	Number of DTA clock cycles in one DTA period.
27:24	clkFreq	RO	-	Frequency of the DTA clock. 0: freq155. 155.52 MHz. 1: freq311. 311.04 MHz.
31	dtaClkFail	RO	-	Asserted if no transitions are seen on the DTA clock.

Name: PMTICK_CFG
Description: PMTICK configuration
Address: 0x0002.0030

Bits	Field	Type	Reset	Description
0	hwCtrl	RW	0x0	Controls whether internal bank swap is driven by activeBankForce or by the hardware pin. 0: pmtickSw. Current bank set by .activeBankForce. 1: pmtickHw. Current bank swaps on G_PMTICK rising edge.
1	activeBankForce	RW	0x0	Selects which counter bank is active, 0 or 1.

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Name: PMTICK_STAT
Description: PMTICK status
Address: 0x0002.0034

Bits	Field	Type	Reset	Description
0	activeBank	RO	-	Current active set.

Name: PMTICK_CHLD
Description: PMTICK status change hold
Address: 0x0002.0038

Bits	Field	Type	Reset	Description
0	activeBank	W1C	-	Current active set.

Name: PMTICK_INTEN
Description: PMTICK interrupt enable
Address: 0x0002.003C

Bits	Field	Type	Reset	Description
0	activeBank	RW	0x0	Set to '1' to enable interrupt.

Name: CORE_INFO
Description: Miscellaneous device information
Address: 0x0002.0040

Bits	Field	Type	Reset	Description
0	reserved	RO	-	Reserved.
1	pmDualEn	RO	-	Dual bank performance counters.
3:2	reserved0	RO	-	Reserved.
7:4	mode	RO	-	Core Mode. 0: modeIn32. IN32.
11:8	reserved1	RO	-	Reserved.
19:12	schedCount	RO	-	System Schedule Length.
20	reserved2	RO	-	Reserved.
21	clnpIncl1p	RO	-	Set if CLNP100G blocks include CL1P blocks.
31:22	reserved3	RO	-	Reserved.
35:32	vcdBus0Devs	RO	-	Number of devices on VCD bus 0.
63:36	reserved4	RO	-	Reserved.
71:64	rateNum100g	RO	-	Number of rates going to each 100G block.
79:72	rateNumPLE	RO	-	Number of rates going to each PLE block.

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Bits	Field	Type	Reset	Description
87:80	pleNum100g	RO	-	Number of 100G system interfaces used per PLE block.
95:88	reserved5	RO	-	Reserved.

4.4.2 Scratch pad RAM

Name: SCR_RAM
Description: Scratch pad RAM
Address: 0x0002.2000
Entries: 1536
Entry size: 0x4
Index: Address

Bits	Field	Type	Reset	Description
31:0	data	RW	-	Data.

4.4.3 DR Profile RAM

Name: DR_RAM
Description: DR Profile RAM
Address: 0x0002.4000
Entries: 512
Entry size: 0x4
Index: XCVR index

Bits	Field	Type	Reset	Description
31:0	pProfileId	RW	-	Previous DR Profile ID.

4.4.4 CLNP100G

Name: CLNP100G
Description: See documentation for the CLNP100G module
Address: 0x0100.0000
Entries: 4
Entry size: 0x0020.0000
Index: CLNP100G number

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5 OTN Overhead

5.1 Features

- Based on ITU-T G.709 and G.798, with optional B100G support.
- Monitoring of incoming overhead.
- Detection of anomalies, defects, BIP errors.
- Consequent actions.
- Insertion of overhead in source direction, including backwards indications like BDI and BEI.
- Test signals generation and checking.
- Delay measurements.
- OTN overhead Add/Drop interface:
    - Physical interface.
    - Allows overhead at any ODUk to be processed by an external device.

5.1.1 Optional features

A few features are optionally implemented in the device. These features are listed in the table below.

Table 5-1 Optional features

Feature	Options	Notes
`dTimOs`	yes/no	As per 6.2.2/G.798, only the SAPI and DAPI fields of the TTI takes part in the dTIM defect detection. With this `dTimOs` feature enabled, the expected TTI includes the last 32 bytes of Operator Specific information, which may then be included in the dTIM detection.
`numStgs`	2, 4, 8	Number of configurable termination stages per termination block: 8: Up to 8 stages, so all 8 levels (PM, TCM1-TCM6, SM) can be configured. 4: Up to 4 stages, may e.g. be configured to three TCM levels plus either PM or SM, or four TCM levels. 2: Up to 2 stages, may e.g. be configured to a single TCM level plus either PM or SM, or two TCM levels.
`numApsLvls`	1, 2, 4, 8	Number of configurable APS levels per termination block.
`b100g`	yes/no	B100G / FlexO support. If included, references to 'ODUk' in this chapter will generally also be applicable to 'ODUCn', where this is relevant.

5.2 Overview

Overhead insertion and extraction blocks perform the overhead processing functions at the appropriate places in the signal path. Each termination block contains overhead extraction ("sink") and insertion (source") functions as shown in **Figure 5-1**.

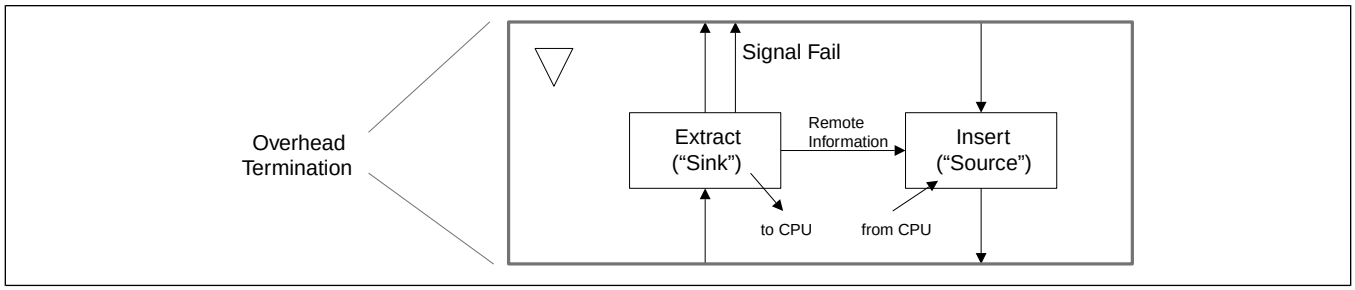


Figure 5-1 Overhead termination process

Remote information communicated from the sink to the source side includes BDI, BEI and DM.

Overhead at any ODUk in the device can additionally be processed by an externally connected device, via the OTN overhead Add/Drop interface. The physical interface is documented in **External Pin Interfaces** on page 486, whereas the registers controlling the add/drop functionality are documented in this chapter.

Generally, all configuration tables should be initialized appropriately following device reset.

5.3 Operation

5.3.1 G.798 terminology and model

In G.798 the overhead termination and fault processing is split into two generic functions: An *adaption function* and a *trail termination* function. G.798 specifies processing models for each of these functions, in the source direction (OH insertion) and in the sink direction (OH extraction), and these models have minor discrepancies depending on their termination level (SM, TCMi, PM).

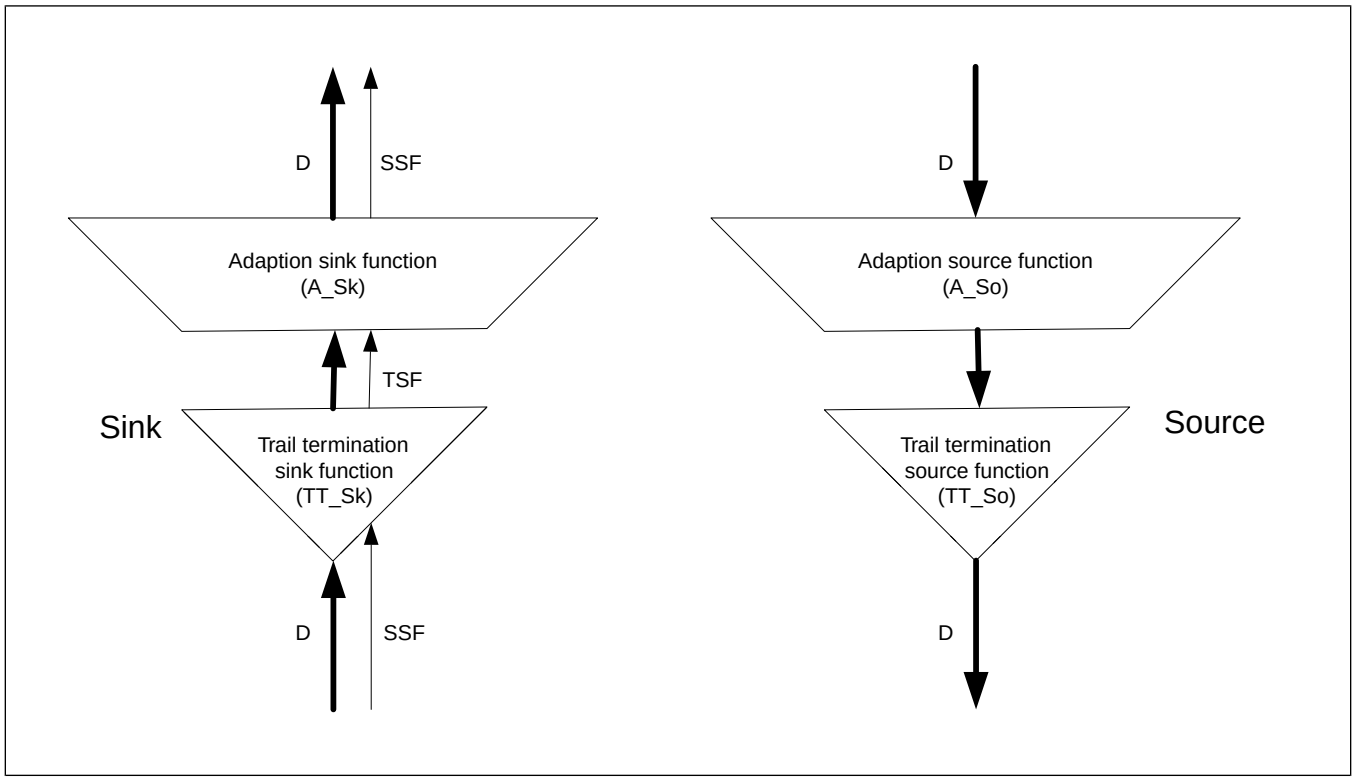


Figure 5-2 Generic layer fault processing (single stage)

In the sink direction, every layer receives a server signal fail (SSF) from its server layer, performs supervision pertaining to the layer, and generates a server signal fail indication to its client layer. Reports of probable causes are made to the management system, based on detected defects.

G.798 describes the overhead processes to be performed by the device, but it does not specify which processes are implemented in hardware and which in software. This manual describes the hardware support.

An *overhead processing block* (in short "OH block") contains up to `numStgs` processing stages with configurable termination levels (PM/TCM/SM, see **Table 5-1**). The level of hardware support differs significantly between the sink and the source side:

- The sink side has hardware support for fault propagation between a hierarchy of up to `numStgs` processing stages. This is needed since a fault detected in one stage can have an impact on the operation of the following stages.
- The source side only has hardware support for basic insert and overwrite functions for each processing stage, so the termination level hierarchy must be implemented fully in software. This is possible since there is no fault detection per stage (so software can compute exactly which bytes to overwrite in the final outgoing overhead).

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The terminology from G.798 is used wherever possible.

5.3.2 OH topology and block indexing

The overhead processing is implemented as a central module and a number of satellite sink and source modules. The central module holds the configuration and is responsible for maintenance interactions. Each satellite contains one or more OH blocks and all satellites are connected to the central module through *routing channels*. A specific overhead processing block is via software assigned a unique *OH block index*. Tables where each entry relates to an entire OH block are indexed using the OH block index. These tables contain _BLK_ in their name. The index may also be referred to as `blkIx`.

The topology is illustrated in **Figure 5-3** below.

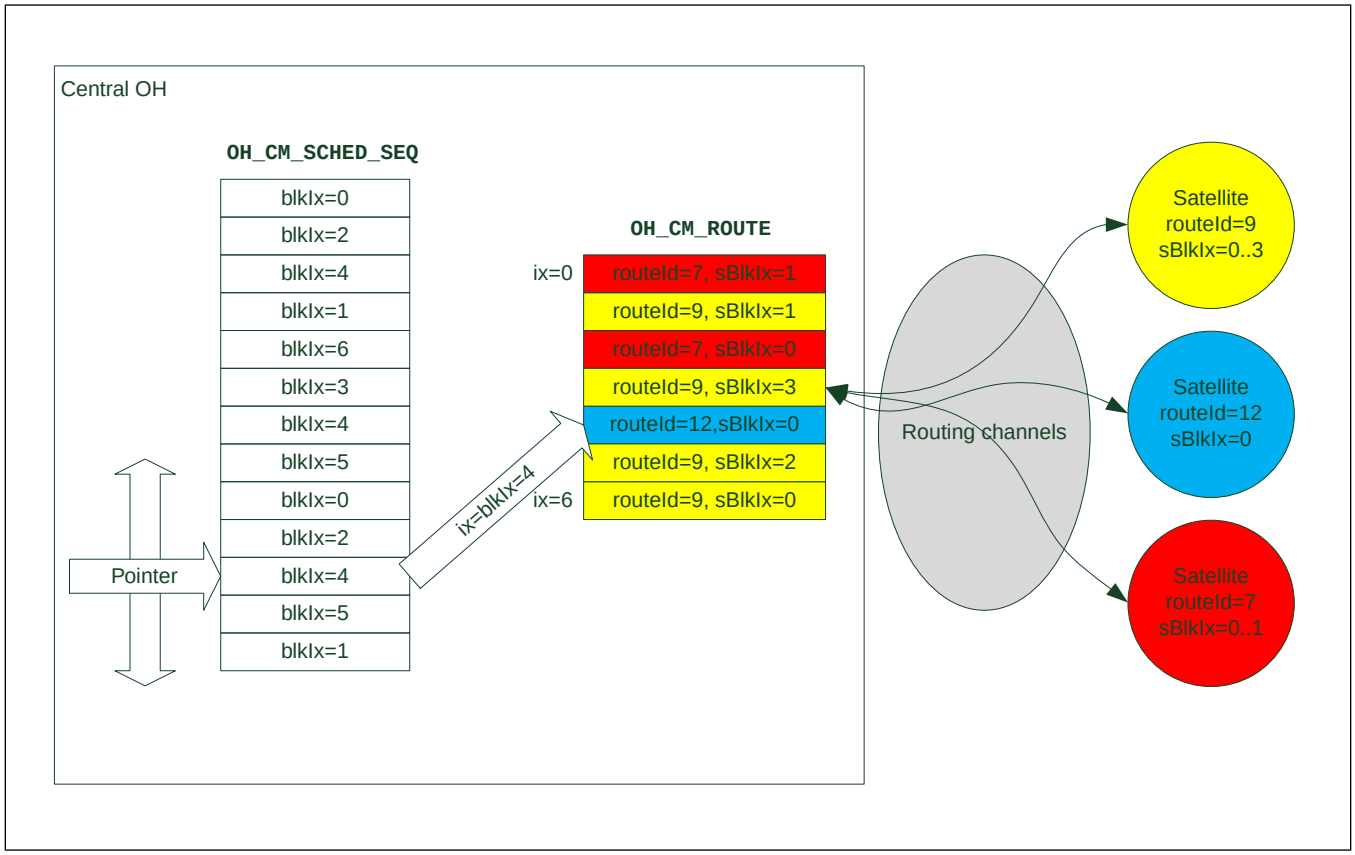


Figure 5-3 OH topology

The figure illustrates a system with three sink/source satellites. Each satellite is uniquely identified by a *routing identifier* called `routeId`. The routing identifier is only used to identify the routing channel between the central OH module and a specific satellite.

A satellite may support more than one overhead processing block. The red satellite for example, supports two blocks and inside the satellite these are referenced as *satellite OH blocks* with a satellite OH block index, `sBlkIx`.

The mapping between the global `blkIx` and a specific satellite OH block is controlled by software. An example is provided below.

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- The blue satellite is uniquely identified with the routing identifier `routeId` = 12. This satellite only has one satellite block; `sBlkIx` = 0. Assume that the OH block in this satellite must be mapped to the global block index `blkIx` = 4. Software will therefore set `OH_CM_ROUTE[4].routeId` = 12 and `OH_CM_ROUTE[4].sBlkIx` = 0. Note that the index into `OH_CM_ROUTE` is the global block index `blkIx`.
- Software must then setup the configuration for `blkIx`=4 in `OH_SK_BLK_CFG[4]`, etc.
- Finally software must configure the scheduling table (`OH_CM_SCHED_SEQ`) so that `blkIx`=4 is scheduled at the rate required. See **Scheduling** on page 399 for details.

The device specific routing identifier formula used to address the OH satellites is found in **OTN overhead indexing** on page 389.

5.3.3 B100G / FlexO support

If `OH_CM_INFO.b100g` is set, support for OTN B100G and FlexO is included. In this case, ODUCn terminations generally function identically to ODUCn, as per G.798. One exception is that for a FlexO group, BIP counters and related functions (BIP, BIE) function on all group members, while the remaining OH termination is only supported on the first group member, as configured in `OH_SK_BLK_CFG[].n1BlkIx`.

Within a FlexO group, interfaces can also be placed in a different order in each direction. This means that for the source and sink satellites within a given FlexO interface, the PID may be different, and therefore backward indications from the sink (BEI/BDI/BIAE) shall be forwarded to the source termination satellite within a different interface.

The mapping of sink to source satellites is configured in the following fields:

- `OH_CM_ROUTE[blkIx].routeId`
- `OH_CM_ROUTE[blkIx].skRouteId`
- `OH_CM_ROUTE[blkIx].skSatIx`
- `OH_SK_BLK_CFG[blkIx].n1BlkIx`

The index of the interface within the FlexO group is given by `skSatIx`. The `n1BlkIx` flag marks the first member #1 of a group. These should be reconfigured if the group ordering changes as may happen during a traffic interruption. The received FlexO PID values indicate the received ordering.

For ODUk (non-FlexO) usage the configuration should be static:

- `OH_CM_ROUTE[blkix].routeId` = routing identifier
- `OH_CM_ROUTE[blkix].skRouteId` = routing identifier
- `OH_CM_ROUTE[blkix].skSatIx` = source satellite index
- `OH_SK_BLK_CFG[blkix].n1BlkIx` = blkIx

5.3.4 Scheduling

The scheduling table controls the frequency at which the OH processing blocks are serviced. I.e. controls the OH data transfer between the OH satellites and the central OH module. There are limited buffer sizes in the OH satellites. The main requirement is to ensure that the all OH data is transfered without data loss. The satellites have 4 buffer entries per `BlkIx` (per ODU), each entry is capable of holding OH data from one ODU frame, first 16 columns, totally 64 byte. One buffer entry is transfered at a time and one entry in the scheduling table `OH_CM_SCHED_CFG` represents the opportunity to transfer one buffer entry. Bandwidth over commitment in the scheduling table is allowed and desired. On average the data transfer rate is one buffer entry per ODU per ODU frame.

The overall arbitration setup is done through `OH_CM_SCHED_CFG` and the actual OH block indexes are placed in `OH_CM_SCHED_SEQ`. The content in `OH_CM_SCHED_SEQ` is divided into two banks, each of the two banks are divided into two parts: a HiSpeed part and a LoSpeed part. Each part of the table is processed in a round-robin fashion. The two separate banks can be selected under software control. The active bank can be swapped by toggling the `OH_CM_SCHED_CFG.use1` field. Bandwidth can be split between the high bandwidth termination blocks and the low bandwidth termination blocks according to the `lowBWfrac` value.

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The Service Time (ST) for one entry in the scheduling table can be read in OH_CM_INFO.scheduleInterval in units of core clock periods. One satellite buffer entry is needed for the satellite internal operation i.e. collecting OH data from all 4 rows of the ODU frame and to allow for timing skew of the data on the datapaths. That is the remaining 3 buffer locations (B=3) or the equivalent of 3 framing periods (FP) skew per ODU is allowed in the scheduling table.

There are three requirements to the scheduling table.

1. The number of entries per ODU in the schedule table part must satisfy the bandwidth requirements of that port.
2. The entries must (ideally) be evenly distributed in the table part. However some unevenness can be tolerated: The sink satellite buffer must not overflow and the source satellite buffer must not underflow. Note this requirement is also relevant when switching between two table banks.
3. After switching bank there must be a hold off period, before the next bank switch. When switching bank the sink satellite buffer level may increase /source satellite buffer level may decrease. Typically the level will increase / decrease up to one during a bank switch. The hold off period allows central module to catch up due the over speed of the service requests, i.e. empty sink satellite buffer / fill source satellite buffer before the next bank switch.

In the following the requirements are rewritten given some assumptions on how the scheduling tables are used:

- When swapping bank the exit point of the present bank and entry point of the next bank are assumed to be at the worst case time, i.e. the fill level of the satellites buffer are furthest away from ideal position.
- The skew allowance (B=3) is divided evenly between the banks, i.e. B/2 per bank.
- lowBWfrac value is set to zero.

The two banks can then be setup and analyzed independently.

One bank is considered at a time and one ODU is considered at time using the following notation:

- ST = OH_CM_INFO.scheduleInterval is the Service Time in core_clk units.
- FP is the Framing Period for the considered ODU in core_clk units.
- HO is the Hold Off time for the considered ODU in core_clk units. This hold off time is sufficient to empty / fill the satellite buffers with B frames of OH, which is the worst case.
- N is the total number of entries in the bank.
- K is the total number of entries for the considered ODU.
- L is the longest distance (the number indexes) between two entries for the considered ODU including wrap around.
- B=3 is the number of buffer locations / number of framing periods skew allowed in the scheduling table.

Given the above assumption and notation the three requirements to the scheduling table can be rewritten:

1. K > ST * N / FP.
2. L < B * FP / 2 / ST.
3. HO > B * FP/(1 - ST * N / K / FP). Where ST * N / K is the average time between two service requests.

Example: ODU2 in a system with 111G bandwidth.

- core_clk period is 2.8 ns, core_clk frequency is 357 MHz.
- ST = 378. This corresponds to a system OTU bandwidth of 123 Gbit/s, 10% more than 111 Gbit/s.
- FP = 4284. An ODU2 period is 12 us.
- N = 80.
- K > 7.1.
- K = 8. The minimum integer the satisfies requirement 1 above.
- L < 17. This value is 70% larger the average L = N / K = 10.
- HO > 0.31 ms.

Example: ODU0 in the same system as above, the requirements are then K >= 1 and HO > 2.2 ms.

5.3.5 Rate generators

The overhead processing blocks may insert test and maintenance signals using the internal rate generators see **Rate and scheduling control** on page 475.

5.3.6 Table indexing

Figure 5-4 and **Figure 5-7** illustrate the OH sink and OH source processing blocks, each with up to `numStgs` processing stages. The processing stages in an OH block are referred to via their *stage offsets* (`stgOfs`) from 0 to `numStgs`-1. The hardware contains tables where each entry pertains to a single processing stage (i.e. handles a single SM/TCMi/PM termination level). The index for these tables is derived from the OH block index (see **OH topology and block indexing** on page 398) concatenated with the processing stage offset (zero extended to three bits if `numStgs` is < 8). These tables contain _STG_ in their name, and the index is referred to as the OH *processing stage index*, or `stgIx`.

So we have

```
stgIx = (blkIx * 8) + stgOfs
```

The tables indexed by stgIx are sparsely populated in case `numStgs` is < 8 (unallocated entries are not writable by software and always reads zeros). Each processing stage may be assigned to a termination level (termLvl) value, according to the table below, and this level is configured in `OH_SK_STG_CFG.termLvl` (for the sink side) and in `OH_SO_STG_CFG.termLvl` (for the source side). The sink and source stages must be configured with identical levels.

Table 5-2 OH termLvl numbers

OH termination level	termLvl value
PM	0
TCM1	1
TCM2	2
TCM3	3
TCM4	4
TCM5	5
TCM6	6
SM	7

The sink side provides hardware support for fault propagation between stages, where the order of the processing stages are software configurable. The source side stages does not support hardware propagation of defects, but it must still be configured identically to the sink side stages with regard to termination levels.

In addition to the processing stages, the hardware also contains resources to handle APS/PCC monitoring on the sink side and insertion on the source side. The maximum number of APS levels that can be handled per OH block is 1, 2, 4 or 8, depending on the device (common for both the sink and source sides). This number can be read in `OH_CM_INFO.numApsLvls`. Most common APS scenarios only require a single APS level to be monitored per OH block, hence most devices are anticipated to be configured with `numApsLvls` = 1.

The APS hardware contains tables where each entry handles a single APS level. The `numApsLvls` APS level resources in an OH block are referred to via their APS offset (`apsOfs`) from 0 to `numApsLvls`-1. The index for the APS related tables is derived from the OH block index (see **OH topology and block indexing** on page 398) concatenated with the APS offset (zero extended to three bits if `numApsLvls` is < 8). These tables contain _APS_ in their name, and the index is referred to as the *APS index*, or `apsIx`.

So we have

```
apsIx = (blkIx * 8) + apsOfs
```

The tables indexed by `apsIx` are sparsely populated in case `numApsLvls` is < 8 (unallocated entries are not writable by software and always reads zeros). So for example in the case of `numApsLvls` = 1, only every eight entry is populated.

The various index types used in the OH processing are summarized in the table below.

Table 5-3 OH index types

OH index long name	OH index short name	Description
Block index	`blkIx`	References one of the OH blocks, see **OH topology and block indexing** on page 398.
Stage offset	`stgOfs`	References one of the `numStgs` processing stages inside an OH block.
Stage index	`stgIx`	References a specific processing stage within a specific OH block: `stgIx = (blkIx*8) + stgOfs.`
APS offset	`apsOfs`	References one of the `numApsLvls` APS level monitoring resources inside an OH block.
APS index	`apsIx`	References a specific APS level monitoring resource within a specific OH block: `apsIx = (blkIx*8) + apsOfs`

5.3.7 OH sink processing (Rx/extraction)

5.3.7.1 Sink direction OH block overview

The sink overhead processing block architecture is shown in **Figure 5-4** below.

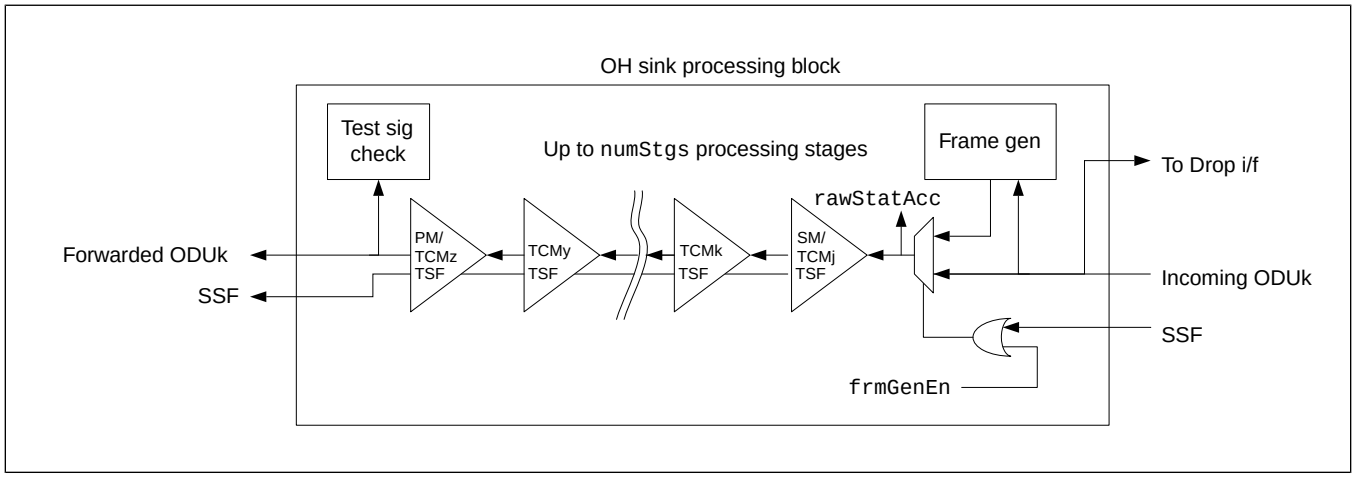


Figure 5-4 Sink OH block

The overhead block contains `numStgs` processing stages, see **Table 5-1**. The first stage entered is normally used for the SM level, the stages in the middle are used for TCM levels, and the last stage at the end of the chain is normally used for the PM level. If `numStgs` is 8 then all 8 levels (SM->TCM6->..->TCM1->PM) can be supported simultaneously, whereas if e.g. `numStgs` is 4 then only a subset of the levels can be supported simultaneously. Unallocated levels are in transparent mode.

The OH block has an SSF input which is equal to the Signal Fail (SF) associated with the incoming datapath. A TSF signal (trail signal fail) may be generated in each of the termination stages, and will be propagated to the following stage, provided that the generating stage does not have MI_AdminState = LOCKED.

5.3.7.2 TCM termination stage hierarchy

The general processing stage indexing is described in **Table indexing** on page 401. For the sink side, the hardware handles fault propagation, so it must be aware of the order of the stages, in order to do this correctly. The processing stages are linked together in a chain as illustrated in **Figure 5-4** above. Each processing stage has an associated `stgOfs`, between 0 and `numStgs`-1. The chain must always end at the stage at `stgOfs` = 0 (typically configured to the PM level), so its fault status is propagated on to the module that follows the OH termination block (this can e.g. be an ODU mapper). For each processing stage, its preceding stage is identified via a preceding stage offset, `preStgOfs`. All stages in use must be linked via the `preStgOfs` index, configured in `OH_SK_STG_CFG.preStgOfs`.

Software thus hooks up the chain of hardware stages starting at the end of the chain at `stgOfs` = 0 and repeatedly uses `preStgOfs` as link to the previous stage, until the first stage of the chain. The first stage must have `preStgOfs` = 0, which indicates that the stage will use the incoming SSF of the OH block as its incoming fail status.

Table 5-4 provides an example of a straight SM -> TCM6 -> TCM5 -> ... -> TCM1 -> PM configuration, in which all six TCM levels are in use (so here `numStgs` is 8).

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Table 5-4 Example of straight SM->TCM6->...->TCM1->PM stage configuration

Stage offset (`stgOfs`)	Preceding Stage Offset (`.preStgOfs`)	`.termLvl` value	OH termination level
0 (SSF out)	1	0	PM
1	2	1	TCM1
2	3	2	TCM2
3	4	3	TCM3
4	5	4	TCM4
5	6	5	TCM5
6	7	6	TCM6
7	0 (SSF in)	7	SM

Table 5-5 provides an example with `numStgs` = 4 where only one TCM levels and the PM level are in use.

Table 5-5 Example of straight TCM4->PM stage configuration

Stage offset (`stgOfs`)	Preceding Stage Offset (`.preStgOfs`)	`.termLvl` value	OH termination level
0 (SSF out)	2	0	PM
1	n/a	n/a	n/a
2	0 (SSF in)	4	TCM4
3	n/a	n/a	n/a

Figure 5-5 illustrates a more complex sink example, in which the basic hierarchy is SM->TCM2->TCM1->PM, and where one of the unused stages (`stgOfs` = 5) is used for non-intrusive monitoring ("NIM") on the PM level after TCM2 (`numStgs` is 8).

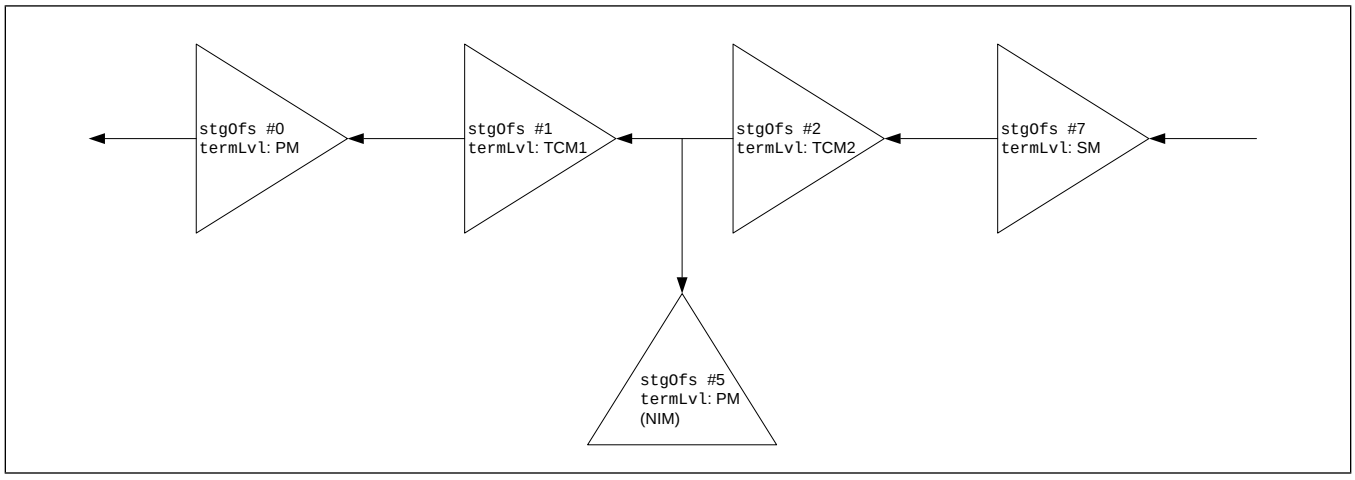


Figure 5-5 Termination stage example with non-intrusive monitor

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The corresponding configurations for `OH_SK_STG_CFG[blkIx*8 + stgOfs].preStgOfs` and `.termLvl` are shown in **Table 5-6**.

Table 5-6 Stage configuration for non-intrusive monitor

Stage offset (`stgOfs`)	Preceding Stage Offset (`.preStgOfs`)	`.termLvl` value	OH termination level
0 (last stage)	1	0	PM
1	2	1	TCM1
2	7	2	TCM2
3	n/a	n/a	n/a
4	n/a	n/a	n/a
5	2	0	PM (NIM)
6	n/a	n/a	n/a
7	0 (first stage)	7	SM

As mentioned in **Table indexing** on page 401, the each sink stage must be configured with the same level as its corresponding source stage. However, in case of a NIM the source stage must be configured as transparent, i.e. with `OH_SO_STG_CFG.lvlEn` = '0'.

A NIM may also be connected directly the input of the OH block, by setting its `.preStgOfs` = 0.

5.3.7.3 Sink termination processing stage overview

Each processing stage may be modelled as consisting of a trail termination function and an adaption function in a fashion similar to G.798, as illustrated in **Figure 5-6** below. The figure shows the functions that are implemented in hardware. Though this model shows a maintenance signal frame generator as part of each termination stage, all eight stages in fact share the same single maintenance signal generator, as shown in **Figure 5-4**. The server signal fail is indicated by CI_SSF, x_D indicates the frame data path, whereas x_CK, x_FS, x_MFS indicate the frame rate and frame synchronization.

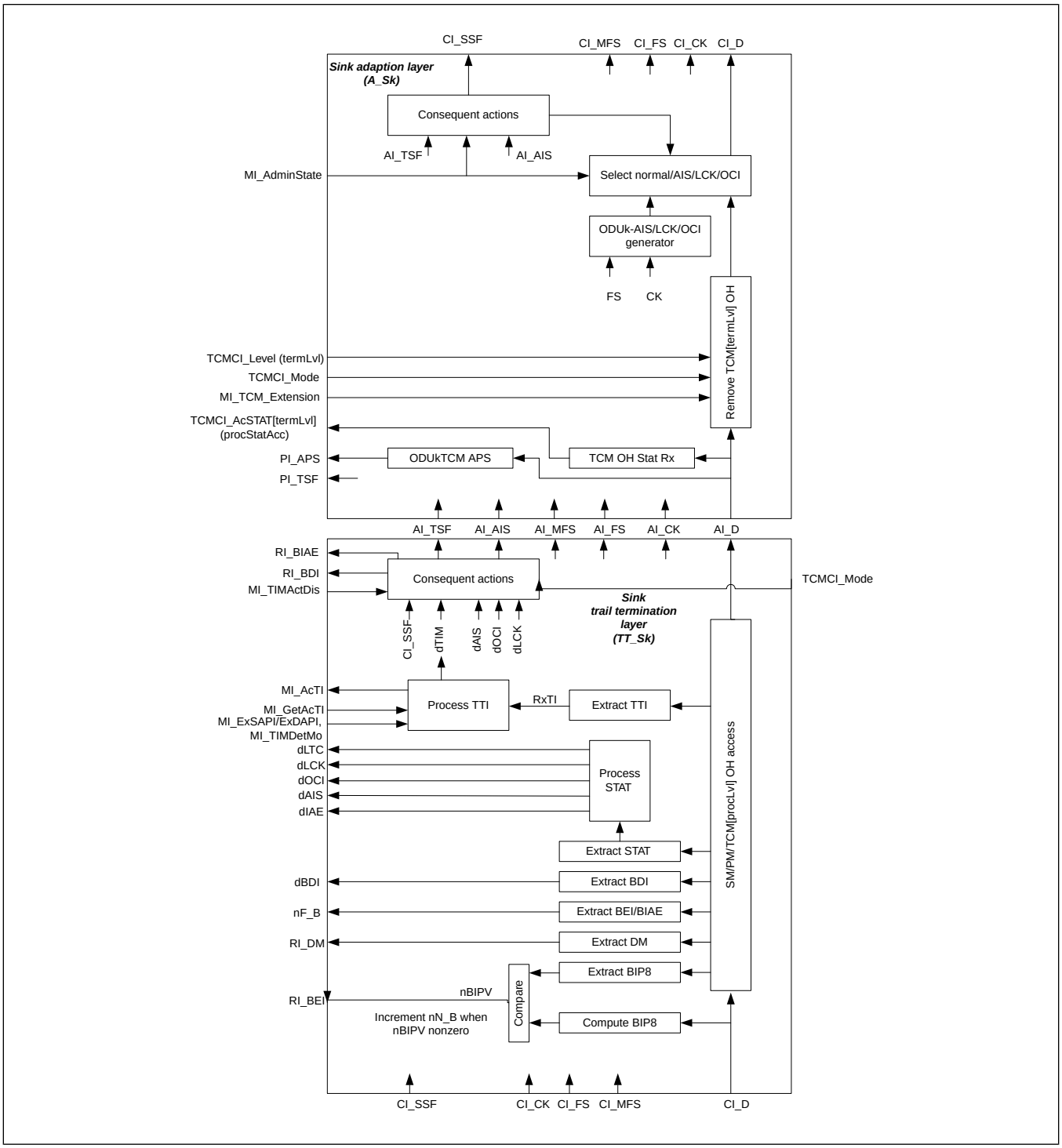


Figure 5-6 Sink termination stage model

Each sink processing stage processes one PM/TCMi/SM level and provides remote information to the companion termination level in the source direction, e.g. for BDI and BEI/BIAE backwards indications, when remote information is enabled.

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5.3.7.4 Modes and interactions

The trail termination sink function (TT_Sk) and adaption sink function (A_Sk) processing is illustrated in **Figure 5-6**.

The implementation provides individual control bits to enable propagation of errors and remote information, rather than implementing the TRANSPARENT, MONITOR and OPERATIONAL sink modes (TCMCI_Mode) of G.798 explicitly in hardware. So the sink mode is a concept handled by the software API, which configures the individual control bits accordingly.

`OH_SK_STG_CFG.riEn` must be set in order to allow remote information (RI_ in **Figure 5-6**) to be send to the companion TT_So function. If `.riEn` is '0' then no information is sent. In case multiple sink stages are configured with identical termination level (`.termLvl`) then only one of these should have `.riEn` set.

`OH_SK_STG_CFG.aEn` must be set in order to enable generation of actions. If `.aEn` is '0' then no signal fail is generated, nor is AIS inserted.

`OH_SK_STG_CFG.mEn` must be set when the stage operates as a non-intrusive monitor (m_TT_Sk NIM).

Table 5-7 provides the relationship between the sink mode (TCMCI_Mode), and the control bits implemented in hardware. Statistics and status is always updated by the hardware, so software should disregard these updates when in TRANSPARENT mode.

Table 5-7 Sink configuration modes

Software API state	Hardware registers			Description
Sink Mode TCMCI_Mode	`.riEn`	`.aEn`	`.mEn`	
TRANSPARENT	0	0	0	No remote information transferred to the source side. No Signal Fail is generated in the sink direction, nor is AIS inserted.
MONITOR	1			Remote information is transferred to the source side. No Signal Fail is generated in the sink direction, nor is AIS inserted.
OPERATIONAL		1		Remote information is transferred to the source side. Signal Fail may be generated in the sink direction, and AIS may be inserted, when applicable.
NON-INTRUSIVE MONITOR (NIM)	0	0	1	AI_TSF is monitored. Remote information is not transferred to the source. No Signal Fail is generated in the sink direction, nor is AIS inserted.

Strictly speaking, the NIM mode is not a sink mode, rather it is its own type of sink (m_TT_Sk), and it should only be configured for stages that are not pre-stage for any of the other stages (e.g. stage 5 in **Figure 5-5**).

When remote information is transferred to the source side, it may or may not be inserted at the source function, depending on the source configuration (i.e. whether the source function is in OPERATIONAL or TRANSPARENT mode).

Regardless of the normal sink mode, it is possible to set a stage in LOCKED mode, via MI_AdminState. When in LOCKED mode, CI_SSF is suppressed and ODUk-LCK is inserted in the sink direction. MI_AdminState = LOCKED is implemented via `OH_SK_STG_CFG.ciSsfClrForce` (suppress signal fail propagation) and `OH_SK_STG_CFG.maintSigForce` = FORCE_LCK.

Table 5-8 MI_Admin state

Software API state	Hardware registers		Description
MI_Adminstate	.ciSsfClrForce	.maintSigForce	
Not LOCKED	0	FORCE_NONE	Normal operation (OPERATIONAL, MONITOR, TRANSPARENT).
LOCKED	1	FORCE_LCK	CI_SSF is forced deasserted and ODUk-LCK is inserted in the sink direction.

The sink processing illustrated in **Figure 5-6** (flow is from bottom to top) and its register interface are described in the following sections.

5.3.7.5 One-second performance monitoring via dual banking

The counters in OH_SK_STG_CNT and status hold in OH_SK_STG_HOLD may be dual banked, in case OH_CM_INFO.dualBank = '1' and the dual bank feature is enabled by software. This allows more precise implementation of one-second performance monitoring than a purely software based method (see also).

5.3.7.6 RI_BEI and near-end errored blocks (nN_B)

The BIP8 is computed over the OPUk and the expected BIP value is extracted from the BIP8 byte position of the relevant field. The computed BIP8 of the second preceding frame is compared with the BIP8 value extracted from the current frame. If there is a mismatch between the two values, a near-end errored block (nN_B) is detected, and the number of BIP violations (nBIPV) is forwarded to the companion TT_So function via RI_BEI, provided that OH_SK_STG_CFG.riEn is '1'. If .riEn is '0' then a fixed value of zero is forwarded to the source side.

Near-end errored blocks are counted in OH_SK_STG_CNT.bip, regardless of OH_SK_STG_CFG.riEn. No counting takes place in case of dAIS, dLCK, dLTC, dOCI and CI_SSF.

In case dual banking is implemented and enabled, then the .bip counter results in the one-second performance monitoring value pN_EBC.

5.3.7.7 RI_DM extraction

For the PM and TCMi stages, the relevant DM bit is extracted from the overhead and forwarded to the companion TT_So function, via RI_DM, provided that OH_SK_STG_CFG.riEn and OH_SK_STG_CFG.riDmEn are both '1'. If either .riEn or .riDmEn is '0' then a fixed value of zero is forwarded to the source side. The .riDmEn bit should be set to '1' when sink mode is OPERATIONAL, and to '0' otherwise (to avoid RI_DM transfer when in MONITOR mode).

The received DM value can also be read in OH_SK_STG_STAT.rxDm.

5.3.7.8 Far-end errored blocks (nF_B)

The BEI bits are extracted from the relevant overhead field. If the BEI value is between 1 and 8 then a far-end errored block (nF_B) has occurred, and the OH_SK_STG_CNT.bei counter is incremented, regardless of OH_SK_STG_CFG.riEn. No counting takes place in case of dAIS, dLCK, dLTC, dOCI and CI_SSF.

In case dual banking is implemented and enabled, then the .bei counter results in the one-second performance monitoring value pF_EBC.

5.3.7.9 dBDI detection

The received BDI is extracted and filtered with X=5 (as per 6.2.6.6.1/G.798). The dBDI is cleared during CI_SSF or dAIS. The filtered dBDI value can be read in `OH_SK_STG_STAT.dBdi`.

5.3.7.10 STAT field extraction and related defects

The incoming STAT bits of the relevant overhead field after filtering with X=3 (as per 8.8/G.798) can be read in `OH_SK_STG_STAT.rawStatAcc`. These values come directly from the field selected by `OH_SK_STG_CFG.termLvl`, i.e. they are "raw" values, they do not reflect any overwriting of the signal at an earlier processing stage, whether by insertion of a maintenance signal or consequent action, as also conceptually indicated in **Figure 5-4**. The processed (overwritten) STAT bits can instead be read in `OH_SK_STG_STAT.procStatAcc`, as indicated in **Figure 5-6**. The processed STAT (`.procStatAcc`) is the main one used for defects, while the raw STAT bits (`.rawStatAcc`) is used to get various AcSTAT[i] in the A_Sk functions. The STAT bits are decoded as indicated in the **Table 5-9** below.

Table 5-9 STAT field decoding

STAT	PM decoding	PM alarm status	TCM decoding	TCM alarm status
0	Reserved	OK	No source TC	dLTC
1	Normal path signal		In use without IAE	OK
2	Reserved		In use with IAE	dIAE
3			Reserved	OK
4				
5	ODUk-LCK	dLCK	ODUk-LCK	dLCK
6	ODUk-OCI	dOCI	ODUk-OCU	dOCI
7	ODUk-AIS	dAIS	ODUk-AIS	dAIS

The corresponding decoded defects can be read in the `.dLtc`, `.dLck`, `.dOci`, `.dAis` and `.dIae` fields in `OH_SK_STG_HOLD`.

5.3.7.11 dTIM detection

The expected TTI SAPI and DAPI fields are configured in `OH_SK_EXTI_CFG` using indirect addressing via `OH_SK_EXTI_CFG_IX`. If `OH_CM_INFO.dTimOs` is '1' then the expected Operator Specific field (32 bytes) can also be configured.

The received TTI is compared against the expected SAPI, DAPI and Operator Specific fields (if implemented), according to the MI_TIMDetMo configured in `OH_SK_STG_CFG.timDetMode`. The criteria for declaring dTIM (X=7) and clearing dTIM (X=3) is based on Appendix IV.2.2/G.798. The dTIM defect status can be read in `OH_SK_STG_STAT.dTim`.

5.3.7.12 RI_BDI

The RI_BDI transmitted to the companion TT_So function for the backwards direction is derived as follows:

```
RI_BDI = (CI_SSF_in || (dLTC||dLCK||dOCI||dAIS||dTIM)) && .riEn
```

The PM and SM stages do not include dLTC. A fixed value of '0' is forwarded to the source function in case `OH_SK_STG_CFG.riEn` is '0'.

5.3.7.13 RI_BIAE

The RI_BIAE transmitted to the companion TT_So function will be set to '1' if dIAE is detected. A fixed value of '0' is forwarded to the source function in case `OH_SK_STG_CFG.riEn` is '0'.

5.3.7.14 AI_TSF

For a normal TT_Sk function, the AI_TSF is determined as follows:

```
AI_TSF = CI_SSF_in ||
         ( ((dTIM && !.timActDis) || dLCK||dOCI||dAIS || (dLTC && .ltcActEn)) &&
           (Sk_mode == OPERATIONAL) || (Sk_mode == NON-INTRUSIVE MONITOR (NIM)) )
```

For an m_TT_Sk function, the sink mode is not applicable, so the last term of the formula above is omitted.

As per **Table 5-7** a normal TT_Sk function will be in OPERATIONAL mode when `.aEn` is '1', and an m_TT_Sk function is configured when mEn is '1'. The AI_TSF reported by hardware is therefore determined as follows:

```
AI_TSF = CI_SSF_in ||
         ( ((dTIM && !.timActDis) || dLCK||dOCI||dAIS || (dLTC && .ltcActEn)) &&
           (aEn || mEn) )
```

The AI_TSF can be can be read in `OH_SK_STG_STAT.aiTsf`.

The TIMActDis option can be implemented using the `OH_SK_STG_CFG.timActDis` field.

The LTCAct_Enable option can be implemented using the `OH_SK_STG_CFG.ltcActEn` field.

5.3.7.15 TCMCI_AcSTAT

As mentioned in **STAT field extraction and related defects** on page 409, the raw STAT field can be read in `OH_SK_STG_STAT.rawStatAcc` and the processed STAT field can be read in `OH_SK_STG_STAT.procStatAcc`, for the referenced level. Software can then construct the AcSTAT for all six TCM levels by using the `.procStatAcc` value for the level of the current stage and for levels of stages processed before the current stage, and use the `.rawStatAcc` value for levels of stages there are after the current stage in the hierarchy.

5.3.7.16 SM/TCM overhead forwarding and erasing

The `OH_SK_STG_CFG.zeroMode` field must be set to '1' to erase the forwarded SM/TCM fields when this is required, according to the TCM extension mode and sink mode as shown **Table 5-10** below.

Table 5-10 TCM extension settings

MI_TCM_Extension	Sink Mode (TCMCI_Mode)	`.zeroMode` 1: erase 0: passThrough
Normal	OPERATIONAL	1
	MONITOR	0
	TRANSPARENT	
Erase	Don't care	1
Pass-through		0

For the SM stage (stgOfs=7), .zeroMode should always be set to '1', unless there is a specific reason to require SM transparency. When .zeroMode is set to '1' for SM, then it implies not only zeroing of the SM bytes, but the entire OTU OH field (columns 8-14).

For PM stages, .zeroMode is don't care since the PM overhead ceases to exist in the next block (which must be a demapper or demultiplexer).

The sink function will not overwrite any other overhead than the TCM fields, except when the entire ODUk is overwritten by a maintenance signal.

The resulting zeroing for each level at the output of the final sink stage is indicated in OH_SK_BLK_STAT.zeroMode.

5.3.7.17 Maintenance signal insertion in sink direction

The incoming frame may be replaced by an ODUk-AIS, ODUk-LCK or ODUk-OCI signal, forced via OH_SK_STG_CFG.maintSigForce, and ODUk-AIS may also be inserted due to aAIS.

Maintenance signals may be inserted at any of the processing stages except PM. The hardware will detect the last active stage which demands signal insertion and set the maintenance signal generator accordingly. The hardware will act as if individual signal generators are present in all stages, e.g. with respect to processed STAT bits.

When a maintenance signal is inserted the entire ODUk is overwritten. In case of ODUk-AIS the FTFL is field is zeroed. The OTU overhead is not overwritten.

An ODUk-AIS signal is automatically inserted if aAIS is declared in any active stage, and the hardware will then ignore any TCM erase operations that logically occurs in any prior stages in the TCM hierarchy.

```
aAIS = (dLCK || dOCI || (dLTC && .ltcActEn) || (dTIM && !.timActDis)) && aEn
```

.aEn is only set when the sink mode is OPERATIONAL, according to **Table 5-7**. The hardware does not include the (dLCK||dOCI || (dLTC && ltcActEn)) term on stage offset 7 (SM level), and dLTC is not included on stage offset 0 (PM level).

The aAIS status can be read in OH_SK_STG_STAT.aAis.

LCK signal insertion should be forced when the software maintained MI_AdminState is LOCKED, according to **Table 5-7**.

Maintenance signal insertion is indicated in OH_SK_BLK_STAT.maintSig, whether it is done automatically (result of aAIS) or by forced by OH_SK_STG_CFG.maintSigForce.

5.3.7.18 Forced frame generation

An internal frame generator is placed before the processing stages, as illustrated in **Sink direction OH block overview** on page 403. It will become enabled if incoming signal fail to the OH block is detected (in which case there is no reliable incoming framing and ODUk-AIS must be generated). The incoming framing will be used again when the incoming signal fail condition is cleared.

The internal frame generator may also be forced active via OH_SK_BLK_CFG.frmGenEn. This should only be done if there is at least one stage with OH_SK_STG_CFG.maintSigForce set to another value than NONE. The last active stage with .maintSigForce different from FORCE_NONE will then win.

When frame generation is active, it is not possible to also monitor the incoming frame data. Therefore, .frmGenEn should normally be set to '0', allowing monitoring of e.g. dTIM at one stage while inserting e.g. LCK at another stage.

Switching from the incoming framing to framing generated by the frame generator is hitless with regard to FAS and MFAS.

5.3.7.19 CI_SSF out of termination function

The CI_SSF from output of adaption sink function is derived as follows:

```
CI_SSF_out = (AI_TSF && (Sk_mode == OPERATIONAL) ||
                 CI_SSF_in && (Sk_mode == NON-INTRUSIVE MONITOR (NIM))
              || .ciSsfSetForce) && !.ciSsfClrForce
```

CI_SSF_out may thus be forced via `OH_SK_STG_CFG.ciSsfSetForce` and may be suppressed via `OH_SK_STG_CFG.ciSsfClrForce`. The ciSsfClrForce bit is used to implement MI_AdminState = LOCKED, according to **Table 5-7**.

The CI_SSF_out reported by hardware is therefore determined as follows:

```
CI_SSF_out = ( CI_SSF_in ||
               ( ((dTIM && !.timActDis) || dLCK||dOCI||dAIS|| (dLTC && .ltcActEn))
                 && aEn)
               || .ciSsfSetForce)
             && !.ciSsfClrForce
```

The output CI_SSF can be read in `OH_SK_STG_STAT.ciSsfOut`.

5.3.7.20 TCM ACT, FTFL, RES and GCC fields

The TCM ACT, FTFL, RES and GCC fields are not accessed by the sink function (unless overwritten by a maintenance signal), but can be accessed via the OTN overhead Add/Drop interface.

5.3.7.21 Accepted OPUk-CSF

The accepted OPUk-CSF bit (filtered using X=3, as per 6.2.10/G.798) can be read in `OH_SK_BLK_HOLD.opuCsf`. Note that this is the raw OPUk-CSF bit, hence software must ignore the bit in cases where it is invalid based on context, e.g. due to OPU sub structuring or encryption.

5.3.7.22 PRBS test signal checking

The OH block contains a test signal checker, which is capable of checking the 2^{31}-1 PRBS inverted signal specified in 17.5.2/G.709. The PRBS checker is always active, and `OH_SK_BLK_HOLD.testErr` will be set if any incoming bit does not match the expected value. On any mismatch, the PRBS checker immediately attempts to lock to a new sequence.

5.3.7.23 Sink side APS monitoring

All APS related tables are addressed using the APS index (`.apsIx`), see **Table 5-3**. Depending on `OH_CM_INFO.numApsLvls` these tables may be sparsely populated.

The APS level for a given APS index is configured in `OH_SK_APS_CFG.apsLvl`. The first three bytes of the four byte APS signal for the configured APS level are filtered with X=3 (as per 8.2/G.873.1) and these accepted APS/PCC bytes can be read in `OH_SK_APS_STAT.aps0`-2. If the filtered APS bytes change, this will be flagged in `OH_SK_APS_CHLD.aps`, and an interrupt may be generated.

5.3.7.24 Moving TTI monitor

A moving monitor allows capture of accepted TTI (AcTI) for a programmable trail. This is used for implementation of the MI_GetAcTI API call.

The moving monitor works as described in the following (same as Appendix IV.2.1/G.798).

A compare and store process uses a 64-byte store, holding the last stored TTI. This can be read via the OH_SK_MON_TTI register.

A TTI capture operation is started by writing an OH block index to OH_SK_MON_CFG.blkIx and a termination level to OH_SK_MON_CFG.termLvl. This will also clear the OH_SK_MON_STAT.stable bit.

Once started, the received TTI byte is compared with the equivalent byte in the store (OH_SK_MON_TTI). After the comparison, the byte is copied into the store. After all 64 bytes have been compared and stored, the total comparison result will be either "equal" (all 64 bytes match) or "unequal" (at least one byte did not match). Now processing continues with the next received TTI.

When the result of the three consecutive TTI comparisons have been "equal", then the OH_SK_MON_STAT.stable bit becomes set, and the TTI capture process is stopped. If the TTI becomes unstable after the store and compare process is stopped, then this will not be visible without restarting the process (via a new write to OH_SK_MON_CFG).

An interrupt will be generated if OH_SK_MON_INTEN.stable is '1' and OH_SK_MON_STAT.stable becomes set.

5.3.8 OH source processing (Tx/insertion)

5.3.8.1 Source direction OH block overview

The source overhead processing block architecture is shown in **Figure 5-7** below.

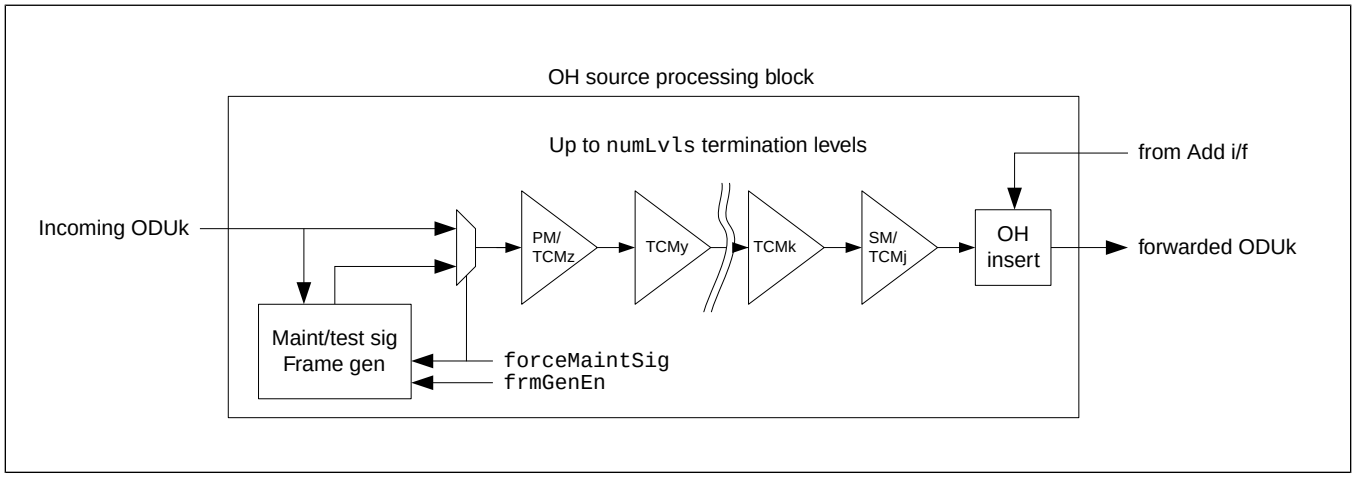


Figure 5-7 Source OH block

As opposed to the overhead sink processing, the overhead source processing does not contain any hardware support for stages to implement the TCM hierarchy. It is the responsibility of software to maintain a representation of the TCM hierarchy, and configure the overhead insert operations of the termination levels accordingly. This is possible since there is no hardware fault detection per level with associated fault propagation in the source direction. The software may implement the TCM hierarchy via software maintained virtual stages, which are then mapped to the termination level insert functions of the hardware.

5.3.8.2 Source termination function overview

Figure 5-8 illustrates the source functions of G.798 that are supported by hardware. It is a conceptual model, so e.g. maintenance signal generators are shown as part of the source function for an individual level, even though only a single generator is present in an overhead block, as also illustrated in **Figure 5-8**. In **Figure 5-7** only an ODUk-LCK generator is illustrated (as only LCK is required in a normal A_So function), however other types of signals may be generated as well. The x_D indicates the frame data path, whereas x_CK, x_FS, x_MFS indicate the frame rate and frame synchronization.

Software must handle the effects of having a maintenance signal inserted at any virtual stage in the TCM hierarchy. So if e.g. the hierarchy is straight PM->TCM1->TCM2->..->TCM6->SM, and an ODUk-LCK signal should be inserted at TCM2, then software must force the shared maintenance signal generator to generate the LCK signal, and then disable the overhead insertion at the PM and TCM1 termination levels, as these are logically overwritten by the LCK signal on TCM2. Since the LCK signal insertion conceptually occurs at the top of the TCM2 source adaption function, TCM2 and the following stages should still have overhead insertion enabled, as they are placed after the signal generator in the TCM hierarchy. The software implementation of the TCM hierarchy must be able to handle the effects of any configuration of the TCM hierarchy, and not just the straight example.

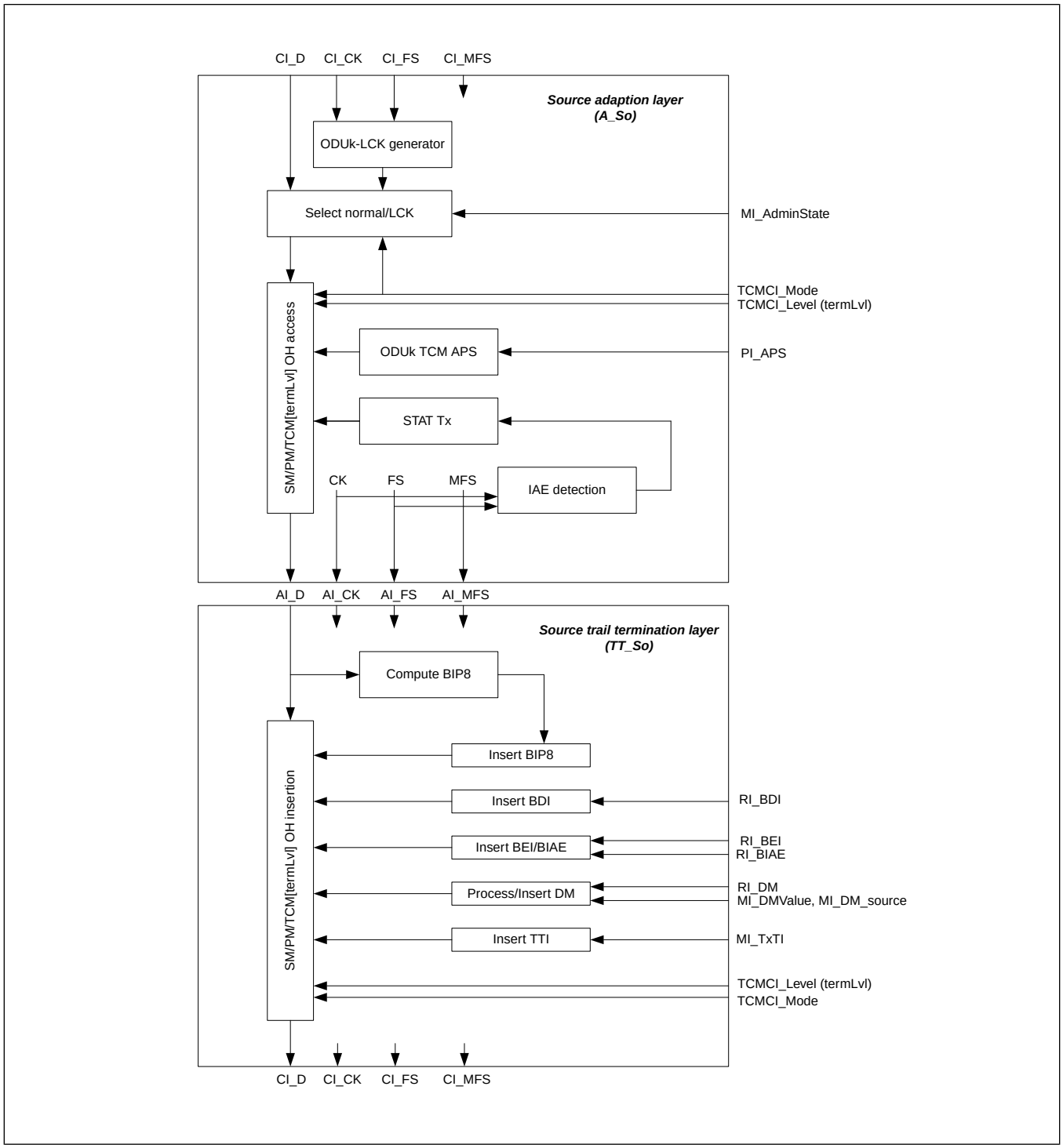


Figure 5-8 Source termination model

Each of the PM/TCMi/SM source termination level receives remote information from its companion sink termination level, e.g. for BDI and BEI/BIAE backwards indications. In case multiple sink stages are configured with identical termination level (termLvl) then only one should have remote information enabled.

5.3.8.3 Status of incoming signal

The general health of the incoming frame can be monitored via the `.sfIn`, `.dataRcvd`, `.fasErr`, `.mfasErr` and `.fstErr` fields of `OH_SO_BLK_HOLD`.

5.3.8.4 Test/maintenance signal insertion and frame generation

At the block level, it is possible to override the entire signal with a test or maintenance signal manually, as selected by `OH_SO_BLK_CFG.maintSigForce`. The generated signal will contain either an ODUk-LCK, ODUk-OCI or ODUk-AIS pattern, or a normal ODUk with a NULL or PRBS pattern (17.5.1-2/G.709) inserted in the OPU area.

In case a test pattern is inserted, the correct BIP will also be inserted at levels that are operational i.e. levels where `OH_SO_STG_CFG.lvlEn` is set, and the payload type (PT) is set according to **Table 5-11**.

Table 5-11 Test pattern payload types

Test pattern	Payload Type
PRBS	0xFE
NULL	0xFD

The frame timing of such inserted signals can either adopt the timing of the incoming ODUk signal, or use a new framing generated internally. When `OH_SO_BLK_CFG.frmGenEn` is set to '1', new framing is generated internally (so the incoming ODUk timing is ignored), otherwise the incoming framing (including multiframing) is adopted.

Switching between the incoming signal and the generated test/maintenance signal is hitless in case `.frmGenEn` = '0'.

5.3.8.5 Termination level operations

For any given PM/TCMi/SM level, the `OH_SO_STG_CFG.lvlEn` enable bit must also be set to '1' to enable overhead insertion at that termination level, otherwise the level is transparent. When the operation is non-transparent, overhead is inserted as follows:

The (TxTI) TTI value is transmitted, using the values configured in `OH_SO_TXTI_CFG` by indirect addressing of the internal TxTI table via `OH_SO_TXTI_CFG_IX`. The TxTI values must be completely configured, including the required leading zero bytes.

The BEI field is set with the value derived from the companion sink termination level, or overridden in the case of IAE. The BDI field is set by the value received from the companion sink termination level. Both the transmitted BDI and BEI fields can be overridden (forced to a given value) by setting fields `.bdiForce` and `.beiForce` in `OH_SO_STG_CFG`.

The Incoming Alignment Error (IAE) can be forced by software at the SM or TCM level by programming the corresponding `OH_SO_STG_CFG.statVal` and setting the `OH_SO_STG_CFG.lvlEn` bit to '1'. Otherwise it is controlled automatically as follows:

- IAE will be set if the OH block detects a misaligned frame. This may be caused by a frame start move in the incoming signal, or by a switch to/from internal frame generator with forced rate. When IAE is triggered by a misaligned frame it is set for 16 multiframes i.e. 4096 frames total.
- For SM the IAE bit is set when a misaligned frame is detected and SM overhead insertion is enabled.
- For the TCMs the status field is set to "In use with IAE" if the software programmed insert value is set to "In use without IAE". Other values of the software programmed status fields is not replaced. TCMi overhead insertion must be enabled to enable any status field insertion in TCMi.

Finally a correct BIP value is calculated and inserted. For test purposes, this value can be forced to an incorrect value with `OH_SO_STG_CFG.bipErrForce`.

5.3.8.6 Source side APS insertion

The APS/PCC bytes are either transparently forwarded, or inserted, per APS level. The OH_SO_APS_CFG controls the APS insertion and is addressed via the APS index, see **Table 5-3**.

The OH_SO_APS_CFG.apsLvl field is used to configure the APS level for the given APS table entry, and the OH_SO_APS_CFG.apsEn controls whether the APS/PCC bytes are inserted at that level, or the incoming APS/PCC bytes are forwarded transparently. The three APS/PCC bytes to be transmitted are configured in the .aps0, .aps1 and .aps2 fields of OH_SO_APS_CFG. The fourth byte (reserved) is always transmitted as zero, when APS insertion is enabled.

5.3.8.7 Schedule monitoring

The OH_SO_BLK_HOLD register contains two status bits that can be used to verify the scheduling. The .ohLost bit is set to indicate that the satellite did not receive OH data from the central module at the required rate, which again points to a problem with the scheduling setup. The .replyRcvd bit is set whenever a reply is received from the satellite. If this bit is never set it indicates a problem with the scheduling and/or routing setup.

5.3.8.8 Delay measurement

The delay measurement function uses the PM & TCM overhead byte to transmit and receive a one-bit DM value per termination level (PM, TCM1-6). The sink transfers the received DM value to the source side, provided that the sink mode is OPERATIONAL, see **RI_DM extraction** on page 408. Received DM values are then transmitted by the source without any processing on all termination levels where overhead insertion is enabled (OH_SO_STG_CFG.lvlEn = '1'). The DM functionality allows delay measurements to be performed on a single termination level. The steps involved in performing a delay measurement are outlined below:

1. Make sure that no DM bits are added from the ADD interface i.e. make sure that OH_AD_ADD_CFG.dmEn = 0 for the block index on which the delay measurement is performed.
2. Program the block index and termination level on which the delay measurement is performed into OH_SO_DM_CFG.blkIx and OH_SO_DM_CFG.termLvl.
3. Read the current received DM bit in OH_SK_STG_STAT.rxDm. Invert this value and program it in OH_SO_DM_CFG.dmVal so an event is created when the bit is send.
4. Now enable the delay measurement operation by setting the OH_SO_DM_CFG.dmEn bit.
5. Wait for OH_SO_DM_STAT.txSubmitted to be set. When this bit is set the delay measurement is submitted in the OH source interface.
6. Wait for OH_SO_DM_STAT.rxDone to be set. When this bit is set the delay measurement is complete and the total round-trip delay is available in OH_SO_DM_CNT which reports the delay in number of 357 MHz clock cycles. If the returned value is all-ones (maximum value) then the measurement failed and the returned value is invalid.
7. Disable the delay measurement state machine by clearing OH_SO_DM_CFG.dmEn.

5.3.8.9 Add/Drop interface

The add/drop interface provides the ability to monitor (drop) and overwrite (add) all overhead bytes from the various ODUs forwarded within the device. For pin interface details, see **Overhead add/drop interface** on page 498. The drop interface is always running, and has no configuration registers to select the contents of the data output. All OH bytes are dropped as received (i.e. insertion of maintenance signals and consequent actions have no effect on the dropped data), with the exception of the BIP fields, where the detected error mask is dropped.

The control of which OH bytes are overwritten and which are not, is done through OH_AD_ADD_CFG. **Table 5-12** below illustrates the granularity for the insertion control. Bytes with no associated notes has a single enable to control insertion. So e.g. RES4 has one enable to control if this byte is added or not.

Table 5-12 Add interface control overview

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
FAS[1]						MFAS[1]	SM[2]			GCC0[3]		RES 0	RES 1	OPU 0	OPU 1
RES 2	RES 3	PM & TCM[4]	TCM ACT	TCM6[2]			TCM5[2]			TCM4[2]			FTFL	OPU 2	OPU 3
TCM3[2]			TCM2[2]			TCM1[2]			PM[2]			EXP 0	EXP 1	OPU 4	OPU 5
GCC1[3]		GCC2[3]		APS/PCC[5]				RES 4	RES 5	RES 6	RES 7	RES 8	RES 9	OPU 6	OPU 7

Notes:

1. FAS and MFAS can not be overwritten from the ADD interface.
2. There is a single enable for the TTI field to control insertion in all multiframes. The enable for the BIP field enables inversion of the transmitted BIP if the value from the ADD interface is non-zero. For the status byte there is a single enable to enable insertion of all fields.
3. For each of GCC0, GCC1 & GCC2 there is one common enable for the two bytes.
4. There are 8 enables for the delay measurement byte. One per bit.
5. There are 8 enables for the APS/PCC bytes. Each enable controls insertion in multiframe N modulo 8. A specific enable controls all four bytes in the multiframe.

If the add interface is unused and left unconnected, it is important to clear all the bits in `OH_AD_ADD_CFG` to disable the add functionality. Further configuration options, mostly for error reporting and debug purposes are found in `OH_AD_CFG`, while related alarms are found in `OH_AD_STAT` and `OH_AD_HOLD`.

5.3.8.9.1 Sample

A single byte can be sampled from either the add or drop data stream. The direction to sample is controlled by the `OH_AD_SAMPLE.sourceSel` field. The index for the OH block to sample is selected in `.blkIx`. and the overhead byte to sample is controlled by `.byteNum`. The collected byte is available in the `.data` field when `OH_AD_HOLD.sampleDataRdy` is set. The `.sampleDataRdy` bit is a hold bit so it must be cleared before a new sampling is issued.

5.3.8.9.2 Loop

An add/drop loop can be enabled using the `OH_AD_CFG.loopEn` field. When the loop is enabled add data from the external device is ignored and instead `OH_AD_CFG.loopAddData` is returned for all add bytes. Drop data are still dropped on the external interface. The loop option is mainly intended as a debug feature.

5.3.8.9.3 Critical errors

The add interface provides status bits `.addLof` and `.addParityErr` in `OH_AD_HOLD` which provides visibility into whether the TPO has successfully locked onto the framing from the external device delivering the add data. If `.addDataLost` is set it indicates that the external device did not deliver the add data requested within the time specified in **Overhead add/drop interface** on page 498. The block index on which data is lost is indicated in `OH_AD_LOST.addBlkIx`. Finally the `.addOflw` indicates that more add request are generated than the external interface can handle.

On the drop side the `.dropDataLost` indicates that drop data was lost due to internal buffer overflow. The index on which data is lost is indicated in `OH_AD_LOST.dropBlkIx`. The `.dropOflw` indicates that more drop requested are issued than the external interface can handle.

All the error bits in `OH_AD_HOLD` mentioned above are critical and should not occur during normal operation, except that `.addDataLost` may get set when there is a jump in the multiframe e.g. due to a switch to/from a conditioning signal.

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5.4 Registers

5.4.1 General registers

Name: OH_CM_INFO
Description: Overhead features information
Address: 0x0020.0000

Bits	Field	Type	Reset	Description
0	dTimOs	RO	-	Set when the Operator Specific 32 bytes of expected TTIs are configurable and can be used in the dTIM criteria.
1	dualBank	RO	-	Set when dual banking of OH_SK_STG_CNT is implemented.
3:2	numStgs	RO	-	Number of configurable sink stages / source insert functions per termination block: 1: stg2. 2 stages. 2: stg4. 4 stages. 3: stg8. 8 stages.
6:4	numApsLvls	RO	-	Number of APS levels per OH block: 0: aps1. 1 APS level. 1: aps2. 2 APS levels. 2: aps4. 4 APS levels. 3: aps8. 8 APS levels.
7	b100g	RO	-	Set when support for OTN B100G and FlexO is included.
15:8	scheduleInterval	RO	-	Number of core_clk periods between two central to satellite service requests.
16	prbsMode	RO	-	PRBS mode: 0: mode0. PRBS inverted. 1: mode1. PRBS noninverted.

5.4.2 Scheduling table registers

Name: OH_CM_SCHED_CFG
Description: OH scheduler control
Address: 0x0020.0004

Bits	Field	Type	Reset	Description
0	use1	RW	0x0	Selects which bank to use: 0: Use table bank 0. 1: Use table bank 1. A switch 0 -> 1 takes place after the entry pointed to by .lastHiSpeed0 has been processed (and vice versa).
41:32	firstHiSpeed0	RW	0x0	First entry used in OH_CM_SCHED_SEQ for high speed part when .use1 = 0.

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Bits	Field	Type	Reset	Description
59:48	lowBWFrac0	RW	0x0	Fraction of slots (out of 0x1000) used for low-speed table part when .use1 = 0. A value of 0 means the low-speed part will not be scheduled at all.
73:64	lastLoSpeed0	RW	0x0	Last entry used in OH_CM_SCHED_SEQ for low speed when .use1 = 0. The first entry used for the low speed part is .lastHiSpeed0 + 1.
89:80	lastHiSpeed0	RW	0x0	Last entry used in OH_CM_SCHED_SEQ for high speed part when .use1 = 0.
105:96	firstHiSpeed1	RW	0x0	First entry used in OH_CM_SCHED_SEQ for high speed part when .use1 = 1.
123:112	lowBWFrac1	RW	0x0	Fraction of slots (out of 0x1000) used for low-speed table part when .use1 = 1. A value of 0 means the low-speed part will not be scheduled at all.
137:128	lastLoSpeed1	RW	0x0	Last entry used in OH_CM_SCHED_SEQ for low speed when .use1 = 1. The first entry used for the low speed part is .lastHiSpeed0 + 1.
153:144	lastHiSpeed1	RW	0x0	Last entry used in OH_CM_SCHED_SEQ for high speed part when .use1 = 1.

Name: OH_CM_SCHED_STAT
Description: OH scheduler status
Address: 0x0020.0024

Bits	Field	Type	Reset	Description
0	using1	RO	-	Active bank table: 0: Table bank 0. 1: Table bank 1.

Name: OH_SK_MS_STAT
Description: OH SK merge status
Address: 0x0020.0028

Bits	Field	Type	Reset	Description
31:0	activeIfs	RO	-	Active interfaces, for previous data collision, valid when OH_SK_MS_HOLD.collision is asserted.

Name: OH_SK_MS_HOLD
Description: OH SK merge status
Address: 0x0020.002C

Bits	Field	Type	Reset	Description
0	collision	W1C	-	Data collision at satellite interface.

Name: OH_SK_RTDLY_CFG
Description: Satellite round trip delay configuration
Address: 0x0020.0030

Bits	Field	Type	Reset	Description
15:0	blkIx	RW	-	Satellite round trip delay performed on blkIx.

Name: OH_SK_SAT_RTDLY
Description: Satellite round trip delay
Address: 0x0020.0034

Bits	Field	Type	Reset	Description
7:0	rtdly	RO	-	Satellite round trip delay with the blkIx programmed in OH_SK_RTDLY_CFG.
23:16	period	RO	-	Satellite service period with the blkIx programmed in OH_SK_RTDLY_CFG (Saturating).

Name: OH_CM_SCHED_SEQ
Description: OH port sequence table
Address: 0x0020.1000
Entries: 1024
Entry size: 0x4
Index: Scheduling index

Bits	Field	Type	Reset	Description
5:0	blkIx	RW	-	OH block index to service.

Name: OH_CM_ROUTE
Description: OH routing configuration table
Address: 0x0020.2000
Entries: 49
Entry size: 0x10
Index: OH block index (blkIx)

Bits	Field	Type	Reset	Description
18:0	routeId	RW	-	Routing ID to SO satellite.
35:32	skSatIx	RW	-	For each SO satellite: The index within the satellite group of the corresponding SK satellite.
67:64	sBlkIx	RW	-	OH termination block index in satellite.
113:96	skRouteId	RW	-	Routing ID to SK satellite. ODUCn: FlexO reorder. ODU0-4: skRouteId = soRouteId.

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5.4.3 OH sink block and stage registers (Rx/extract)

Name: OH_SK_EXTI_CFG
Description: Sink expected TTI configuration
Address: 0x0022.0000
Entries: 392
Index: Indirectly addressed via OH_SK_EXTI_CFG_IX

Bits	Field	Type	Reset	Description
127:0	sapi	RW	-	Expected SAPI (TTI bytes 0 to 15).
255:128	dapi	RW	-	Expected DAPI (TTI bytes 16 to 31).
511:256	os	RW	-	Expected Operator Specific field (TTI bytes 32 to 63).

Name: OH_SK_EXTI_CFG_IX
Description: Indirect index register for OH_SK_EXTI_CFG
Address: 0x0022.0080

Bits	Field	Type	Reset	Description
8:0	stgIx	RW	-	Processing stage index.

Name: OH_SK_BLK_CFG
Description: Sink overhead configuration per overhead block
Address: 0x0022.2000
Entries: 49
Entry size: 0x4
Index: OH block index (blkIx)

Bits	Field	Type	Reset	Description
0	frmGenEn	RW	-	This field controls the framing source when any active OH_SK_STG_CFG.maintSigForce field is different from forceNone: 0: No new frame timing is generated (the overhead block will overwrite the existing frame). 1: The framing is generated by the rate generator. New frame timing may also be generated in case of incoming signal fail.
3:1	outStgOfs	RW	-	This field points to the output stage. This stage drives the SSF out of the termination function.
31:26	n1BlkIx	RW	-	ODUC: OH block index of #1 member. ODU0-4: blkIX.

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Name: OH_SK_BLK_HOLD
Description: Sink overhead status hold per overhead block
Address: 0x0022.4000
Entries: 49
Entry size: 0x4
Index: OH block index (blkIx)

Bits	Field	Type	Reset	Description
0	fstErr	W1C	-	Set when the incoming frame start is not at the expected frame start position. For debug purposes only.
1	fasErr	W1C	-	Set when the frame start byte is not 0xF6. For debug purposes only.
2	mfasErr	W1C	-	Set when the received MFAS differs from the expected value. For debug purposes only.
3	dataRcvd	W1C	-	Set when data is received on the input to the overhead source block. For debug purposes only.
4	testErr	W1C	-	Error in PRBS test data seen.
5	opuCsf	W1C	-	OPUk-CSF. Software must ignore this bit, when the context makes it invalid.

Name: OH_SK_BLK_STAT
Description: Sink overhead status per ODUk
Address: 0x0022.6000
Entries: 49
Entry size: 0x4
Index: OH block index (blkIx)

Bits	Field	Type	Reset	Description
7:0	zeroMode	RO	-	Indicates whether OTUk/TCM has been zeroed after the final sink stage.
9:8	maintSig	RO	-	Indicates if a maintenance signal has been inserted in the sink direction, whether this is done automatically or by configuration. 0: norm. Normal mode. 1: oduLck. ODU-LCK. 2: oduOci. ODU-OCI. 3: oduAis. ODU-AIS.

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Name: **OH_SK_STG_HOLD**
Description: **Sink overhead status hold per stage**
Address: **0x0023.0000**
Entries: **392**
Entry size: **0x4**
Index: **OH stage index (stgIx)**

Bits	Field	Type	Reset	Description
0	aAis	W1C	-	aAIS declared.
1	ciSsfOut	W1C	-	CI_SSF output forwarded from this stage.
2	aiTsf	W1C	-	AI_TSF generated at this stage.
3	ciSsfIn	W1C	-	CI_SSF input to this stage.
4	dTim	W1C	-	dTIM: TTI mismatch defect indication.
5	dBiae	W1C	-	dBIAE: Backward Incoming Alignment Error (SM & TCM levels only).
6	dBdi	W1C	-	dBDI: Backward Defect Indication. Filtered value.
7	dIae	W1C	-	dIAE: Incoming Alignment Error (SM & TCM levels only).
28	dLtc	W1C	-	dLTC detected via STAT field.
29	dLck	W1C	-	dLCK detected via STAT field.
30	dOci	W1C	-	dOCI detected via STAT field.
31	dAis	W1C	-	dAIS detected via STAT field.

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Name: OH_SK_STG_CFG
Description: Sink overhead configuration per stage
Address: 0x0024.0000
Entries: 392
Entry size: 0x4
Index: OH stage index (stgIx)

Bits	Field	Type	Reset	Description
2:0	timDetMode	RW	-	The 64 byte TTI is divided into three fields: 0-15: SAPI. 16-31: DAPI. 32-63: Operator Specific. The three bits indicate which fields of the expected TTI that is included in the TTI mismatch detection (dTIM generation). So each bit may be set individually. The OS bit may only be set if the expected TTI includes the Operator Specific TTI field, as indicated in `OH_CM_INFO.dTimOs`. This field is set by the MI_TIMDetMo management interface API call. 1: `timSapiBit`. SAPI (bit 0). 2: `timDapiBit`. DAPI (bit 1). 4: `timOsBit`. OS (bit 2).
3	timActDis	RW	-	Set to `1' to disable consequent action on dTIM for this level. Allows implementation of the TIMActDis option.
4	ltcActEn	RW	-	Set to `1' to enable consequent action on dLTC for this level. Allows implementation of the LTCAct_Enable option.
5	ciSsfSetForce	RW	-	Set to `1' to force CI_SSF output from this level.
6	ciSsfClrForce	RW	-	Set to `1' to suppress/clear CI_SSF output from this level.
18:16	termLvl	RW	-	OH termination level. 0: `termPm`. PM. 1: `termTcm1`. TCM1. 2: `termTcm2`. TCM2. 3: `termTcm3`. TCM3. 4: `termTcm4`. TCM4. 5: `termTcm5`. TCM5. 6: `termTcm6`. TCM6. 7: `termSm`. SM.
21:19	preStgOfs	RW	-	Preceding stage offset.
23:22	maintSigForce	RW	-	Maintenance signal to be forced in sink direction. The last active stage in the TCM hierarchy will win. 0: `forceNone`. No force of maintenance signal. 1: `forceLck`. Force ODUk-LCK. 2: `forceOci`. Force ODUk-OCI. 3: `forceAis`. Force ODUk-AIS.
27	zeroMode	RW	-	Set to `1' to enable SM/TCM overhead zeroing.
28	riDmEn	RW	-	Set to `1' to enable DMti output to RI_DM.
29	mEn	RW	-	Non-intrusive monitor enable.

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Bits	Field	Type	Reset	Description
30	aEn	RW	-	Actions enable.
31	riEn	RW	-	Remote information enable.

Name: OH_SK_STG_STAT
Description: Sink overhead status per stage
Address: 0x0025.0000
Entries: 392
Entry size: 0x4
Index: OH stage index (stgIx)

Bits	Field	Type	Reset	Description
0	aAis	RO	-	aAIS declared.
1	ciSsfOut	RO	-	CI_SSF output forwarded from this stage.
2	aiTsf	RO	-	AI_TSF generated at this stage.
3	ciSsfIn	RO	-	CI_SSF input to this stage.
4	dTim	RO	-	dTIM: TTI mismatch defect indication.
5	dBiae	RO	-	dBIAE: Backward Incoming Alignment Error (SM & TCM levels only).
6	dBdi	RO	-	dBDI: Backward Defect Indication. Filtered value.
7	dIae	RO	-	dIAE: Incoming Alignment Error (SM & TCM levels only).
25	rxDm	RO	-	Received value of DM bit (unfiltered).
28:26	procStatAcc	RO	-	Accepted STAT field (TCM &n PM, byte 3), processed values.
31:29	rawStatAcc	RO	-	Accepted STAT field (TCM &n PM, byte 3), raw values.

Name: OH_SK_STG_CHLD
Description: Sink overhead change hold per stage
Address: 0x0026.0000
Entries: 392
Entry size: 0x4
Index: OH stage index (stgIx)

Bits	Field	Type	Reset	Description
0	aAis	W1C	-	aAIS declared.
1	ciSsfOut	W1C	-	CI_SSF output forwarded from this stage.
2	aiTsf	W1C	-	AI_TSF generated at this stage.
3	ciSsfIn	W1C	-	CI_SSF input to this stage.
4	dTim	W1C	-	dTIM: TTI mismatch defect indication.

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Bits	Field	Type	Reset	Description
5	dBiae	W1C	-	dBIAE: Backward Incoming Alignment Error (SM & TCM levels only).
6	dBdi	W1C	-	dBDI: Backward Defect Indication. Filtered value.
7	dIae	W1C	-	dIAE: Incoming Alignment Error (SM & TCM levels only).
28	dLtc	W1C	-	dLTC detected via STAT field.
29	dLck	W1C	-	dLCK detected via STAT field.
30	dOci	W1C	-	dOCI detected via STAT field.
31	dAis	W1C	-	dAIS detected via STAT field.

Name: OH_SK_STG_INTEN
Description: Sink overhead interrupt enable per stage
Address: 0x0027.0000
Entries: 392
Entry size: 0x4
Index: OH stage index (stgIx)

Bits	Field	Type	Reset	Description
0	aAis	RW	-	aAIS declared.
1	ciSsfOut	RW	-	CI_SSF output forwarded from this stage.
2	aiTsf	RW	-	AI_TSF generated at this stage.
3	ciSsfIn	RW	-	CI_SSF input to this stage.
4	dTim	RW	-	dTIM: TTI mismatch defect indication.
5	dBiae	RW	-	dBIAE: Backward Incoming Alignment Error (SM & TCM levels only).
6	dBdi	RW	-	dBDI: Backward Defect Indication. Filtered value.
7	dIae	RW	-	dIAE: Incoming Alignment Error (SM & TCM levels only).
28	dLtc	RW	-	dLTC detected via STAT field.
29	dLck	RW	-	dLCK detected via STAT field.
30	dOci	RW	-	dOCI detected via STAT field.
31	dAis	RW	-	dAIS detected via STAT field.

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Name: **OH_SK_STG_CNT**
Description: **Sink overhead counters per stage**
Address: **0x0028.0000**
Entries: **2, 392**
Entry size: **0x20000, 0x8**
Index: **PMTICK bank number, OH stage index (stgIx)**

Bits	Field	Type	Reset	Description
21:0	bip	RW	-	Near-end errored block (nN_B or pN_EBC) detected via BEI.
53:32	bei	RW	-	Far-end errored block count (nF_B or pF_EBC) detected via BIP8.

5.4.4 OH sink APS registers

Name: **OH_SK_APS_CFG**
Description: **Sink overhead APS level configuration**
Address: **0x002C.0000**
Entries: **392**
Entry size: **0x4**
Index: **OH APS index (apsIx)**

Bits	Field	Type	Reset	Description
2:0	apsLvl	RW	-	APS level: 0: apsPath. APS Path. 1: apsTcm1. APS TCM1. 2: apsTcm2. APS TCM2. 3: apsTcm3. APS TCM3. 4: apsTcm4. APS TCM4. 5: apsTcm5. APS TCM5. 6: apsTcm6. APS TCM6. 7: apsServer. APS Server.
31	apsEn	RW	-	APS extraction enable.

Name: **OH_SK_APS_STAT**
Description: **Sink overhead APS status**
Address: **0x002D.0000**
Entries: **392**
Entry size: **0x4**
Index: **OH APS index (apsIx)**

Bits	Field	Type	Reset	Description
7:0	aps0	RO	-	Accepted APS/PCC byte from column 5.
15:8	aps1	RO	-	Accepted APS/PCC byte from column 6.
23:16	aps2	RO	-	Accepted APS/PCC byte from column 7.

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Name: **OH_SK_APS_CHLD**
Description: **Sink overhead APS change hold**
Address: **0x002E.0000**
Entries: **392**
Entry size: **0x4**
Index: **OH APS index (apsIx)**

Bits	Field	Type	Reset	Description
0	aps	W1C	-	Change of accepted APS value.

Name: **OH_SK_APS_INTEN**
Description: **Sink overhead APS interrupt enable**
Address: **0x002F.0000**
Entries: **392**
Entry size: **0x4**
Index: **OH APS index (apsIx)**

Bits	Field	Type	Reset	Description
0	aps	RW	-	Change of accepted APS value.

5.4.5 OH sink TTI monitor registers

Name: **OH_SK_MON_CFG**
Description: **Sink overhead moving monitor control**
Address: **0x0022.0100**

Bits	Field	Type	Reset	Description
2:0	termLvl	RW	-	PM/TCM/SM termination level.
8:3	blkIx	RW	-	OH block index.

Name: **OH_SK_MON_STAT**
Description: **Sink overhead moving monitor status**
Address: **0x0022.0104**

Bits	Field	Type	Reset	Description
0	stable	RO	-	Cleared when an OH level index is written to OH_SK_MON_CFG. Becomes set if three consecutive identical TTIs have been received.

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Name: OH_SK_MON_INTEN
Description: Sink overhead moving monitor interrupt enable
Address: 0x0022.0108

Bits	Field	Type	Reset	Description
0	stable	RW	-	Interrupt mask for bit(s) in OH_SK_MON_STAT.

Name: OH_SK_MON_TTI
Description: Sink received TTI from moving monitor
Address: 0x0022.0140

Bits	Field	Type	Reset	Description
127:0	sapi	RO	-	Received SAPI (TTI bytes 0 to 15).
255:128	dapi	RO	-	Received DAPI (TTI bytes 16 to 31).
511:256	os	RO	-	Received Operator Specific field (TTI bytes 32 to 63).

5.4.6 OH source registers (Tx/insert)

Name: OH_SO_TXTI_CFG
Description: Source transmitted TTI (indirectly addressed)
Address: 0x0030.0000
Entries: 392
Index: Indirectly addressed via OH_SO_TXTI_CFG_IX

Bits	Field	Type	Reset	Description
127:0	sapi	RW	-	Transmitted SAPI (TTI bytes 0 to 15).
255:128	dapi	RW	-	Transmitted DAPI (TTI bytes 16 to 31).
511:256	os	RW	-	Transmitted Operator Specific field (TTI bytes 32 to 63).

Name: OH_SO_TXTI_CFG_IX
Description: Indirect index register for OH_SO_TXTI_CFG
Address: 0x0030.010C

Bits	Field	Type	Reset	Description
8:0	stgIx	RW	-	Processing stage index.

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Name: OH_SO_BLK_CFG
Description: Source configuration per overhead block
Address: 0x0030.8000
Entries: 49
Entry size: 0x4
Index: OH block index (blkIx)

Bits	Field	Type	Reset	Description
2:0	maintSigForce	RW	-	Test and maintenance signal insertion control: 0: `forceNone`. No force of maintenance signal. 1: `forceLck`. Force ODUk-LCK. 2: `forceOci`. Force ODUk-OCI. 3: `forceAis`. Force ODUk-AIS. 4: `forcePrbs`. Force ODUk-PRBS. 5: `forceNull`. Force ODUk-NULL.
4	frmGenEn	RW	-	This field controls the framing source when the `.maintSigForce` field is different from forceNone: 0: No new frame timing is generated. The overhead block will overwrite the existing frame. 1: The framing is generated by the rate generator.

Name: OH_SO_BLK_HOLD
Description: Source status hold per overhead block
Address: 0x0030.A000
Entries: 49
Entry size: 0x4
Index: OH block index (blkIx)

Bits	Field	Type	Reset	Description
0	fstErr	W1C	-	Set when the incoming frame start is not at the expected frame start position. Will lead to IAE. For debug purposes only.
1	fasErr	W1C	-	Set when the frame start byte is not 0xF6. For debug purposes only.
2	mfasErr	W1C	-	Set when the received MFAS differs from the expected value. For debug purposes only.
3	ohLost	W1C	-	Set when overhead data for a frame was not available. Indicates a problem with the scheduling of overhead data from the central overhead storage.
4	sfIn	W1C	-	Set when signal fail is present on the input to the overhead source block. Note that if the overhead source block generates a maintenance signal then `.sfIn` is always cleared.
5	dataRcvd	W1C	-	Set when data is active on the output of the overhead source block. For debug purposes only.

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Bits	Field	Type	Reset	Description
6	schedSyncErr	W1C	-	Set when one of the following events occur in a multiport satellite: The scheduling period of the rate differs from the scheduling period of data. The port sequence in the rate differs from the port sequence in data. Enable is asserted without port enable in either rate or data signal.
8	replyRcvd	W1C	-	Set when satellite has replied on the request from the central module. If not set it indicates either that the satellite is not scheduled or the routing is mis-configured. For debug purposes only.

Name: OH_SO_STG_STAT
Description: Source status per termination level
Address: 0x0031.0000
Entries: 392
Entry size: 0x4
Index: OH stage index (stgIx)

Bits	Field	Type	Reset	Description
3:0	riBei	RO	-	Current RI_BEI value from companion sink termination level.
4	riBdi	RO	-	Current RI_BDI value from companion sink termination level.
5	riBiae	RO	-	Current RI_BIAE value from companion sink termination level.

Name: OH_SO_STG_CFG
Description: Source overhead configuration per termination level
Address: 0x0032.0000
Entries: 392
Entry size: 0x4
Index: OH stage index (stgIx)

Bits	Field	Type	Reset	Description
7	bipErrForce	RW	-	Set to `1' to force insertion of a wrong BIP-8 value. The inserted value the correct BIP-8 XOR'ed 0xFF (i.e. the BIP-8 is inverted to force errors).
11:8	beiVal	RW	-	Forced BEI(/BIAE for TCMi) value. Inserted when both `.lvlEn` and `.beiForce` are set.
12	beiForce	RW	-	Set to `1' to force insertion of beiVal value instead of value from sink side.
13	bdiVal	RW	-	Forced BDI value. Inserted when both `.lvlEn` and `.bdiForce` are set.
14	bdiForce	RW	-	Set to `1' to force insertion of `.bdiVal` value insstead of value from sink side.

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Bits	Field	Type	Reset	Description
18:16	statVal	RW	-	STAT value. In case .termLvl selects the SM level (.termLvl = 7) then .statVal controls the IAE/RES bits.
30:28	termLvl	RW	-	OH termination level.
31	lvlEn	RW	-	Enable SM/PM/TCM insertion for this level. Must be set to use forced values as well.

5.4.7 OH source APS registers

Name: OH_SO_APS_CFG
Description: Source overhead APS configuration
Address: 0x0033.0000
Entries: 392
Entry size: 0x4
Index: OH APS index (apsIx)

Bits	Field	Type	Reset	Description
7:0	aps0	RW	-	APS/PCC byte in column 5 to transmit.
15:8	aps1	RW	-	APS/PCC byte in column 6 to transmit.
23:16	aps2	RW	-	APS/PCC byte in column 7 to transmit.
26:24	apsLvl	RW	-	APS level for insertion: 0: apsPath. APS Path. 1: apsTcm1. APS TCM1. 2: apsTcm2. APS TCM2. 3: apsTcm3. APS TCM3. 4: apsTcm4. APS TCM4. 5: apsTcm5. APS TCM5. 6: apsTcm6. APS TCM6. 7: apsServer. APS Server.
31	apsEn	RW	-	APS insertion enable. Only insert if `1' (otherwise APS level is transparent).

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5.4.8 Delay measurement registers

Name: OH_SO_DM_CFG
Description: Delay Measurement control
Address: 0x0030.0100

Bits	Field	Type	Reset	Description
2:0	termLvl	RW	0x0	Termination level to perform delay measurement on: 0: termPm. PM. 1: termTcm1. TCM1. 2: termTcm2. TCM2. 3: termTcm3. TCM3. 4: termTcm4. TCM4. 5: termTcm5. TCM5. 6: termTcm6. TCM6. 7: termSm. SM.
8:3	blkIx	RW	-	Overhead block index to perform delay measurement.
17	dmVal	RW	0x0	Sets value of DM bit to transmit.
18	dmEn	RW	0x0	If OH_SO_STG_CFG.lvlEn = '1' then this bit enables DM processing for the OH indicated by blkIx and termLvl. All other levels will have the DM bit looped back from sink to source. If OH_SO_STG_CFG.lvlEn = '0' the OH block is transparent and none of the DM bits are modified.

Name: OH_SO_DM_STAT
Description: Delay Measurement status
Address: 0x0030.0104

Bits	Field	Type	Reset	Description
0	txSubmitted	RO	-	Set to '1' when .dmVal is submitted for transmission.
1	rxDone	RO	-	Set to '1' when a transition is detected on the .rxDm bit. The delay measured will then be available in the OH_SO_DM_CNT register.

Name: OH_SO_DM_CNT
Description: Delay value
Address: 0x0030.0108

Bits	Field	Type	Reset	Description
31:0	count	RO	-	Counts system clock cycles between transmitted and received transition on DMx bit. Becomes valid when OH_SO_DM_STAT.rxDone is read as '1'. If the value read is all-ones, then the delay measurement has failed.

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5.4.9 Satellite ID registers

Name: OH_SO_SAT_ID_RID
Description: Satellite identifier string retrieval
Address: 0x0030.0110

Bits	Field	Type	Reset	Description
17:0	routeId	RW	-	Set to the routing ID to a satellite and the satellite identifier string is presented in OH_SO_SAT_ID.

Name: OH_SO_SAT_ID
Description: Satellite identifier string
Address: 0x0030.0114

Bits	Field	Type	Reset	Description
63:0	id	RO	-	Satellite identifier string for satellite with the routing ID programmed in OH_SO_SAT_ID_RID.

5.4.10 OH add/drop registers

Name: OH_AD_CFG
Description: Overhead add/drop configuration
Address: 0x003F.0000

Bits	Field	Type	Reset	Description
0	scrambleDis	RW	0x0	Disable scrambling of overhead drop frame (outgoing frame).
1	descrambleDis	RW	0x0	Disable descrambling of overhead add frame (incoming frame).
2	loopEn	RW	0x0	Enables internal loop. When loop is enabled the drop data stream is analyzed for add requests and all add requests are answered on the add interface with all overhead bytes set to the value defined in the .loopAddData field. Drop data is still transmitted on the external interface but add data from the external interface is ignored.
3	dropParityInv	RW	0x0	Enables parity inversion in drop frames.
15:8	loopAddData	RW	0x0	Overhead data returned for all bytes on the add interface in loop mode i.e. when .loopEn='1'.

Name: OH_AD_STAT
Description: Overhead add status
Address: 0x003F.0004

Bits	Field	Type	Reset	Description
0	addLof	RO	-	LOF on the add interface.
1	addOflw	RO	-	Too many requests for overhead add data issued.

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Bits	Field	Type	Reset	Description
2	addDataLost	RO	-	Add data was not returned from external device in time. The OH_AD_LOST.addBlkIx field is updated with the index for the OH block where data is lost.
3	dropOflw	RO	-	Too many request to drop overhead data issued.
4	dropDataLost	RO	-	Drop data could not be sent fast enough. The OH_AD_LOST.dropBlkIx field is updated with the index for the OH block where data is lost.
5	addParityErr	RO	-	Parity error detected in Add data frame.
6	sampleDataRdy	RO	-	Set when sample data is available in OH_AD_SAMPLE.data.

Name: OH_AD_HOLD
Description: Overhead add status hold
Address: 0x003F.0008

Bits	Field	Type	Reset	Description
0	addLof	W1C	-	LOF on the add interface.
1	addOflw	W1C	-	Too many requests for overhead add data issued.
2	addDataLost	W1C	-	Add data was not returned from external device in time. The OH_AD_LOST.addBlkIx field is updated with the index for the OH block where data is lost.
3	dropOflw	W1C	-	Too many request to drop overhead data issued.
4	dropDataLost	W1C	-	Drop data could not be sent fast enough. The OH_AD_LOST.dropBlkIx field is updated with the index for the OH block where data is lost.
5	addParityErr	W1C	-	Parity error detected in Add data frame.
6	sampleDataRdy	W1C	-	Set when sample data is available in OH_AD_SAMPLE.data.
7	addDataRcvd	W1C	-	Add data received.

Name: OH_AD_SAMPLE
Description: Overhead sample
Address: 0x003F.000C

Bits	Field	Type	Reset	Description
5:0	blkIx	RW	0x0	Index for OH block to sample.
17:12	byteNum	RW	0x0	Byte number to sample.
20	sourceSel	RW	0x0	Selects the stream to sample from: 0: selDrop. Get sample from drop interface. 1: selAdd. Get sample from add interface.
31:24	data	RO	-	The byte collected from either add or drop stream. Valid when OH_AD_HOLD.sampleDataRdy is set.

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Name: OH_AD_LOST
Description: Index for add/drop data lost
Address: 0x003F.0010

Bits	Field	Type	Reset	Description
5:0	addBlkIx	RO	-	Index for OH block where data is lost. Valid when OH_AD_HOLD.addDataLost is set.
21:16	dropBlkIx	RO	-	Index for OH block where data is lost. Valid when OH_AD_HOLD.dropDataLost is set.

Name: OH_AD_ADD_CFG
Description: Overhead add enable configuration
Address: 0x003F.8000
Entries: 49
Entry size: 0x10
Index: OH block index (blkIx)

Bits	Field	Type	Reset	Description
0	opu1En	RW	-	Add interface OPU1 insertion enable.
1	opu0En	RW	-	Add interface OPU0 insertion enable.
23	res1En	RW	-	Add interface RES1 insertion enable.
24	res0En	RW	-	Add interface RES0 insertion enable.
25	gcc0En	RW	-	Add interface GCC0 insertion enable.
31:29	smEn	RW	-	Add interface SM insertion enable: 1: statBit. Enable stat field (bit 0). 2: bipBit. Enable XOR'ing of calculated BIP (bit 1). 4: ttiBit. Enable TTI (bit 2).
32	opu3En	RW	-	Add interface OPU3 insertion enable.
33	opu2En	RW	-	Add interface OPU2 insertion enable.
37	ftflEn	RW	-	Add interface FTFL insertion enable.
42:40	tcm4En	RW	-	Add interface TCM4 insertion enable: 1: statBit. Enable stat field (bit 0). 2: bipBit. Enable XOR'ing of calculated BIP (bit 1). 4: ttiBit. Enable TTI (bit 2).
47:45	tcm5En	RW	-	Add interface TCM5 insertion enable: 1: statBit. Enable stat field (bit 0). 2: bipBit. Enable XOR'ing of calculated BIP (bit 1). 4: ttiBit. Enable TTI (bit 2).
52:50	tcm6En	RW	-	Add interface TCM6 insertion enable: 1: statBit. Enable stat field (bit 0). 2: bipBit. Enable XOR'ing of calculated BIP (bit 1). 4: ttiBit. Enable TTI (bit 2).
53	tcmActEn	RW	-	Add interface TCM ACT insertion enable.

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Bits	Field	Type	Reset	Description
61:54	dmEn	RW	-	Add interface DM insertion enable. The leftmost bit enables the leftmost bit in the DM byte, etc.
62	res3En	RW	-	Add interface RES3 insertion enable.
63	res2En	RW	-	Add interface RES2 insertion enable.
64	opu5En	RW	-	Add interface OPU5 insertion enable.
65	opu4En	RW	-	Add interface OPU4 insertion enable.
74	exp1En	RW	-	Add interface EXP1 insertion enable.
75	exp0En	RW	-	Add interface EXP0 insertion enable.
80:78	pmEn	RW	-	Add interface PM insertion enable: 1: statBit. Enable stat field (bit 0). 2: bipBit. Enable XOR'ing of calculated BIP (bit 1). 4: ttiBit. Enable TTI (bit 2).
85:83	tcm1En	RW	-	Add interface TCM1 insertion enable: 1: statBit. Enable stat field (bit 0). 2: bipBit. Enable XOR'ing of calculated BIP (bit 1). 4: ttiBit. Enable TTI (bit 2).
90:88	tcm2En	RW	-	Add interface TCM2 insertion enable: 1: statBit. Enable stat field (bit 0). 2: bipBit. Enable XOR'ing of calculated BIP (bit 1). 4: ttiBit. Enable TTI (bit 2).
95:93	tcm3En	RW	-	Add interface TCM3 insertion enable: 1: statBit. Enable stat field (bit 0). 2: bipBit. Enable XOR'ing of calculated BIP (bit 1). 4: ttiBit. Enable TTI (bit 2).
96	opu7En	RW	-	Add interface OPU7 insertion enable.
97	opu6En	RW	-	Add interface OPU6 insertion enable.
112	res9En	RW	-	Add interface RES9 insertion enable.
113	res8En	RW	-	Add interface RES8 insertion enable.
114	res7En	RW	-	Add interface RES7 insertion enable.
115	res6En	RW	-	Add interface RES6 insertion enable.
116	res5En	RW	-	Add interface RES5 insertion enable.
117	res4En	RW	-	Add interface RES4 insertion enable.
125:118	apsEn	RW	-	Add interface APS insertion enable. Bit #7 (left most) enables insertion when multiframe mod 8 = 7. Bit #6 enables insertion when multiframe mod 8 = 6 etc.
126	gcc2En	RW	-	Add interface GCC2 insertion enable.
127	gcc1En	RW	-	Add interface GCC1 insertion enable.

6 Filter

6.1 Features

- PI filter processor
- Multiple ports
- Optional IIR filtering
- Diagnostic interface

6.2 Overview

The filter is a central module in the device which services filter requests from other modules which require these functions. It functions as a loop controller, performing proportional plus integral (PI) processing on the input samples, and providing one output sample back to the requesting module per input sample received.

Each port is configured as required - documentation of this is not in the scope of this manual, and reference should be made to the associated API code.

Optionally, the output samples on any port can be passed through a (configurable) IIR filter, typically implemented as a low-pass filter in order to reduce high frequency noise.

6.3 Operation

6.3.1 Configuration

The central filter module holds the configuration for all satellites, which is set up in `FILT_CFG` and `FILT_IIR_CFG`. For details, refer to the API code.

6.3.2 Diagnostics

In order to analyze the filter operation, software can monitor the operation of the filter "live" via the `FILT_DIAG` and `FILT_DIAG_CFG` registers.

6.3.3 Filter topology and control

The filter processing is implemented as a central filter module and a number of filter satellites. All satellites are connected to the central modul through routing channels.

The topology is illustrated in the figure below.

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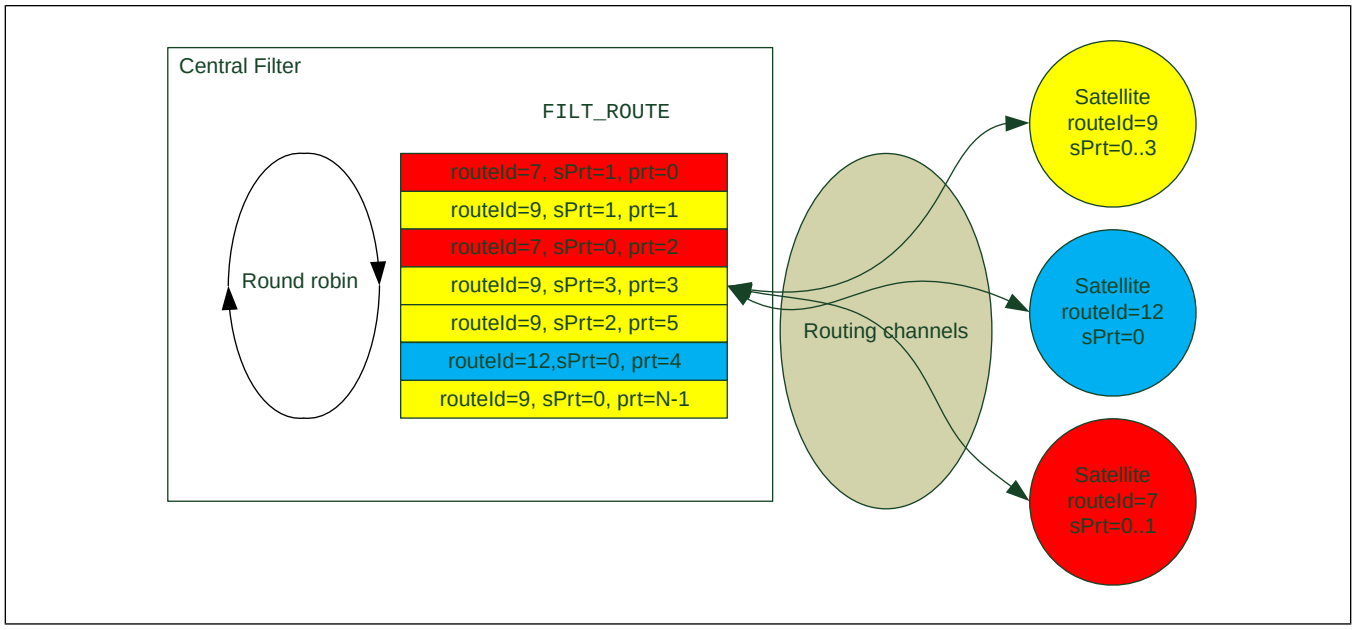


Figure 6-1 Filter topology

The figure illustrates a system with the one central filtermodule which has N ports (identified by the Filter Port Number shown as `prt`) and three filter satellites. Each satellite is uniquely identified by a routing identifier called `routeId`. The routing identifier is only used to identify the routing channel between the central filter module and a specific satellite.

A satellite may support more than one filter client, where each client is associated with a port number in the satellite called a satellite filter port, or just `sPrt`. The red satellite for example, supports two ports (`sPrt`=0 and `sPrt`=1).

Seen from the central filter modul a specific port in a specific satellite is uniquely identified by the `sPrt` and the `routeId`. This is programmed into the `FILT_ROUTE` table to get the port serviced. The index in `FILT_ROUTE` where a specific client is mapped is freely selectable by by specifying the global filter port number `prt`. All filter ports are serviced by a round robin scheduler as shown in the figure.

An example:

- The blue satellite is uniquely identified with the routing identifier `routeId`=12. This satellite only has one client; `sPrt`=0. Assume that `routeId`, `sPrt` and `prt` for this client are programmed into `FILT_ROUTE[5]` as `FILT_ROUTE[5].routeId`=12, `FILT_ROUTE[5].sPrt`=0 and `FILT_ROUTE[5].prt`=4. The client is therefore associated with global port 4.
- Software must then setup the configuration in `FILT_CFG[5]` to match the desired client filtering requirements.

6.4 Registers

Name: FILT_HOLD
Description: Filter status hold
Address: 0x0003.0000

Bits	Field	Type	Reset	Description
0	overwrite	W1C	0x0	Indicates overlapping service request.
32	overrun	W1C	-	Filter interface overrun. This bit is set whenver a FIFO overrun in one of the filter satellites occurs. Can be caused by a design error or scheduling configuration error.
52:48	overrunPort	RO	-	Indicates FIFO overruns in the satellite which the `.overrunPort` is belonged to. Whenever an overrun occurs, this field is updated (at the same time `.overrun` is set) to the failing port number.

Name: FILT_DIAG_CFG
Description: Filter diagnostics configuration
Address: 0x0003.0020

Bits	Field	Type	Reset	Description
4:0	port	RW	-	Index of selected port identical to the index used to index into `FILT_CFG`.
22	forceAll	RW	-	Force all. When asserted, integrator value force is performed for all filter ports. Otherwise force is only performed for port matching `.port` field.
23	forceEn	RW	-	Force enable. Force of integrator value is enabled.
31	en	RW	0x0	Monitor enable.
63:32	igt	RW	-	igt value to be forced.

Name: FILT_DIAG
Description: Filter diagnostics
Address: 0x0003.0040

Bits	Field	Type	Reset	Description
31:0	input	RO	-	Input to filter.
95:32	igt	ROA	-	Integrator signed value.
127:96	frontOut	RO	-	Output of filter front.
159:128	iirOut	RO	-	Output of IIR filter.
189:160	output	RO	-	Output of filter.
191:190	cnt	RO	-	Increments by 1 for each new value written into output.

Name: FILT_DIAG_INPUT
Description: Filter diagnostics of the mininum and maximum input value
Address: 0x0003.0060

Bits	Field	Type	Reset	Description
31:0	inputMin	RO	-	Minimum input value seen since last write of FILT_DIAG_CFG.
63:32	inputMax	RO	-	Maximum input value seen since last write of FILT_DIAG_CFG.

Name: FILT_ROUTE
Description: Filter routing configuration
Address: 0x0003.1000
Entries: 18
Entry size: 0x4
Index: Filter port number

Bits	Field	Type	Reset	Description
9:0	routeId	RW	-	Routing ID for satellite.
20:16	sPrt	RW	-	Port number within satellite.
28:24	prt	RW	-	Port number (same as clkSyncIx).
31	en	RW	-	Enable routing entry for the filter port. Software must configure all the routing entries.

Name: FILT_CFG
Description: Filter configuration
Address: 0x0003.4000
Entries: 18
Entry size: 0x10
Index: Filter port number

Bits	Field	Type	Reset	Description
5:0	propMult	RW	-	Proportional multiplier. Signed number.
17:12	propShift	RW	-	Shift proportional multiplier result .propShift bits down. Signed number.
29:26	decimate	RW	-	Only one in 2^ decimate filter results is forwarded after the IIR filter. The rest of the filter results are discarded before reaching the delta sigma function. Note that all filter results reach the filter functions before delta sigma.
30	iirEn	RW	-	Enables the IIR filter. The IIR filter is configured in FILT_IIR_CFG.

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Bits	Field	Type	Reset	Description
31	dsEn	RW	-	Delta Sigma enable. Delta sigma function is enabled when .dsEn is asserted, otherwise disabled. Delta Sigma function has a gain of 2^25.
43:32	igtMult	RW	-	Integrator multiplier. Signed number.
48:44	igtShift	RW	-	Integrator shift down. Integrator value is shifted down .igtShift bits before use. Unsigned number.
54:50	igtSat	RW	-	Integrator saturate. Integrator is saturated as a signed number .igtSat+8 bits (0 means 40 bits). Unsigned number.
59:55	outSat	RW	-	Output saturate. Output from filter frontend is saturated as a signed number of .outSat bits (0 means 32 bits). Unsigned number.
67:64	limitAdd	RW	-	Limit add. See .limitTh. Unsigned number.
71:68	limitTh	RW	-	Limit threshold. If filter input (after offset) is larger than or equal to (3/2)*2^.limitTh or smaller than -(3/2)*2^.limitTh then 2^(1+.limitTh+.limitAdd)-1 is added or 2^(1+.limitTh+.limitAdd) is subtracted to the integrator instead of the filter input. Limit function is disabled when .limitTh is 0. Unsigned number.
95:80	inputOffset	RW	-	Input offset. Before a filter input is processed, it is added with offset, signed number.
123:96	outputOffset	RW	-	Output offset. Added to the filter output, signed number.

Name: FILT_IIR_CFG
Description: Filter IIR configuration
Address: 0x0003.8000
Entries: 18
Entry size: 0x20
Index: Filter port number

Bits	Field	Type	Reset	Description
31:0	a0	RW	-	Multiplier used by IIR filter. Signed value where 30 LSBs is the fractional value.
63:32	b1	RW	-	
95:64	a1	RW	-	
127:96	b2	RW	-	
159:128	a2	RW	-	

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7 DCXO Interface

7.1 Features

- Control external DCXO devices through SPI interface
- Provides full CPU access to DCXO registers for configuration and monitoring
- Generates regular frequency updates for frequency tracking
- Controls multiple devices with multiple ports per device

7.2 Overview

The DCXO interface is used to implement frequency tracking by performing regular frequency updates. This is used to control the external transmit reference clock generator.

The only supported DCXO device is the SiLabs Si5345 clock generator, see Silicon Labs Si5345 Data Sheet.

Via the CPU interface, the DCXO registers can be programmed for the nominal frequency for each port. The device(s) can also be polled for internal status, to ensure circuit integrity and that communication is possible.

Under normal operation, the central filter sends regular updates as a requested deviation (+/-) from nominal to the DCXO interface, which performs some limiting and conversion operations before forwarding the data over the SPI interface to update the appropriate frequency adjustment registers inside the DCXO.

All this can be implemented following the description in the data sheet for the DCXO, but can be handled entirely transparently using the supplied API software package, which it is strongly recommended to use.

7.3 Operation

The following two sections describes the general aspects of using the DCXO interface block to control external DCXO devices for frequency tracking and for configuration and monitoring. This followed by section(s) describing the specific configuration used for the current device.

7.3.1 DCXO update control

Under normal operation, the clock generator device registers are constantly updated with the values output from the filter block. All operations are performed through the `CLKSYNC_CFG` register. This process must be enabled by setting `.updateEn` to '1'. Note that `.updateEn` must never be written to 0 while any traffic is active on the given port as this will cause data corruption as the clocks are no locked.

To prevent the frequency from leaving the valid frequency band `.limit` should be set to the maximum deviation permitted. This should be set in relation to the permitted frequency range of the protocol used. For example, if the protocol allows a ±100 ppm offset, then limit could be set to e.g. (frequency × 150 ppm) to give a suitable margin allowing for reference clock inaccuracies up to ±50 ppm. This is important as the transmit reference clock is also used for the receive reference clock, so it should not deviate too greatly from nominal. The limit is applied after the shift operation, so the calculated limit should be shifted by the same amount before it is configured.

Before the value is sent to the external device the input can be negated by setting `.negate`, and shifted up to 16 positions to the right using `.shift`. For test purposes the value can be forced to 0 using `.fixed`. If needed a constant offset can also be added using `.offset`.

The mapping from SerDes interface number to device should be configured in `.bus` and `.device`. The bus fields is only used in devices with more then one external interface, and is used to select which bus the device is placed on. The selection of device would normally be a straightforward mapping with device #0 controlling interfaces 0-4 on channels 0-4. The mapping to a specific channel is device specific.

Each update of the external device can consist of up to three accesses: a pre-write, the actual access, and a post-write. The pre- and post-write can be enabled separately when needed (using `.preEn` and `.postEn`), and are intended to be used for page/channel selection in devices that requires this. The address programmed for each of these accesses contains all 8 bits that are transmitted on the external interface, including device specific control bits. The length of the access is programmed in `.len`. The interface sends MSB first unless `.little` is set, and the external devices shall be setup to use the same setting. When needed an extra control byte can be prepended the data for each of the three accesses.

The clock used towards the external device is the core clock divided by 4, then scaled down using the value programmed in `.clkDiv`. The value should be chosen such that the resulting frequency is no higher than the maximum supported by the device used.

7.3.2 CPU access

The CPU can access registers in the external device using `CLKSYNC_CPU_CFG`. Most fields in the register match the fields in `CLKSYNC_CFG`, and the function is as described in **DCXO update control** on page 445, however `.limit` and `.offset` are not available. For CPU accesses a second pre-write operation can be used, configured using the `.pre2*` fields. For some devices this allows more efficient updates form the filter, while still allowing full access form the CPU.

When reading from or writing to the external device, the data used is located in `.data`. When the operation is completed `.en` will be cleared.

7.3.3 Si5345 Clock Generator

For the Si5345 device the "addr" fields are used to select one of the six SPI commands, and the "ctrl" fields can then be used for the address when this is required by the command. Write operations should always be performed using the "Burst Write" command, while reads are performed using the "Set Address" command followed by the "Read" command.

The registers used for controlling the frequency are all located on page 3 in the Si5345 device, and to make most efficient use of the bandwidth on the SPI interface, the page shall be set to 3 unless a CPU access is performed. This means that frequency updates can be performed using a single burst write command. When performing a CPU access while frequency updates are in progress, the pre-write is used to set the page for the register, and the post_write is used to set the page back to 3. For read operations the second pre-write operation is used for the "Set Address" command, while the normal access is used for the "Read" command. For writes the second pre-write is not used, and the normal access is performed as a "Burst Write".

Channel selection is performed by programming the address of the register for the selected channel in `.addr`.

For the Si5345 both `.little` and `.negate` shall be set, while `.upd` and `.longIdle` is zero and `.shift` is unused. For this device `.clkDiv` shall be set to 5 for clocks under 400MHz and 6 for over.

Following a reset, a number of register in the Si5345 needs to be set up for the device to function as intended. The steps then needed to setup a specific clock frequency depends on the choice made during the initial setup. It is recommended than an offset is chosen that minimizes the number of bytes that has to be written for the frequency updates.

7.3.4 VCXO Replacement Agilex

When using the internal PLLs to replace the external clocking devices, this is indicated by setting `.bus` to the value after the last external bus (e.g. in a device with two external busses, the values 0 and 1 are used for the external busses, and 2 is used for the internal PLLs) and set `.updateEn` to 0x1 to enable the giving xcvr port.

In this mode the following fields has to be set to match the corresponding xcvr location:

`.offsetSide` = Side.

`.offsetTile` = Tile.

`.offsetGroup` = Group.

`.offsetResource` = Resource ID.

`.offsetLane` = Lane number.

`.offsetPLL` = PLL type (Slow, Medium or Fast). These addresses can be found in the xcvr register documentation.

`.offset` must also be set to the initial K-value.

All others fields in the row must be set to zero

7.4 Registers

Name: `CLKSYNC_CFG`
Description: Clock sync configuration
Address: `0x0002.1000`
Entries: `18`
Entry size: `0x20`
Index: Port number

Bits	Field	Type	Reset	Description
7:0	device	RW	-	Select external device to use.
9:8	bus	RW	-	SPI bus number. If only one SPI bus is employed, set to 0.
16	updateEn	RW	0x0	Set to enable updates to be sent to external device.
17	fixed	RW	-	If set input is forced to be 0.
18	negate	RW	-	Negate input.
19	little	RW	-	Set to send data LSB first.
27:24	shift	RW	-	Number of bits to shift left the input before it is used.
39:32	preData	RW	-	Pre-write data.
47:40	preCtrl	RW	-	Pre-write control.
55:48	preAddr	RW	-	Pre-write address.
62	preCtrlEn	RW	-	Pre-write control enable.
63	preEn	RW	-	Set to enable pre-write.
71:64	clkDiv	RW	-	Clock towards external device is core clock/4/`.clkDiv`.
80	upd	RW	-	Set to toggle upd pin after access is performed.
81	longIdle	RW	-	Extend idle period between CS asserted, and first clock pulse.
127:96	limit	RW	-	Maximum deviation from 0 of input.
159:128	offset	RW	-	Offset to the filter input.
171:160	offsetPLL	RW	-	XCVR PLL select fast, medium or slow. The register address shifted down 8 bits. For example, to specify `SRDS_FGT_SLOW_DIV0` with address offset 0x44000, use value 0x440.
181:178	offsetLane	RW	-	Serdes_sel lane number.
185:182	offsetResource	RW	-	Serdes_sel resource ID.
188:186	offsetGroup	RW	-	Serdes_sel group.
190:189	offsetTile	RW	-	Serdes_sel tile.
191	offsetSide	RW	-	Serdes_sel side.
199:192	addr	RW	-	Address to use towards the external device, including control bits.

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Bits	Field	Type	Reset	Description
207:200	ctrl	RW	-	Control.
210:208	len	RW	-	Length of register to read/write in bytes, minus 1. I.e. a value of 2 results in 3 bytes being written.
222	ctrlEn	RW	-	Control enable.
231:224	postData	RW	-	Post-write data.
239:232	postCtrl	RW	-	Pre-write control.
247:240	postAddr	RW	-	Post-write address.
254	postCtrlEn	RW	-	Post-write control enable.
255	postEn	RW	-	Set to enable post-write.

Name: CLKSYNC_CPU_CFG
Description: Clock sync CPU configuration
Address: 0x0002.1240

Bits	Field	Type	Reset	Description
7:0	device	RW	-	Select external device to use.
9:8	bus	RW	-	SPI bus number. If only one SPI bus is employed, set to 0.
19	little	RW	-	Set to send data LSB first.
30	wr	RW	-	Set if CPU access is a write.
31	en	RW	0x0	Set to start CPU access.
39:32	preData	RW	-	Pre-write data.
47:40	preCtrl	RW	-	Pre-write control.
55:48	preAddr	RW	-	Pre-write address.
62	preCtrlEn	RW	-	Pre-write control enable.
63	preEn	RW	-	Set to enable pre-write.
71:64	clkDiv	RW	-	Clock towards external device is core clock/4/.clkDiv.
80	upd	RW	-	Set to toggle upd pin after access is performed.
81	longIdle	RW	-	Extend idle period between CS asserted, and first clock pulse.
103:96	pre2Data	RW	-	Pre-write data.
111:104	pre2Ctrl	RW	-	Pre-write control.
119:112	pre2Addr	RW	-	Pre-write address.
126	pre2CtrlEn	RW	-	Pre-write control enable.
127	pre2En	RW	-	Set to enable pre-write.
191:128	data	RW	-	Data read from or written to the device.
199:192	addr	RW	-	Address to use towards the external device, including control bits.

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Bits	Field	Type	Reset	Description
207:200	ctrl	RW	-	Control.
210:208	len	RW	-	Length of register to read/write in bytes, minus 1. I. e. a value of 2 results in 3 bytes being written.
222	ctrlEn	RW	-	Control enable.
231:224	postData	RW	-	Post-write data.
239:232	postCtrl	RW	-	Pre-write control.
247:240	postAddr	RW	-	Post-write address.
254	postCtrlEn	RW	-	Post-write control enable.
255	postEn	RW	-	Set to enable post-write.

8 Private Line Emulation (PLE)

8.1 Features

- Mapping between system bus clients and PLE clients.
- Packetization of client bit-streams into PLE packets.
- PLE packet encapsulation with timestamps based on a common clock.
- Per PLE client 1+1 protection.
- Injection of control messages.
- Line-side Ethernet MAC interface.
- Identification of PLE packets and control messages.
- Extraction of control messages.
- PLE packet descapsulation with extraction of header information, including timestamps.
- PLE payload de-jitter buffering with reconstruction of client timing based on timestamps.

8.2 Overview

TODO

8.3 Operation

8.3.1 PLE Packet Encapsulation

The PLE packets carry 1024 bytes of client data. The header consists of a flexible Ethernet/MPLS header, the PLE control word and the RTP header.

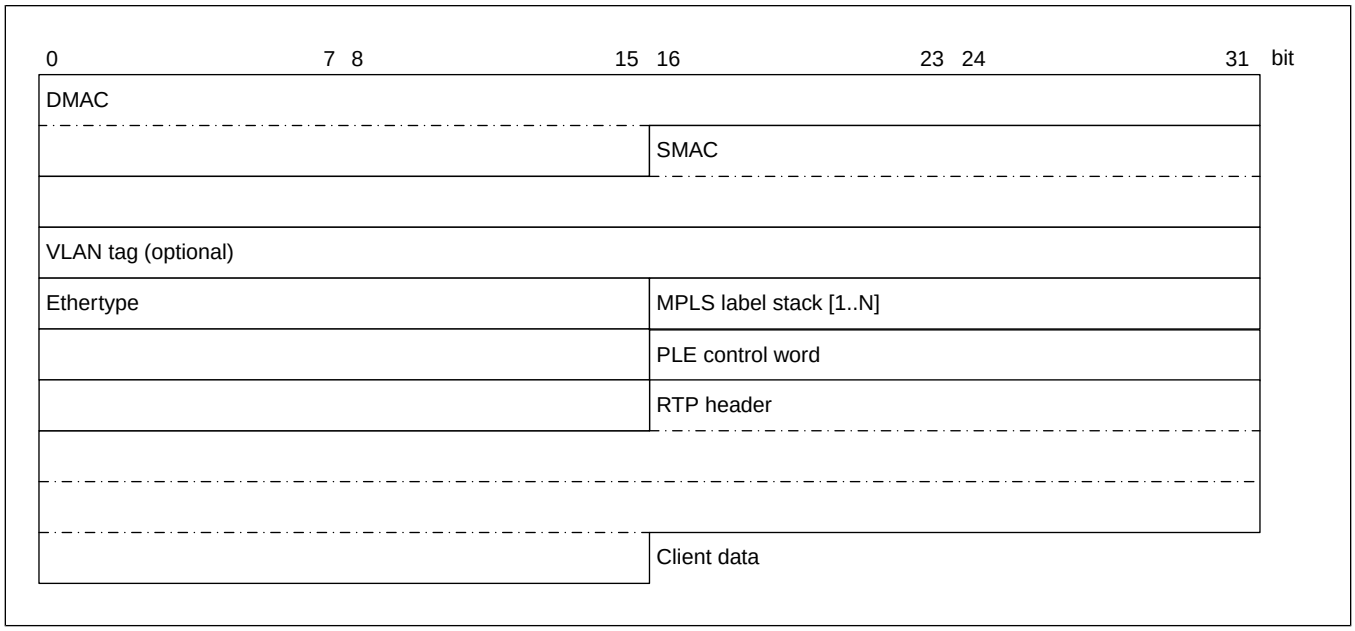


Figure 8-1 PLE encapsulation

The details of the MPLS label is illustrated below.

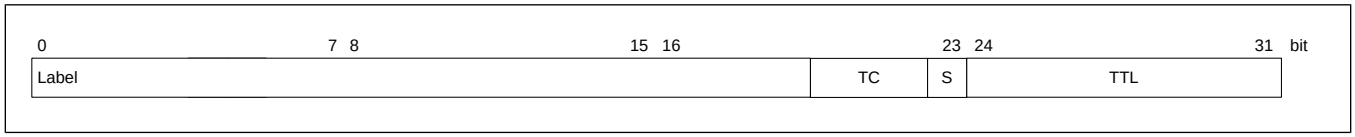


Figure 8-2 MPLS label

The details of the PLE control word is illustrated below.

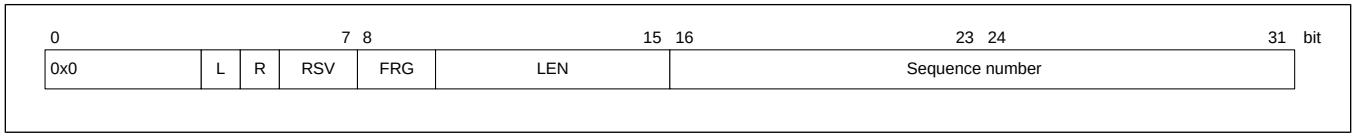


Figure 8-3 PLE control word

The details of the RTP header is illustrated below.

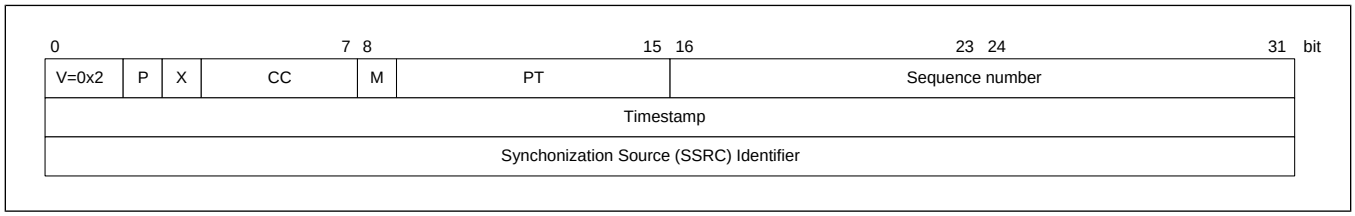


Figure 8-4 RTP header

The added Ethernet/MPLS header is fully configurable per PLE client per line interface via PLE_EG_ENCAP0_CFG. The supported sizes are 14 to 30 bytes in increments of two.

The fields of the PLE encapsulation, which are dynamic, are described in the following table.

Table 8-1 Dynamic PLE encapsulation fields

Field	Description
PLE - L	Local fault based on the signal fail indication from the client
PLE - R	Remote fault indicating packet loss (PLOS) from the PSN
PLE - Sequence number	Incrementing sequence number from the PLE IWF
RTP - Sequence number	Incrementing sequence number from the PLE IWF (same as PLE - Sequence number)
RTP - Timestamp	Incrementing timestamp from the PLE IWF based on the 125MHz common clock

The PLE line interface module receives client data, timestamp, sequence number, local fault and remote fault indications from the PLE IWF module and adds the encapsulation in the PSN-facing direction. The header is stripped by the PLE line interface module in the NSP-facing direction, before passing the client bit stream payload to the PLE IWF module, along with the timestamp, sequence number, local fault and remote fault indications extracted from the header.

The MAC interfacing to the PSN handles preamble and FCS addition and removal, as well as Ethernet PCS layer functions.

TODO - cfg for static PLE control word and RTP header fields.

8.3.2 PLE and Control Packet Identification

The headers of incoming packets from the PSN are matched against a filter. The lower entries of the filter are used to identify packets matching the Ethernet/MPLS encapsulation of PLE clients. The upper entries are used to match control packets for the SGMII message interface. The header and mask values of the filter are configured per line interface in PLE_IG_ENCAP0_CFG.

Packets matching the Ethernet/MPLS encapsulation of a PLE client are subject to additional inspection. Based on the configured Ethernet/MPLS header size (PLE_IG_ENCAP1_CFG.hdrByteSize) the first nibble of the PLE control word is extracted. Packet with a first nibble of the PLE control word with the value of zero are idenfified as PLE packets carrying client data. Packets with a non-zero first nibble are identified as control packets for extraction to the SGMII message interface. An optional MPLS TTL value check is performed (PLE_IG_ENCAP1_CFG.mplsTtlPos and PLE_IG_ENCAP1_CFG.mplsTtlChkEn). For clients with the check enabled, packets with a TTL value less than two are discarded. The event is flagged in PLE_IG_CL_HOLD.ttlErr.

8.3.3 Line Interface Statistics

Incoming packets matching a filter entry are counted in PLE_IG_FILTER_CNT.

Packets not matching any of the filter entries are counted in PLE_IG_GLOBAL_CNT.igPktNoMatch.

Packets matching a PLE client filter entry that are identified as control packets are counted in PLE_IG_PLE_CNT.igPktMsg.

For client N, the number of PLE payload packets can be calculated as PLE_IG_FILTER_CNT[N].igPktMatch minus PLE_IG_PLE_CNT[N].igPktMsg, provided that no packets are discarded due to failing MPLS TTL check.

In case the control packet buffer in the line interface module overflows, resulting in control packet drops, these packets are counted in PLE_IG_GLOBAL_CNT.igPktMsgDrop.

The outgoing PLE payload packets are counted in PLE_EG_PLE_CNT.

8.3.4 Line Interface Diagnostics

It is possible to capture the header of the last packet that did not match any of the filter entries or that matched a specific filter entry through PLE_IG_LAST_CFG and PLE_IG_LAST_STAT.

TBD - loopbacks.

8.3.5 1+1 Protection

Transmission of PLE clients to the line interfaces is enabled through PLE_1P1_EG_CFG.en.

The active line interface is selected per PLE client through PLE_1P1_IG_CFG.lineSel.

Enable/disable and switching will be made on PLE payload boundaries in hardware to prevent packet corruption during transitions.

In case of network misconfiguration, incoming PLE payload sizes may differ from the expected payload size. If the incoming payload is too large, the payload will be discarded and the event flagged in PLE_1P1_CL_HOLD.pktOversized. If the incoming payload is too small, the payload will be discarded and the event flagged in PLE_1P1_CL_HOLD.pktUndersized.

Protocol violations, which should never occur during operation, are flagged in PLE_1P1_DBG_HOLD.pktNoSop and PLE_1P1_DBG_HOLD.pktNoEop.

While long and short payload detection only affects the misconfigured client or clients, the effects of protocol violations may hit multiple clients.

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8.4 Registers

Name: PLE_INFO
Description: PLE Information
Address: 0x0040.0000

Bits	Field	Type	Reset	Description
15:0	dwIwfPsnIg	RO	-	Datawidth of the PLE_IWF PSN Ingres Interface.
31:16	dwIwfPsnEg	RO	-	Datawidth of the PLE_IWF PSN Egres Interface.
47:32	dwIwfNsp	RO	-	Datawidth of the PLE_IWF NSP Interfaces.
55:48	numIwfNsp	RO	-	Number of PLE_IWF NSP Interfaces.
63:56	numLif	RO	-	Number of PLE Line Interfaces.
64	lif1p1Present	RO	-	The PLE Line Interface 1+1 Protection module is present.
73:65	numClients	RO	-	Number of PLE clients.
89:74	plePayloadBytes	RO	-	PLE packet payload bytes. The length refers to the ODUk or PCS payload size without encapsulation.
127:96	igBufkBytes	RO	-	Ingress (de-jitter) buffer size in kbytes.

8.4.1 PLE_IWF

8.4.1.1 PLE_IWF_P2N

Name: PLE_IWF_P2N_MAP_CFG
Description: PLE IWF PSN to NSP Mapping Configuration
Address: 0x0050.0000
Entries: 512
Entry size: 0x4
Index: System bus and port number. Entries 0-80 map the port numbers on system bus 0 to PLE client numbers, entries 128-207 map the port numbers on system bus 1 to PLE client numbers, etc

Bits	Field	Type	Reset	Description
8:0	pleClient	RW	0x0	Use this PLE client number for the corresponding system bus and port number.

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Name: PLE_IWF_BUF_STAT
Description: PLE IWF Client Buffer Status
Address: 0x0050.1000
Entries: 83
Entry size: 0x4
Index: PLE client number

Bits	Field	Type	Reset	Description
9:0	minFillLvl	RO	-	Ingress buffer minimum fill level since last cleared. Clear with PLE_IWF_P2N_CFG.clrFillLvl. The value indicates the level in 1/1024th of the total buffer capacity.
19:10	maxFillLvl	RO	-	Ingress buffer maximum fill level since last cleared. Clear with PLE_IWF_P2N_CFG.clrFillLvl. The value indicates the level in 1/1024th of the total buffer capacity.
29:20	currFillLvl	RO	-	Ingress buffer current fill level. The value indicates the level in 1/1024th of the total buffer capacity.

Name: PLE_IWF_P2N_CFG
Description: PLE IWF PSN to NSP Configuration
Address: 0x0050.9000

Bits	Field	Type	Reset	Description
15:0	plosTimeUs	RW	0x3e8	Packet Loss (PLOS) state is declared for a PLE client when no ingress PLE packets have been accepted for this amount of time in microseconds.
16	clrFillLvl	RW	0x0	Ingress buffer minimum/maximum fill level clear for the PLE client configured in .clrClient. Write '1' to clear PLE_IWF_BUF_STAT.minFillLvl and PLE_IWF_BUF_STAT.maxFillLvl. This configuration will be cleared by HW then the clearing of the fill level status has completed.
25:17	clrClient	RW	0x0	PLE client for which to clear the ingress buffer minimum/maximum fill levels.

Name: PLE_IWF_CL_STAT
Description: PLE IWF Client Status
Address: 0x0050.A000
Entries: 83
Entry size: 0x4
Index: PLE client number

Bits	Field	Type	Reset	Description
0	igOflw	RO	-	Ingress buffer overflow.
1	igUflw	RO	-	Ingress buffer underflow.

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Bits	Field	Type	Reset	Description
2	seqNumErr	RO	-	Buffer initialization/flush triggered by consequtive sequence numbers violating PLE_IWF_CL_CFG.seqNumLimit.
3	tsErr	RO	-	Buffer initialization/flush triggered by consequtive packets passing sequence number validation but with decreasing timestamp values.
4	sfOut	RO	-	Signal Fail indication at the output towards the NSP.
5	plos	RO	-	Ingress Packet Loss (PLOS) state.
6	igL	RO	-	PLE control word L-bit of last received PLE packet. Indicates that the other end experiences an attachment circuit fault.
7	igR	RO	-	PLE control word R-bit of last received PLE packet. Backward fault indication showing that the other end is in the PLOS state.

Name: PLE_IWF_CL_HOLD
Description: PLE IWF Client Hold Status
Address: 0x0050.B000
Entries: 83
Entry size: 0x4
Index: PLE client number

Bits	Field	Type	Reset	Description
0	igOflw	W1C	-	Ingress buffer overflow.
1	igUflw	W1C	-	Ingress buffer underflow.
2	seqNumErr	W1C	-	Buffer initialization/flush triggered by consequtive sequence numbers violating PLE_IWF_CL_CFG.seqNumLimit.
3	tsErr	W1C	-	Buffer initialization/flush triggered by consequtive packets passing sequence number validation but with decreasing timestamp values.
4	sfOut	W1C	-	Signal Fail indication at the output towards the NSP.
5	plos	W1C	-	Ingress Packet Loss (PLOS) state.
6	igL	W1C	-	PLE control word L-bit of last received PLE packet. Indicates that the other end experiences an attachment circuit fault.
7	igR	W1C	-	PLE control word R-bit of last received PLE packet. Backward fault indication showing that the other end is in the PLOS state.

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Name: PLE_IWF_CL_CHLD
Description: PLE IWF Client Change Hold Status
Address: 0x0050.C000
Entries: 83
Entry size: 0x4
Index: PLE client number

Bits	Field	Type	Reset	Description
0	igOflw	W1C	-	Ingress buffer overflow.
1	igUflw	W1C	-	Ingress buffer underflow.
2	seqNumErr	W1C	-	Buffer initialization/flush triggered by consequtive sequence numbers violating PLE_IWF_CL_CFG.seqNumLimit.
3	tsErr	W1C	-	Buffer initialization/flush triggered by consequtive packets passing sequence number validation but with decreasing timestamp values.
4	sfOut	W1C	-	Signal Fail indication at the output towards the NSP.
5	plos	W1C	-	Ingress Packet Loss (PLOS) state.
6	igL	W1C	-	PLE control word L-bit of last received PLE packet. Indicates that the other end experiences an attachment circuit fault.
7	igR	W1C	-	PLE control word R-bit of last received PLE packet. Backward fault indication showing that the other end is in the PLOS state.

Name: PLE_IWF_CL_INTEN
Description: PLE IWF Client Change Hold Status
Address: 0x0050.D000
Entries: 83
Entry size: 0x4
Index: PLE client number

Bits	Field	Type	Reset	Description
0	igOflw	RW	-	Ingress buffer overflow.
1	igUflw	RW	-	Ingress buffer underflow.
2	seqNumErr	RW	-	Buffer initialization/flush triggered by consequtive sequence numbers violating PLE_IWF_CL_CFG.seqNumLimit.
3	tsErr	RW	-	Buffer initialization/flush triggered by consequtive packets passing sequence number validation but with decreasing timestamp values.
4	sfOut	RW	-	Signal Fail indication at the output towards the NSP.
5	plos	RW	-	Ingress Packet Loss (PLOS) state.
6	igL	RW	-	PLE control word L-bit of last received PLE packet. Indicates that the other end experiences an attachment circuit fault.

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Bits	Field	Type	Reset	Description
7	igR	RW	-	PLE control word R-bit of last received PLE packet. Backward fault indication showing that the other end is in the PLOS state.

8.4.1.1.1 PLE_IWF_P2N_CL_CFG

Name: PLE_IWF_CL_CFG
Description: PLE IWF Client Configuration
Address: 0x0051.0000
Entries: 83
Entry size: 0x10
Index: PLE client number

Bits	Field	Type	Reset	Description
9:0	initFillLvl	RW	0x0	Ingress buffer initialization fill level. The buffer must reach this fill level after initialization/flush before readout starts. The value specifies the level in 1/1024th of the total buffer capacity.
31:16	seqNumLimit	RW	0x2	Maximum allowed increment between two incoming sequence numbers without triggering buffer initialization/flush.
41:32	tsAdjHighWm	RW	0x3ff	Ingress buffer high watermark. When the buffer fill level is above this watermark, local timestamp clock adjustments will be in effect to make the clock run faster. The value specifies the level in 1/1024th of the total buffer capacity.
51:42	tsAdjLowWm	RW	0x0	Ingress buffer low watermark. When the buffer fill level is below this watermark, local timestamp clock adjustments will be in effect to make the clock slower faster. The value specifies the level in 1/1024th of the total buffer capacity.
79:64	tsAdjPeriod	RW	0x0	Local timestamp clock adjustment back-off period. After an adjustment has been made to speed up or slow down the local timestamp clock, this number of common clock cycles need to pass before another ajustment is allowed.
95:80	bufSize	RW	0x0	Size of the de-jitter buffer allocated to the client in number of PLE packets.

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8.4.1.2 PLE_IWF_N2P

Name: PLE_IWF_N2P_MAP_CFG
Description: PLE IWF NSP to PSN Mapping Configuration
Address: 0x0052.0000
Entries: 512
Entry size: 0x4
Index: System bus and port number. Entries 0-80 map the port numbers on system bus 0 to PLE client numbers, entries 128-207 map the port numbers on system bus 1 to PLE client numbers, etc

Bits	Field	Type	Reset	Description
8:0	pleClient	RW	0x0	Use this PLE client number for the corresponding system bus and port number.

Name: PLE_IWF_N2P_CFG
Description: PLE IWF NSP to PSN Configuration
Address: 0x0052.0800

Bits	Field	Type	Reset	Description
3:0	lossBurstSize	RW	0x0	Number of packets in a burst of lost packets triggered by .lossBurstTrig.
5	lossBurstTrig	RW	0x0	Write '1' to trigger loss of a burst of .lossBurstSize packets for PLE client .lossClient in the store and forward buffer. This configuration will be cleared by HW then the loss action has completed.
14:6	lossClient	RW	0x0	The PLE client subject to packet loss.

8.4.1.3 PLE shared flexible RAM

Name: PLE_IWF_XTRABITS_CFG
Description: Lookup table returning number of extra address bits for each client
Address: 0x0054.0000
Entries: 83
Entry size: 0x4
Index: Client number

Bits	Field	Type	Reset	Description
1:0	extraBits	RW	0x0	The number of extra address bits for a client.

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Name: PLE_IWF_CHUNKIDX_CFG
Description: Lookup table returning index (pointer) to a chunk group in the memory chunk address lookup table
Address: 0x0054.0800
Entries: 83
Entry size: 0x4
Index: Client number

Bits	Field	Type	Reset	Description
9:0	chunkIdx	RW	0x0	The index (pointer) to a chunk group for a client.

Name: PLE_IWF_CHUNKADR_CFG
Description: Lookup table returning base addresses of memory chunks used by chunk groups
Address: 0x0054.1000
Entries: 1024
Entry size: 0x4
Index: Client number

Bits	Field	Type	Reset	Description
9:0	chunkAddr	RW	0x0	The base address of the memory chunk.

8.4.1.4 PLE_IWF_NSP

Name: PLE_IWF_NSP
Description:
Address: 0x0058.0000
Entries: 4
Entry size: 0x8000
Index: System bus number

Register Name	Entries	Description
PLE_IWF_NSP_CL_CFG	80	PLE IWF NSP Client Configuration
PLE_IWF_NSP_CL_STAT	80	PLE IWF NSP Client Alignment Status
PLE_IWF_NSP_CL_HOLD	80	PLE IWF NSP Client Alignment Hold Status
PLE_IWF_NSP_CL_CHDL	80	PLE IWF NSP Client Alignment Change Hold Status
PLE_IWF_NSP_CL_INTEN	80	PLE IWF NSP Client Alignment Interrupt Enable

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Name: PLE_IWF_NSP.PLE_IWF_NSP_CL_CFG
Description: PLE IWF NSP Client Configuration
Address: 0x0000.0000
Entries: 80
Entry size: 0x4
Index: System bus port number

Bits	Field	Type	Reset	Description
2:0	clientType	RW	0x0	Client type of the corresponding system bus port number. 0: none. Unknown. 1: ge1. 1G Ethernet (1GE LAN). 2: ge10. 10G Ethernet (10GE LAN). 3: ge100. 100GE Ethernet (100GE LAN). 4: odu. ODU.

Name: PLE_IWF_NSP.PLE_IWF_NSP_CL_STAT
Description: PLE IWF NSP Client Alignment Status
Address: 0x0000.0200
Entries: 80
Entry size: 0x4
Index: System bus port number

Bits	Field	Type	Reset	Description
0	notLocked1g	RO	-	No 1GE PCS symbol alignment on the extracted PLE payload.
1	notBlLock10g	RO	-	No 10G PCS block lock on the extracted PLE payload.
2	notBlLock100g	RO	-	No 100G PCS block lock on extracted PLE payload.
3	oduOof	RO	-	ODU frame alignment is in the Out-of-frame state.

Name: PLE_IWF_NSP.PLE_IWF_NSP_CL_HOLD
Description: PLE IWF NSP Client Alignment Hold Status
Address: 0x0000.0400
Entries: 80
Entry size: 0x4
Index: System bus port number

Bits	Field	Type	Reset	Description
0	notLocked1g	W1C	-	No 1GE PCS symbol alignment on the extracted PLE payload.
1	notBlLock10g	W1C	-	No 10G PCS block lock on the extracted PLE payload.
2	notBlLock100g	W1C	-	No 100G PCS block lock on extracted PLE payload.
3	oduOof	W1C	-	ODU frame alignment is in the Out-of-frame state.

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Name: **PLE_IWF_NSP.PLE_IWF_NSP_CL_CHDL**
Description: **PLE IWF NSP Client Alignment Change Hold Status**
Address: **0x0000.0600**
Entries: **80**
Entry size: **0x4**
Index: **System bus port number**

Bits	Field	Type	Reset	Description
0	notLocked1g	W1C	-	No 1GE PCS symbol alignment on the extracted PLE payload.
1	notBlLock10g	W1C	-	No 10G PCS block lock on the extracted PLE payload.
2	notBlLock100g	W1C	-	No 100G PCS block lock on extracted PLE payload.
3	oduOof	W1C	-	ODU frame alignment is in the Out-of-frame state.

Name: **PLE_IWF_NSP.PLE_IWF_NSP_CL_INTEN**
Description: **PLE IWF NSP Client Alignment Interrupt Enable**
Address: **0x0000.0800**
Entries: **80**
Entry size: **0x4**
Index: **System bus port number**

Bits	Field	Type	Reset	Description
0	notLocked1g	RW	0x0	No 1GE PCS symbol alignment on the extracted PLE payload.
1	notBlLock10g	RW	0x0	No 10G PCS block lock on the extracted PLE payload.
2	notBlLock100g	RW	0x0	No 100G PCS block lock on extracted PLE payload.
3	oduOof	RW	0x0	ODU frame alignment is in the Out-of-frame state.

8.4.1.5 PLE_IWF_STAT

Name: **PLE_IWF_CNT**
Description: **PLE IWF Counters**
Address: **0x005C.0000**
Entries: **2, 83**
Entry size: **0x14c0, 0x40**
Index: **PMTICK bank number, PLE client number**

Bits	Field	Type	Reset	Description
31:0	egPkt	RW	-	Egress PLE packet payloads.
63:32	igPkt	RW	-	Ingress PLE packet payloads.
95:64	igPktLost	RW	-	Ingress PLE packet payloads lost. Replacement payload has been generated in place of the lost payload.

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8.4.2 PLE_LIF

Name: PLE_LIF
Description:
Address: 0x0060.0000
Entries: 2
Entry size: 0x100000
Index: Line interface number

Register Name	Entries	Description
PLE_LIF_CFG	-	PLE Line Interface Configuration
PLE_IG_ENCAP0_CFG	120	PLE Line Interface Ingress Packet Encapsulation Configuration, Header Upto PLE Control Word
PLE_IG_ENCAP1_CFG	83	PLE Line Interface Packet Encapsulation Configuration
PLE_IG_CL_STAT	83	PLE Line Interface Ingress Status
PLE_IG_CL_HOLD	83	PLE Line Interface Ingress Hold Status
PLE_IG_CL_CHLD	83	PLE Line Interface Ingress Change Hold Status
PLE_IG_CL_INTEN	83	PLE Line Interface Ingress Interrupt Enable
PLE_IG_LAST_CFG	-	PLE Line Interface Ingress Last Header Capture Configuration
PLE_IG_LAST_STAT	-	PLE Line Interface Last Incoming Packet Encapsulation Status
PLE_IG_STAT	-	PLE Line Interface Ingress Status
PLE_IG_HOLD	-	PLE Line Interface Ingress Hold Status
PLE_IG_CHLD	-	PLE Line Interface Ingress Change Hold Status
PLE_IG_INTEN	-	PLE Line Interface Ingress Interrupt Enable
PLE_IG_FILTER_CNT	2, 120	PLE Line Interface Ingress filter statistics counters
PLE_IG_PLE_CNT	2, 83	PLE Line Interface Ingress PLE statistics counters
PLE_IG_GLOBAL_CNT	2	PLE Line Interface Ingress global statistics counters
PLE_EG_STAT	-	PLE Line Interface Egress Status
PLE_EG_HOLD	-	PLE Line Interface Egress Status Hold
PLE_EG_CHLD	-	PLE Line Interface Egress Status Change Hold
PLE_EG_INTEN	-	PLE Line Interface Egress Status Interrupt Enable
PLE_EG_ENCAP0_CFG	512	PLE Line Interface Egress Packet Encapsulation Configuration, Header Upto PLE Control Word
PLE_EG_ENCAP1_CFG	512	PLE Line Interface Egress Packet Encapsulation Configuration, PLE Control Word and RTP Header
PLE_EG_PLE_CNT	2, 83	PLE Line Interface Egress PLE Statistics Counters

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Name: PLE_LIF.PLE_LIF_CFG
Description: PLE Line Interface Configuration
Address: 0x0000.0000

Bits	Field	Type	Reset	Description
0	lbPortIg2Eg	RW	0x0	Set to enable port side loopback ingress-to-egress. Always reads 0 when port loops disabled.
1	lbPortEg2Ig	RW	0x0	Set to enable port side loopback egress-to-ingress. Always reads 0 when port loops disabled.

8.4.2.1 PLE_LIF_IG

Name: PLE_LIF.PLE_IG_ENCAP0_CFG
Description: PLE Line Interface Ingress Packet Encapsulation Configuration, Header Upto PLE Control Word
Address: 0x0004.0000
Entries: 120
Entry size: 0x40
Index: Match entry

Bits	Field	Type	Reset	Description
255:0	hdr	RW	0x0	DMAC, SMAC, ...
511:256	hdrMask	RW	0x0	Mask for DMAC, SMAC, ... For bit positions with '1' in .hdr_mask, the corresponding bit in the .hdr field must match that position in the packet. For bit positions with '0' in .hdr_mask, the value of the corresponding bit position in the packet is ignored. Bit positions beyond the configured .hdrByteSize must be set to '0'.

Name: PLE_LIF.PLE_IG_ENCAP1_CFG
Description: PLE Line Interface Packet Encapsulation Configuration
Address: 0x0004.9000
Entries: 83
Entry size: 0x4
Index: PLE client number

Bits	Field	Type	Reset	Description
4:0	hdrByteSize	RW	0x0	Number of header bytes upto the PLE Control Word. An even number of bytes from 14 to 30 is supported.
9:5	mplsTtlPos	RW	0x0	Position of the MPLS TTL byte. Values up to 29 are supported.
10	mplsTtlChkEn	RW	0x0	Enable MPLS TTL check.

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Name: **PLE_LIF.PLE_IG_CL_STAT**
Description: **PLE Line Interface Ingress Status**
Address: **0x0004.A000**
Entries: **83**
Entry size: **0x4**
Index: **PLE client number**

Bits	Field	Type	Reset	Description
0	ttlErr	RO	-	MPLS TTL check failed.

Name: **PLE_LIF.PLE_IG_CL_HOLD**
Description: **PLE Line Interface Ingress Hold Status**
Address: **0x0004.A800**
Entries: **83**
Entry size: **0x4**
Index: **PLE client number**

Bits	Field	Type	Reset	Description
0	ttlErr	W1C	-	MPLS TTL check failed.

Name: **PLE_LIF.PLE_IG_CL_CHLD**
Description: **PLE Line Interface Ingress Change Hold Status**
Address: **0x0004.B000**
Entries: **83**
Entry size: **0x4**
Index: **PLE client number**

Bits	Field	Type	Reset	Description
0	ttlErr	W1C	-	MPLS TTL check failed.

Name: **PLE_LIF.PLE_IG_CL_INTEN**
Description: **PLE Line Interface Ingress Interrupt Enable**
Address: **0x0004.B800**
Entries: **83**
Entry size: **0x4**
Index: **PLE client number**

Bits	Field	Type	Reset	Description
0	ttlErr	RW	0x0	MPLS TTL check failed.

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Name: PLE_LIF.PLE_IG_LAST_CFG
Description: PLE Line Interface Ingress Last Header Capture Configuration
Address: 0x0004.C000

Bits	Field	Type	Reset	Description
0	lastHold	RW	0x0	Hold the last packet header capture status. Write '1' to hold the current values in PLE_IG_LAST_STAT, allowing SW to read an atomic capture. Write '0' to allow HW updates of PLE_IG_LAST_STAT.
1	lastMatch	RW	0x0	Hold the last packet header of a packet with a match at lookup table entry .lastMatchEntry. If '0', the last header with no match will be captured.
8:2	lastMatchEntry	RW	0x0	Lookup table entry when .lastMatch is '1'.

Name: PLE_LIF.PLE_IG_LAST_STAT
Description: PLE Line Interface Last Incoming Packet Encapsulation Status
Address: 0x0004.C010

Bits	Field	Type	Reset	Description
255:0	hdr	RO	0x0	DMAC, SMAC, ...

Name: PLE_LIF.PLE_IG_STAT
Description: PLE Line Interface Ingress Status
Address: 0x0004.D000

Bits	Field	Type	Reset	Description
0	igPktNoMatch	RO	-	Ingress packets not matching any of the filter entries. These packets are discarded.
1	igPktMsgDrop	RO	-	Ingress message packets dropped due to buffer full condition.

Name: PLE_LIF.PLE_IG_HOLD
Description: PLE Line Interface Ingress Hold Status
Address: 0x0004.D010

Bits	Field	Type	Reset	Description
0	igPktNoMatch	W1C	-	Ingress packets not matching any of the filter entries. These packets are discarded.
1	igPktMsgDrop	W1C	-	Ingress message packets dropped due to buffer full condition.

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Name: PLE_LIF.PLE_IG_CHLD
Description: PLE Line Interface Ingress Change Hold Status
Address: 0x0004.D020

Bits	Field	Type	Reset	Description
0	igPktNoMatch	W1C	-	Ingress packets not matching any of the filter entries. These packets are discarded.
1	igPktMsgDrop	W1C	-	Ingress message packets dropped due to buffer full condition.

Name: PLE_LIF.PLE_IG_INTEN
Description: PLE Line Interface Ingress Interrupt Enable
Address: 0x0004.D030

Bits	Field	Type	Reset	Description
0	igPktNoMatch	RW	0x0	Ingress packets not matching any of the filter entries. These packets are discarded.
1	igPktMsgDrop	RW	0x0	Ingress message packets dropped due to buffer full condition.

8.4.2.1.1 PLE_LIF_IG_STATS_FILTER

Name: PLE_LIF.PLE_IG_FILTER_CNT
Description: PLE Line Interface Ingress filter statistics counters
Address: 0x0005.0000
Entries: 2, 120
Entry size: 0x1e0, 0x4
Index: PMTICK bank number, ingress packet filter entry

Bits	Field	Type	Reset	Description
31:0	igPktMatch	RW	-	Ingress packets matching the filter entry. Entries zero to number of PLE clients minus one are used to match PLE clients.

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8.4.2.1.2 PLE_LIF_IG_STATS_PLE

Name: PLE_LIF.PLE_IG_PLE_CNT
Description: PLE Line Interface Ingress PLE statistics counters
Address: 0x0006.0000
Entries: 2, 83
Entry size: 0x14c, 0x4
Index: PMTICK bank number, PLE client number

Bits	Field	Type	Reset	Description
31:0	igPktMsg	RW	-	Ingress packets that match a PLE client filter entry with a non-zero first nibble after the Ethernet/MPLS encapsulation, which pass the MPLS TTL validation. These packets are sent to the SGMII Message Interface.

8.4.2.1.3 PLE_LIF_IG_STATS_GLOBAL

Name: PLE_LIF.PLE_IG_GLOBAL_CNT
Description: PLE Line Interface Ingress global statistics counters
Address: 0x0007.0000
Entries: 2
Entry size: 0x8
Index: PMTICK bank number

Bits	Field	Type	Reset	Description
31:0	igPktNoMatch	RW	-	Ingress packets not matching any of the filter entries. These packets are discarded.
63:32	igPktMsgDrop	RW	-	Ingress message packets dropped due to buffer full condition.

8.4.2.2 PLE_LIF_EG

Name: PLE_LIF.PLE_EG_STAT
Description: PLE Line Interface Egress Status
Address: 0x0008.0000

Bits	Field	Type	Reset	Description
0	egOflw	RO	-	Egress buffer overflow.

Name: PLE_LIF.PLE_EG_HOLD
Description: PLE Line Interface Egress Status Hold
Address: 0x0008.0010

Bits	Field	Type	Reset	Description
0	egOflw	W1C	-	Egress buffer overflow.

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Name: PLE_LIF.PLE_EG_CHLD
Description: PLE Line Interface Egress Status Change Hold
Address: 0x0008.0020

Bits	Field	Type	Reset	Description
0	egOflw	W1C	-	Egress buffer overflow.

Name: PLE_LIF.PLE_EG_INTEN
Description: PLE Line Interface Egress Status Interrupt Enable
Address: 0x0008.0030

Bits	Field	Type	Reset	Description
0	egOflw	RW	0x0	Egress buffer overflow.

8.4.2.2.1 PLE_LIF_EG_ENCAP

Name: PLE_LIF.PLE_EG_ENCAP0_CFG
Description: PLE Line Interface Egress Packet Encapsulation Configuration, Header Upto PLE Control Word
Address: 0x0008.8000
Entries: 512
Entry size: 0x20
Index: PLE client number

Bits	Field	Type	Reset	Description
239:0	hdr	RW	0x0	DMAC, SMAC, ...
244:240	hdrByteSize	RW	0x0	Number of header bytes upto the PLE Control Word. An even number of bytes from 14 to 30 is supported.

Name: PLE_LIF.PLE_EG_ENCAP1_CFG
Description: PLE Line Interface Egress Packet Encapsulation Configuration, PLE Control Word and RTP Header
Address: 0x0008.C000
Entries: 512
Entry size: 0x8
Index: PLE client number

Bits	Field	Type	Reset	Description
31:0	rtpSsrc	RW	0x0	RTP header, synchronization source identifier.
38:32	rtpPt	RW	0x0	RTP header, payload type.
39	rtpM	RW	0x0	RTP header, marker.
43:40	rtpCc	RW	0x0	RTP header, CSRC count.

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Bits	Field	Type	Reset	Description
44	rtpX	RW	0x0	RTP header, header extension.
45	rtpP	RW	0x0	RTP header, padding.
47:46	rtpV	RW	0x2	RTP header, version.
53:48	pleLen	RW	0x0	PLE control word, LEN.
55:54	pleFrg	RW	0x0	PLE control word, FRG.
57:56	pleRsv	RW	0x0	PLE control word, RSV.
63:60	pleFirstNibble	RW	0x0	PLE control word, first nibble.

8.4.2.2.2 PLE_LIF_EG_STATS

Name: PLE_LIF.PLE_EG_PLE_CNT
Description: PLE Line Interface Egress PLE Statistics Counters
Address: 0x0009.0000
Entries: 2, 83
Entry size: 0x14c, 0x4
Index: PMTICK bank number, PLE client number

Bits	Field	Type	Reset	Description
31:0	egPkt	RW	-	Egress PLE packets.

8.4.3 PLE_LIF_1P1

Name: PLE_1P1_IG_CFG
Description: PLE Line Interface 1+1 Protection Ingress Configuration
Address: 0x00C0.0000

Bits	Field	Type	Reset	Description
511:0	lineSel	RW	0x0	Line interface selected as active per PLE client.

Name: PLE_1P1_EG_CFG
Description: PLE Line Interface 1+1 Protection Egress Configuration
Address: 0x00C0.0400
Entries: 2
Entry size: 0x40
Index: Line interface number

Bits	Field	Type	Reset	Description
511:0	en	RW	0x0	Enable client transmission per PLE client.

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Name: PLE_1P1_CL_STAT
Description: PLE Line Interface 1+1 Protection Client Hold Status
Address: 0x00C0.0500
Entries: 2
Entry size: 0x80
Index: Line interface number

Bits	Field	Type	Reset	Description
511:0	pktOversized	RO	-	Incoming payload was oversized and was discarded.
1023:512	pktUndersized	RO	-	Incoming payload was undersized and was discarded.

Name: PLE_1P1_CL_HOLD
Description: PLE Line Interface 1+1 Protection Client Hold Status
Address: 0x00C0.0600
Entries: 2
Entry size: 0x80
Index: Line interface number

Bits	Field	Type	Reset	Description
511:0	pktOversized	W1C	-	Incoming payload was oversized and was discarded.
1023:512	pktUndersized	W1C	-	Incoming payload was undersized and was discarded.

Name: PLE_1P1_CL_CHLD
Description: PLE Line Interface 1+1 Protection Client Change Hold Status
Address: 0x00C0.0700
Entries: 2
Entry size: 0x80
Index: Line interface number

Bits	Field	Type	Reset	Description
511:0	pktOversized	W1C	-	Incoming payload was oversized and was discarded.
1023:512	pktUndersized	W1C	-	Incoming payload was undersized and was discarded.

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Name: PLE_1P1_CL_INTEN
Description: PLE Line Interface 1+1 Protection Client Interrupt Enable
Address: 0x00C0.0800
Entries: 2
Entry size: 0x80
Index: Line interface number

Bits	Field	Type	Reset	Description
511:0	pktOversized	RW	0x0	Incoming payload was oversized and was discarded.
1023:512	pktUndersized	RW	0x0	Incoming payload was undersized and was discarded.

Name: PLE_1P1_DBG_HOLD
Description: Miscellaneous debug status hold. For internal use only
Address: 0x00C0.0900
Entries: 2
Entry size: 0x4
Index: Line interface number

Bits	Field	Type	Reset	Description
0	pktNoSop	W1C	-	Incoming payload without SOP.
1	pktNoEop	W1C	-	Incoming payload with SOP without preceeding EOP.

8.4.4 PLE message interface

Name: PLE_SGMII_IVLAN_CFG
Description: IVLAN Configuration
Address: 0x00D0.0000
Entries: 2
Entry size: 0x8
Index: Line interface

Bits	Field	Type	Reset	Description
11:0	ivlanEg	RW	0x0	IVLAN for egress packets from the associated line interface.
15:12	ivlanTagEg	RW	0x0	IVLAN Priority Code and Drop Indicator for egress packets from the associated line interface.
27:16	ivlanIg	RW	0x0	IVLAN for ingress packets to the associated line interface. The packets with a IVLAN doesn't match with this value will be dropped.

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Name: PLE_SGMII_CNT
Description: Line Interface Egress and Ingress Counters
Address: 0x00D0.0100
Entries: 2
Entry size: 0x10
Index: PMTICK bank number

Bits	Field	Type	Reset	Description
31:0	egPkt0	RW	-	Egress PLE packets drop count from line interface index 0 due to error indication and overflow.
63:32	egPkt1	RW	-	Egress PLE packets drop count from line interface index 1 due to error indication and overflow.
95:64	igPkt	RW	-	Ingress PLE packets drop count due to IVLAN mismatch.

Name: PLE_SGMII_RATE
Description: Line Interface Output Rate Control
Address: 0x00D0.0200

Bits	Field	Type	Reset	Description
23:0	egPktRate	RW	0x0	Egress PLE Packets Output Rate can be configured to X kbps here.

Name: PLE_SGMII_RD
Description: Integrity Check for Packets to be read into Message Interface
Address: 0x00D0.0300

Bits	Field	Type	Reset	Description
0	egInte	RO	0x0	Egress PLE Packets Output Integrity Check for data comes out of the FIFO.

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9 Rate and scheduling control

9.1 Features

- Sources data rate and scheduling information for use in datapaths throughout the device
- Sources data rate information for signal timing

9.2 Overview

9.2.1 Endpoints

The Rate / Schedule module generates rate and scheduling information for use at various locations in the device (termed endpoints):

- Sourcing test signals, e.g. PRBS
- Sourcing maintenance signals, e.g. ODUk-AIS
- OTU/ODU frame generation, e.g. within a multiplexer or mapper function

It has a number of Rate Schedule Interfaces (RSI) to connect to the endpoints. One RSI may drive all the rates needed in one region of the device, i.e. more than one endpoint may be attached to a given RSI. **Figure 9-1** below shows a generic example of how a rate generator can be connected to such endpoints.

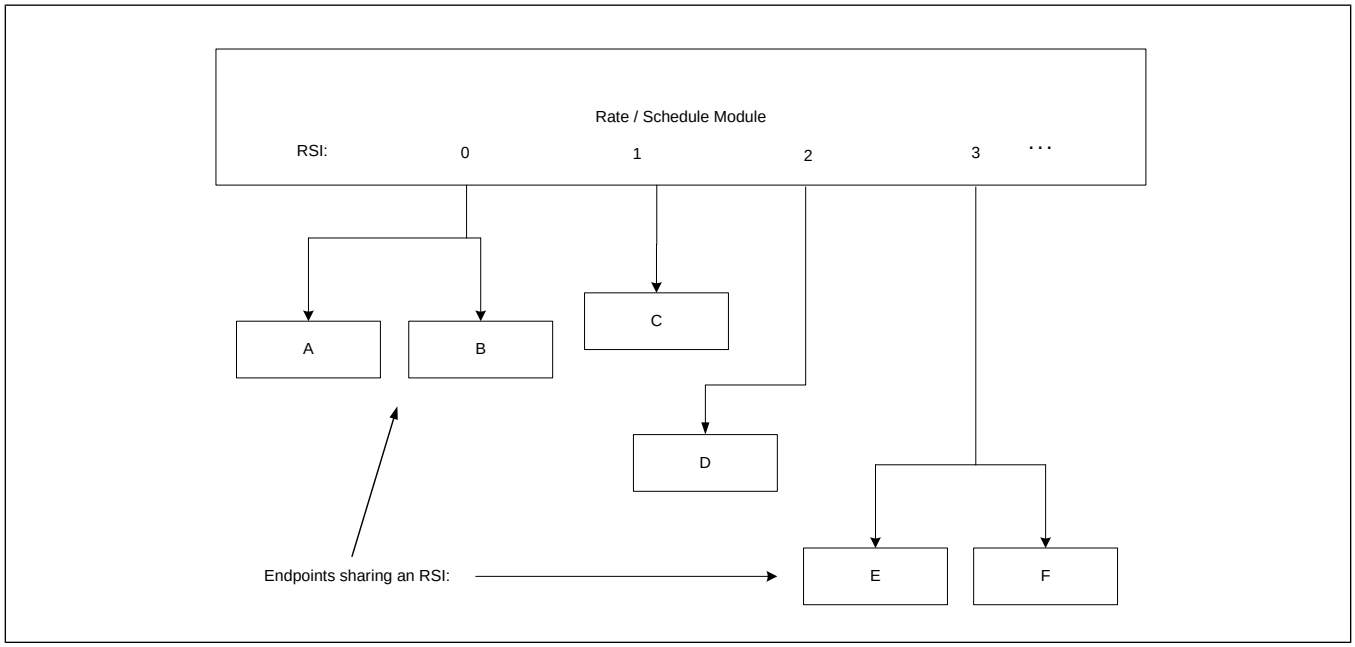


Figure 9-1 Rate generator endpoints (generic example)

The number of RSI and the connection of these rate and schedule signals to the endpoints are device specific

Each endpoint in the device can have different combinations of the following attributes:

- `maxDataWidth`: This is the width of the widest datapath within the endpoint that will be using the rate/scheduling information.
- `numPorts`: This is the number of individual ports sharing a datapath at the endpoint.
- `maxScheduleLength`: This is the longest scheduling period to be used within the endpoints.
- `paddingFactor`: This is the factor by which datapath data is padded relative to the nominal data rate.

When provisioning a specific port the rate and scheduling configuration depends upon:

- `rsIfIx`: The index number for the RSI.
- `portNumber`: Identifying the port within the endpoint.
- `dataWidth` for the relevant datapath.
- `scheduleLength` for the relevant datapath.
- `numTs` is the number of time slots in use for this port.
- `paddingFormat` as defined in **Padded signals** on page 479.
- `clientBitrate` as defined in **Deterministic Timing Architecture (DTA)** on page 385.
- `coreClk`: the device core clock frequency.

9.2.2 Internal architecture

Figure 9-2 shows a generic example of the internal modules of the rate generator. The generator consists of:

1. Rate generation modules which compute the individual rates. There are two types, Common (which can supply rates to a number of RSI) and Dedicated (which supplies rates to a single RSI). A given device may have any number, or none, of each of these two types.

2. Scheduling modules, one per RSI.

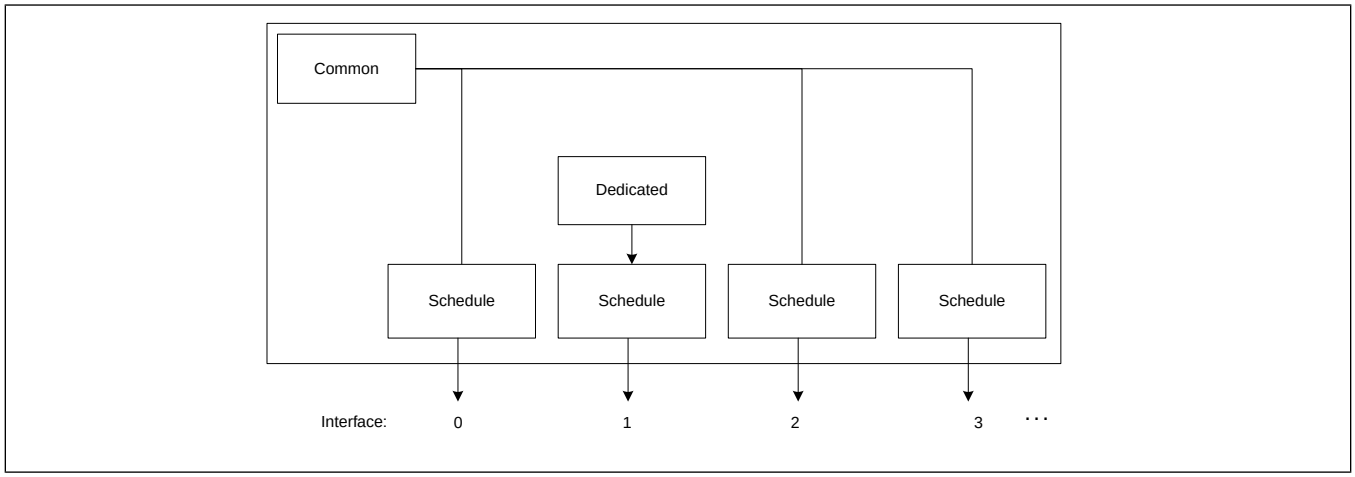


Figure 9-2 Rate / scheduling module (generic example)

9.2.3 Remapper blocks and scheduling

Where a remapper block (not described here - may be a component of some device functional blocks) is connected to a multi-port datapath, the scheduling applicable to that datapath must be reflected in the configuration of the remapper. See the example below:

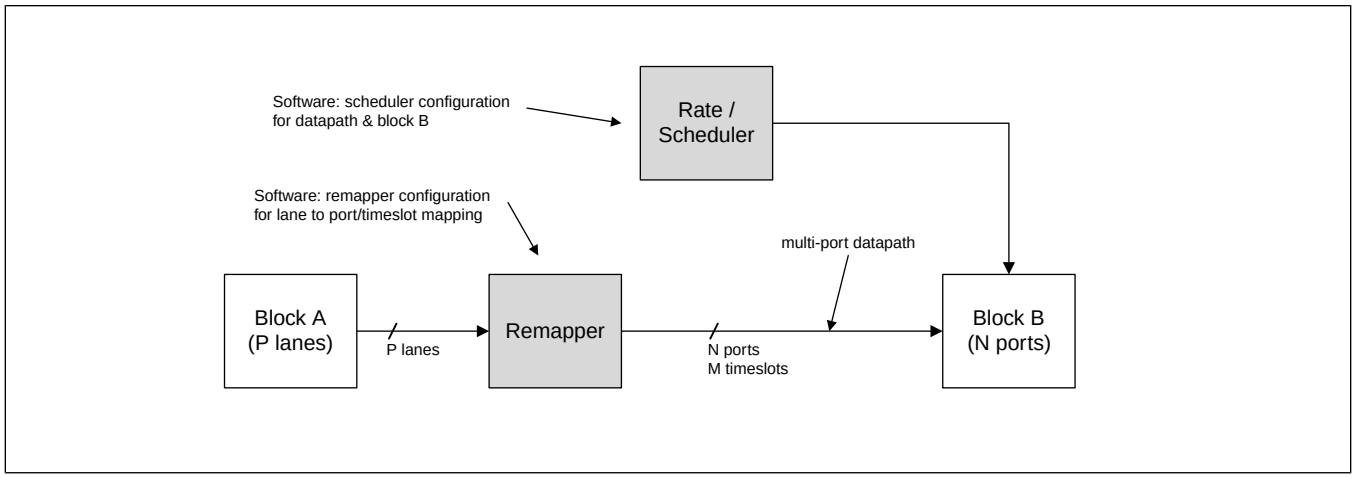


Figure 9-3 Remapper and scheduled datapath

Here there is a multi-port datapath which supports N ports with a scheduling over M timeslots, connected to the remapper. The Rate / Scheduler that controls any blocks connected to that datapath (in this case block B) is configured to distribute bandwidth between the various ports by allocating a number of timeslots to each port in use.

The remapper maps between a number (P) of physical lanes on one side and the N ports of the scheduled datapath on the other. It is set up using a standard configuration function in the API. For each port number in use, the input to this function consists of:

- The desired lane(s) to be mapped to the port
- The datapath timeslot numbers allocated to the port

For traffic to flow between the Remapper and Block B on a given port, the timeslots allocated to that port number must be identical in both the Rate / Scheduler and the Remapper.

9.3 Operation

9.3.1 Global device parameters

`maxScheduleLengthPerIf` is the largest `maxScheduleLength` on any RSI rounded up to nearest power of 2.

`maxPortsPerIf` is the largest `number of ports` per interface rounded up to nearest power of 2.

`.commonPeriod` is read from `RS_INFO`.

`coreClk` is the internal device core clock frequency.

9.3.2 Scheduling Control

The internal scheduling of client data must be configured using `RS_SCHED_CFG`. For each client and its container ODU, sufficient bandwidth must be allocated. The granularity of this control is defined by the length of the scheduling sequence which must be set for each RSI in `RS_SCHED_IF_CFG.scheduleLength`. This gives a resolution in units of `1/.scheduleLength` of the total bandwidth at the endpoint. Note: Internally there is an overhead to this bit rate, so calculations need not add additional margin.

When adding new clients/ODUs, entries allocated to existing ports should not be changed. When a client/ODU is de-provisioned, all entries for that client must be disabled. There is no requirement to distribute the entries in any way -

one algorithm that can be used is to map table entries to the utilized timeslots in the ODUs once mapped. By default, the simplest algorithm possible whereby free entries are used in a consecutive sequence, starting from the lowest available, should be used.

For each rate needed, a number of entries in RS_SCHED_CFG must have `.en` set to '1' and `.port` programmed with the port number within the given endpoint. This number of entries is referred to as `numTs` and equates to the number of timeslots occupied by this client.

The index into the RS_SCHED_CFG is based on the following formula:

```
schedIx = (rsIfIx * maxScheduleLengthPerIf) + scheduleTimeSlot
```

where `scheduleTimeSlot` is the range of entries to be filled out, i.e. from 0 to `numTs` - 1, giving `numTs` values of `schedIx`.

The above sets an approximate rate. The precise rate is then configured in RS_RATE_COM and RS_RATE_DED, see **Rate control** on page 478.

The `.dsPeriod` parameter must also be set and is computed as follows for each RSI:

```
.dsPeriod = .commonPeriod * numTs / .scheduleLength
```

Note the `.dsPeriod` value must not be rounded.

9.3.3 Rate control

The common and dedicated rate generators are almost identical, the main difference being configuration of the `dataWidth` parameter, which must be configured per RSI in RS_RATE_CFG`.dataWidth` for Dedicated generators and in RS_RATE_COM`.cdw` for the Common generator.

One entry is required per client rate per data width in the Common rate generator. I.e. if two clients use the same rate but have different data widths, two entries will be needed.

The index into the RS_RATE_DED table is based on the following formula:

```
rateDedIx = (rsIfIx * maxPortsPerIf) + portNumber
```

The content of RS_RATE_CFG and RS_RATE_COM are computed using the following formulas:

```
.dtaRateExp = 24 - ceiling(log2(dtaBitcount))

.dtaRateInt = floor(dtaBitcount * (1 << .dtaRateExp) )
```

Where `dtaBitcount` is as specified for the device in **Deterministic Timing Architecture (DTA)** on page 385. The `.dtaRateExp/.dtaRateInt` thus define the exact rate generated.

Additionally, `.dedCoreRate` and `.comCoreRate` must be set as follows:

```
.dedCoreRate = floor( (1 << 14) * clientBitrate * .scheduleLength / numTs / dataWidth
                      / (coreClk*1023/1024) )

.comCoreRate = floor( (1 << (14-ceiling(log2(commonPeriod+1))) )
                      * commonPeriod * clientBitrate / dataWidth / coreClk )
```

The `clientBitrate` value is the actual bitrate of the signal for which a rate is being provided.

9.3.3.1 Padded signals

The device datapath for which a rate is being generated may have a padded format, for example an ODUk may be transported in an "OTU" frame which is 4080 columns long. This affects the values to be configured.

In general, when computing the content of RS_RATE_CFG and RS_RATE_COM:

1. Set `.paddingControl` to the selected padding factor.
2. `clientBitrate` (and `dtaBitcount`) are set to their nominal values multiplied by the padding factor (which can be 1 or greater).

9.3.4 Sparsely populated tables and fields

The populated register entries are:

- RS_RATE_CFG: Only entries corresponding to Dedicated generators are populated.
- RS_RATE_COM: All entries are populated if the Common generator is present.
- RS_RATE_DED: Valid entry ranges are given if any Dedicated generators are present.
- RS_SCHED_IF_CFG: Entries are valid for RSI where `maxScheduleLength` > 1.
- RS_SCHED_CFG: Valid entries depend on the number of possible entries per RSI. See the indexing formula for `schedIx` in **Scheduling Control** on page 477 for details.

Unused table entries cannot be written.

Within the following bit fields the following bits are populated depending on the attributes of the given RSI:

- RS_RATE_CFG.`dataWidth`: The width of this field is limited to the minimum number of bits needed to hold the maximum `dataWidth`.
- RS_SCHED_IF_CFG.`scheduleLength`: The width of this field is limited to the minimum number of bits needed to hold the `maxScheduleLength.`
- RS_SCHED_CFG.`commonIndex`: The width of this field is limited to the minimum number of bits needed to index RS_RATE_COM.
- RS_SCHED_CFG.`port`: The width of this field is limited to the minimum number of bits needed to index the number of ports. For RSI with only one port, this field is non-existent.

9.3.5 Provisioning examples

This section contains some examples of how to compute the content of the rate and scheduling setup registers. The formulas for the computations are given in **Operation** on page 477 and **Scheduling Control** on page 477.

The device in this example has 4 interfaces indexed by `rsIfIx`, see **Table 9-1**.

Table 9-1 Rate generation indexing (Generic example)

rsIfIx	Endpoint Block	max DataWidth	max scheduleLength	Number of ports	PaddingFactor	Type
0	OTU4	320	1	1	1	Common
1	LO ODU	320	80	10	3824/4080	Dedicated
2	STM64/OC192	32	1	1	1	Common
3	ODU2 MAP	32	1	1	3824/4080	Common

Table 9-1 shows the hardware setup, the columns `type` and `rsIfIx` are fixed. The columns `maxDataWidth` and `maxScheduleLength` also depend upon hardware setup, but the active values may change as part of a partial

reconfiguration (PR) process. The maximum active values for `.dataWidth` and `.scheduleLength` are listed in **Table 9-1**.

The number of common rates supported is assumed to be 16 in this example.

Derived from **Table 9-1**:

```
maxPortPerIf = 16


maxScheduleLengthPerIf = 128
```

In the example given above the interfaces 0, 2 and 3 are sourced by the common generator and port 1 has it's own dedicated generator. The destination point for interface 0 is a OTU4 line interface termination point and interface 1 is the associated LO embedded in the OTU4. In this example the OTU4 multiplexer function has 80 tributary slots, but only supports 10 ports. The destination for the interface 3 is a OTU2 mapping function associated with the SONET/SDH termination point at interface 2. A graphical representation of this example is given in the figures in **Overview** on page 475.

The chip configurations is:

```
commonPeriod = 80
maxPortsPerIf =  128
maxScheduleLengthPerIf = 16
coreClk = 357 MHz.
```

In the first two examples below, there is only one port and thus no scheduling to be setup, in the third example, scheduling setup is included.

9.3.5.1 STM64/OC192

The entry in the common generator can be selected arbitrarily:

```
rateComIx = 4
```

The following setup is from **Table 9-1**.

```
numTs = 1
dataWidth = 32
scheduleLength = 1
```

STM64/OC192 bitrate:

```
clientBitrate = 9953280000 bits/s
```

Register setup:

```
RS_RATE_COM[4].cdw = 32
RS_RATE_COM[4].comCoreRate = 8921
RS_RATE_COM[4].dtaRateExp = 8
RS_RATE_COM[4].dtaRateInt = 15925248
RS_RATE_COM[4].paddingControl = 0

rsIfIx = 2, from Table 9-1.
schedIx = 256
RS_SCHED_CFG[256].port = 0
RS_SCHED_CFG[256].commonIndex = 4
RS_SCHED_CFG[256].first = 1
RS_SCHED_CFG[256].enable = 1
```

9.3.5.2 ODU2 mapper

The entry in the common generator can be selected arbitrarily:

```
rateComIx = 5
```

The following setup is from **Table 9-1**.

```
numTs = 1
dataWidth = 32
scheduleLength = 1
```

ODU2 bitrate:

```
ODU2 bitrate = 239/237 * 953280000 bits/s
```

However in this example the internal format / frame that carries the ODU is assumed to be padded to 4080 columns. It is this padded bit rate that must be used in the computations:

```
clientBitrate = 4080/3824 * ODU2 bitate
clientBitrate = 255/237 * 953280000 bits/s
```

Register setup:

```
RS_RATE_COM[5].cdw = 32
RS_RATE_COM[5].ComCoreRate = 9599
RS_RATE_COM[5].dtaRateExp = 7
RS_RATE_COM[5].dtaRateInt = 8567380
RS_RATE_COM[5].paddingControl = 1

rsIfIx = 3, from Table 9-1.
schedIx = 384
RS_SCHED_CFG[384].port = 0
RS_SCHED_CFG[384].commonIndex = 5
RS_SCHED_CFG[384].first = 1
RS_SCHED_CFG[384].enable = 1
```

9.3.5.3 LO ODU2

In this example a dedicated generator is used, the following setup is from **Table 9-1**.

```
rsIfIx = 1
RS_RATE_CFG[1].dataWidth = 320
RS_SCHED_IF_CFG[1].scheduleLength = 80
```

When multiplexing an ODU2 to an OPU4, 8 tributary slots are used:

```
numTs = 8
```

ODU2 bitrate:

```
ODU2 bitrate = 239/237 * 953280000 bits/s
```

However in this example the internal format / frame that carries the ODU is assumed to be padded to 4080 columns. It is this padded bit rate that must be used in the computations:

```
clientBitrate = 4080/3824 * ODU2 bitate
clientBitrate = 255/237 * 953280000 bits/s
```

The local port number is set to 6:

```
portNumber = 6
rateDedIx = 22
```

Register setup:

```
RS_RATE_DED[22].dedCoreRate = 14409
RS_RATE_DED[22].dtaRateExp = 8
RS_RATE_DED[22].dtaRateInt = 16059638
RS_RATE_DED[22].paddingControl = 1

RS_SCHED_CFG[128].first = 1
For all schedIx in [128,138,148,158,168,178,188,198]
  RS_SCHED_CFG[schedIx].port = 6
  RS_SCHED_CFG[schedIx].enable = 1
```

If the generator type is assumed to be common (i.e. not dedicated as above) then the following setup can be used:

The entry in the common generator can be selected arbitrarily:

```
rateComIx = 7
```

Register setup:

```
RS_RATE_COM[7].cdw = 320
RS_RATE_COM[7].comCoreRate = 899
RS_RATE_COM[7].dtaRateExp = 8
RS_RATE_COM[7].dtaRateInt = 16059638
RS_RATE_COM[7].paddingControl = 1

RS_SCHED_CFG[128].first = 1
For all schedIx in [128,138,148,158,168,178,188,198]
  RS_SCHED_CFG[schedIx].port = 6
  RS_SCHED_CFG[schedIx].dsPeriod = 8
  RS_SCHED_CFG[schedIx].commonIndex = 7
  RS_SCHED_CFG[schedIx].enable = 1
```

9.4 Registers

Name: RS_RATE_CFG
Description: Dedicated datapath data width configuration
Address: 0x0004.0000
Entries: 20
Entry size: 0x4
Index: rsIfIx

Bits	Field	Type	Reset	Description
11:0	dataWidth	RW	-	Dedicated datapath data width.

Name: RS_SCHED_IF_CFG
Description: Schedule length configuration
Address: 0x0004.4000
Entries: 20
Entry size: 0x4
Index: rsIfIx

Bits	Field	Type	Reset	Description
6:0	scheduleLength	RW	-	Length of schedule period. For dedicated generator: Minimum value is 8.

Name: RS_RATE_COM
Description: Common signal rate configuration
Address: 0x0004.8000
Entries: 18
Entry size: 0x8
Index: rateComIx

Bits	Field	Type	Reset	Description
23:0	dtaRateInt	RW	-	Integer part, DTA rate.
25:24	paddingControl	RW	-	Exponent for DTA rate. 0: padNone. No padding. Padding factor: 1. 1: pad3824. Padding factor: 4080/3824. 2: padRes1. Reserved. 3: padRes2. Reserved.
31:28	dtaRateExp	RW	-	Exponent for DTA rate.
45:32	comCoreRate	RW	-	Common core data rate.
63:48	cdw	RW	-	Data width.

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Name: RS_INFO
Description: Rate schedule configuration
Address: 0x0004.C000

Bits	Field	Type	Reset	Description
9:0	commonPeriod	RO	-	commonPeriod.

Name: RS_HOLD
Description: Rate schedule configuration
Address: 0x0004.E000

Bits	Field	Type	Reset	Description
0	dtaOflw	W1C	-	DTA serialize FIFO overflow.
19:1	schedOflw	W1C	-	Common rate scheduler overflow.

Name: RS_SCHED_CFG
Description: Scheduler configuration
Address: 0x0005.0000
Entries: 2560
Entry size: 0x4
Index: schedIx

Bits	Field	Type	Reset	Description
6:0	port	RW	-	Port number.
19:8	dsPeriod	RW	-	dsPeriod = commonPeriod*numTs/scheduleLength. (Value must not be rounded). Not present, when generator type is dedicated. Not present, when maxScheduleLength=1.
24:20	commonIndex	RW	-	Common rate generator index, when generator type is common. Not present, when generator type is dedicated.
30	first	RW	-	Set to '1' to indicate the first timeslot per port.
31	en	RW	-	Set to '1' to allocate one timeslot of bandwidth unit to this port.

Name: RS_RATE_DED
Description: Dedicated signal rate configuration
Address: 0x0006.0000
Entries: 1600
Entry size: 0x8
Index: rateDedIx

Bits	Field	Type	Reset	Description
23:0	dtaRateInt	RW	-	Integer part DTA rate.

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Bits	Field	Type	Reset	Description
25:24	paddingControl	RW	-	Exponent for DTA rate. 0: padNone. No padding. Padding factor: 1. 1: pad3824. Padding factor: 3824/4080. 2: padRes1. Reserved. 3: padRes2. Reserved.
31:28	dtaRateExp	RW	-	Exponent for DTA rate.
45:32	dedCoreRate	RW	-	Dedicated core data rate.

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10 External Pin Interfaces

10.1 Introduction

This chapter describes all the external interfaces on the device. The first section lists all signals sorted into functional groups, including a short description of each signal.

The subsequent sections of this chapter then go into further details with each external interface, and also includes all AC timing specifications.

Please refer to the document "Intel® Agilex™ Device Family Pin Connection Guidelines" from www.intel.com for details not found in this chapter.

All signals and associated AC timing are referenced to the rising edge of the associated clock, unless it is specifically mentioned otherwise. All setup, hold, output delay and other timing numbers provided in this manual are referenced at 50% of the full signal swing, e.g. for 1.2 V LVCMOS the reference point is 0.6 V.

10.2 Functional pinout

The following nomenclature has been used in the I/O direction and type columns in the functional pinout tables below:

Table 10-1 Abbreviations used in pin tables

Abbreviation	Explanation
I, O, I/O	Describes the IO direction as input, output or bidirectional.
LC12	LVCMOS 1.2 V.
LC18	LVCMOS 1.8 V.
LCxx PU	LVCMOS with weak on-chip pull-up. Typically 25 kΩ.
LCxx BH	LVCMOS with bus hold circuitry (only used on bidirectional busses).
LVDS	LVDS.
LVDS IT	LVDS input with internal 100 Ω termination.
LVPECL25 IT	2.5V LVPECL with internal termination.
CML	Current Mode Logic.
SerDes	High speed serial interface.

Throughout the tables below, positive logic is generally applied. Signals which are inverted (asserted when low) either end with a "_N" or a "(n)" postfix.

LVDS and SerDes differential signals are only listed using the logical signal name, not both physical signals making up the differential pair.

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Table 10-2 Global pins

Pin Name	IO dir.	IO type	Drive [mA]	Description
G_CLK0,1,2,3	I	LVDS IT		These clocks directly drive the internal core logic clocks. There are 4 of these. Frequencies are specified in the Clock AC specifications table.
G_RESET_N	I	LC12		Global chip reset.
G_PMTICK	I	LC12 PU		Signal for control of PMTICK functionality.
G_SPARE[7:0]	I/O	LC12 BH	8	Spare pins. Leave unconnected but it is recommended to make some of these accessible for connecting a measurement probe.

Table 10-3 Interface auxiliary pins[1]

Pin Name	IO dir.	IO type	Drive [mA]	Description
xxx_LOS	I	LC12 PU		Loss of signal input. Active high.
xxx_TXDIS	O	LC12	8	Transmit disable output. Active high.
IF_REC_CLK[1:0]	O	LVDS		Output clocks derived from one of the SerDes interface lane recovered clocks (selectable by CSR). Frequencies will depend on SerDes data rate and lane width.
SYNCE_REC_CLK[1:0]	O	LVDS		SyncE recovered clocks derived from either the line DDQ ports, or 10GE clients, depending on the variant.
SYNCE_CORE_CLK	I	LVDS IT		SyncE synchronization clock for internal FPGA core use.
PLE_COMMON_CLK	I	LVDS IT		PLE common clock for internal FPGA core use.
SGMII_REFCLK	I	LVDS IT		SGMII 125MHz reference clock.
SAR_CLK	I	LC12		SAR synchronization 8 kHz frame clock.
EPPS_CLK	I	LC12		IEEE 1588 EPPS clock. Reserved for future use.

1. The xxx prefixes are one of the following:

- Q[1..4], S[1..16], DDQ[1..2]

Table 10-4 Clock synchronization interface for Si5394[1]

Pin Name	IO dir.	IO type	Drive [mA]	Description
VCDx_CLK[0]	O	LC12	8	Serial clocks. Connect to SCLK on Silabs device.
VCD0_CS_N[0]	O	LC12	8	Chip select. Connect to A0_CSb.
VCD0_SI	O	LC12	8	Serial data output. Connect to SDA_SDIO.
VCD0_SO	I	LC12 PU		Serial data input. Connect to A1_SDO.

1. There is 1 DXCO device per interface and one interface.

Table 10-5 Time of Day interface

Pin Name	IO dir.	IO type	Drive [mA]	Description
TOD_CLK	I	LC12		Time of Day Clock input.
TOD_SYNC	I	LC12		Time of Day Sync input.
TOD_DATA	I	LC12		Time of Day ToD data input.

Table 10-6 LED Debugging interface

Pin Name	IO dir.	IO type	Drive [mA]	Description
LED0	O	LC12		Red debug LED. Indicates 1 second heartbeat.
LED1	O	LC12		Green debug LED. Indicates PCIe CPU activity.

Table 10-7 OTN overhead Full add/drop interface.

Pin Name	IO dir.	IO type	Drive [mA]	Description
AD_DROP_CLK	O	LVDS		700 MHz output clock edge-aligned with AD_DROP_DATA.
AD_DROP_DATA[7:0]	O	LVDS		Drop data. 1.4Gbit/s.
AD_ADD_CLK	I	LVDS		Input clock for add data. Must be frequency identical to AD_DROP_CLK, but the phase relationship to AD_DROP_CLK is not important.
AD_ADD_DATA[7:0]	I	LVDS IT		Add data. 1.4Gbit/s.
AD_REFCLK	I	LVDS IT		125MHz Reference clock for OH add/drop interface.

Table 10-8 SerDes interfaces

Pin Name	IO dir.	IO type	Drive [mA]	Description
PCIE_TX[3:0]	O	SerDes	N/A	PCIe SerDes transmit and receive pairs. The SerDes inputs must be AC coupled.
PCIE_RX[3:0]	I	SerDes		
PCIE_REFCLK	I	SerDes		PCIe SerDes reference clock. Must be AC coupled.
xxx_TX[*]	O	SerDes	N/A	SerDes transmit and receive pairs. The SerDes inputs must be AC coupled.
xxx_RX[*]	I	SerDes		

Pin Name	IO dir.	IO type	Drive [mA]	Description
TILE_*_REFCLK	I	See description		SerDes reference clocks. F-Tile reference clocks are CML. Can be AC or DC coupled. Clock driver must be compatible with input requirement in DC coupling case. See "**F-Tile Transceiver Reference Clock Specifications**" for full details.

1. The xxx prefixes are one of the following:

- Q[1..4], S[1..16], DDQ[1..2], ILK1, SGMII

Table 10-9 Technology specific signals

Pin Name	IO dir.	IO type	Description
~ALTERA_AVST_CLK~	I/O	LC12	See Agilex Pin Connection Guidelines for details.
~ALTERA_AVST_DATA0-15~	I	LC12	See Agilex Pin Connection Guidelines for details.
~ALTERA_AVST_VALID~	I	LC12	See Agilex Pin Connection Guidelines for details.
~ALTERA_AVST_READY~	O	LC12	See Agilex Pin Connection Guidelines for details.
~CONF_DONE~	I/O	LC18	See Agilex Pin Connection Guidelines for details.
NC			No Connect. This pin has no internal connection to the device.
DNU			Do Not Use. This pin MUST NOT be connected.
VCCINT			Dedicated power pin, which MUST be connected to VCC (VID2).
VCCIO			Dedicated power pin, which MUST be connected to specific voltage of its bank.
GND			GND.
GND+			Unused input pin. It can also be used to report unused dual-purpose pins. This pin should be connected to GND.
RESERVED			Unused I/O pin, which MUST be left unconnected.
RESERVED_INPUT			Pin is tri-stated and should be connected to the board.
RESERVED_INPUT_WITH_WEAK_PULLUP			Pin is tri-stated with internal weak pull-up resistor.
RESERVED_INPUT_WITH_BUS_HOLD			Pin is tri-stated with bus-hold circuitry.
RESERVED_OUTPUT_DRIVEN_HIGH			Pin is output driven high.
GXB_GND*			Unused GXB/GXF Receiver pin or unused GXB/GXF dedicated reference clock pin. This pin must be connected directly to GND.
GXB_1KGND*			Unused GXP/GXR Receiver pin or unused GXE/GXP/GXR dedicated reference clock pin. This pin must be connected to GND through a 1K ohm resistor.

Pin Name	IO dir.	IO type	Description
GXB_NC			Unused GXB/GXE/GXP/GXR/GXF Transmitter pin or unused GXE Receiver pin. This pin must not be connected.
~INIT_DONE~	O	LC18	See Agilex Pin Connection Guidelines for details.
~ALTERA_MSEL*~			See Agilex Pin Connection Guidelines for details.
nCONFIG			See Agilex Pin Connection Guidelines for details.
nSTATUS			See Agilex Pin Connection Guidelines for details.
~ALTERA_OSC_CLK_1~	I	LC18	Fixed 125 MHz clock used for device initialization. See Agilex Pin Connection Guidelines for details.
PCIE_nPERST	I	LC18	PCIe controller reset. Drive from board level PERST# signal.
RREF_SDM			Pin is reference resister input for the PLLs of the SDM interface. See Agilex Pin Connection Guidelines for details.
RREF			Transceiver reference resistors. Connect 2k (1%) resistor to GND. See Agilex Pin Connection Guidelines for details.
SDM_IO{13,14,15}			Unused SDM pins. Connect pair-wise to NM resistors for access if needed.
~SEU_ERROR~			See Agilex Pin Connection Guidelines for details.
altera_reserved_{tck, tdi, tdo, tms}			JTAG signals. Operating at 1.8 V (powered by VCCIO_SDM). It is recommended to connect these to a suitable pin header as they can be useful for system diagnostic purposes. The JTAG-readable USERCODE will read the same 32-bit value as the `CHIP_VER` register. See Agilex Pin Connection Guidelines for more details.
TEMPDIODE0-6n TEMPDIODE0-6p			Temperature measurement diodes for the main fabric and six transceiver tiles. Must be used for temperature monitoring and over-temperature shutdown protection. See Agilex Pin Connection Guidelines for details.
VCC, VCCP			0.85 V. Core and periphery circuitry power supply.
VCCADC			1.8 V. ADC voltage sensor analog power supply. See Agilex Pin Connection Guidelines for details.
VCCA_PLL			1.2 V. PLL analog power supply. See Agilex Pin Connection Guidelines for details.
VCCBAT			1.8 V. Battery back-up power supply. Connect to normal 1.8 V supply.
VCCPLLDIG_SDM			0.8 V. Embedded memory and SDM PLL digital supply.
VCCFUSEWR_SDM			1.8 V. Fuse block writing power supply. Leave unconnected if unused. Do not tie to GND. See Agilex Pin Connection Guidelines for details.

Pin Name	IO dir.	IO type	Description
VCCIO_SDM			1.8 V. IO buffer supply.
VCCLSENSE, GNDSENSE			Differential remote sense lines for VCC. See Agilex Pin Connection Guidelines for details.
VCCPLL_SDM			1.8 V. PLL analog power supply. See Agilex Pin Connection Guidelines for details.
VCCPT			1.8 V. Programmable technology and IO pre-driver power supply.
VREFN_ADC, VREFP_ADC			ADC external reference. See Agilex Pin Connection Guidelines for details. If no external reference is supplied, always connect the pin to GND.
VSIGN_0/1, VSIGP_0/1			ADC external inputs. See Agilex Pin Connection Guidelines for details.
DTS_CALIB			Pins are used for Calibration of the temperature sensor during manufacturing. They can be left open on board when not in use.
DTS_CALIB			Pins are used for Calibration of the temperature sensor during manufacturing. They can be left open on board when not in use.

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10.3 Clocking and reset

The G_CLK[0-2] inputs generate internal core clocks that drive functional logic, and are unrelated to external timing references. These clocks must be driven from a common clock generator, so they can be derived from a common source with fixed ratios.

G_CLK3 provides a 311.04 MHz timing reference, often synchronized across multiple devices or line cards. G_CLK3 provides a reference for internally sourced OTN and client signals. For this reason, G_CLK3 must obey the +/-20 ppm limits. External OTN interface rates are controlled by the appropriate interface reference clock(s).

In addition to these main clocks, there are a number of other interface-related clocks as well as SerDes reference clocks. These clocks are described in the relevant interface sections later in this chapter.

Each of the clocks listed in the tables below is categorized into one of the following types:

Table 10-10 Input clock presence requirement categories

Clock category	Clock presence requirements
A	The clock is mandatory. It must be present and valid prior to FPGA download commencing.
B	The clock is optional. It can be either used or not used in a specific design. If a design uses the particular feature or interface to which the clock is associated, the clock must be present and valid prior to FPGA download commencing.
C	The clock can for specific reasons not be valid prior to FPGA download commencing. There are special initialization requirements which must be performed once the clock is known to be stable. Clocks in this category can be either optional or mandatory.

Table 10-11 Input clock quality categories

Quality category	Clock quality definition
1	Stable timing reference. Used as reference for OTN signals where the device is the originator, and thus the long term stability of these inputs will define the stability of the output. Jitter and phase noise not critical as input is filtered.
2	High-speed optical interface SerDes reference clocks. These require good jitter and phase noise characteristics, as they affect the jitter and phase noise of the optical transmitted signals. If driven directly from the Si5345 they will meet the required specifications.
3	Interface reference clocks, non-critical quality. These clocks are used as references for lower-speed interfaces (<2Gbit/s) where minimal jitter is not a requirement.
4	Internal logic clocks. These clock internal logic and must be continuous and glitch-free, but good jitter and frequency stability is not a requirement.

Reset is controlled through the global G_RESET_N pin, which can be asynchronously asserted and deasserted.

10.3.1 Clock AC specifications

The tables below lists the AC specifications for all clocks on the device. The SerDes interface reference clocks are separated into their own table.

Table 10-12 Input clock specifications, all clocks except SerDes optical interface reference clocks

Clock name	Min. clock frequency [MHz]	Typ. clock frequency [MHz]	Max. clock frequency [MHz]	Duty cycle	On-chip PLL	Presence category	Quality category
PCIE_REFCLK	-100ppm	100	+100ppm	40-60	Yes	A	3
OSC_CLK_1	-100ppm	125	+100ppm	40-60	Yes	A	4
G_CLK0	-100 ppm	357	+100 ppm	40-60	Yes	A	4
G_CLK1	-100 ppm	413	+100 ppm	40-60	Yes	A	4
G_CLK2	-100 ppm	439	+100 ppm	40-60	Yes	A	4
G_CLK3	-20 ppm	311.04	+20 ppm	40-60	Yes	A	1
AD_REFCLK	-100ppm	125	+100ppm	40-60	Yes	B	3
SAR_REFCLK	-100ppm	0.008	+100ppm	40-60	Yes	B	3
SYNCE_CORE_CLK	-100ppm	125	+100ppm	40-60	Yes	B	1
PLE_COMMON_CLK	-100ppm	125	+100ppm	40-60	Yes	B	1
xxx_REFCLK SerDes ref. clocks	-	Protocol dependent	-	40-60	Yes	A	2

10.3.2 AC timing

Table 10-13 Global interface AC timing specifications

Pin Name	IO dir.	Ref. clock	Output			Input	
			Load [pF]	T_{co} min [ns]	T_{co} max [ns]	T_{su} min [ns]	T_h min [ns]
G_RESET_N	I	ASYNC					

Table 10-14 G_RESET_N AC timing specification

Name	Min	Max	Unit	Description
T_{reset}	1		ms	Minimum reset assertion time.
$T_{cpuaccess}$	256		ms	Minimum delay after reset deassertion before first CPU access.

Reset is controlled through the global G_RESET_N pin, which can be asynchronously asserted and deasserted. Reset must be asserted for a minimum time of T_{reset} to guarantee that the reset is performed correctly.

Following the reset deassertion, the device is ready to accept the first CPU access on the CPU interface after the time $T_{cpuaccess}$ has elapsed.

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10.4 TODO - SerDes interfaces

The SerDes interfaces are grouped with their reference clocks as illustrated in the following tables:

10.4.1 UTM2C card

Table 10-15 SerDes interfaces and tile mappings (UTM2C)

Interface	Faceplate#	Tile - Channels	Notes
DDQ0	TBD	12C (R1) - Q3(3,2,1,0);Q2(3,2,1,0)	Lanes in reverse order
DDQ1	TBD	12A (R0) - Q3(3,2,1,0);Q2(3,2,1,0)	Lanes in reverse order
QSFP0	TBD	13C (L1) - Q0(3,2,1,0)	Lanes in reverse order
QSFP1	TBD	13C (L1) - Q1(3,2,1,0)	Lanes in reverse order
QSFP2	TBD	13A (L0) - Q0(3,2,1,0)	Lanes in reverse order
QSFP3	TBD	13A (L0) - Q1(3,2,1,0)	Lanes in reverse order
SFP0	TBD	13C (L1) - Q2(0)	
SFP1	TBD	13C (L1) - Q2(1)	
SFP2	TBD	13C (L1) - Q2(2)	
SFP3	TBD	13C (L1) - Q2(3)	
SFP4	TBD	13C (L1) - Q3(0)	
SFP5	TBD	13C (L1) - Q3(1)	
SFP6	TBD	13C (L1) - Q3(2)	
SFP7	TBD	13C (L1) - Q3(3)	
SFP8	TBD	13A (L0) - Q2(0)	
SFP9	TBD	13A (L0) - Q2(1)	
SFP10	TBD	13A (L0) - Q2(2)	
SFP11	TBD	13A (L0) - Q2(3)	
SFP12	TBD	13A (L0) - Q3(0)	
SFP13	TBD	13A (L0) - Q3(1)	
SFP14	TBD	13A (L0) - Q3(2)	
SFP15	TBD	13A (L0) - Q3(3)	

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Table 10-16 Clock generator to reference clock mappings (UTM2C)

VCD bus#	VCD dev#	Out#	Tile	refclk#	Pin name	Notes
0	0	0	12C (R1)	1	TILE_12C_FGT1_REFCLK	
		1	12C (R1)	3	TILE_12C_FGT3_REFCLK	
		2	12C (R1)	4	TILE_12C_FGT4_REFCLK	
		3	12C (R1)	5	TILE_12C_FGT5_REFCLK	
		4	12C (R1)	6	TILE_12C_FGT6_REFCLK	
		5	12A (R0)	1	TILE_12A_FGT1_REFCLK	
		6	12A (R0)	3	TILE_12A_FGT3_REFCLK	
		7	12A (R0)	4	TILE_12A_FGT4_REFCLK	
		8	12A (R0)	5	TILE_12A_FGT5_REFCLK	
		9	12A (R0)	6	TILE_12A_FGT6_REFCLK	
	1	0	13C (L1)	1	TILE_13C_FGT1_REFCLK	
		1	13C (L1)	3	TILE_13C_FGT3_REFCLK	
		2	13C (L1)	4	TILE_13C_FGT4_REFCLK	
		3	13C (L1)	5	TILE_13C_FGT5_REFCLK	
		4	13C (L1)	6	TILE_13C_FGT6_REFCLK	
		5	13A (L0)	1	TILE_13A_FGT1_REFCLK	
		6	13A (L0)	3	TILE_13A_FGT3_REFCLK	
		7	13A (L0)	4	TILE_13A_FGT4_REFCLK	
		8	13A (L0)	5	TILE_13A_FGT5_REFCLK	
		9	13A (L0)	6	TILE_13A_FGT6_REFCLK	

10.4.2 Auxiliary interface signals

Each interface has a related Loss Of Signal input, xxx_LOS[], intended to be driven directly from the corresponding optics where available. If not available, this pin should be driven inactive (low).

Similarly, a Tx (laser) disable output pin is also provided for each interface, xxx_TXDIS[], which is intended to directly drive the corresponding pin on the optical module. Both of these signals will normally need to be level converted on the PCB.

10.5 DCXO interface

The device provides a mandatory mechanism to control the transmit reference clocks for the SerDes interfaces, using one or more Silicon Labs Si5345 devices. The synthesizer devices are controlled by a serial digital interface consisting of dedicated clock and chip select lines, and shared unidirectional write and read lines. The interface is a fully synchronous interface.

If more than one DCXO interface is present on a given device, the pins are numbered VCDx_ and each interface can control up to five Si5345 devices.

10.5.1 Connecting the Si5345 devices

The figure below shows the interface to a single Si5345 device.

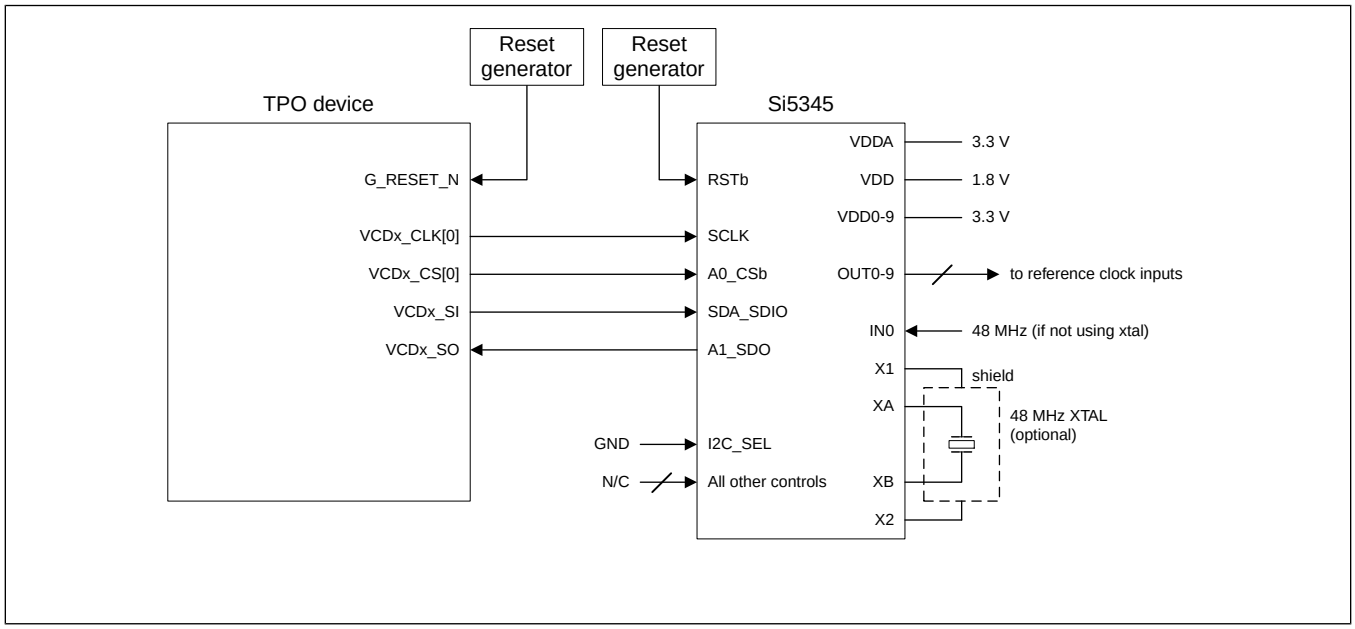


Figure 10-1 DCXO interface to Si5345

The figure does not show the required serial termination resistors on the interface signals, nor does it show the discrete components required around the XTAL or the power supply filtering components.

Si5345 outputs that go to E-Tile reference clocks must have LVPECL output standards configured and be DC coupled. The FPGA input pins have on-die LVPECL terminations.

Si5345 outputs that go to H-Tile reference clocks must have LVDS output standards configured and be AC coupled. The FPGA input pins have 100Ω on-die parallel terminations.

The clock synchronization interface uses a serial interface consisting of 4 pins to write and read clock control information to/from the Si5345 registers. This is a fully synchronous interface. If there are multiple Si5345 devices connected to an interface, then each Si5345 must be driven by an individual VCD_CLK[] signal as well as an individual VCD_CS[] signal, while the VCD_SI and VCD_SO lines are shared by all devices on that particular interface.

The Si5345 48 MHz reference clock comes from either the IN0 input, or from a directly attached crystal. Even though the frequency of this crystal is not directly used to generate any client or line signals (e.g. replacement signals), the frequency accuracy still needs to be ±30 ppm, including all tolerance temperature and aging effects, to allow the frequency synthesis and DSP filters to operate within their defined range. The recommended crystal is TXC 7M48072002 (48 MHz, 8 pF load capacitance). This crystal has been specifically selected and tested to provide optimal performance together with the

Si5345 device. As the Si5345 device includes configurable loading capacitors, no discrete loading caps are required on the PCB when using this crystal. See the Silicon Labs documentation for details.

The Si5345 RSTb pins must be controlled separately from the TPO G_RESET_N pin. Si5345 RSTb must be asserted briefly and released before commencing FPGA initial configuration download, to ensure the default transceiver reference clocks are running.

All other Si5345 pins can be left floating as the functions are controlled through device registers.

10.5.2 AC timing

VCD_CLK runs intermittently at a frequency of up to 20 MHz. The frequency is software configurable, but this is the recommended value to be used with the Si5345 devices. The output pins VCD_CS and VCD_SI change state approximately simultaneously with a falling edge on VCD_CLK. This ensures ample setup/hold margin at the Si5345 inputs, which are sampled on the rising edge. The SDO output from Si5345 changes after the falling edge of VCD_CLK, and is sampled by the TPO device at the next falling edge of the clock.

Table 10-17 Clock synchronization interface AC timing specifications

Pin Name	IO dir.	Ref. clock	Output			Input	
			Load [pF]	T_{co} min [ns]	T_{co} max [ns]	T_{su} min [ns]	T_h min [ns]
VCD_CS	O	VCD_CLK	5	45	55		
VCD_SI	O	VCD_CLK	5	45	55		
VCD_SO	I	VCD_CLK				15	0

10.5.3 PCB layout and termination guidelines

Serial terminations should be used on the point-to-point VCD control interface pins in both directions: 33 Ω placed as close as possible to the driver end. The length of the bus VCD_SI and VCD_SO lines should be kept short to avoid excessive reflections.

10.5.4 Terminating an unused interface

If one or more of the provided DCXO control interfaces are not used, the pins associated with those interfaces can be left unconnected.

10.6 Overhead add/drop interface

The OTN overhead add/drop interface allows OH data from ODUs processed at various points inside the TPO device to be extracted for processing in an external device, and for OH data to be similarly inserted into the ODUs in the transmit direction. The figure below shows how to connect the external device to the overhead add/drop interface.

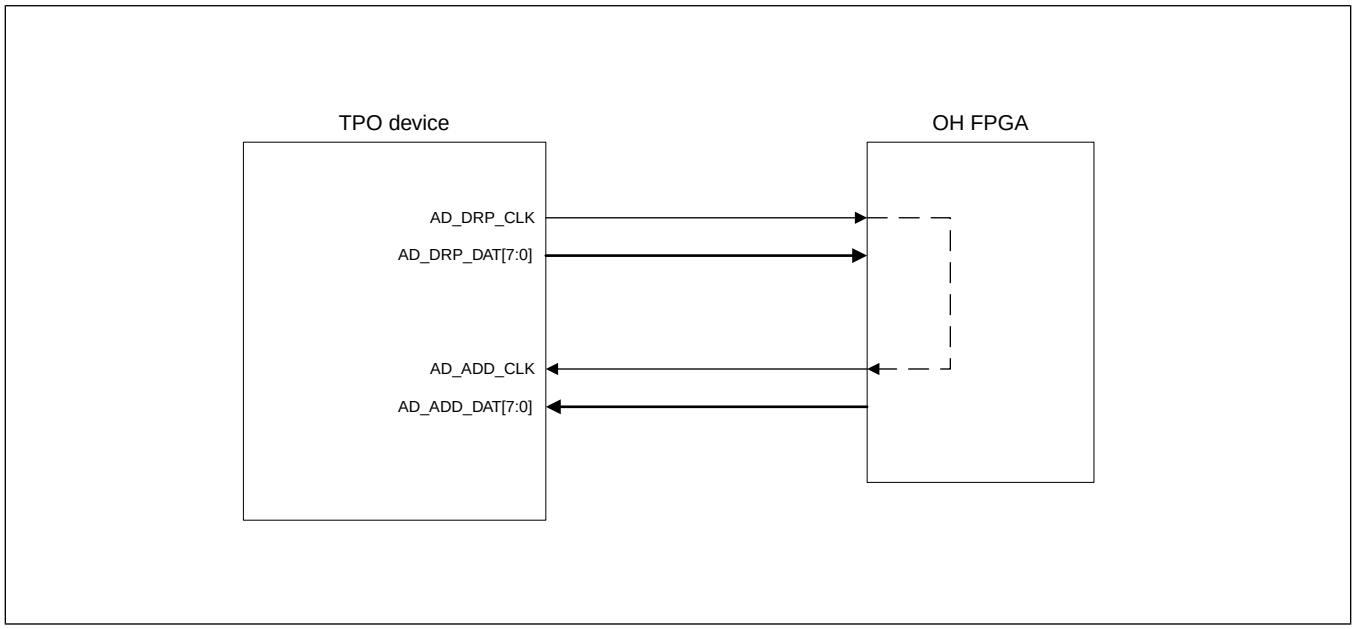


Figure 10-2 Overhead add/drop connection to external device

The drawing does not show any termination resistors, which might be required at the external device. In both directions the interface consists of a byte-wide databus with an accompanying source-synchronous clock.

The drop direction of the interface drops all the overhead bytes received using a fixed 64-byte frame structure, described below. The interface operates at overspeed, so individual frames delivered on the interface are either full or empty.

The add interface is similar, but fully controlled from the TPO device. The source synchronous clock in this direction shall be a looped copy of AD_DROP_CLK (phase relationship is not important). The framing structure is identical to that of the drop direction, but whether a specific add frame contains overhead data for a ODUk or not, is fully controlled through request bytes in the frames sent in the drop direction. In this way, it is ultimately the framers inside the TPO device which, individually per ODUk, control the pace of the overhead add bytes being delivered across the add interface.

Because of the clocking structure, the AD_ADD_CLK input clock cannot be present and valid prior to the TPO FPGA download beginning. Therefore, once AD_ADD_CLK is stable, the ADD interface PLL must be reset by toggling the `PLL_RST.ad` bit.

10.6.1 Framing structure

Both the drop and add interfaces uses a fixed 64-byte framing structure, consisting of 6 header bytes and 58 overhead bytes. The framing structure and header formats for both directions is shown in the figure below.

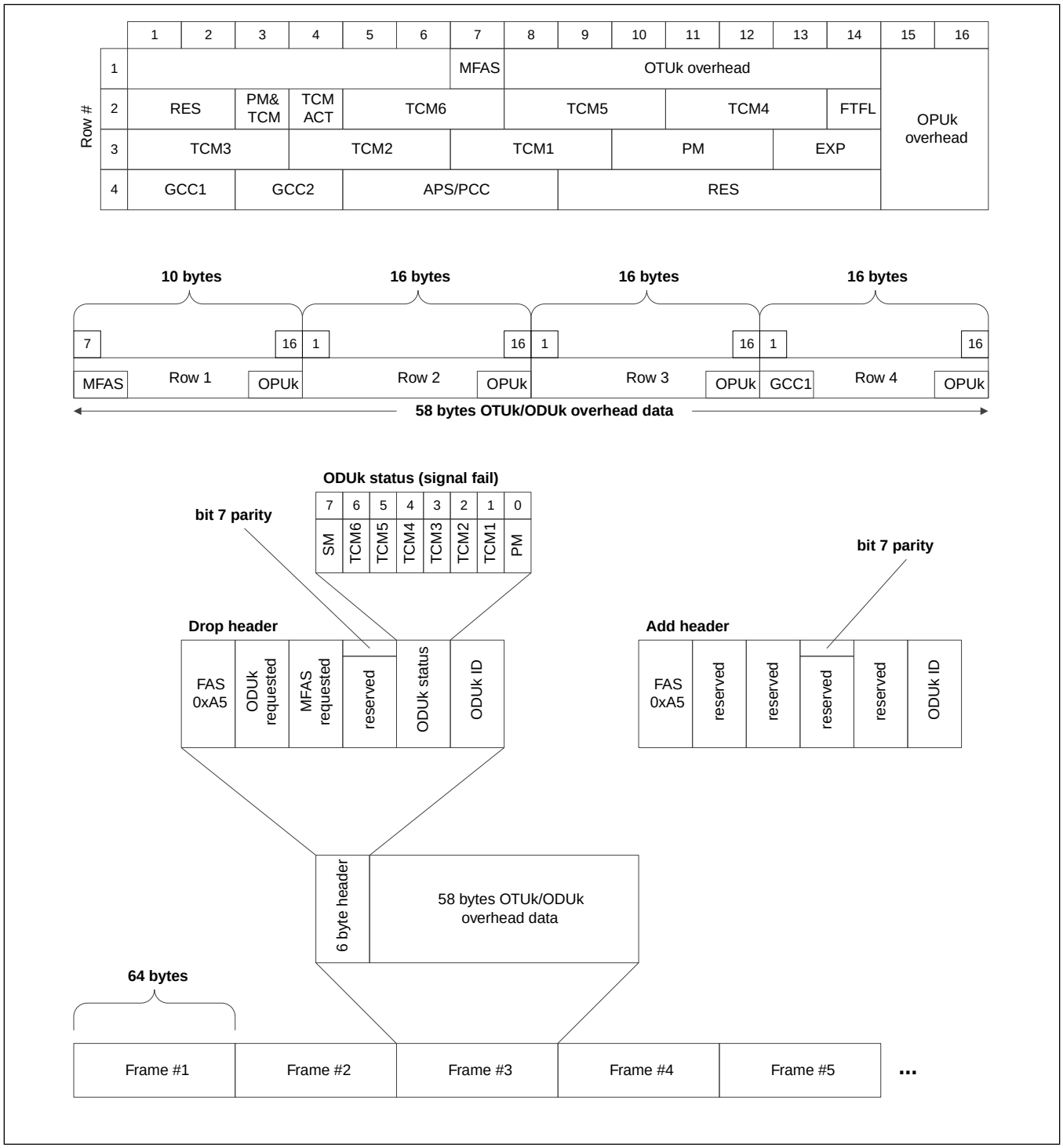


Figure 10-3 Framing structure and header formats

The first byte in the frame (both directions) is a fixed FAS byte, which must be used by the receiver to obtain framing. The remaining 63 bytes of the frame (in both directions) is scrambled using the GFP self-synchronizing scrambler with the polynomial $1 + x^{43}$.

In both the drop and add directions, the ODUk ID field indicates which ODU the overhead data belongs to. The ODUk ID field corresponds to the OH block index in the OTN Overhead chapter. A value of 0xFF is used to indicate an empty frame without overhead bytes (the bytes are present in the frame, but their contents are meaningless).

A single parity bit in both the Add and Drop directions is calculated as an XOR of all 512 bits in the previous frame on the interface.

The ODUk status field contains a signal fail indication ('1' means signal fail) per monitoring level.

The two remaining fields in the 6-byte drop header (ODUk requested, MFAS requested) are used to issue requests for overhead data to be delivered over the add bus.The OH FPGA shall only deliver add data when requested to do so, it never sends add data uninvited.

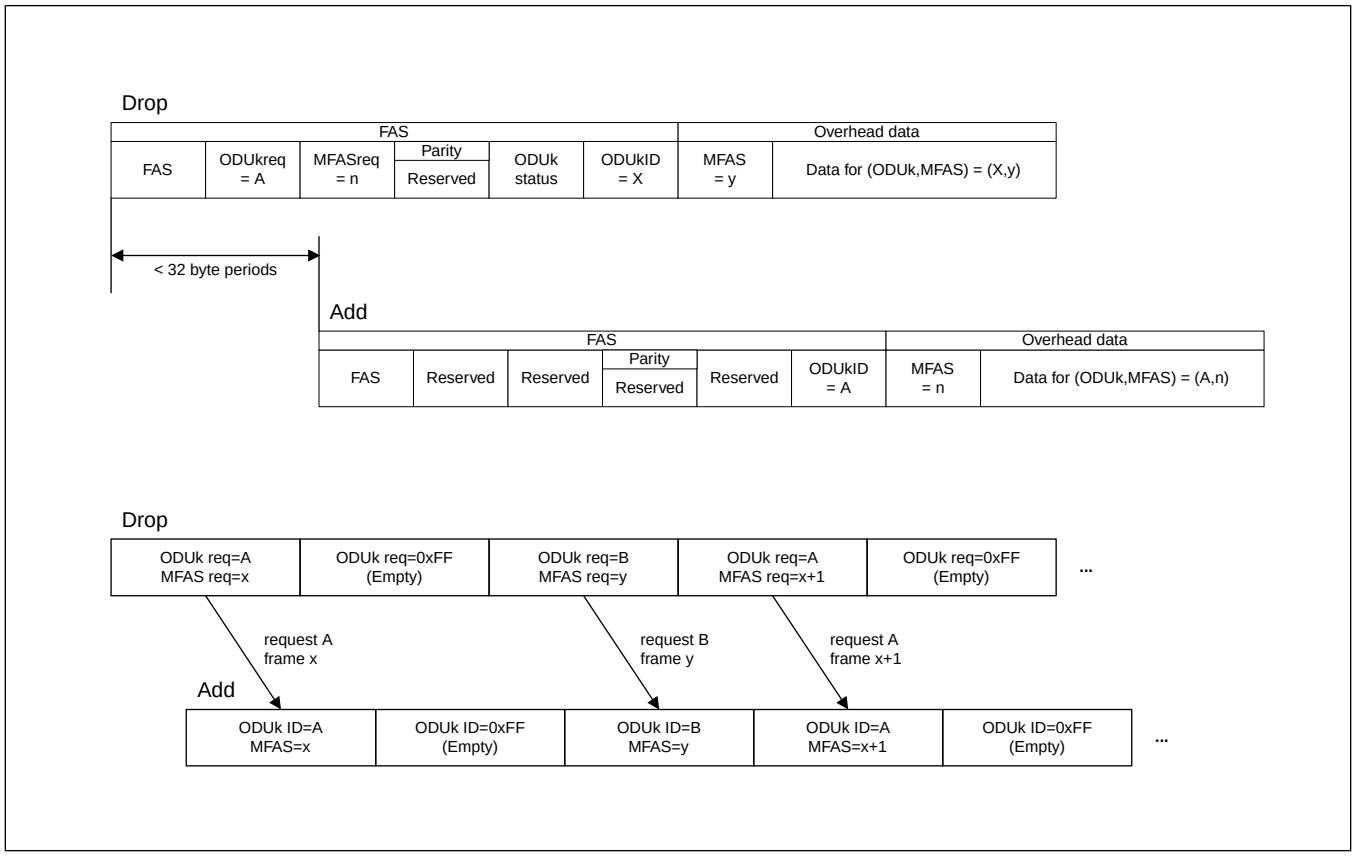


Figure 10-4 Overhead drop (request) to add (response) relationship

The figure above shows the relationship between the requests issued in the drop header and the mandatory response on the add bus. Note that the two request bytes are valid in all drop frame headers, even if the ODUk ID field is set to 0xFF (no drop overhead data present in this frame). Also as shown above, the (fixed) delay from the drop request frame FAS word to the add response frame FAS word must be less than 32 byte periods (16 clock periods). As the add bus clock is a copy (with some fixed phase delay) of the drop bus clock, the drop and add framing structures are always in lockstep with a fixed delay between them.

10.6.2 AC timing (DDR interface)

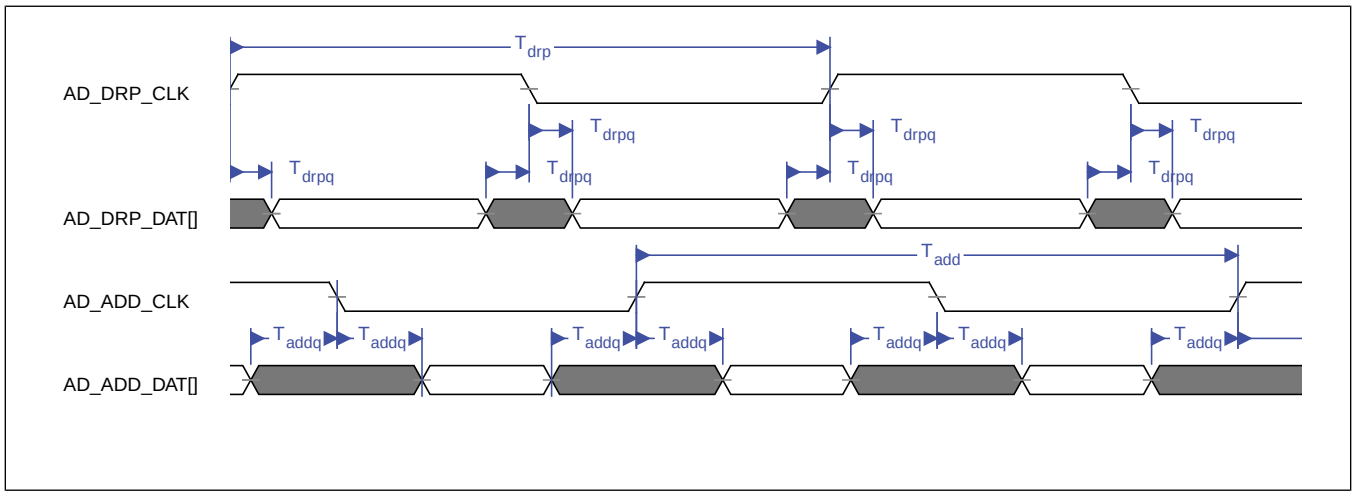


Figure 10-5 Overhead add/drop waveforms

Table 10-18 Overhead add/drop interface AC timing specification[1]

Name	Min	Max	Unit	Description
T_{drp}	2.49	2.51	ns	Drop clock cycle time.
$Duty_{drp}$	45	55	%	Drop clock duty cycle.
T_{drpq}		0.25	ns	Drop data invalid time before and after AD_DROP_CLK edge. The same number applies for both clock edges.
C_{farend}		4	pF	Far end transmission line load for AD_DROP_CLK and AD_DROP_DATA[].
T_{add}	2.49	2.51	ns	Add clock cycle time. Must be frequency identical to AD_DROP_CLK.
$Duty_{add}$	45	55	%	Add clock duty cycle.
T_{addq}		0.39	ns	Permitted add data invalid time before and after AD_ADD_CLK edge. The same number applies for both clock edges.

1. Values above are for an interface running at 800 Mbps

10.6.3 PCB layout and termination guidelines

Each of the two databusses should be routed with equal length as the source-synchronous clock belonging to the bus. The maximum recommended skew between the clock and any of the 8 data lines is 250 ps. The HSTL add inputs are terminated inside the TPO device. The HSTL drop outputs should be terminated at the destination device, either using discrete resistors (parallel termination), or ODT (On-Die-Termination) if available. HSTL termination details are described in more detail later in this chapter. It is strongly recommended that the actual layout and termination scheme chosen is simulated prior to implementation.

10.6.4 Terminating an unused interface

If the add/drop interface is unused, then outputs should be left unconnected, while the unused input pins should be driven to a legal level and not left floating.

11 Revision History

Date	Revision	Notes
2025-01-24	0.01	Initial draft version
2025-01-27	0.02	Updated clocks in external pin interfaces
2025-xx-yy	0.03	Work in progress...